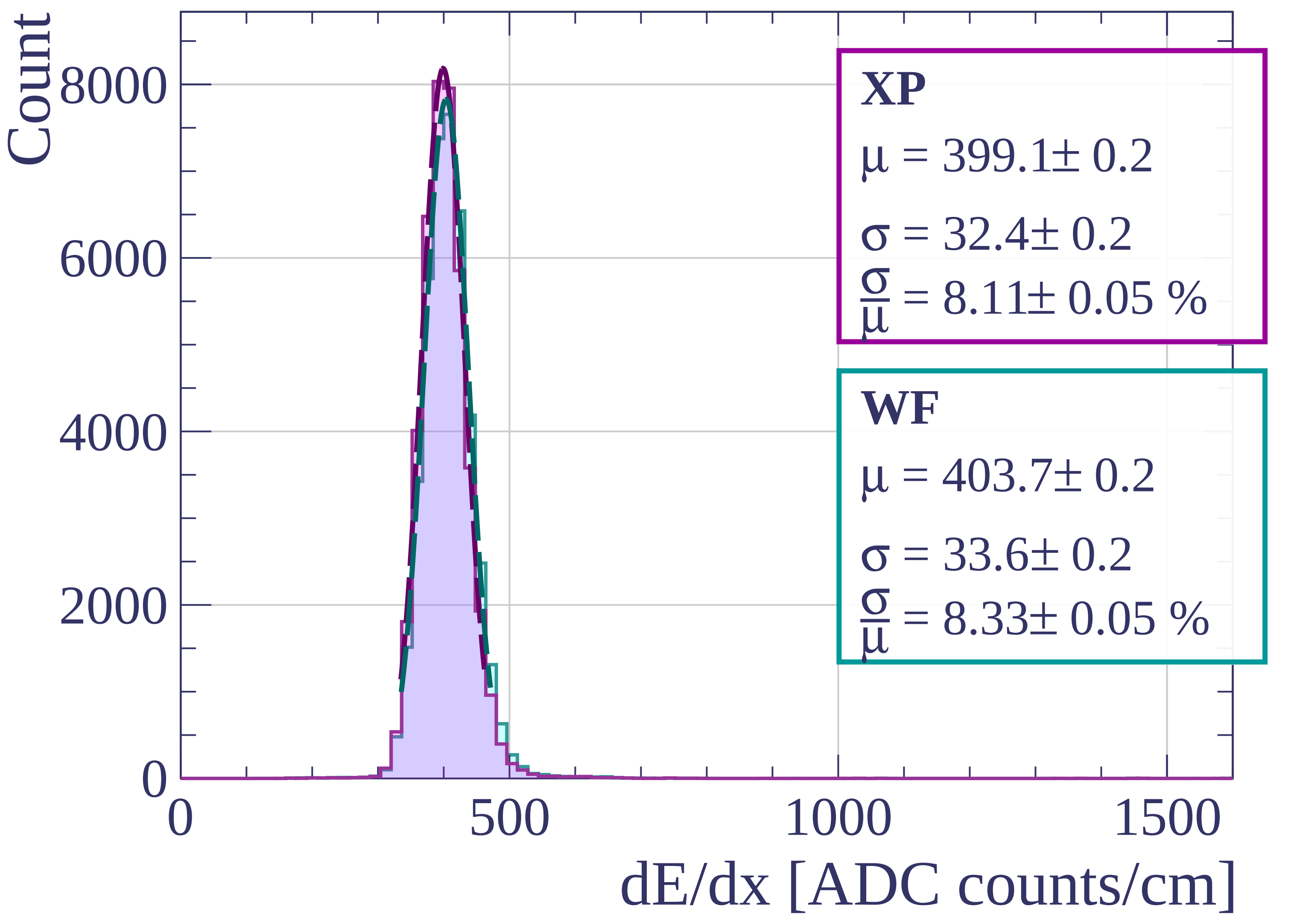
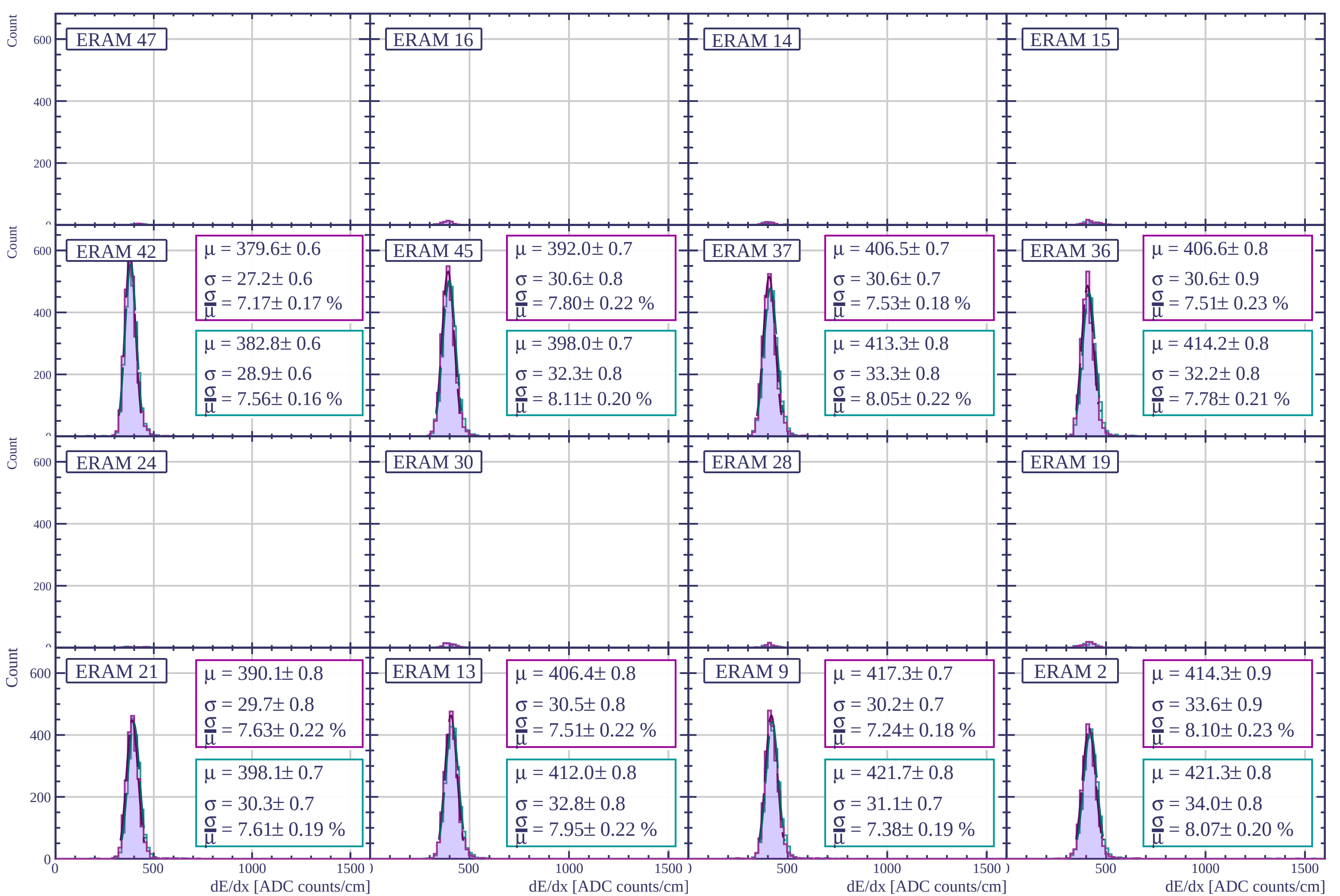
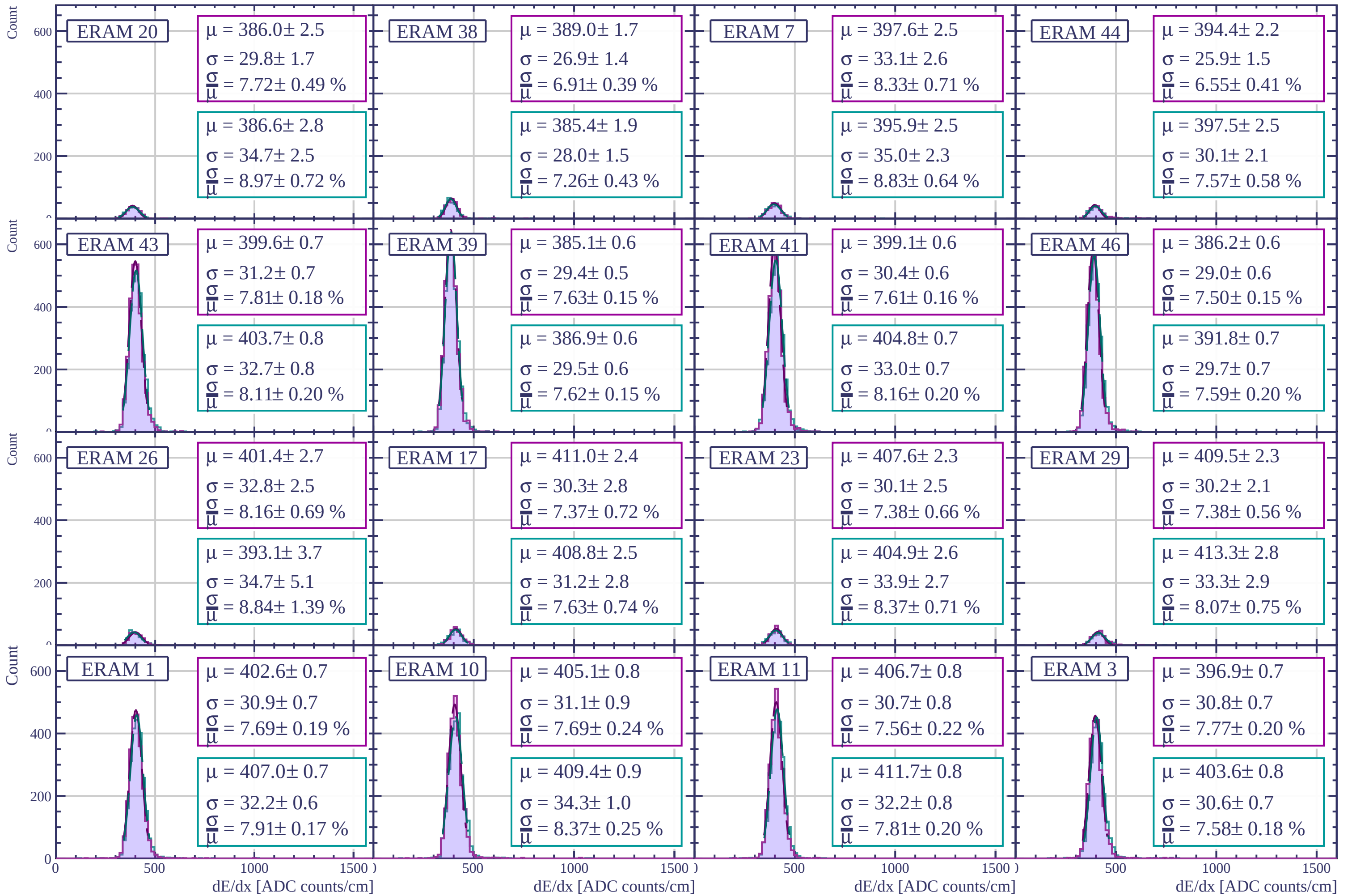


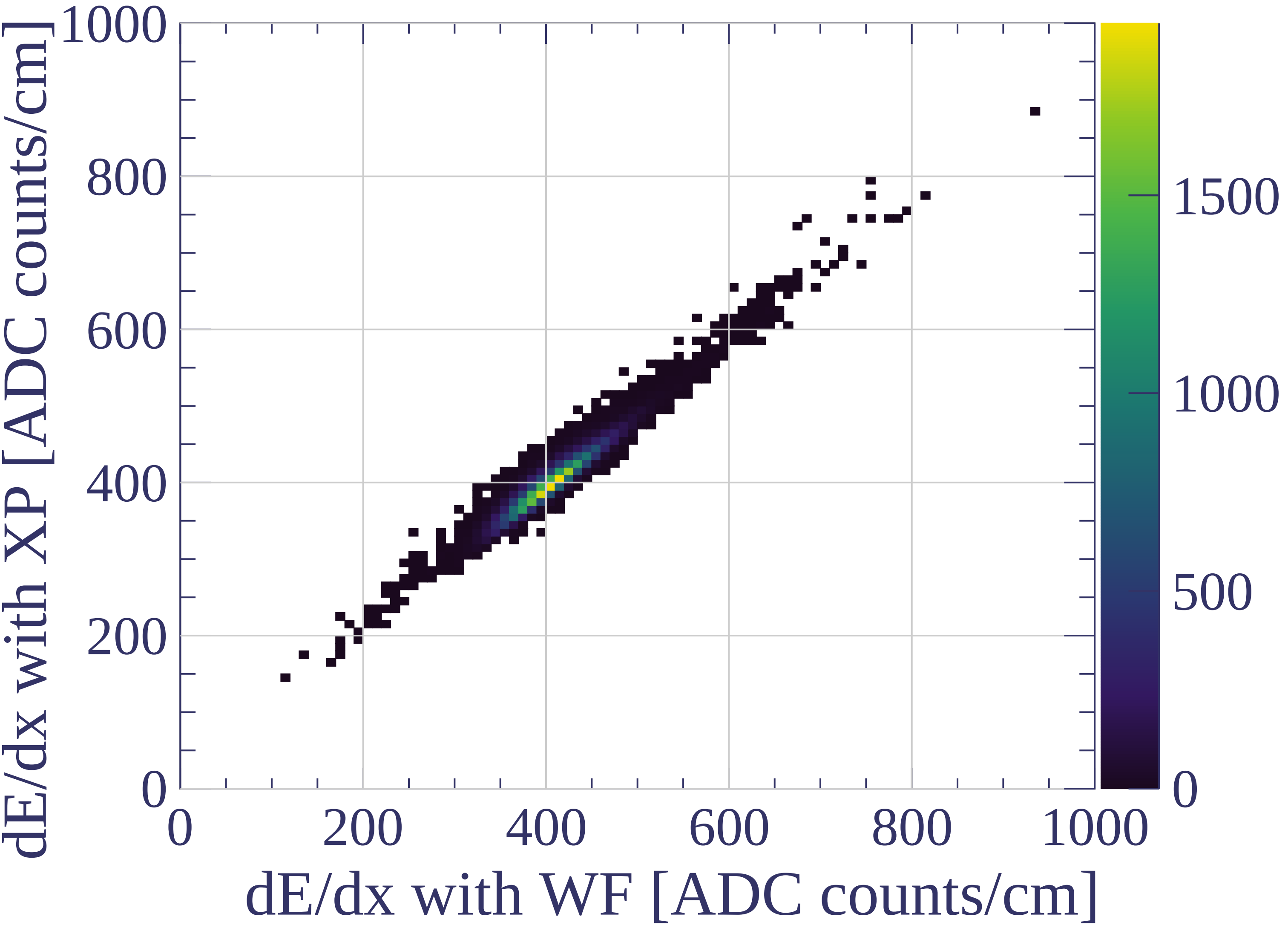
Count
8000
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 399.1± 0.2
σ = 32.4± 0.2
σ/μ = 8.11± 0.05 %
WF
μ = 403.7± 0.2
σ = 33.6± 0.2
σ/μ = 8.33± 0.05 %



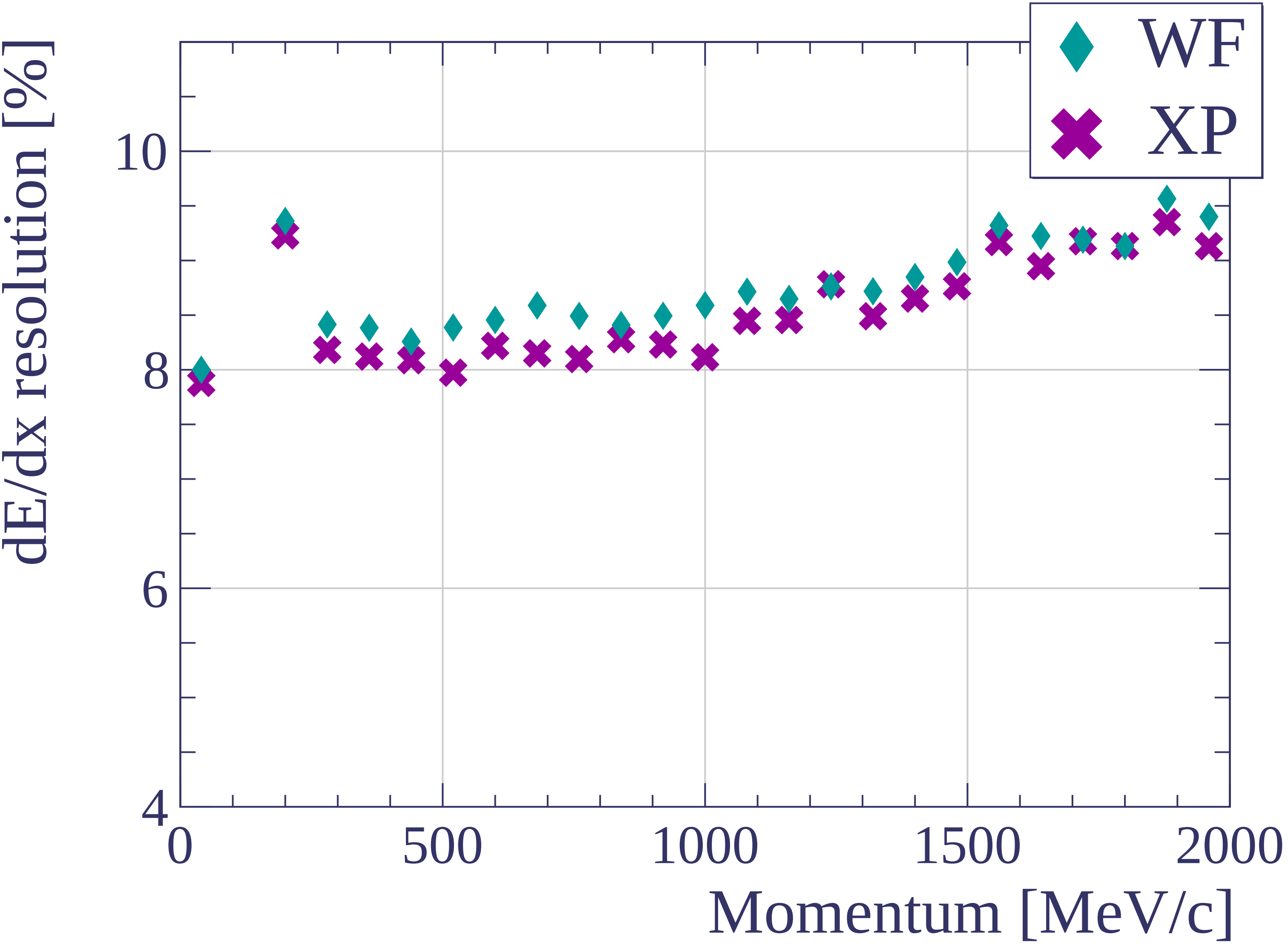
ERAM 47
ERAM 16
ERAM 14
ERAM 15
ERAM 42
μ = 379.6± 0.6
σ = 27.2± 0.6
σ/μ = 7.17± 0.17 %
μ = 382.8± 0.6
σ = 28.9± 0.6
σ/μ = 7.56± 0.16 %
ERAM 45
μ = 392.0± 0.7
σ = 30.6± 0.8
σ/μ = 7.80± 0.22 %
μ = 398.0± 0.7
σ = 32.3± 0.8
σ/μ = 8.11± 0.20 %
ERAM 37
μ = 406.5± 0.7
σ = 30.6± 0.7
σ/μ = 7.53± 0.18 %
μ = 413.3± 0.8
σ = 33.3± 0.8
σ/μ = 8.05± 0.22 %
ERAM 36
μ = 406.6± 0.8
σ = 30.6± 0.9
σ/μ = 7.51± 0.23 %
μ = 414.2± 0.8
σ = 32.2± 0.8
σ/μ = 7.78± 0.21 %
ERAM 24
ERAM 30
ERAM 28
ERAM 19
ERAM 21
μ = 390.1± 0.8
σ = 29.7± 0.8
σ/μ = 7.63± 0.22 %
μ = 398.1± 0.7
σ = 30.3± 0.7
σ/μ = 7.61± 0.19 %
ERAM 13
μ = 406.4± 0.8
σ = 30.5± 0.8
σ/μ = 7.51± 0.22 %
μ = 412.0± 0.8
σ = 32.8± 0.8
σ/μ = 7.95± 0.22 %
ERAM 9
μ = 417.3± 0.7
σ = 30.2± 0.7
σ/μ = 7.24± 0.18 %
μ = 421.7± 0.8
σ = 31.1± 0.7
σ/μ = 7.38± 0.19 %
ERAM 2
μ = 414.3± 0.9
σ = 33.6± 0.9
σ/μ = 8.10± 0.23 %
μ = 421.3± 0.8
σ = 34.0± 0.8
σ/μ = 8.07± 0.20 %
Count
0
200
400
600
500
1000
1500
dE/dx [ADC counts/cm]



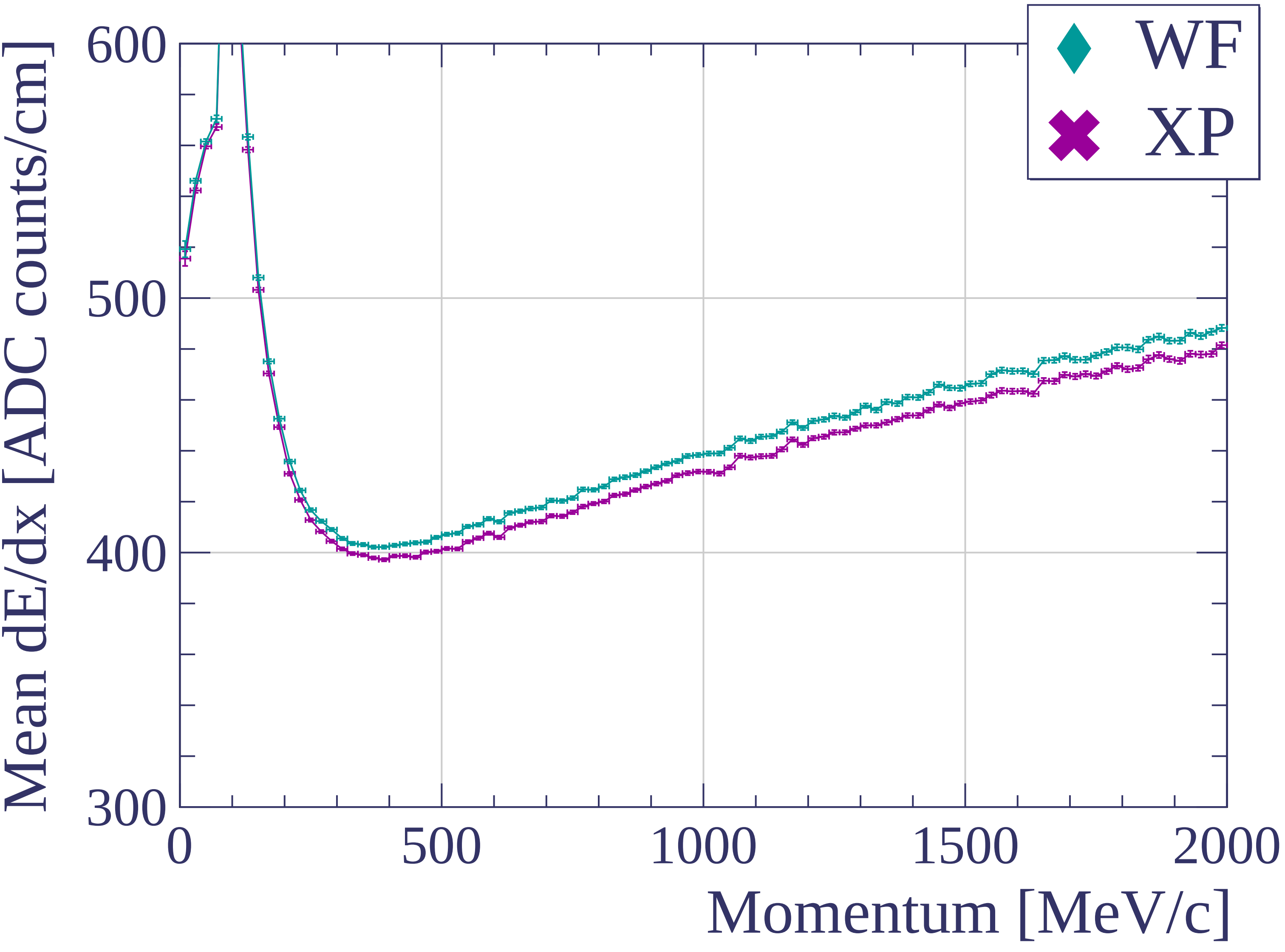
ERAM 20
μ = 386.0± 2.5
σ = 29.8± 1.7
σ/μ = 7.72± 0.49 %
μ = 386.6± 2.8
σ = 34.7± 2.5
σ/μ = 8.97± 0.72 %
ERAM 38
μ = 389.0± 1.7
σ = 26.9± 1.4
σ/μ = 6.91± 0.39 %
μ = 385.4± 1.9
σ = 28.0± 1.5
σ/μ = 7.26± 0.43 %
ERAM 7
μ = 397.6± 2.5
σ = 33.1± 2.6
σ/μ = 8.33± 0.71 %
μ = 395.9± 2.5
σ = 35.0± 2.3
σ/μ = 8.83± 0.64 %
ERAM 44
μ = 394.4± 2.2
σ = 25.9± 1.5
σ/μ = 6.55± 0.41 %
μ = 397.5± 2.5
σ = 30.1± 2.1
σ/μ = 7.57± 0.58 %
ERAM 43
μ = 399.6± 0.7
σ = 31.2± 0.7
σ/μ = 7.81± 0.18 %
μ = 403.7± 0.8
σ = 32.7± 0.8
σ/μ = 8.11± 0.20 %
ERAM 39
μ = 385.1± 0.6
σ = 29.4± 0.5
σ/μ = 7.63± 0.15 %
μ = 386.9± 0.6
σ = 29.5± 0.6
σ/μ = 7.62± 0.15 %
ERAM 41
μ = 399.1± 0.6
σ = 30.4± 0.6
σ/μ = 7.61± 0.16 %
μ = 404.8± 0.7
σ = 33.0± 0.7
σ/μ = 8.16± 0.20 %
ERAM 46
μ = 386.2± 0.6
σ = 29.0± 0.6
σ/μ = 7.50± 0.15 %
μ = 391.8± 0.7
σ = 29.7± 0.7
σ/μ = 7.59± 0.20 %
ERAM 26
μ = 401.4± 2.7
σ = 32.8± 2.5
σ/μ = 8.16± 0.69 %
μ = 393.1± 3.7
σ = 34.7± 5.1
σ/μ = 8.84± 1.39 %
ERAM 17
μ = 411.0± 2.4
σ = 30.3± 2.8
σ/μ = 7.37± 0.72 %
μ = 408.8± 2.5
σ = 31.2± 2.8
σ/μ = 7.63± 0.74 %
ERAM 23
μ = 407.6± 2.3
σ = 30.1± 2.5
σ/μ = 7.38± 0.66 %
μ = 404.9± 2.6
σ = 33.9± 2.7
σ/μ = 8.37± 0.71 %
ERAM 29
μ = 409.5± 2.3
σ = 30.2± 2.1
σ/μ = 7.38± 0.56 %
μ = 413.3± 2.8
σ = 33.3± 2.9
σ/μ = 8.07± 0.75 %
ERAM 1
μ = 402.6± 0.7
σ = 30.9± 0.7
σ/μ = 7.69± 0.19 %
μ = 407.0± 0.7
σ = 32.2± 0.6
σ/μ = 7.91± 0.17 %
ERAM 10
μ = 405.1± 0.8
σ = 31.1± 0.9
σ/μ = 7.69± 0.24 %
μ = 409.4± 0.9
σ = 34.3± 1.0
σ/μ = 8.37± 0.25 %
ERAM 11
μ = 406.7± 0.8
σ = 30.7± 0.8
σ/μ = 7.56± 0.22 %
μ = 411.7± 0.8
σ = 32.2± 0.8
σ/μ = 7.81± 0.20 %
ERAM 3
μ = 396.9± 0.7
σ = 30.8± 0.7
σ/μ = 7.77± 0.20 %
μ = 403.6± 0.8
σ = 30.6± 0.7
σ/μ = 7.58± 0.18 %
Count
dE/dx [ADC counts/cm]



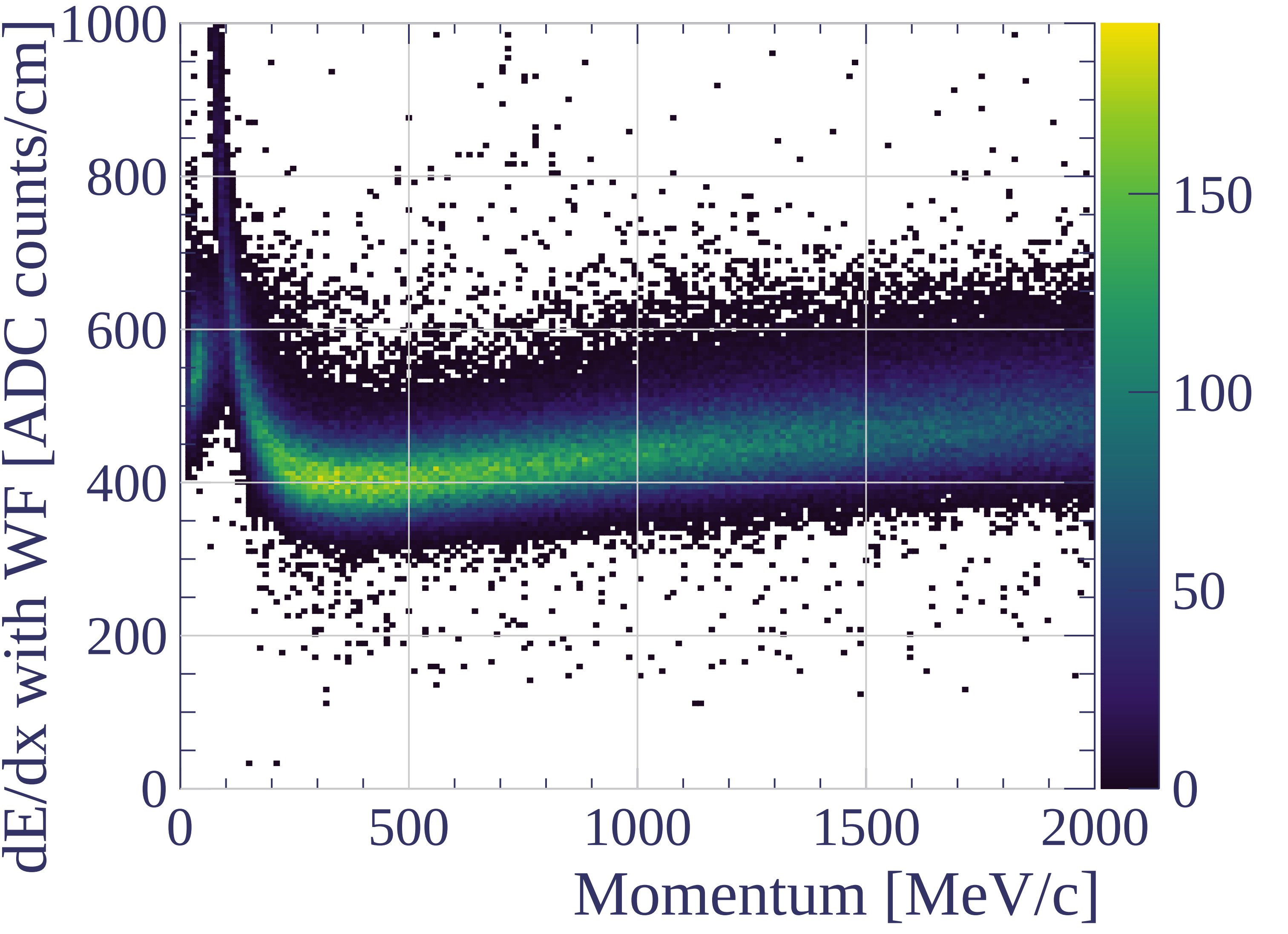
dE/dx with XP [ADC counts/cm]
dE/dx with WF [ADC counts/cm]
0
200
400
600
800
1000
0
500
1000
1500



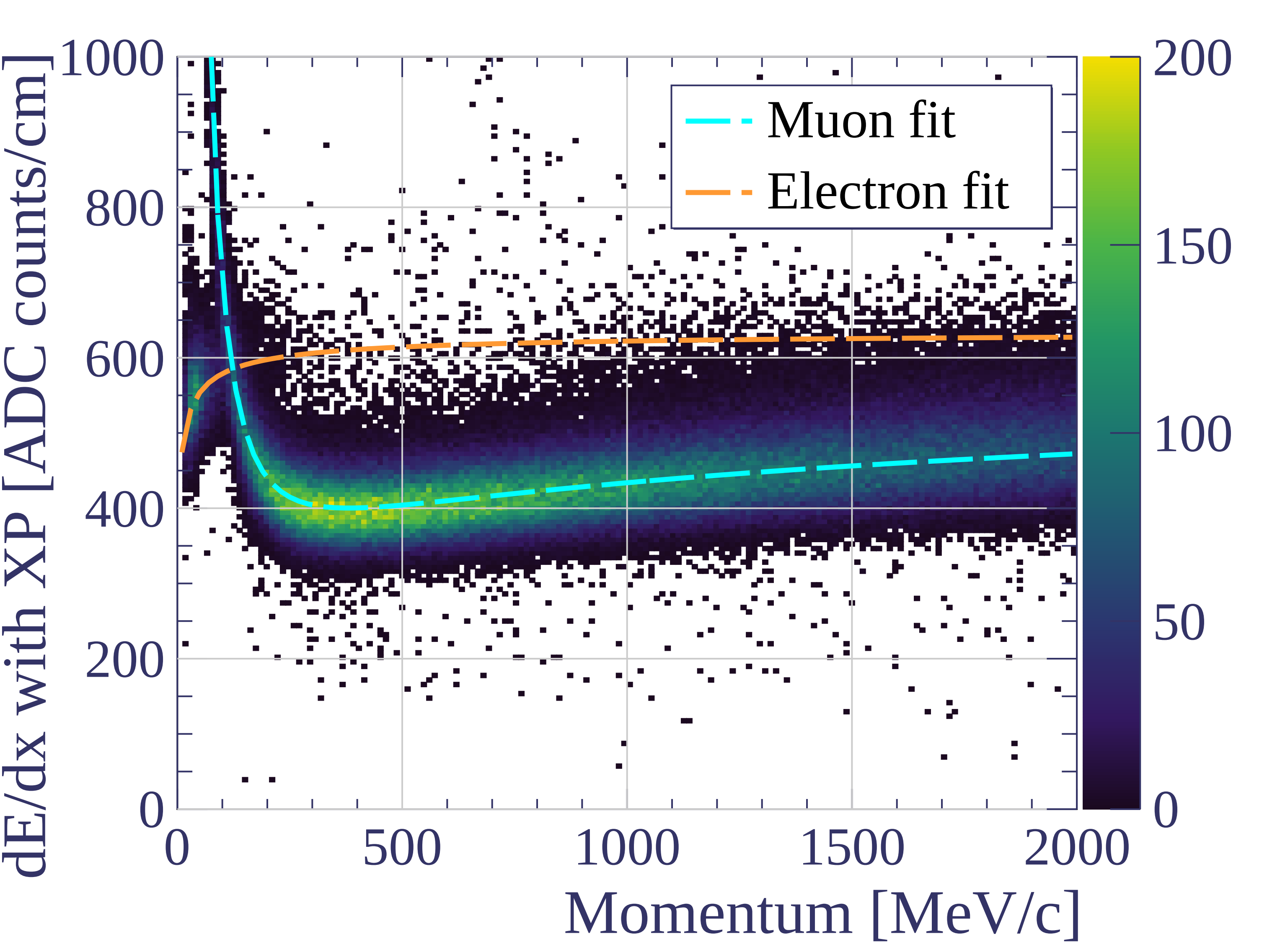
WF
XP
dE/dx resolution [%]
4
6
8
10
0
500
1000
1500
2000
Momentum [MeV/c]



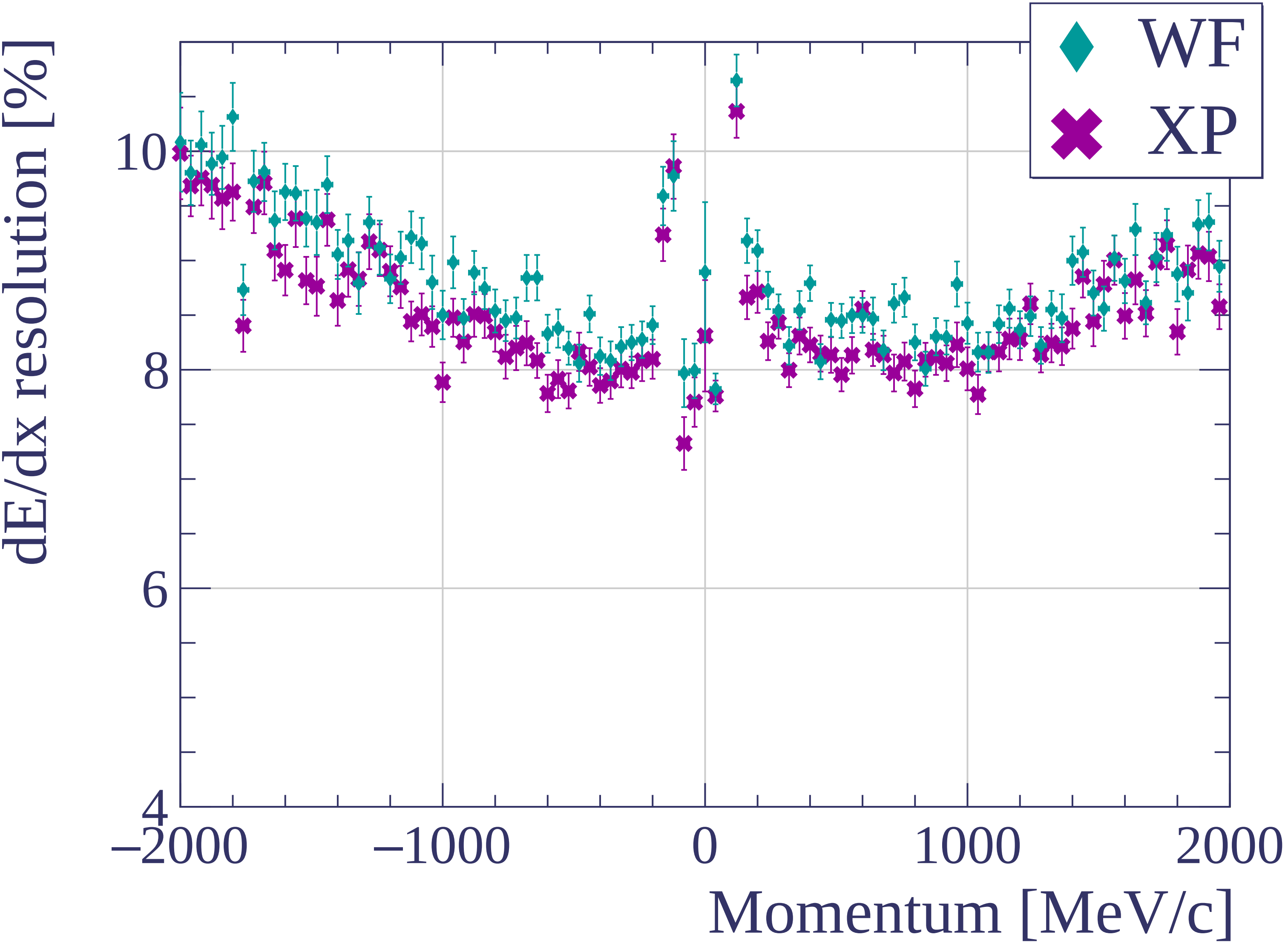
WF
XP
600
500
400
300
Mean dE/dx [ADC counts/cm]
0
500
1000
1500
2000
Momentum [MeV/c]



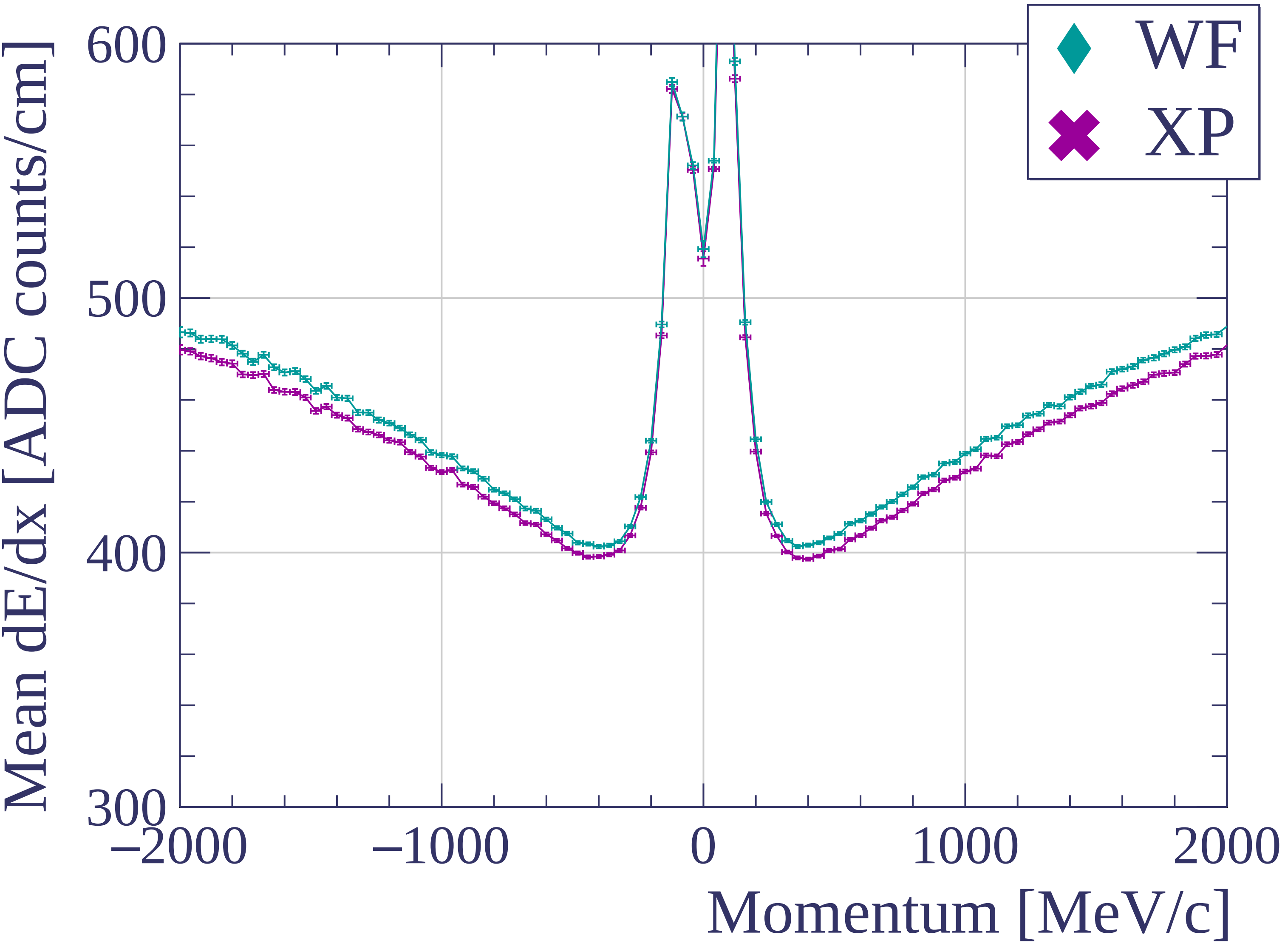
dE/dx with WF [ADC counts/cm]
1000
800
600
400
200
0
0
500
1000
1500
2000
Momentum [MeV/c]
150
100
50
0



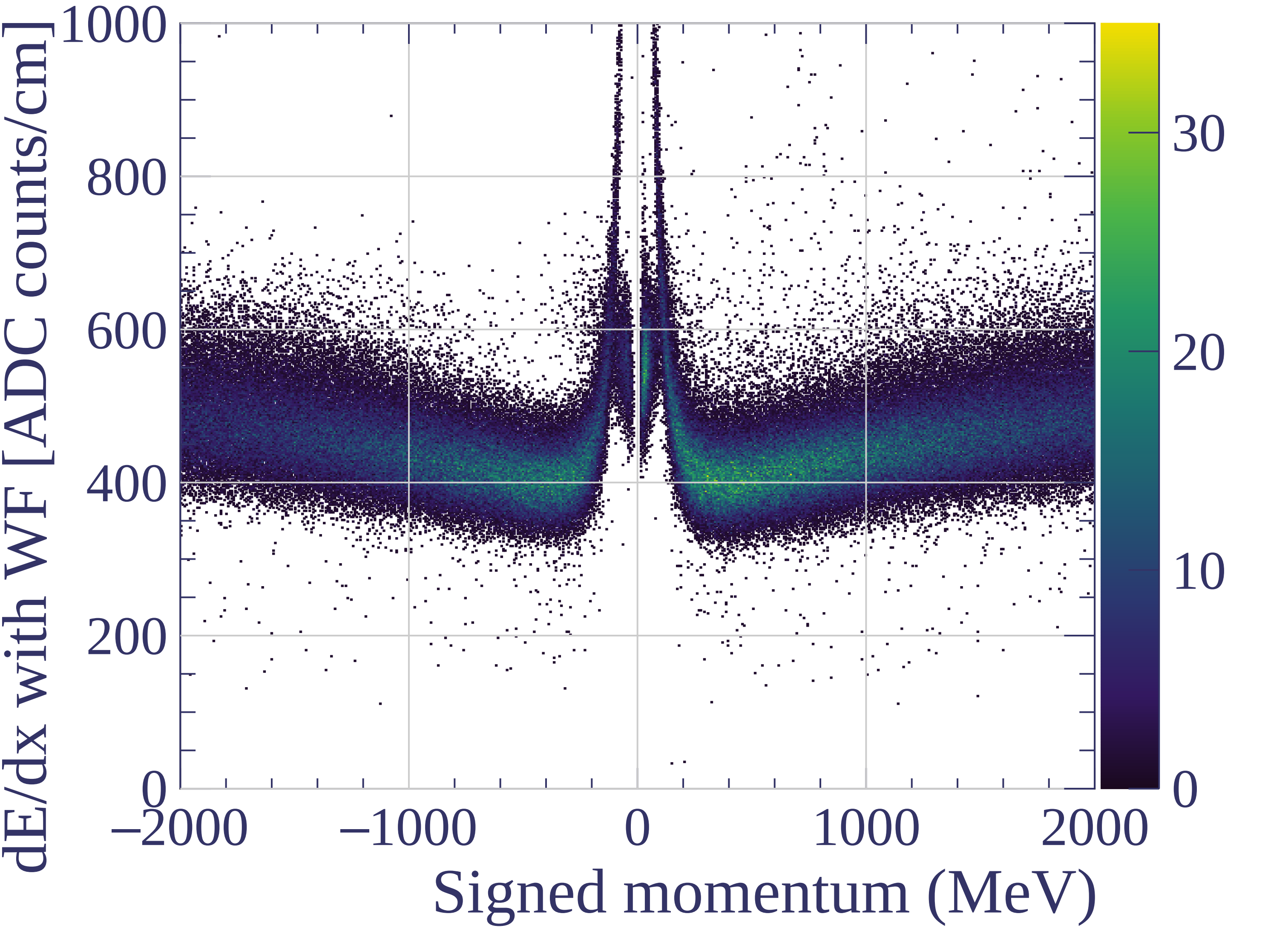
dE/dx with XP [ADC counts/cm]
Momentum [MeV/c]
Muon fit
Electron fit
0
200
400
600
800
1000
0
500
1000
1500
2000
0
50
100
150
200



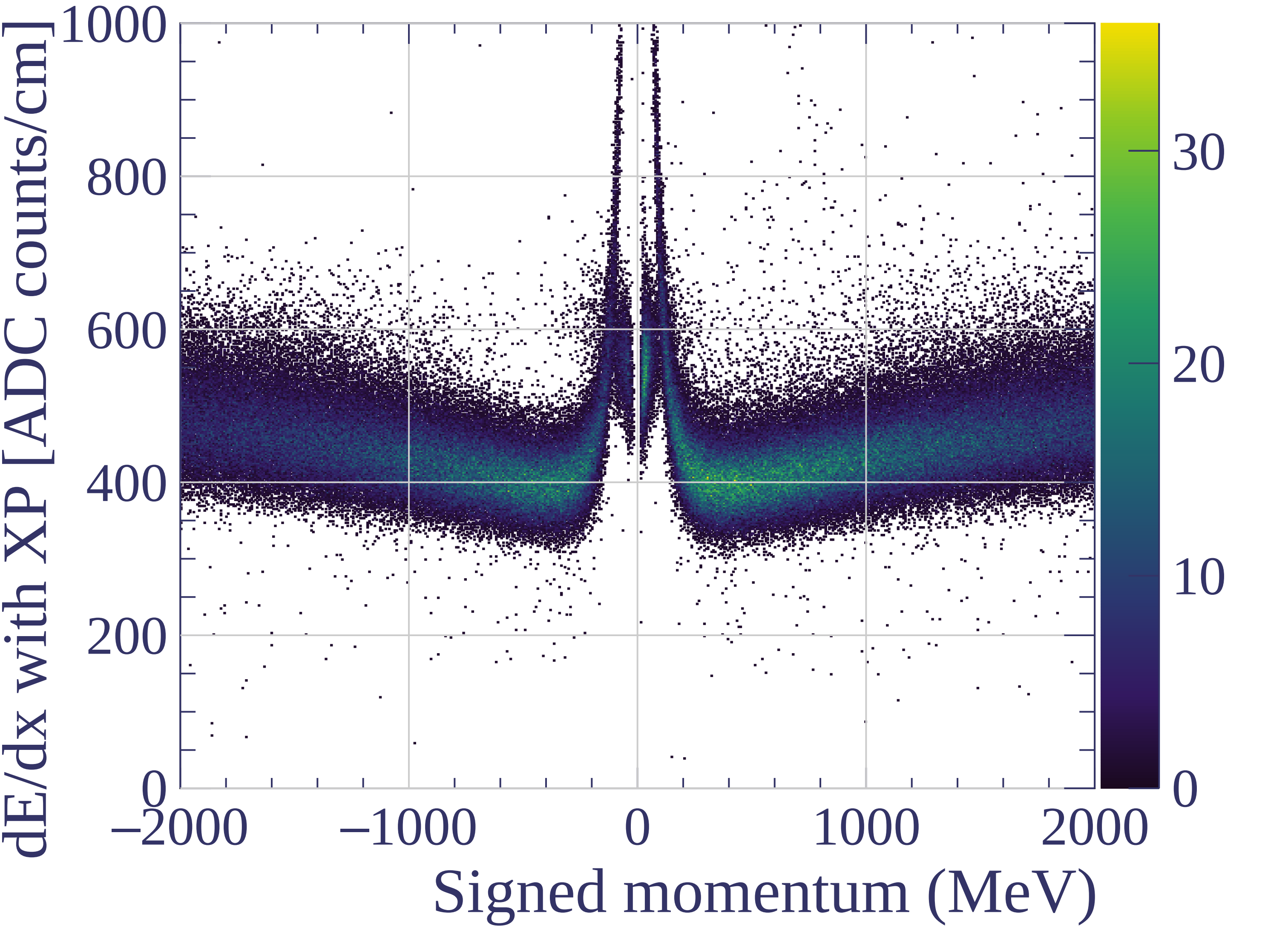
WF
XP
dE/dx resolution [%]
10
8
6
4
−2000
−1000
0
1000
2000
Momentum [MeV/c]



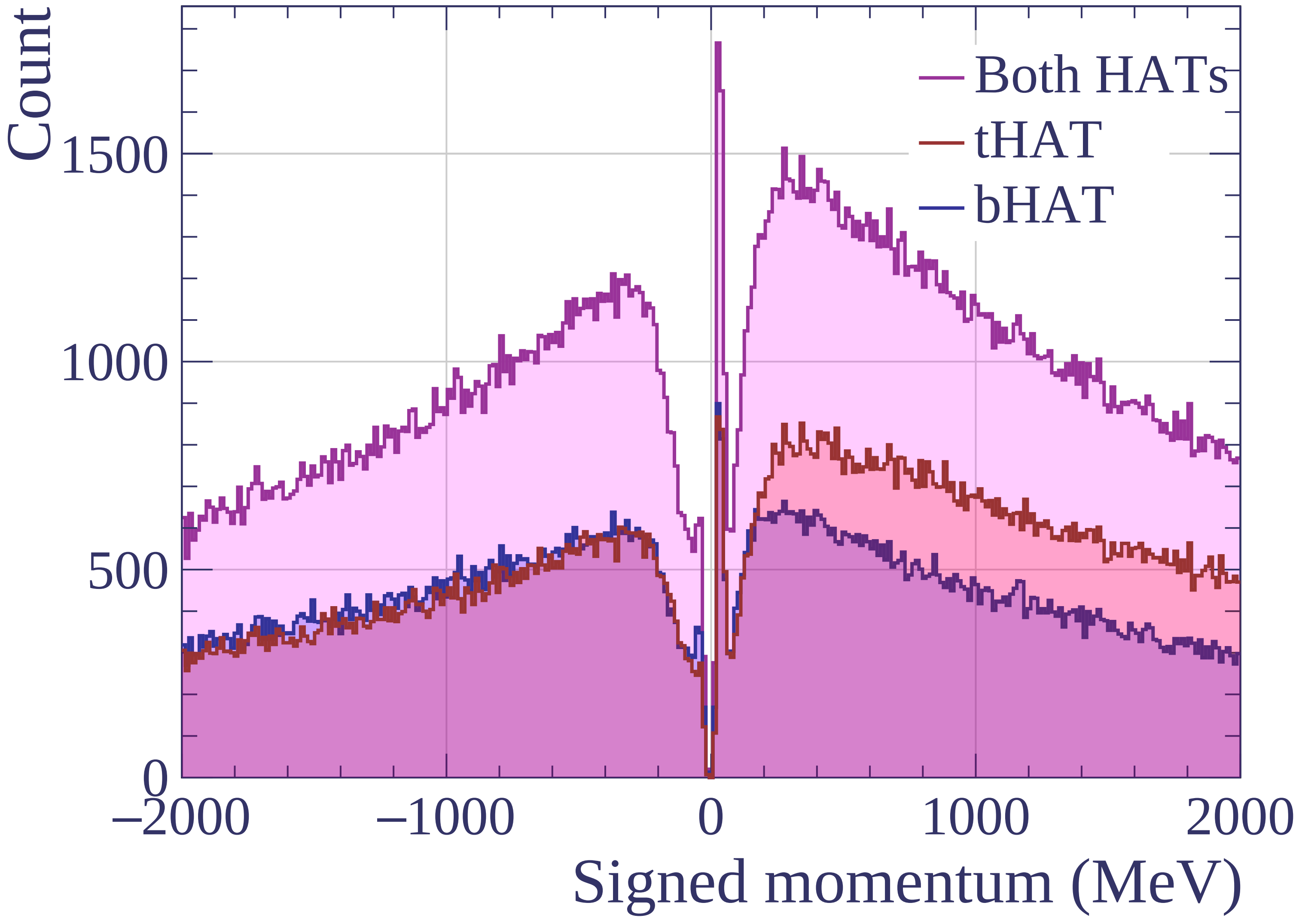
WF
XP
600
500
400
300
−2000
−1000
0
1000
2000
Mean dE/dx [ADC counts/cm]
Momentum [MeV/c]

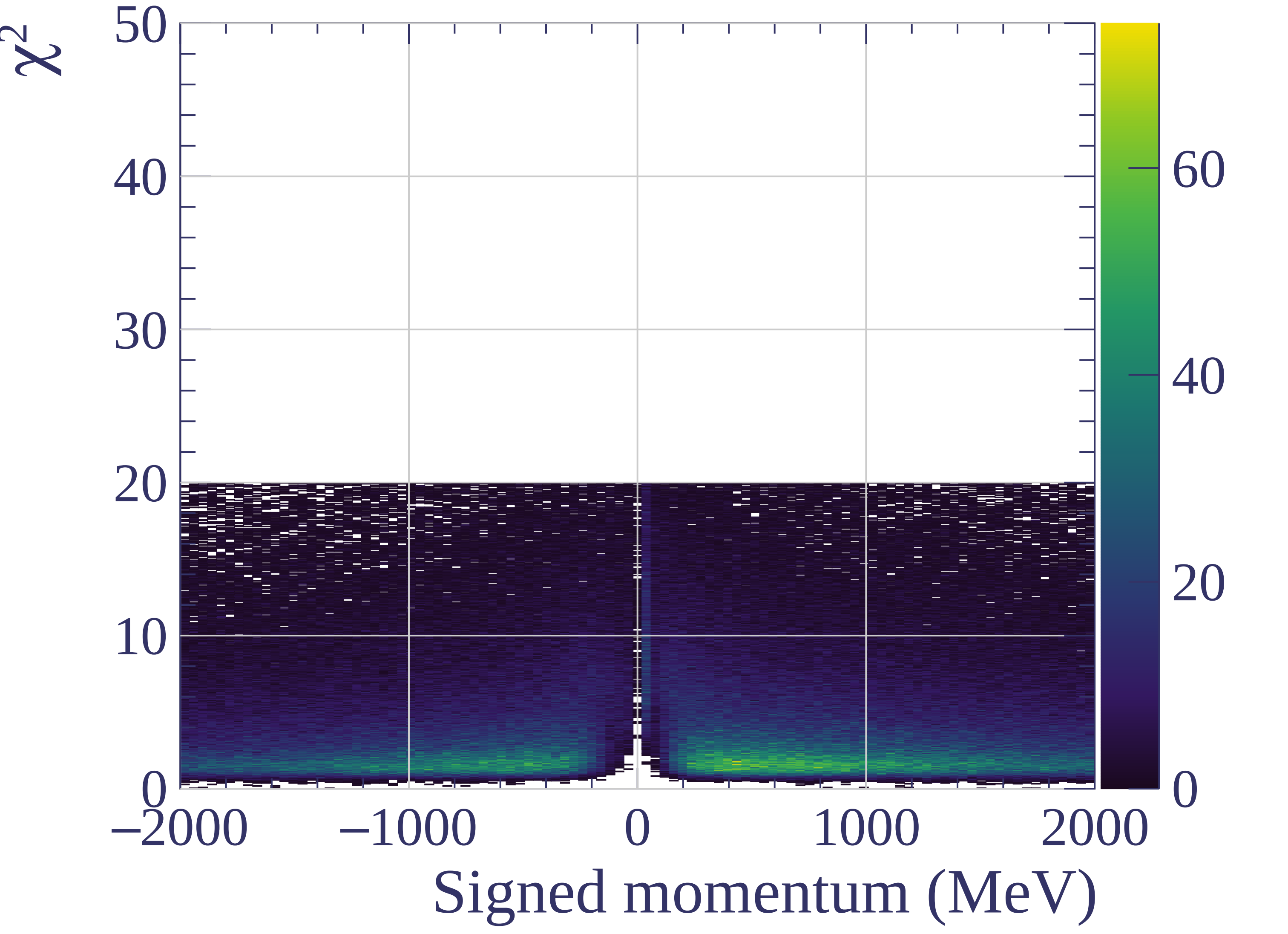


dE/dx with WF [ADC counts/cm]
1000
800
600
400
200
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
30
20
10
0

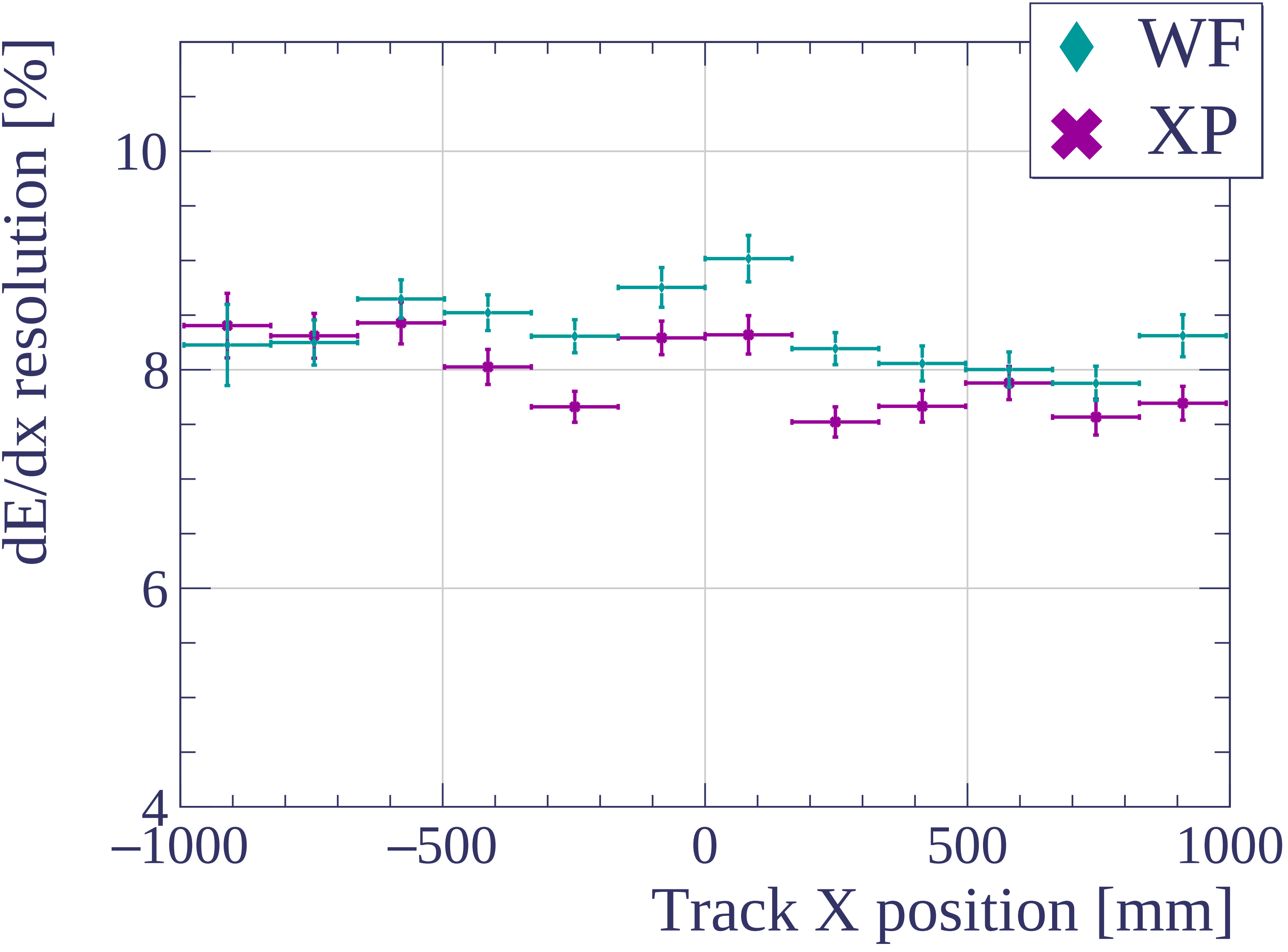


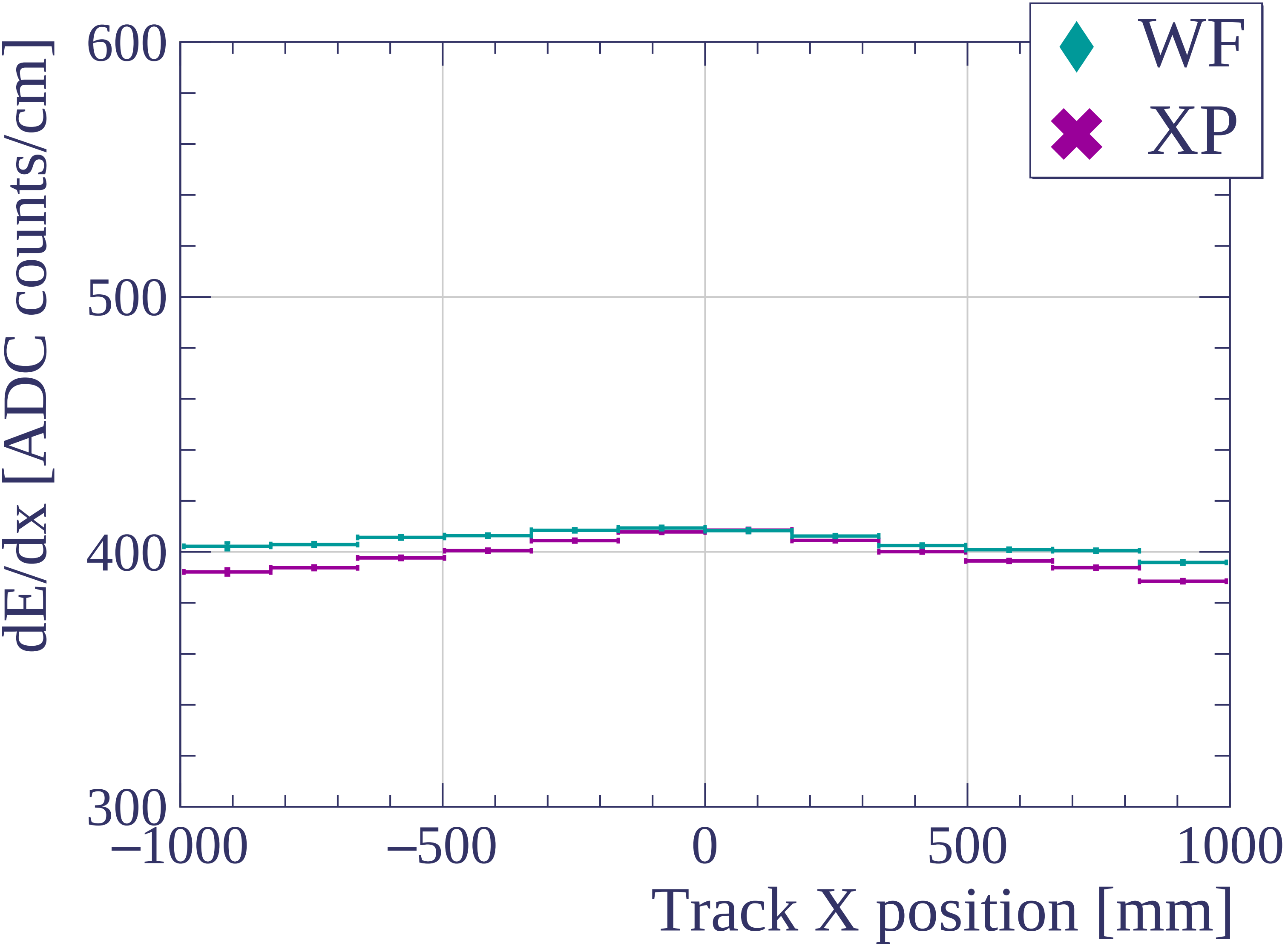
dE/dx with XP [ADC counts/cm]
Signed momentum (MeV)
1000
800
600
400
200
0
−2000
−1000
0
1000
2000
30
20
10
0

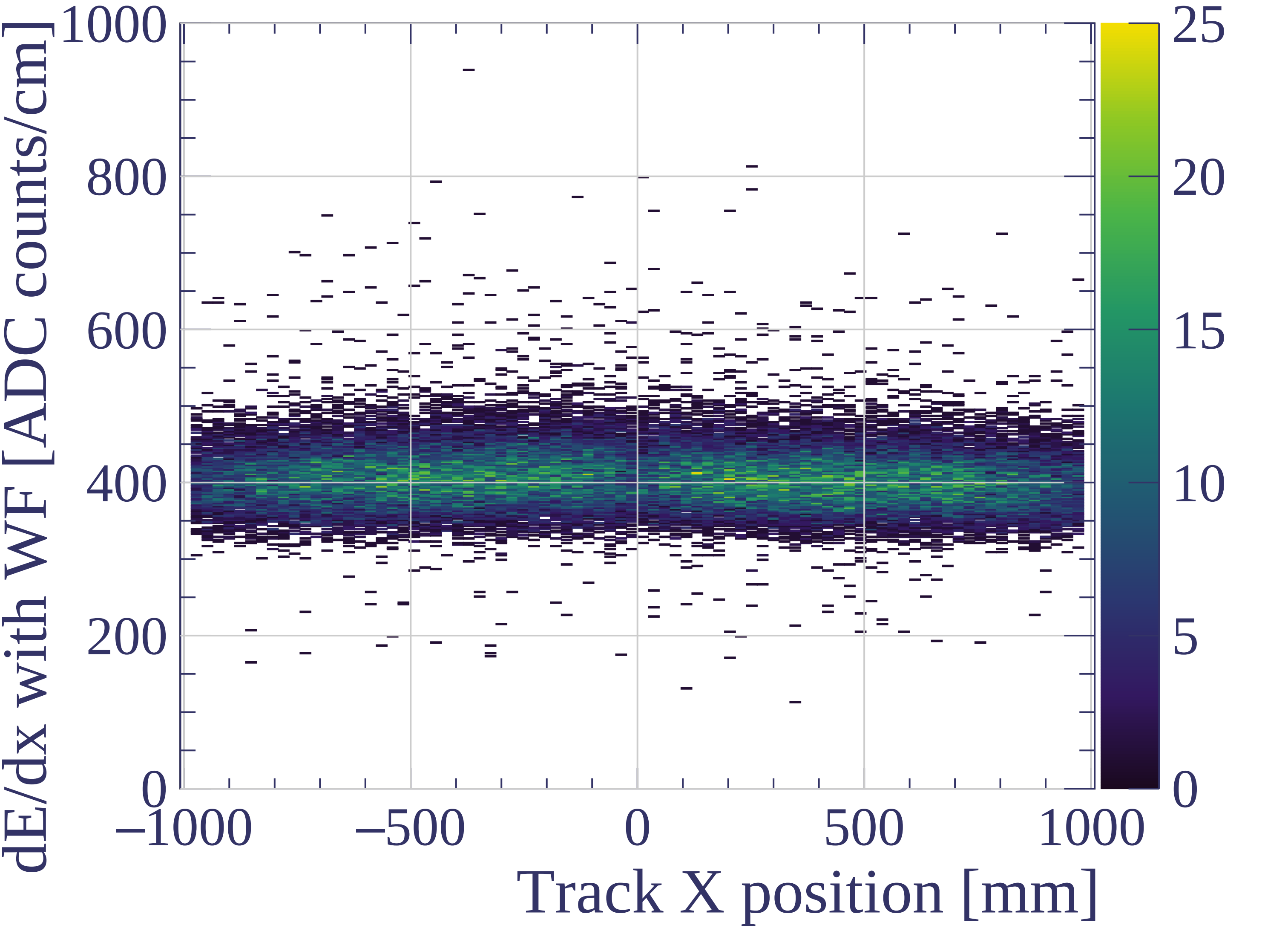

Count
Signed momentum (MeV)
Both HATs
tHAT
bHAT
0
500
1000
1500
−2000
−1000
0
1000
2000



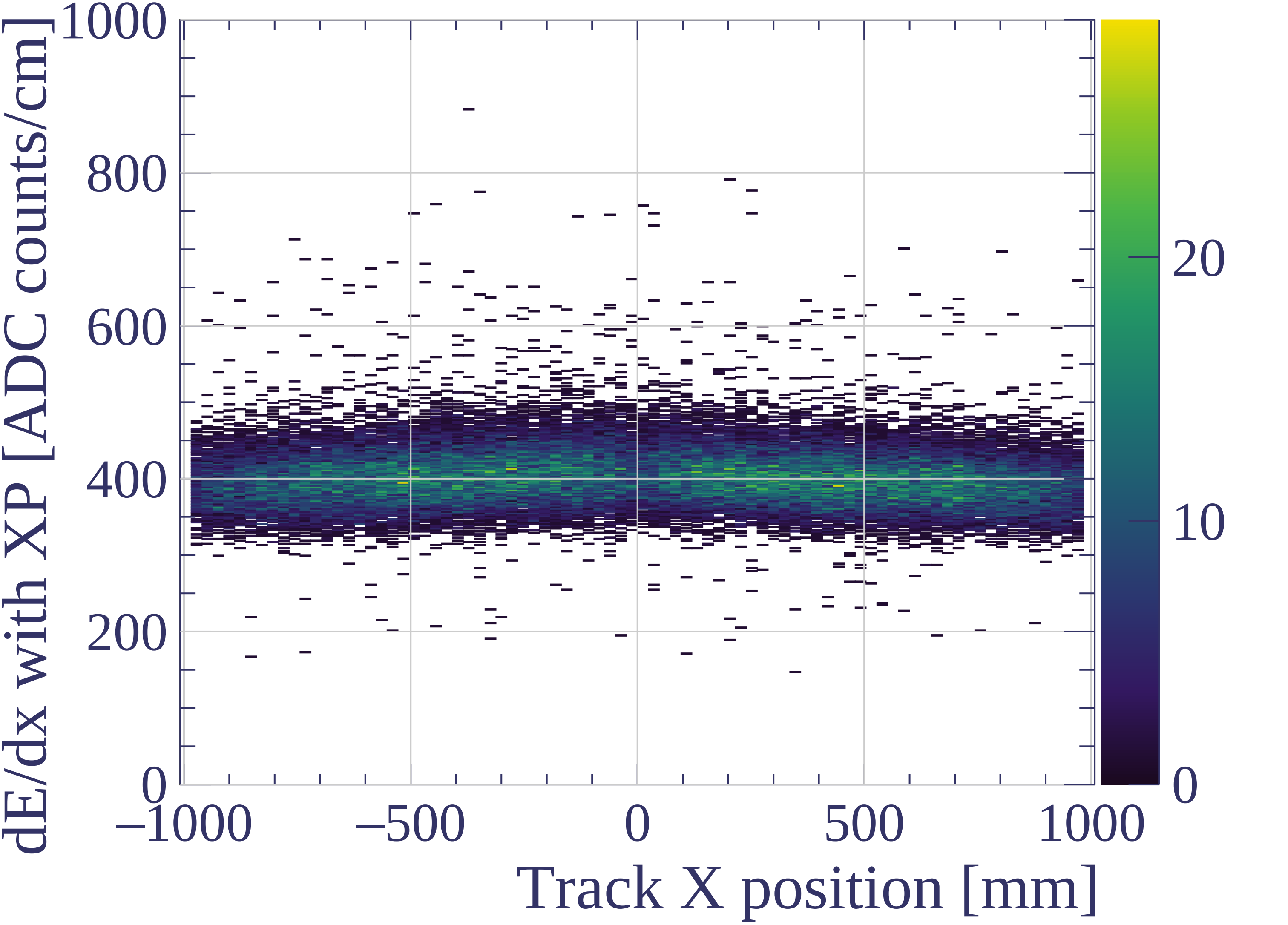
χ^2
50
40
30
20
10
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
60
40
20
0

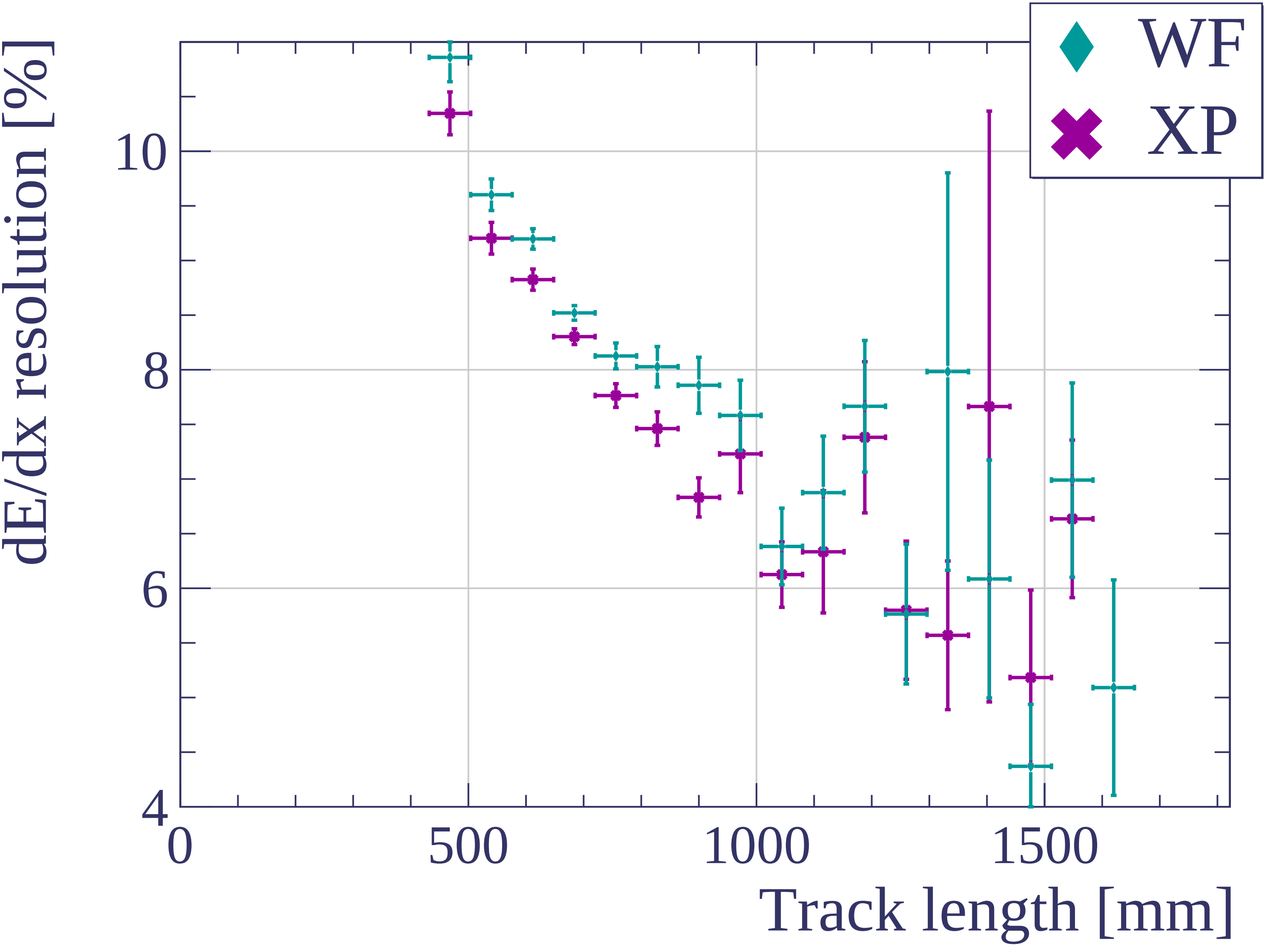

WF
XP
dE/dx resolution [%]
10
8
6
4
−1000
−500
0
500
1000
Track X position [mm]

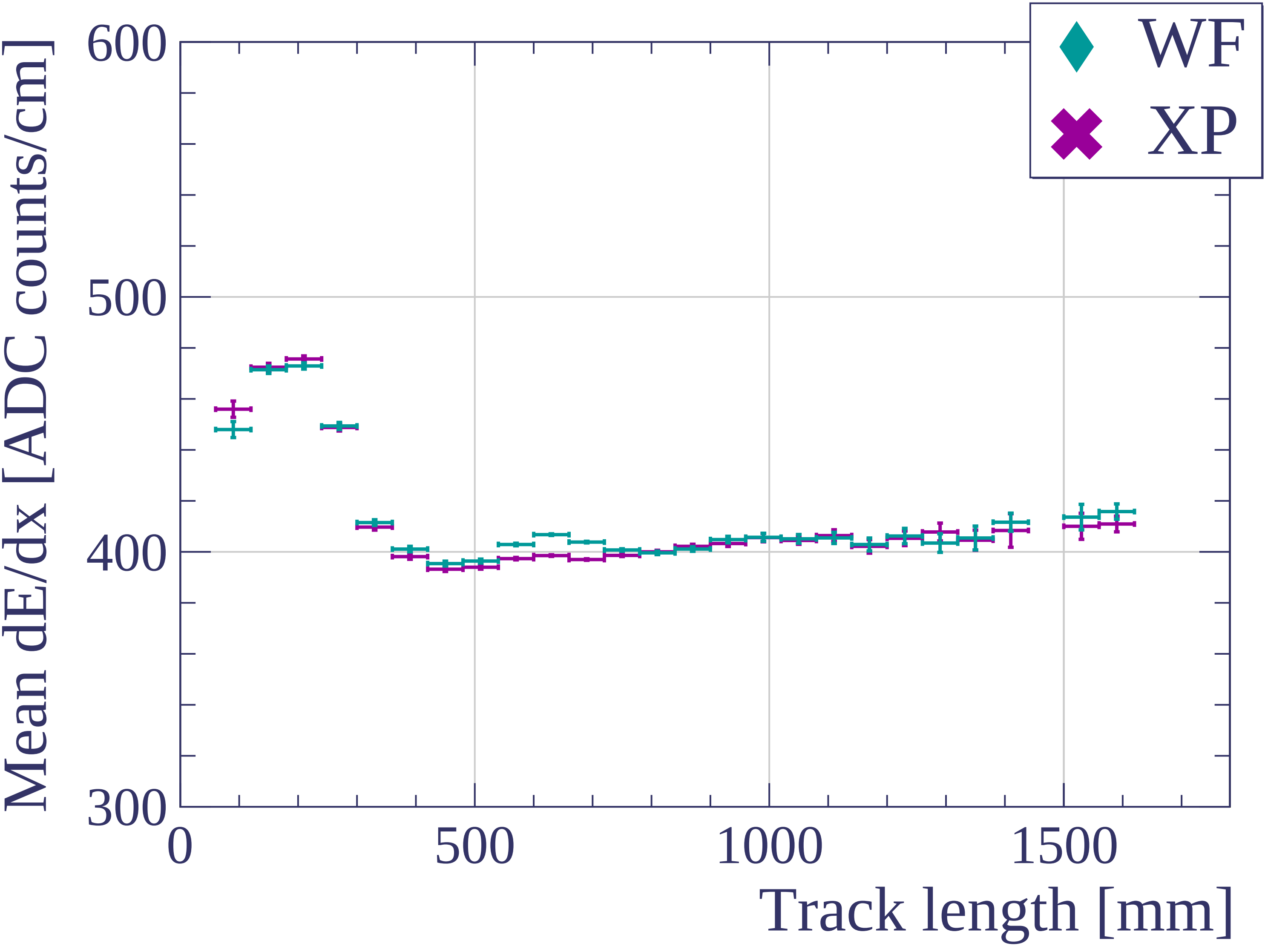

WF
XP
dE/dx [ADC counts/cm]
600
500
400
300
−1000
−500
0
500
1000
Track X position [mm]



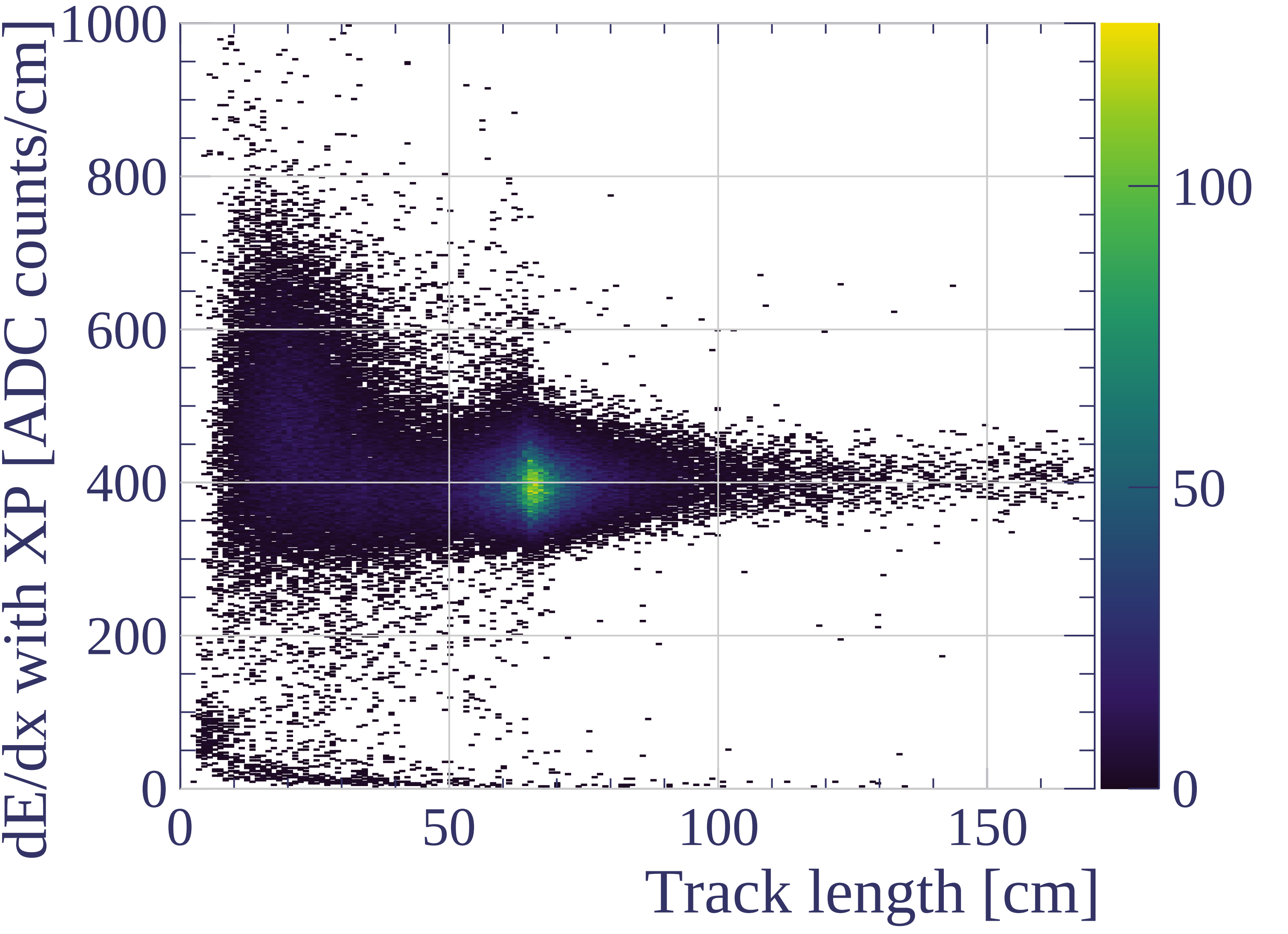
dE/dx with WF [ADC counts/cm]
Track X position [mm]
1000
800
600
400
200
0
−1000
−500
0
500
1000
25
20
15
10
5
0


dE/dx with XP [ADC counts/cm]
Track X position [mm]
1000
800
600
400
200
0
−1000
−500
0
500
1000
20
10
0

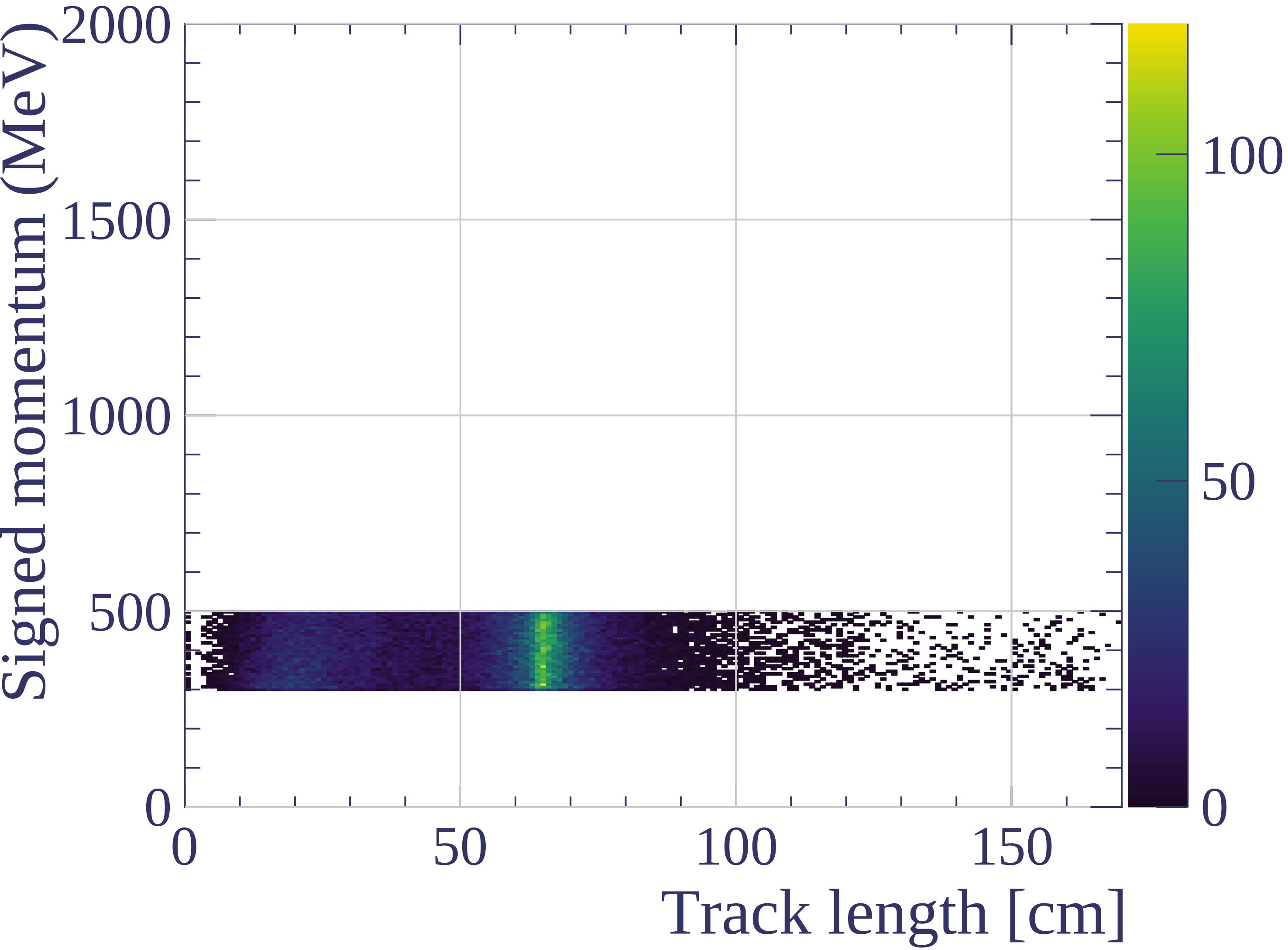

WF
XP
dE/dx resolution [%]
Track length [mm]
4
6
8
10
0
500
1000
1500

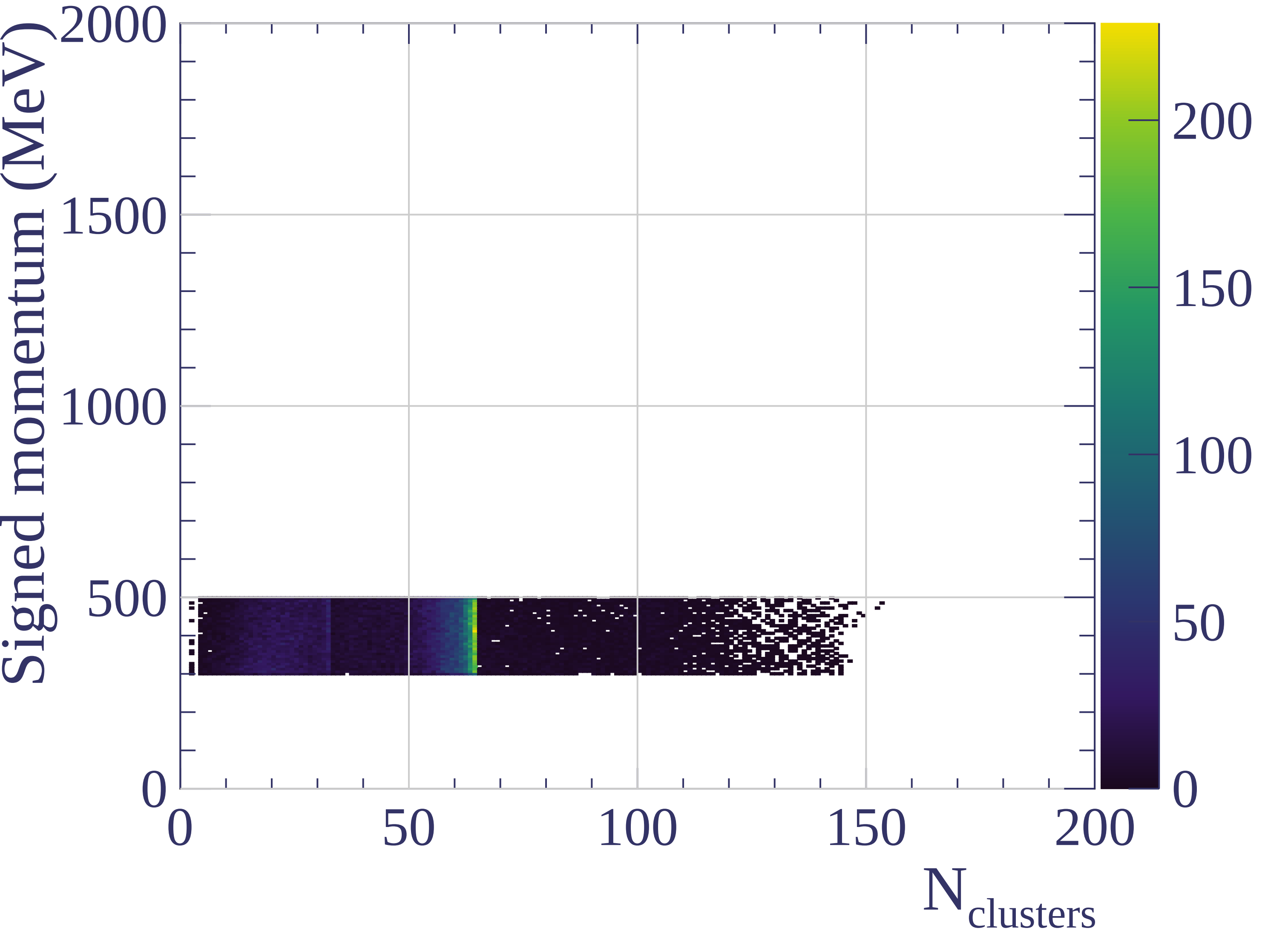


WF
XP
Mean dE/dx [ADC counts/cm]
600
500
400
300
0
500
1000
1500
Track length [mm]

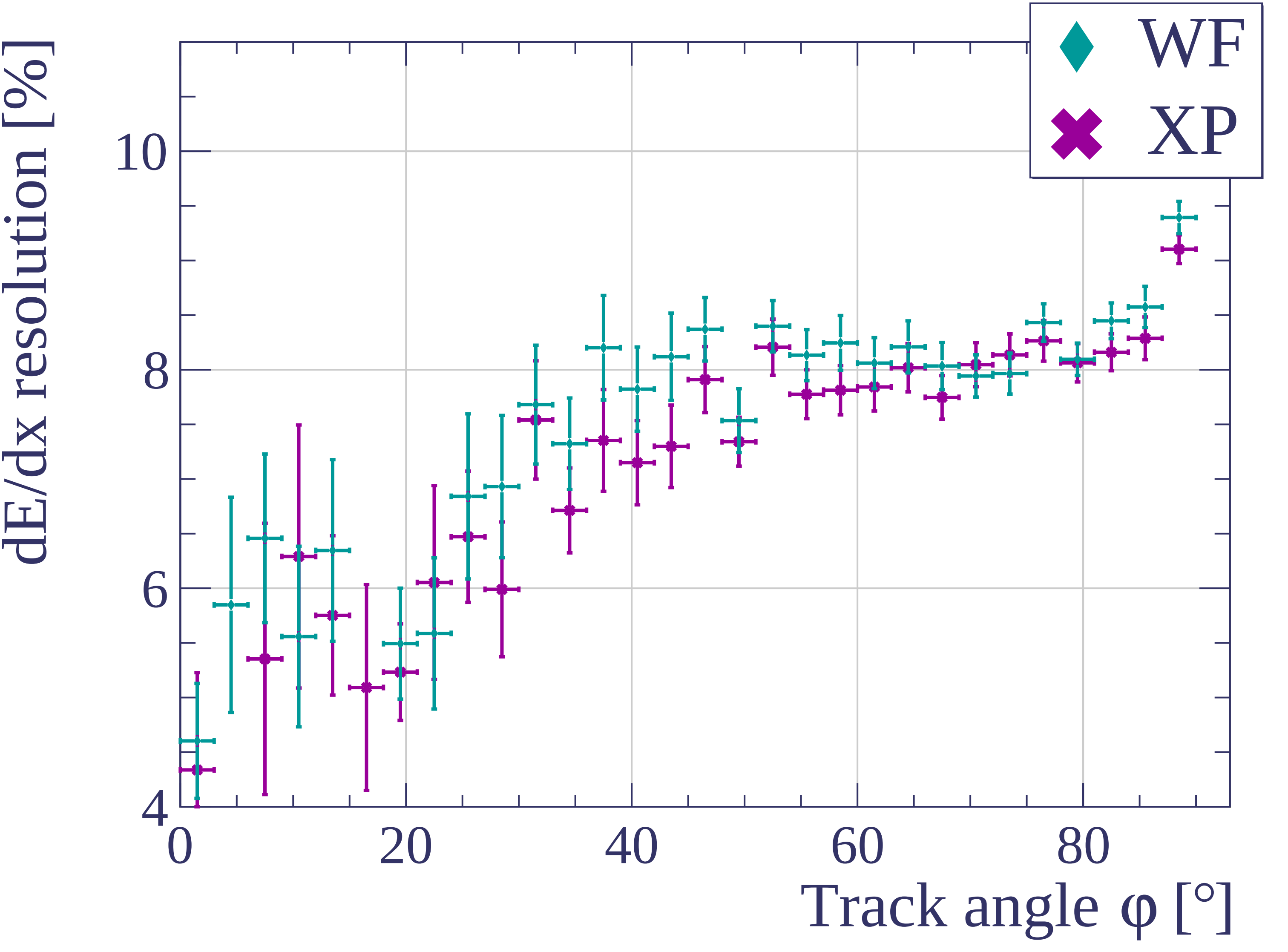


dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
0
50
100
150
Track length [cm]
100
50
0

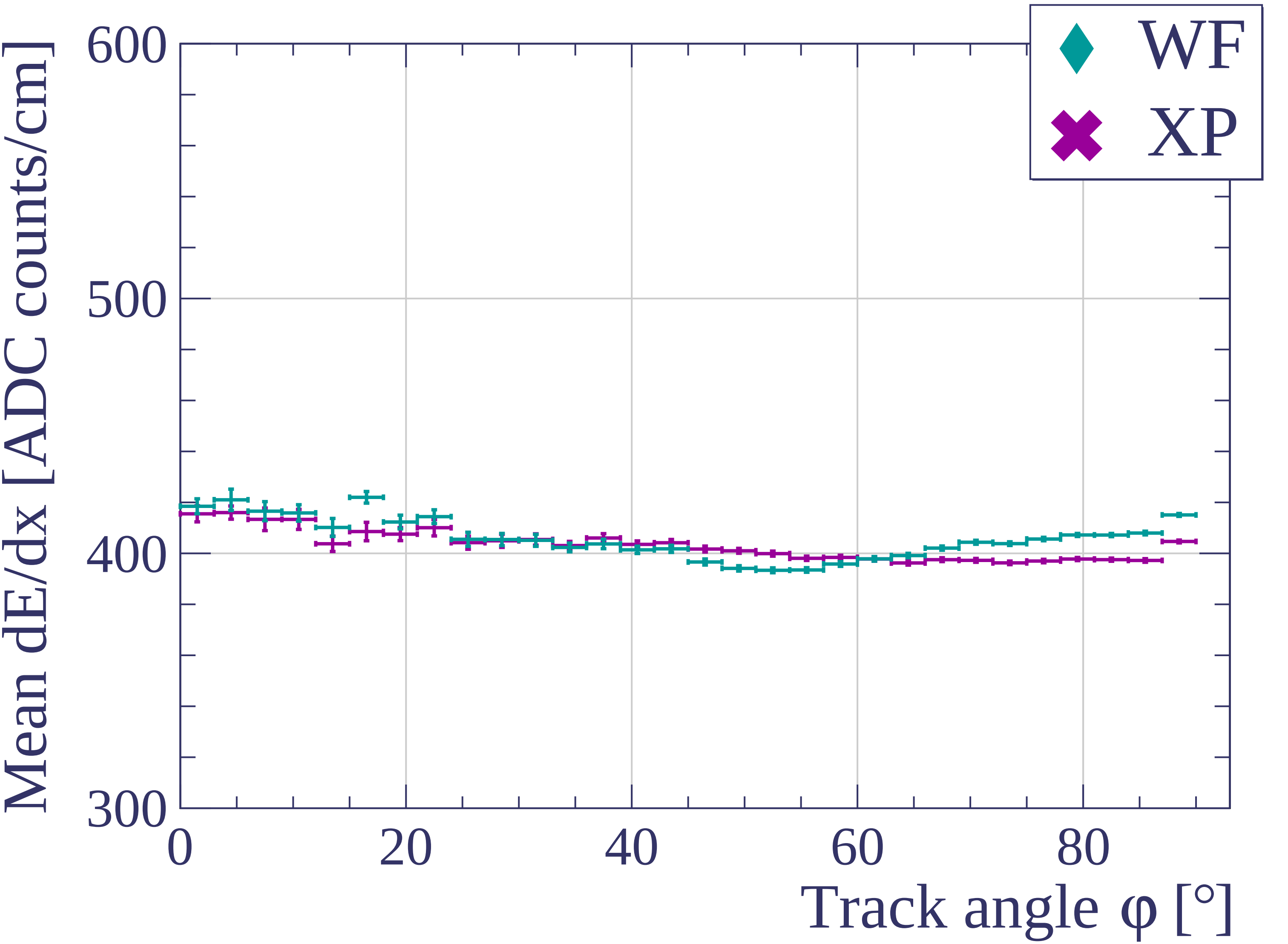

Signed momentum (MeV)
2000
1500
1000
500
0
0
50
100
150
Track length [cm]
100
50
0



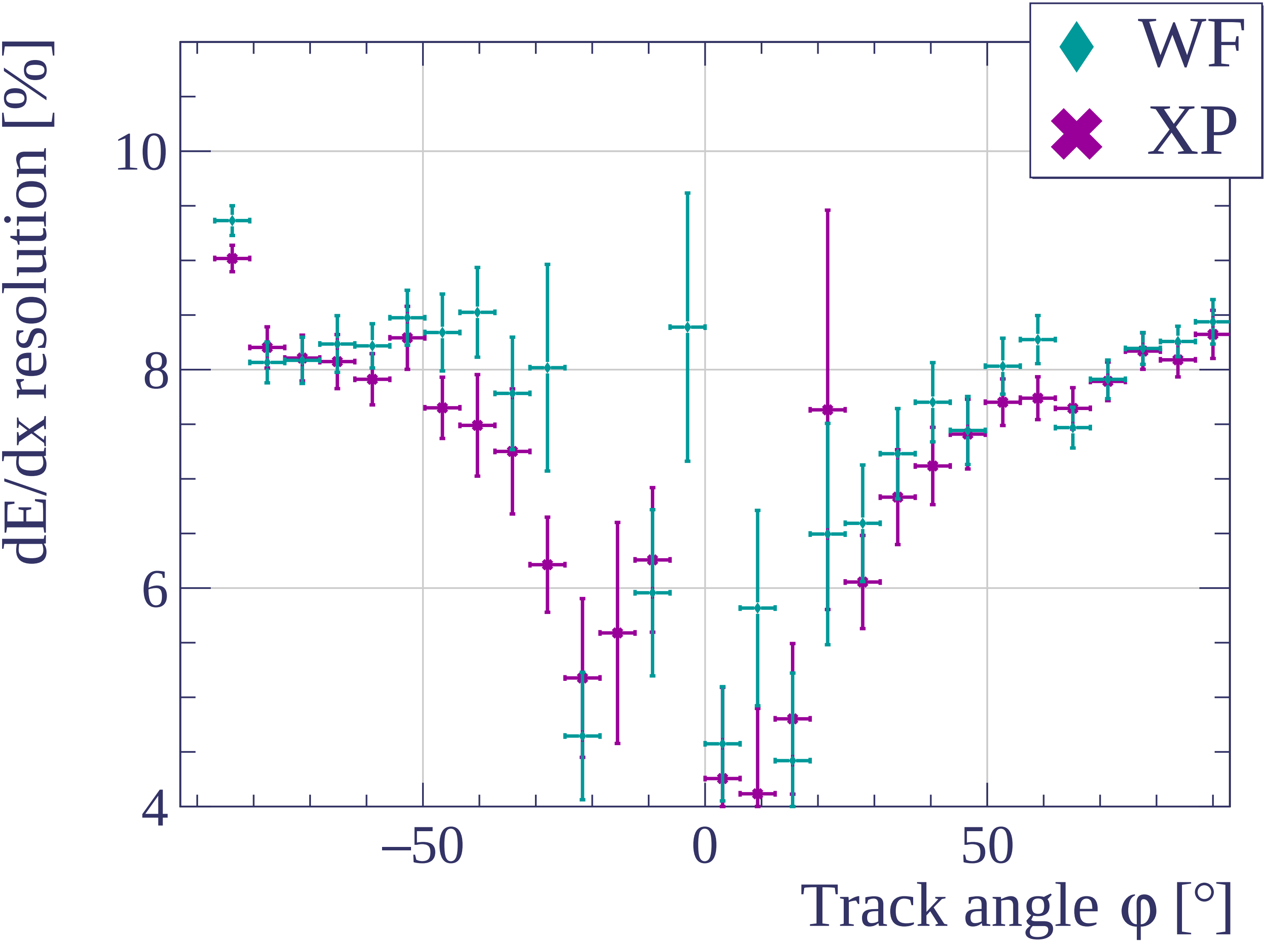
Signed momentum (MeV)
2000
1500
1000
500
0
0
50
100
150
200
$N_{clusters}$
200
150
100
50
0



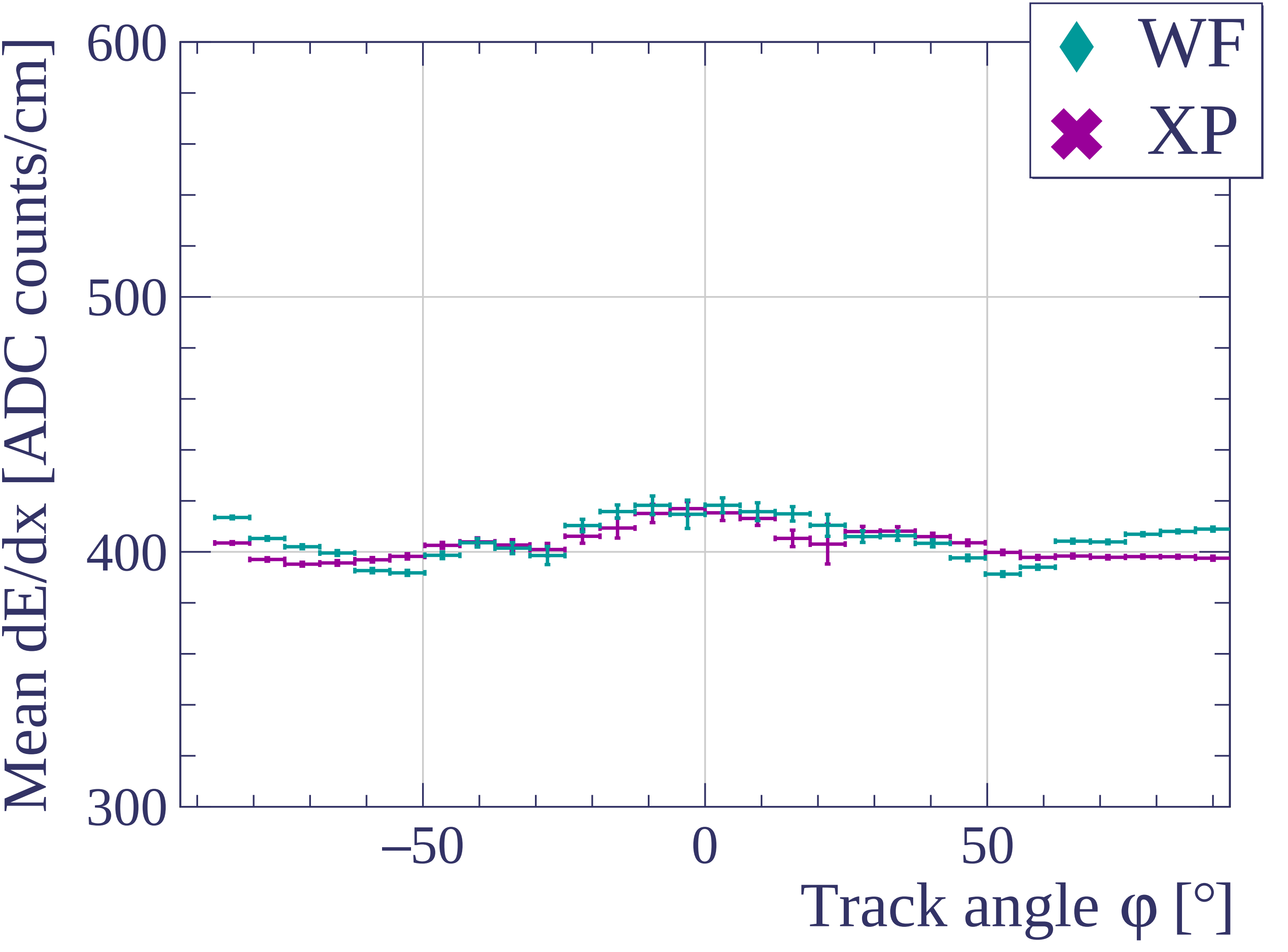
WF
XP
dE/dx resolution [%]
10
8
6
4
0
20
40
60
80
Track angle φ [°]



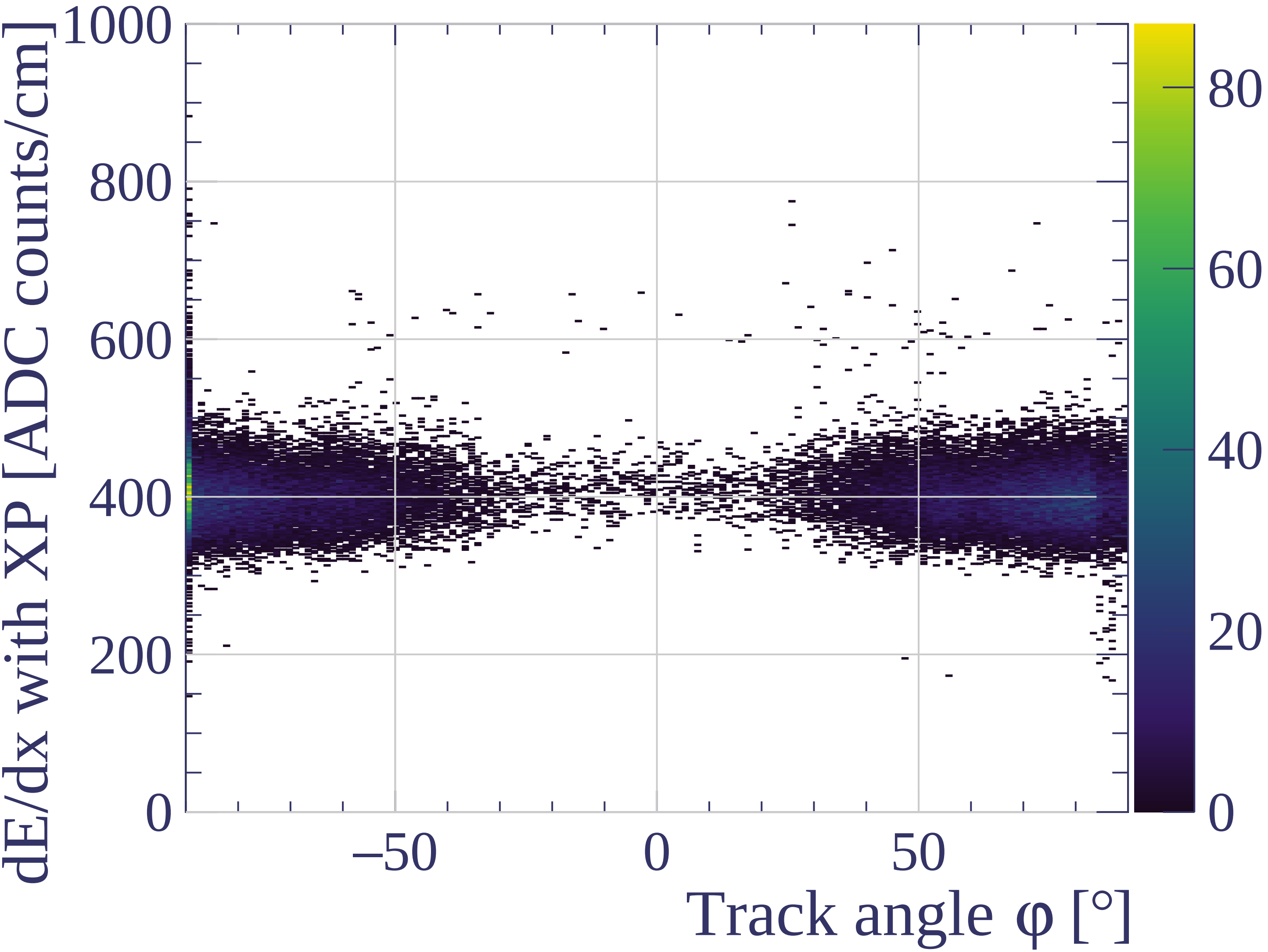
WF
XP
600
500
400
300
0
20
40
60
80
Mean dE/dx [ADC counts/cm]
Track angle φ [°]

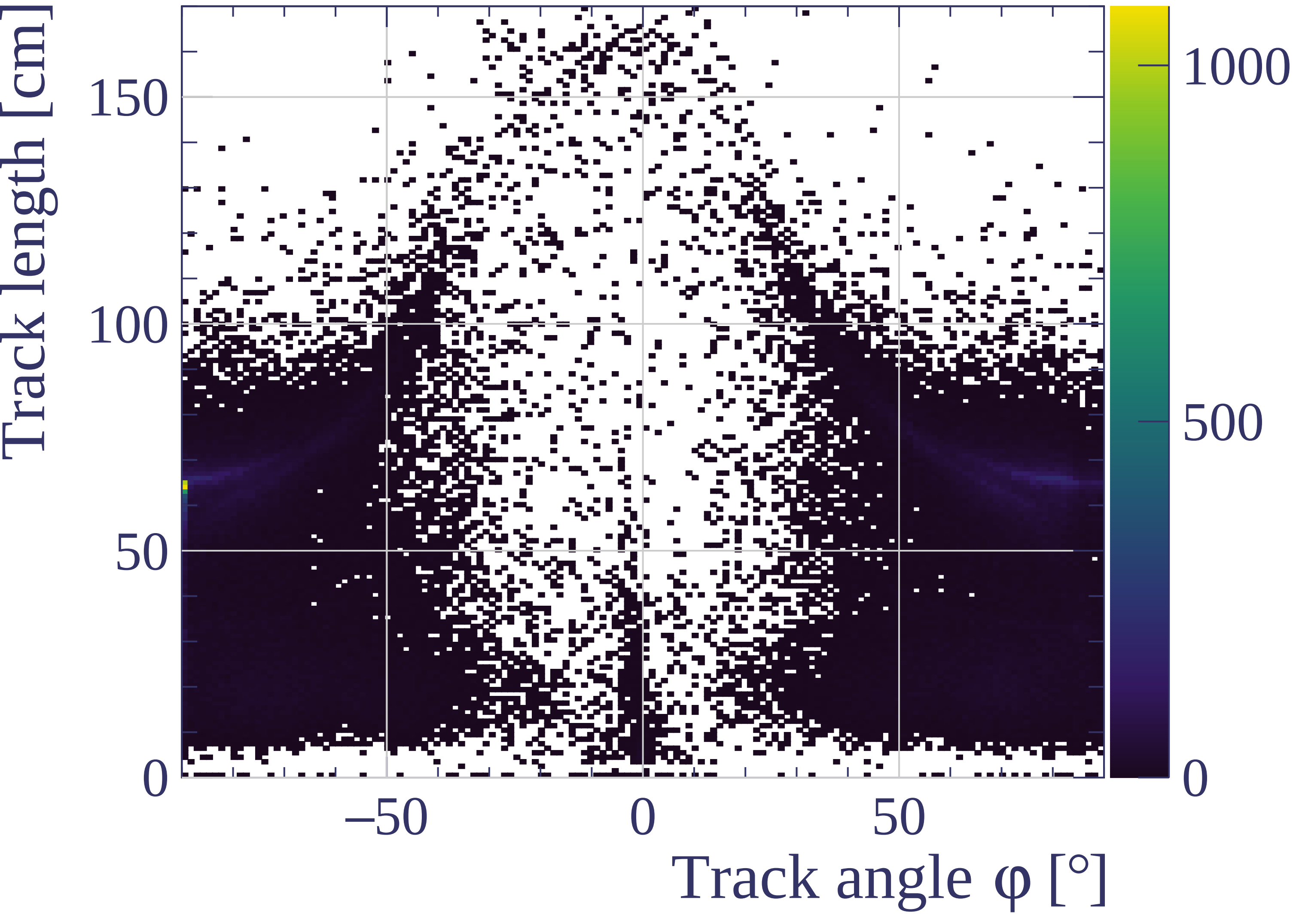


WF
XP
dE/dx resolution [%]
10
8
6
4
−50
0
50
Track angle φ [°]

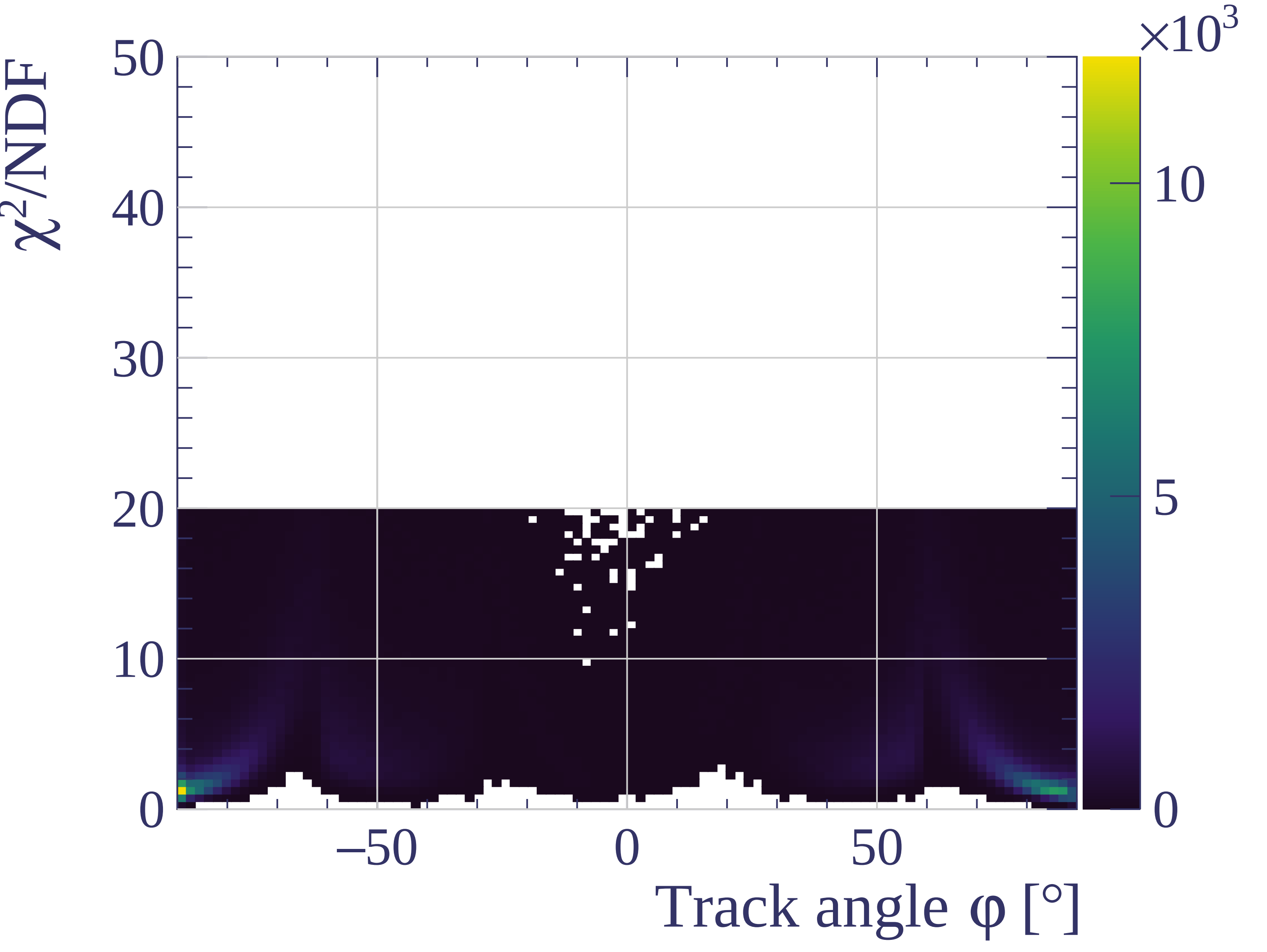


WF
XP
Mean dE/dx [ADC counts/cm]
600
500
400
300
−50
0
50
Track angle φ [°]

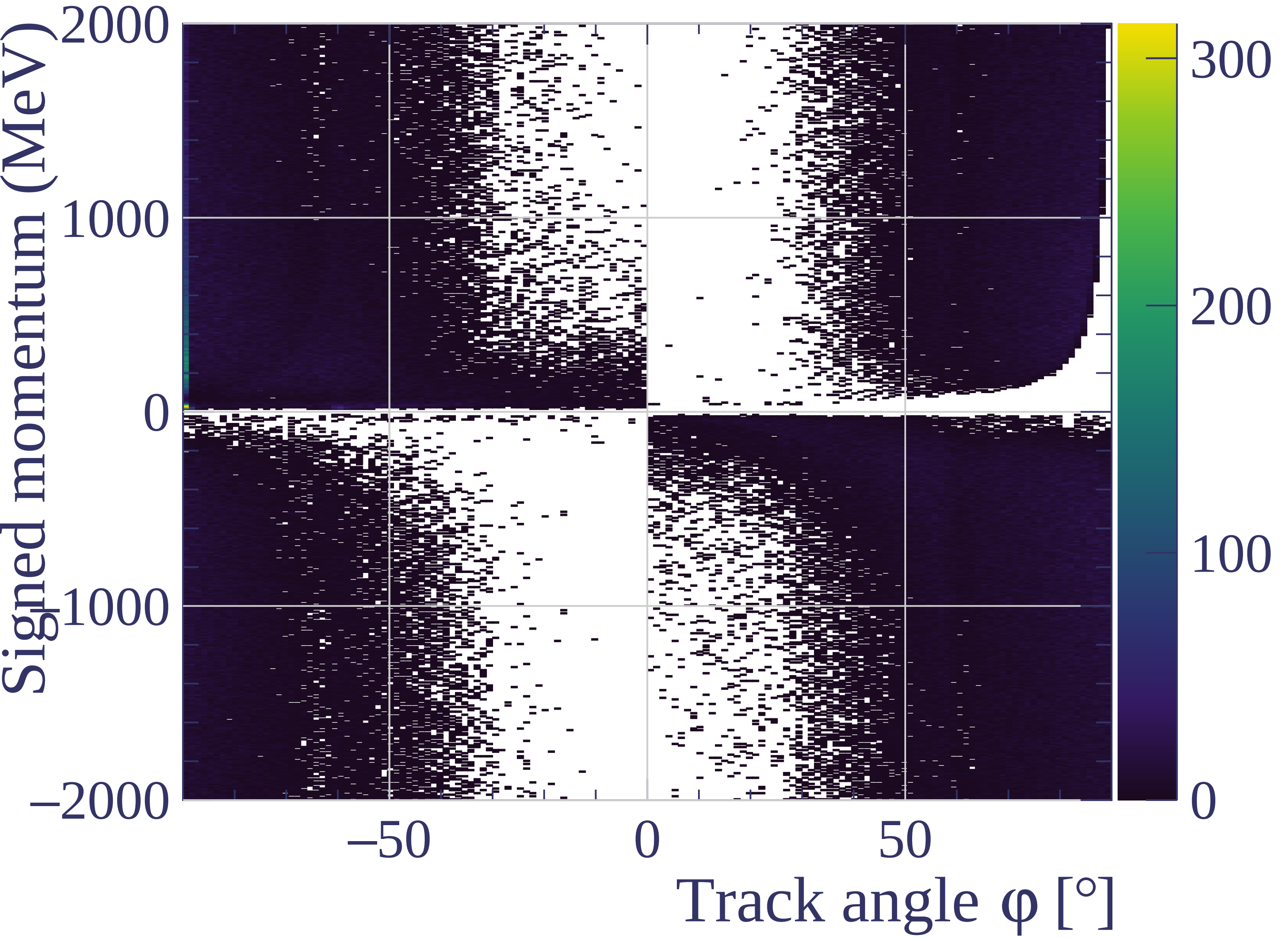

dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−50
0
50
Track angle φ [°]
80
60
40
20
0



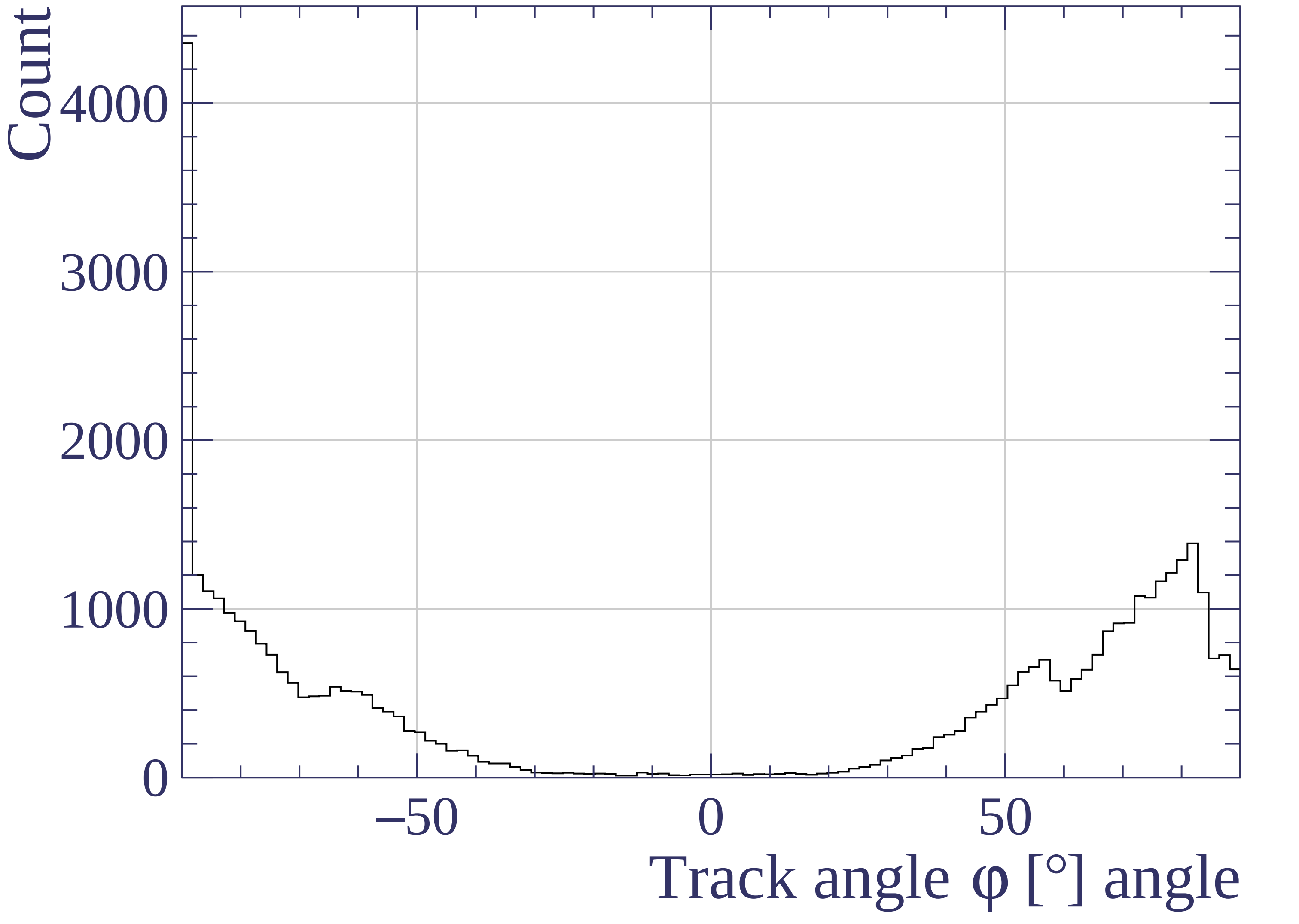
Track length [cm]
150
100
50
0
−50
0
50
Track angle φ [°]
1000
500
0



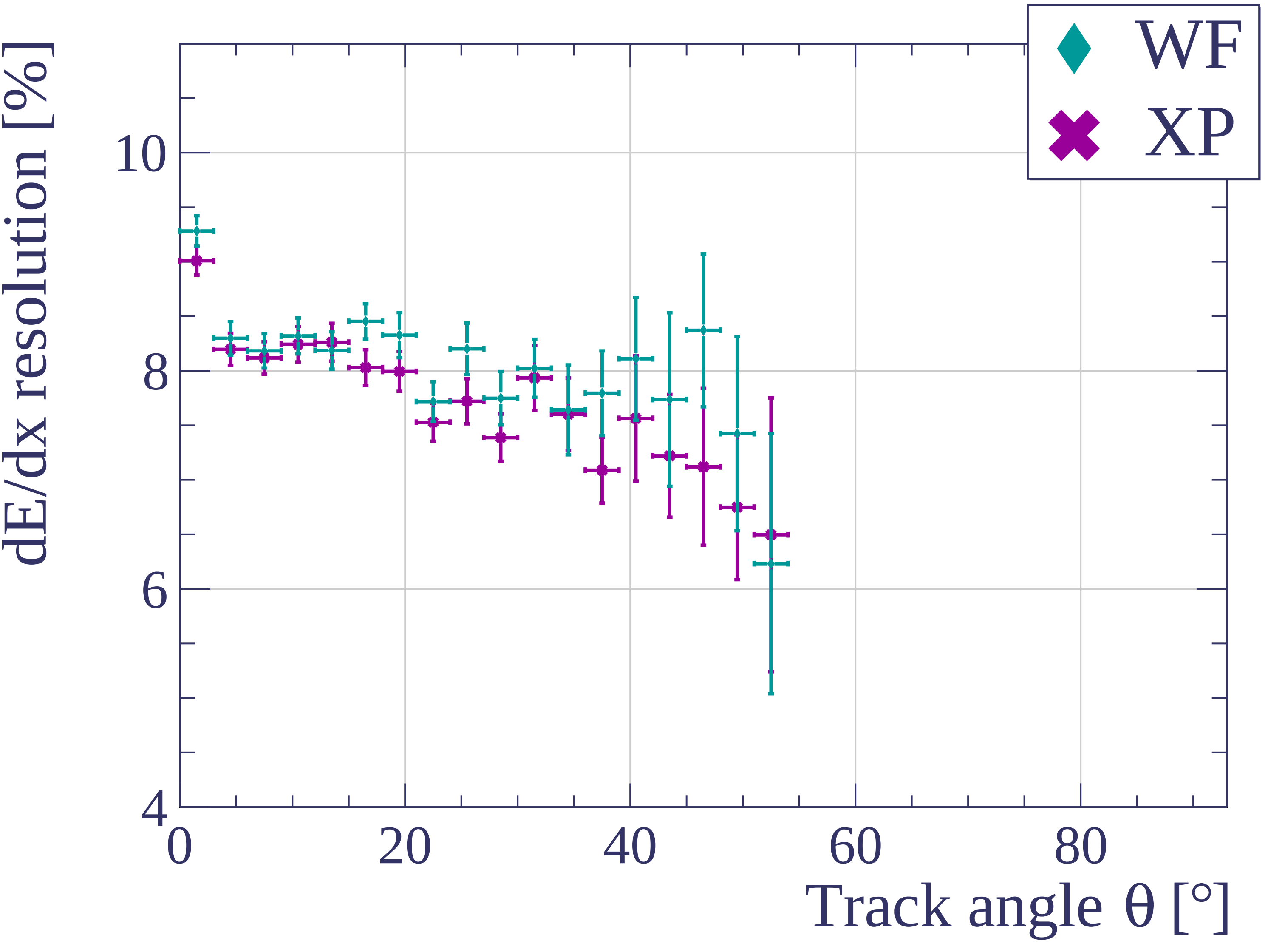
×10^3
50
40
30
20
10
0
χ²/NDF
−50
0
50
Track angle φ [°]
10
5
0



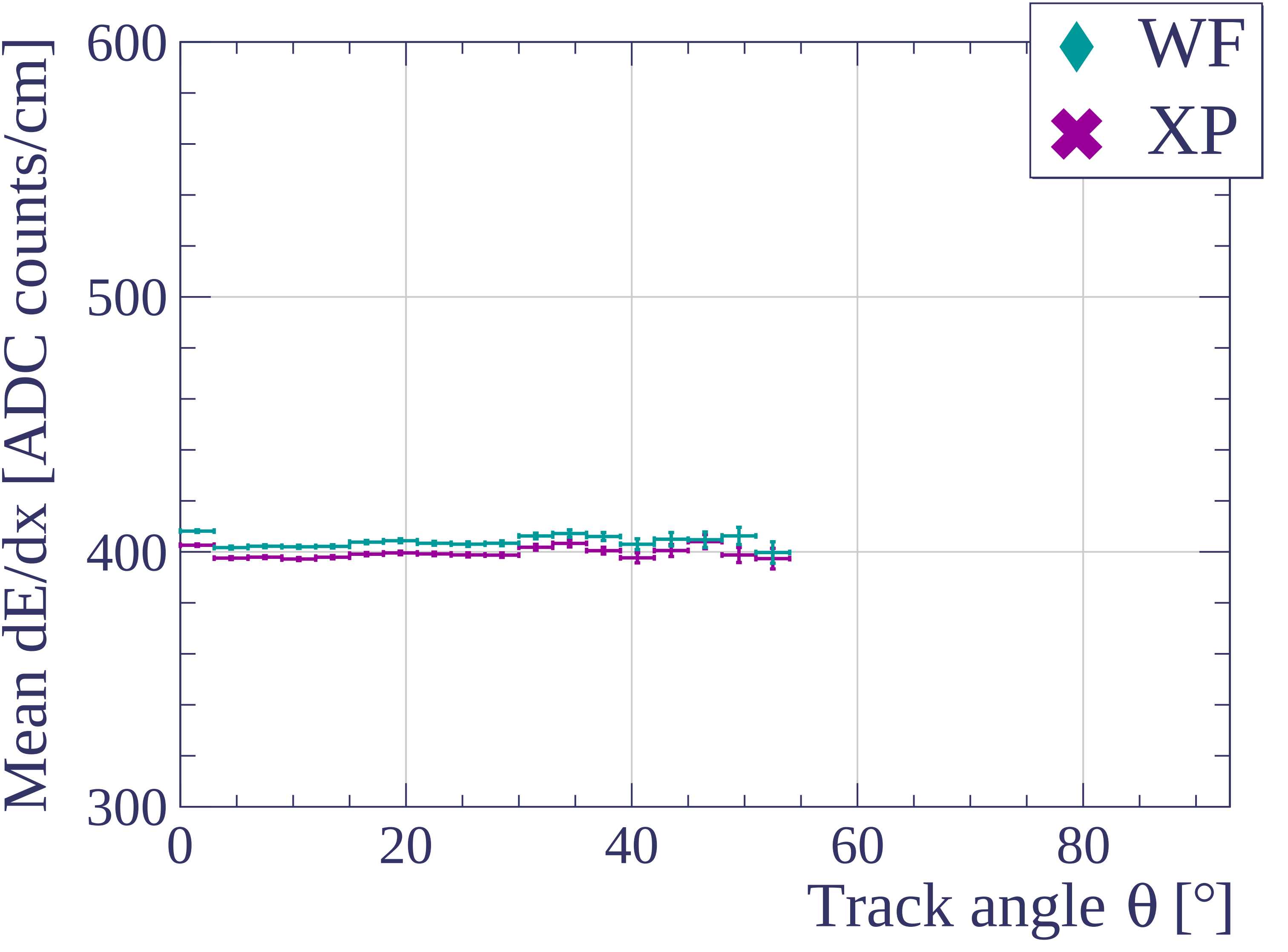
Signed momentum (MeV)
2000
1000
0
−1000
−2000
−50
0
50
Track angle φ [°]
300
200
100
0

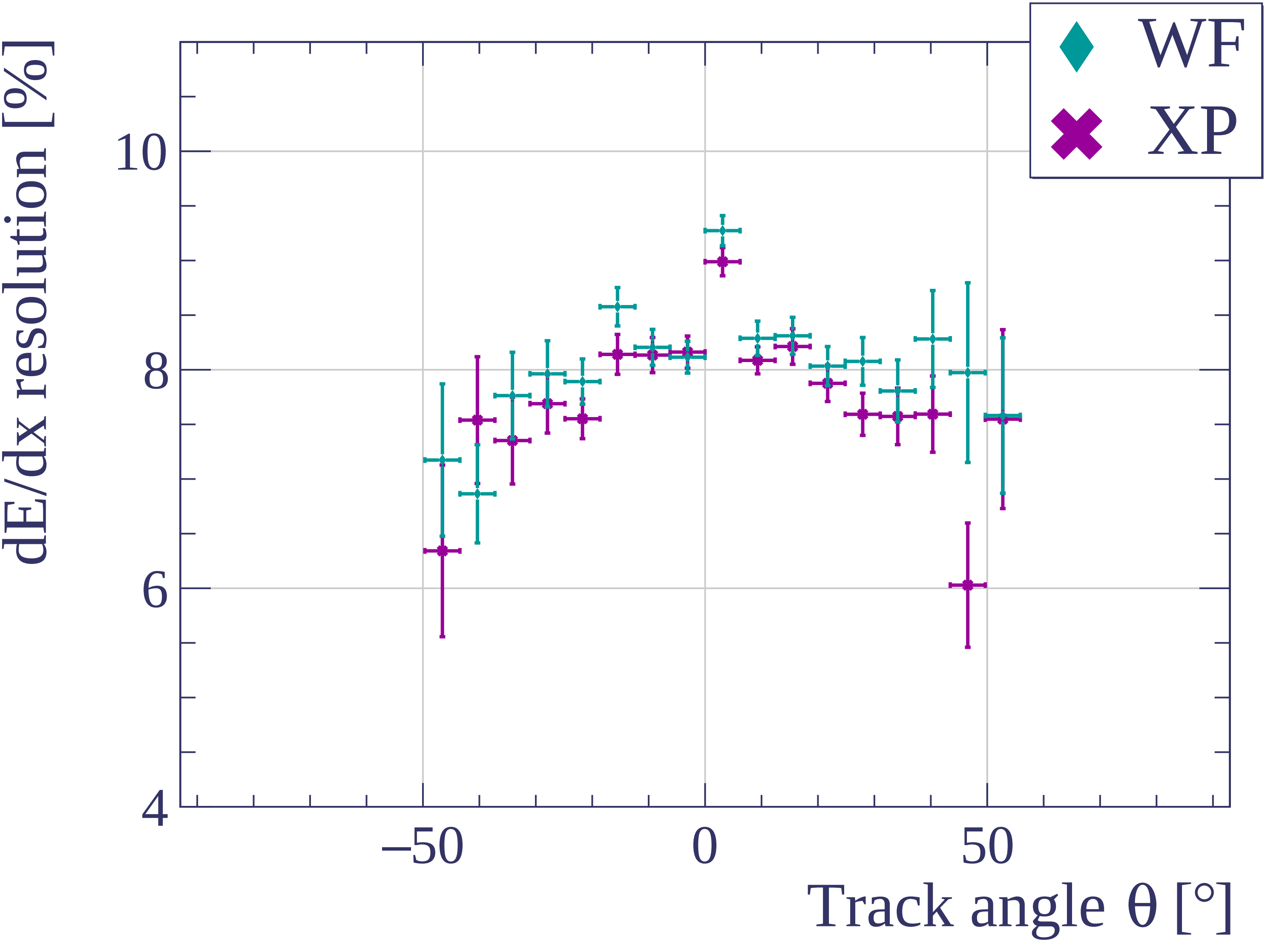


Count
4000
3000
2000
1000
0
−50
0
50
Track angle φ [°] angle

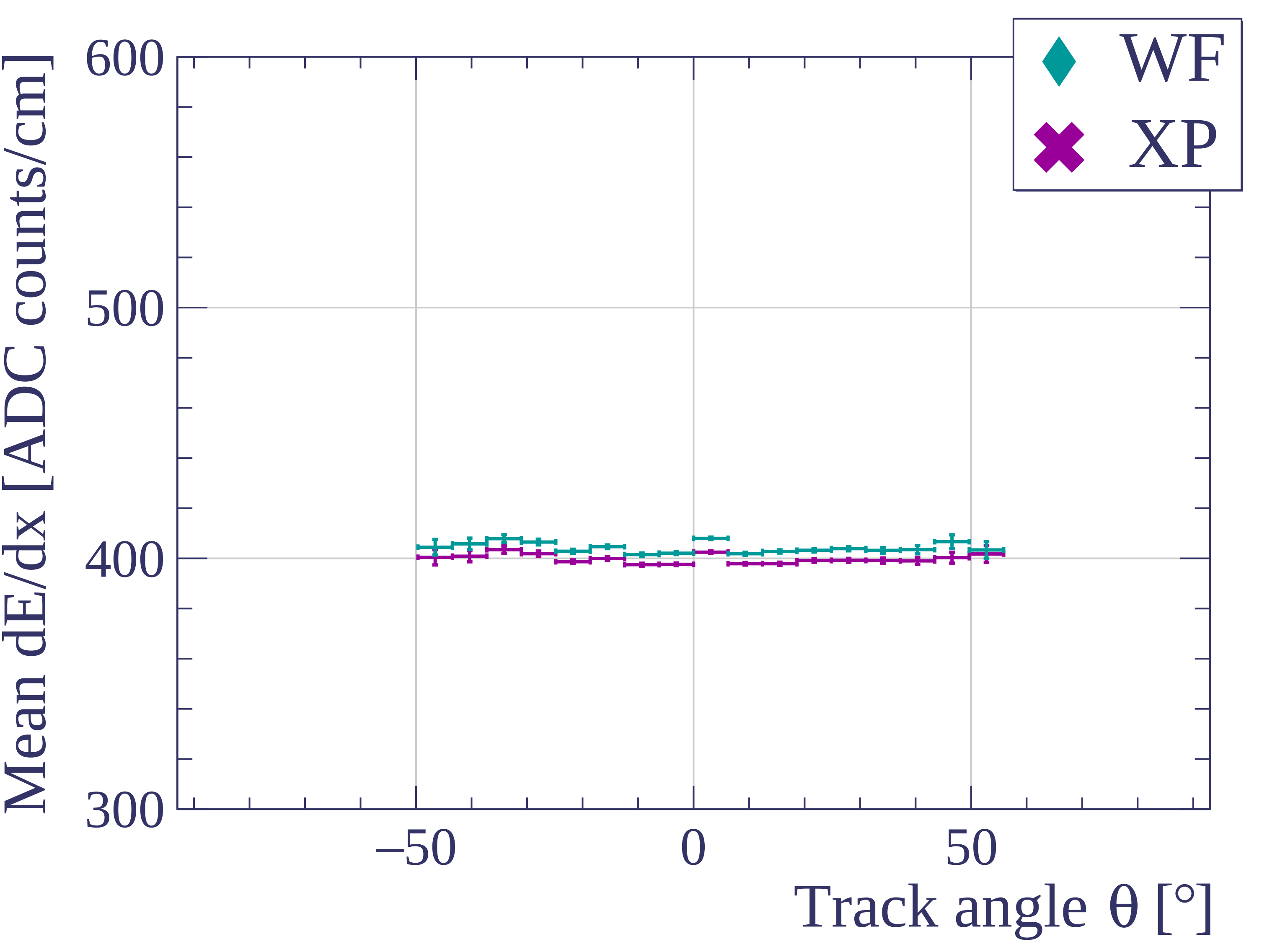


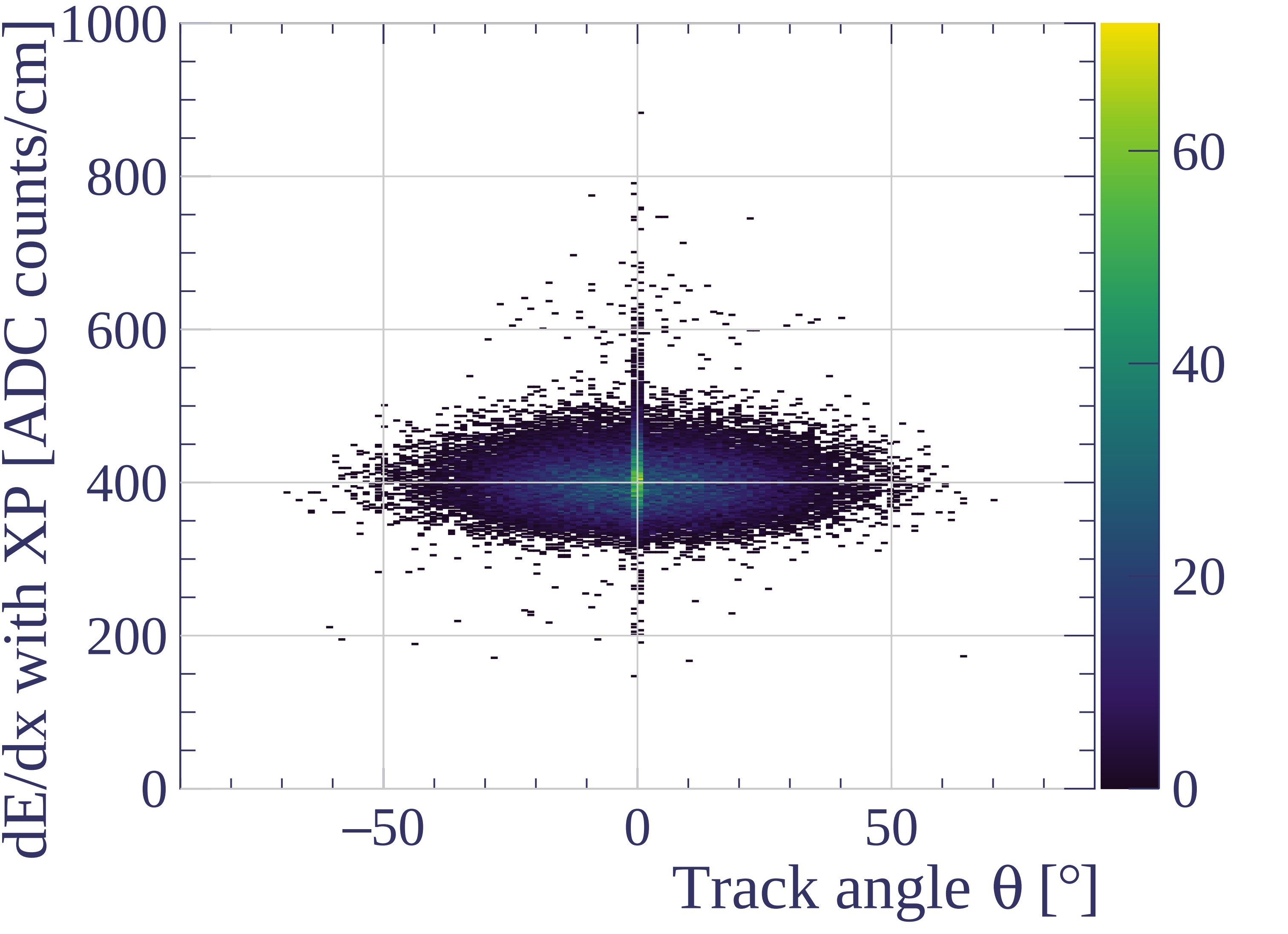
WF
XP
dE/dx resolution [%]
10
8
6
4
0
20
40
60
80
Track angle θ [°]


WF
XP
Mean dE/dx [ADC counts/cm]
600
500
400
300
0
20
40
60
80
Track angle θ [°]

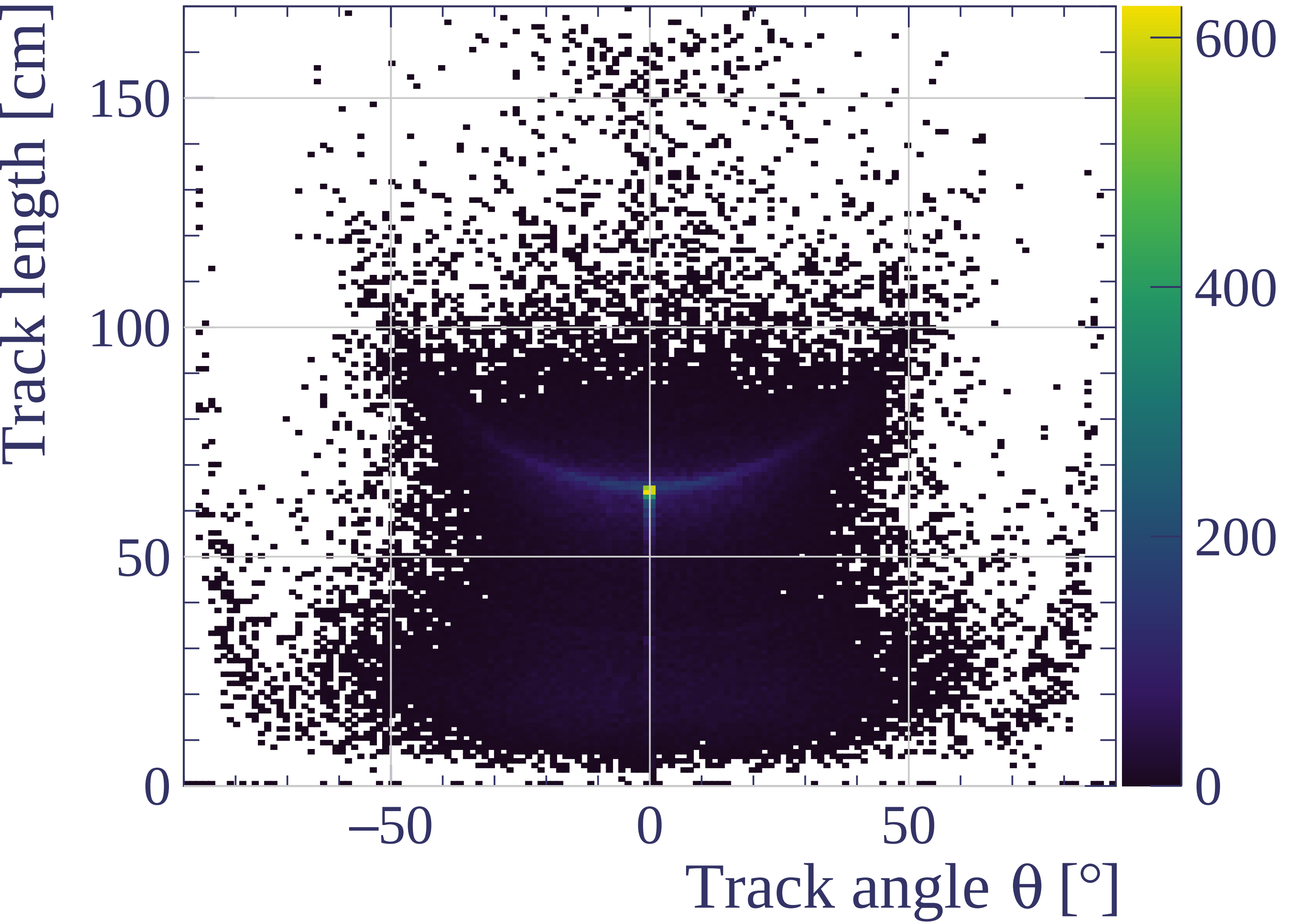


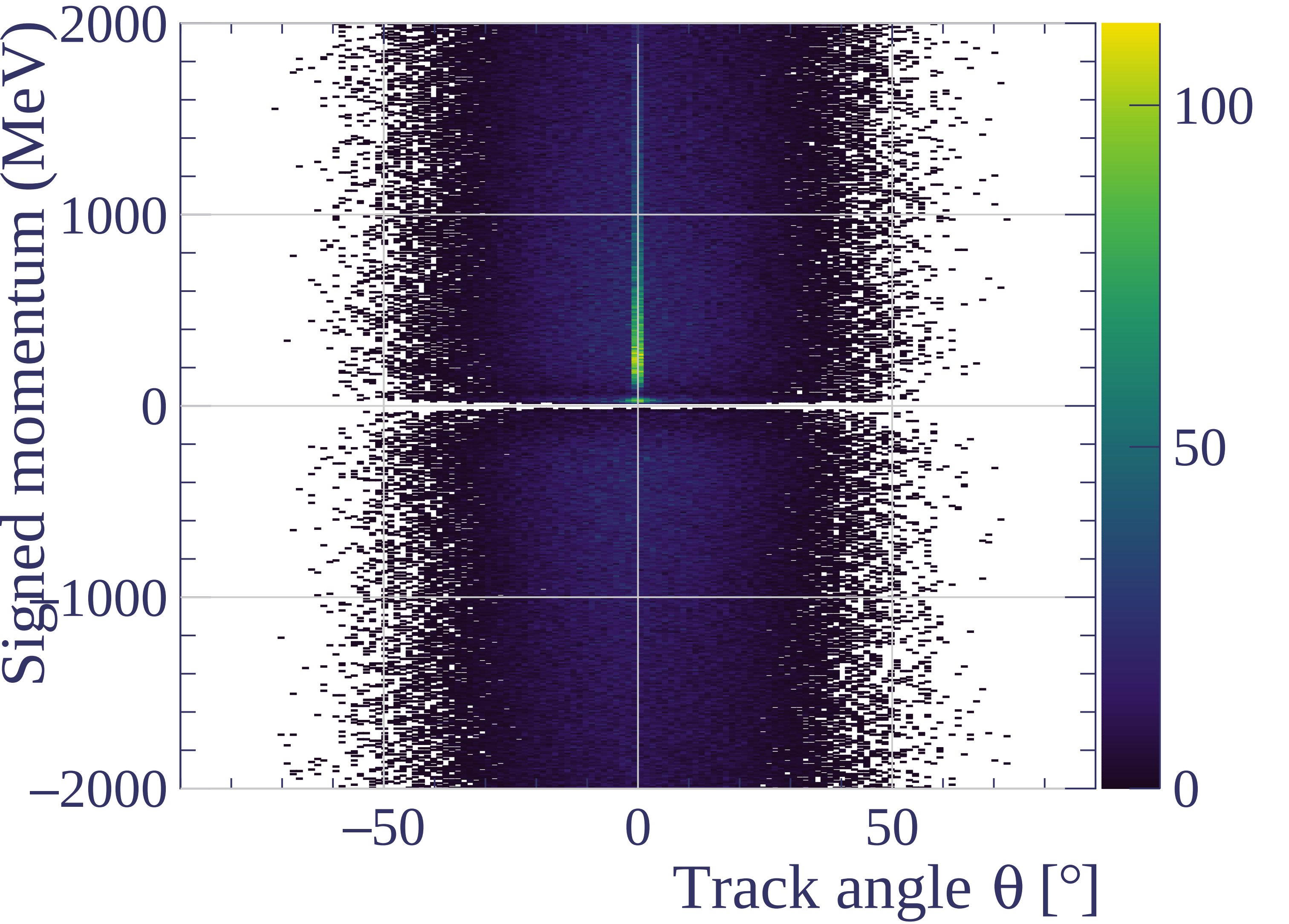
WF
XP
dE/dx resolution [%]
10
8
6
4
−50
0
50
Track angle θ [°]

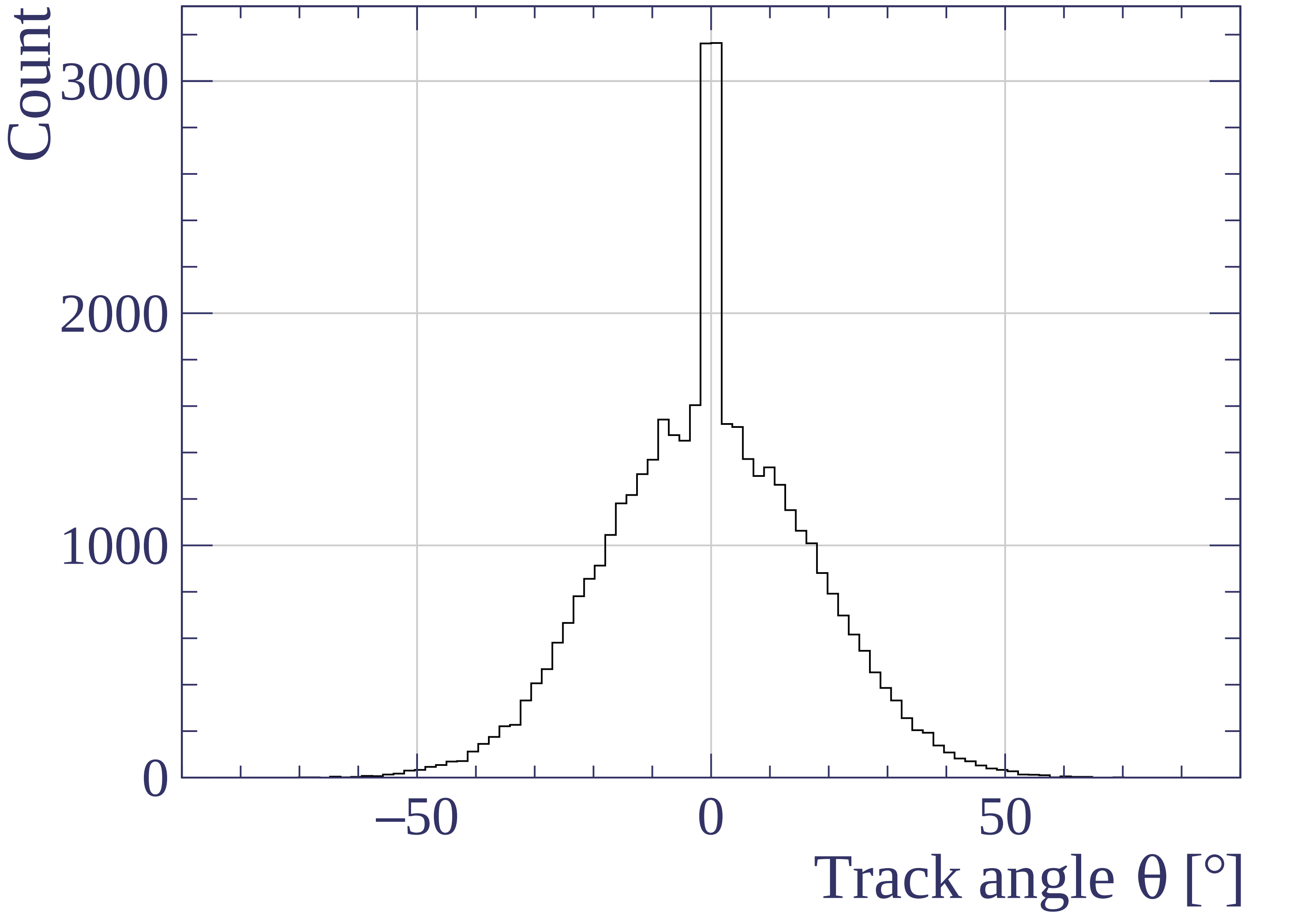

WF
XP
600
500
400
300
Mean dE/dx [ADC counts/cm]
−50
0
50
Track angle θ [°]

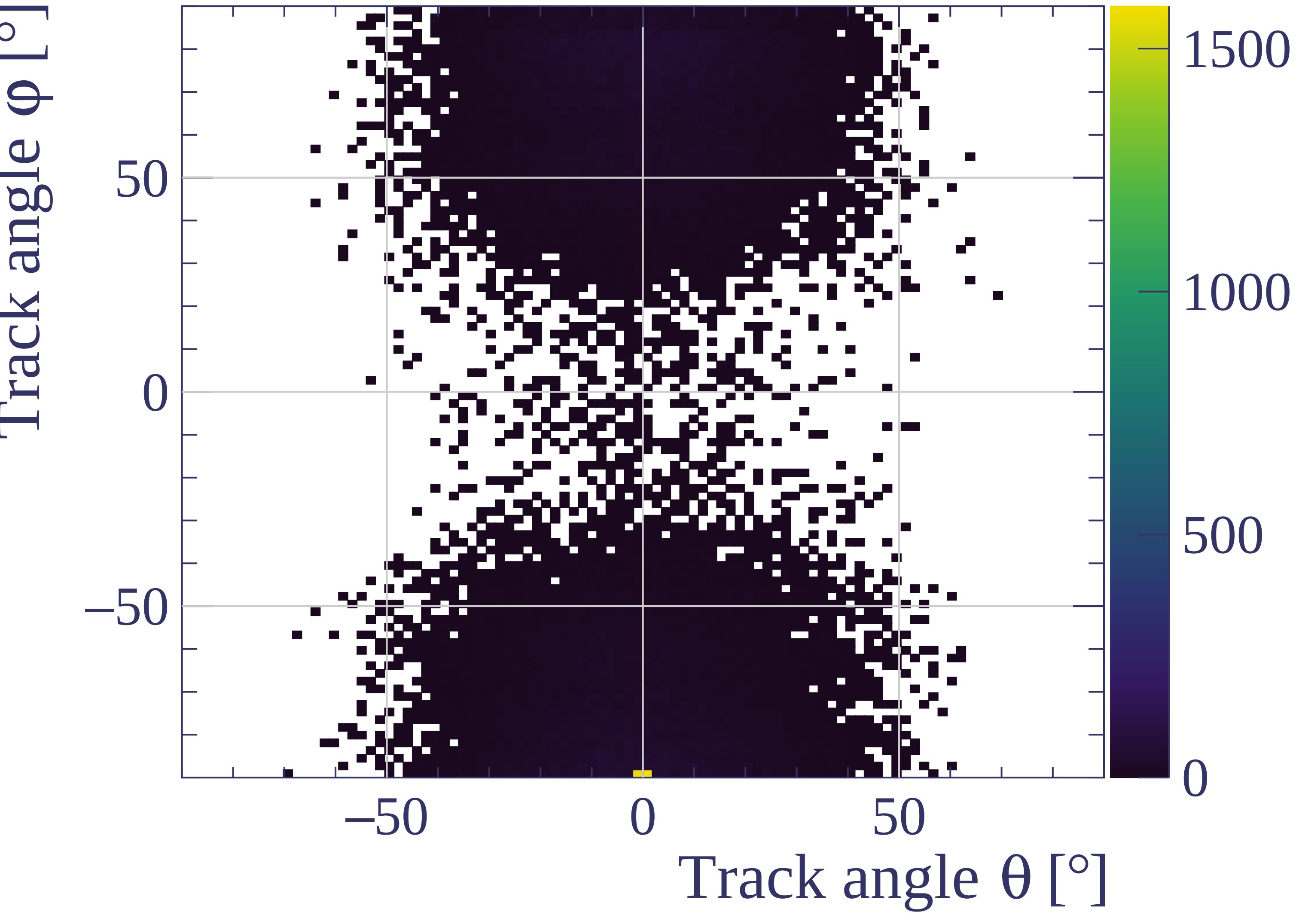


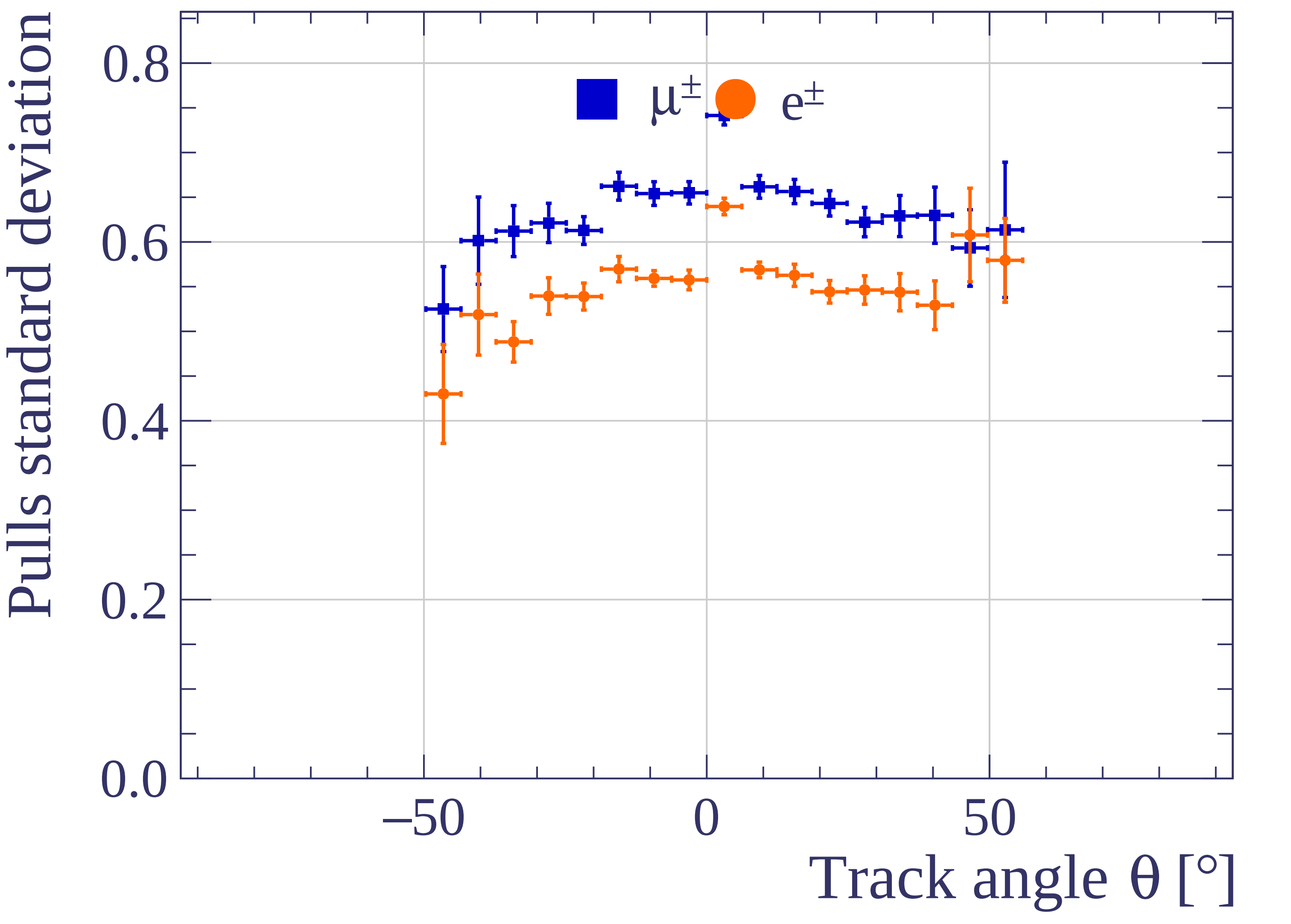
dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−50
0
50
Track angle θ [°]
60
40
20
0


Track length [cm]
150
100
50
0
−50
0
50
Track angle θ [°]
600
400
200
0

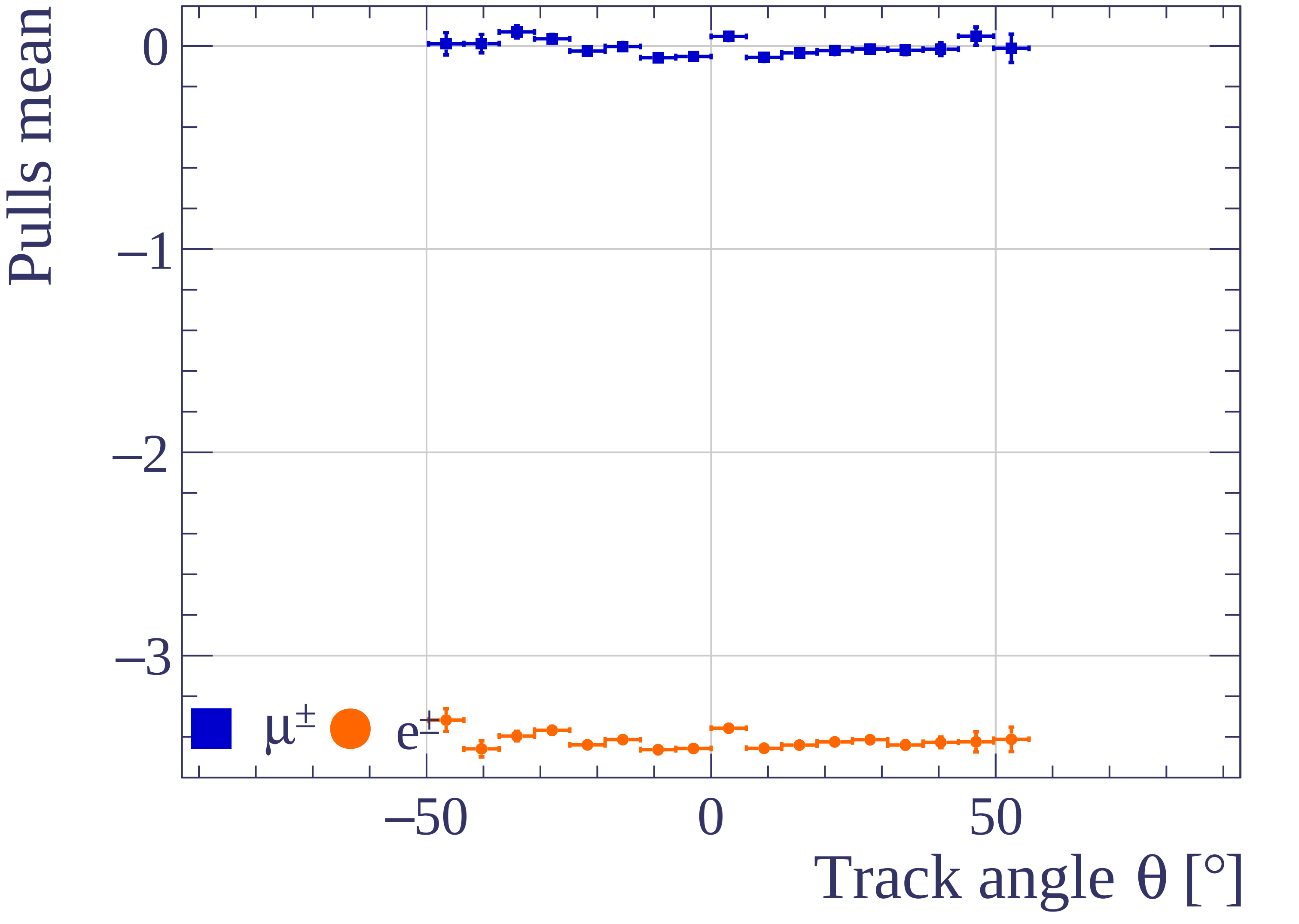

Signed momentum (MeV)
2000
1000
0
−1000
−2000
−50
0
50
Track angle θ [°]
100
50
0


Count
3000
2000
1000
0
−50
0
50
Track angle θ [°]

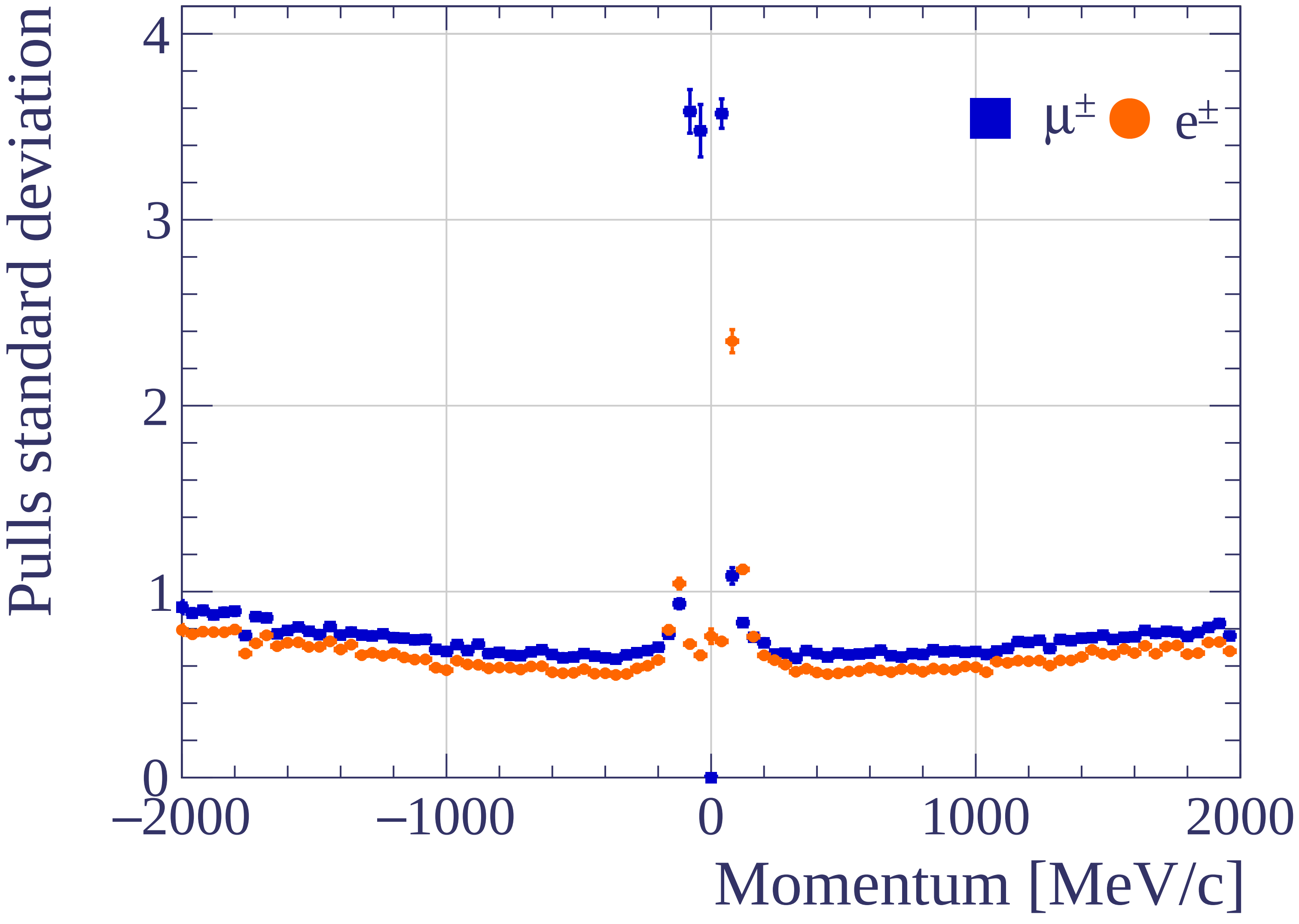

Track angle φ [°]
50
0
−50
−50
0
50
Track angle θ [°]
1500
1000
500
0



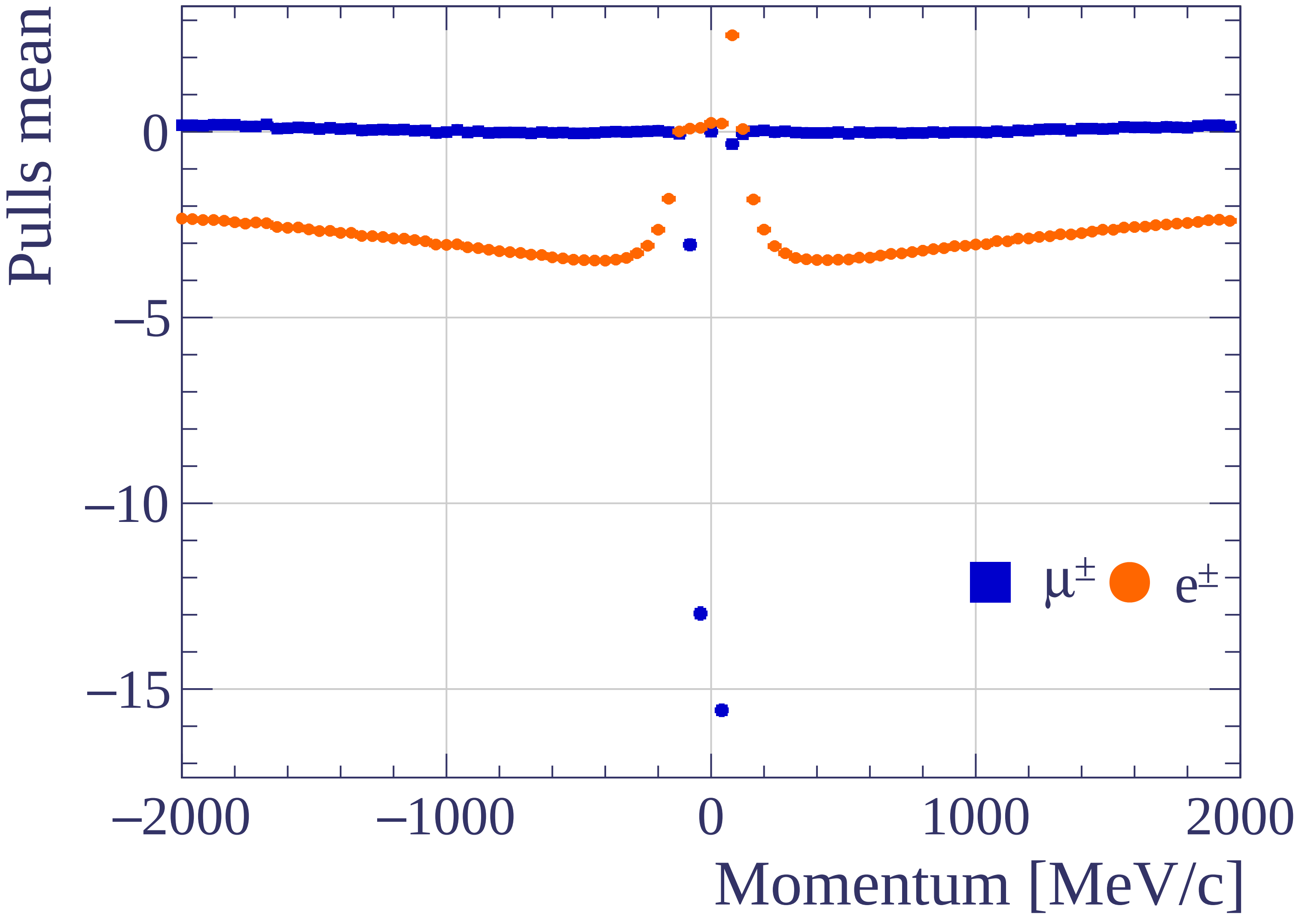
Pulls standard deviation
0.0
0.2
0.4
0.6
0.8
−50
0
50
Track angle θ [°]
μ±
e±

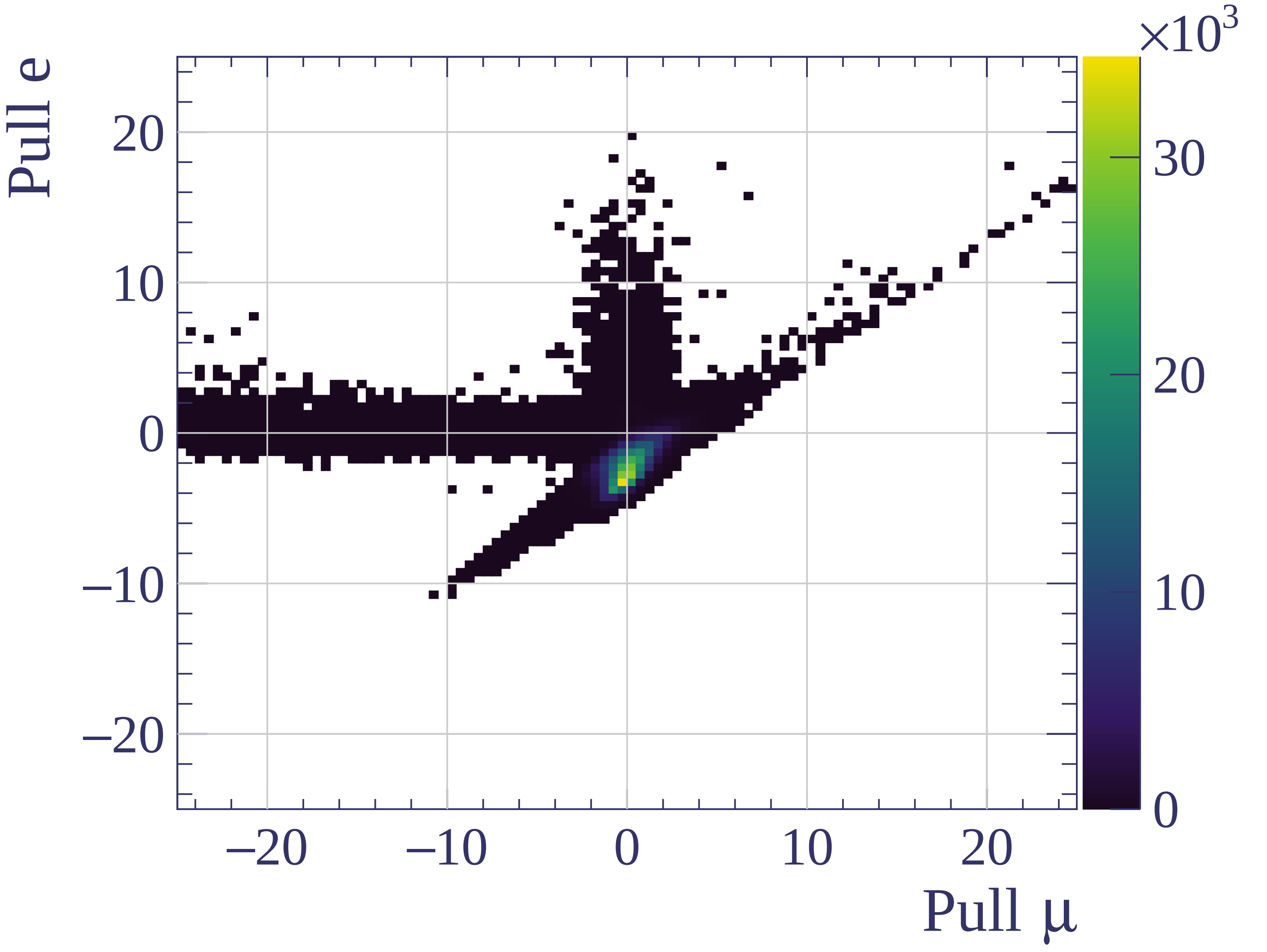


Pulls mean
0
−1
−2
−3
μ±
e±
−50
0
50
Track angle θ [°]

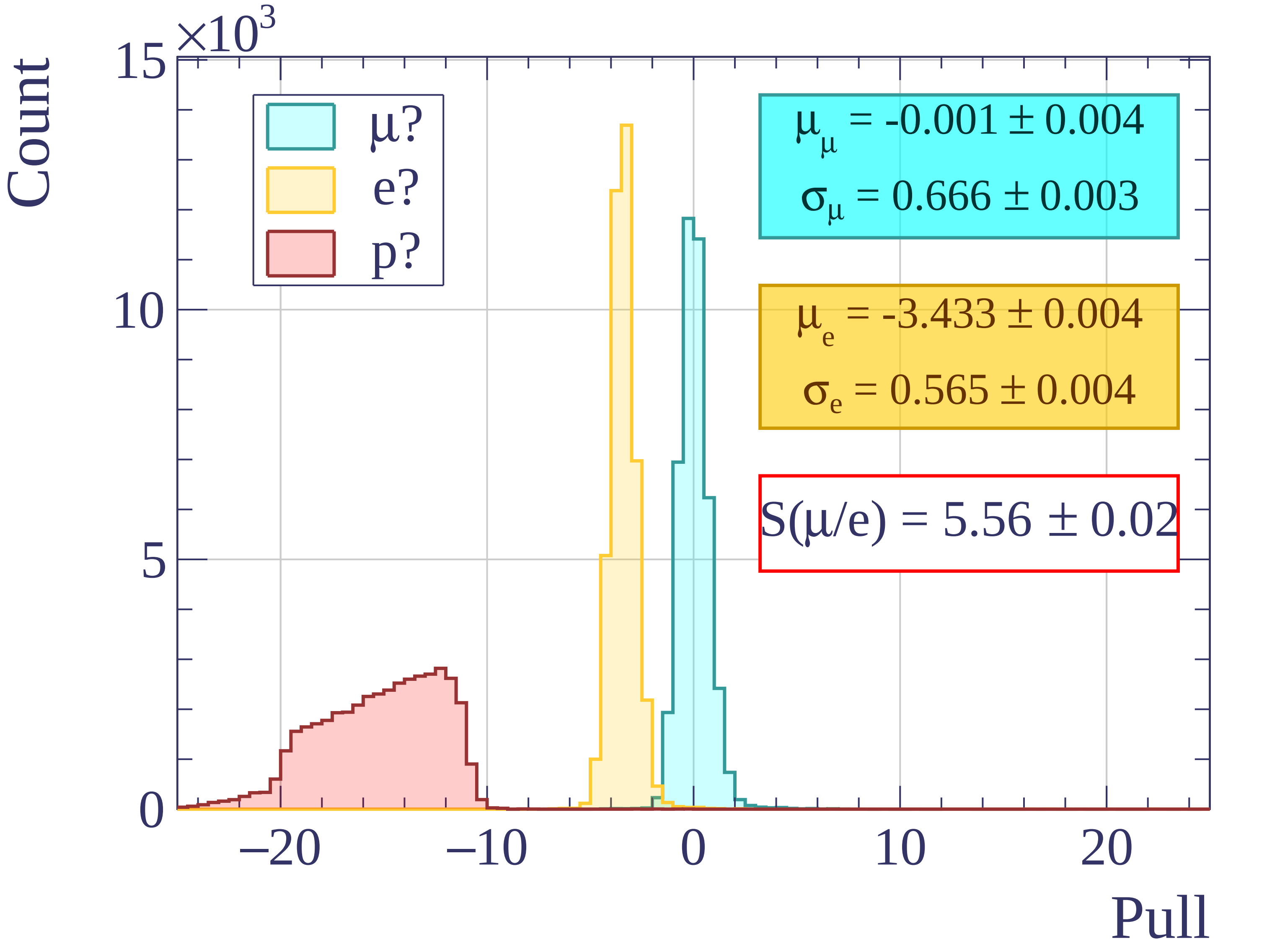


Pulls standard deviation
Momentum [MeV/c]
μ±
e±
4
3
2
1
0
−2000
−1000
0
1000
2000

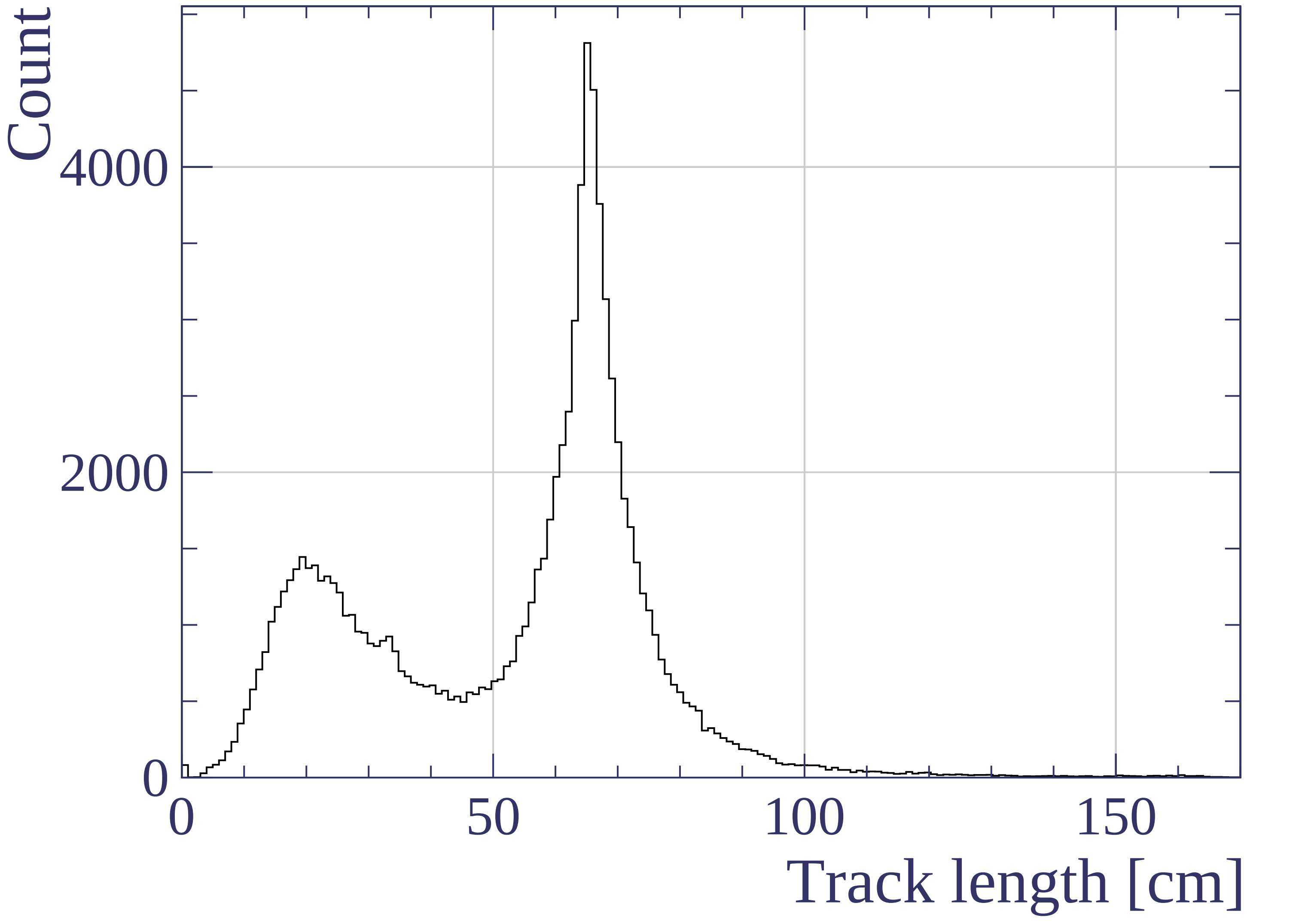

Pulls mean
0
−5
−10
−15
−2000
−1000
0
1000
2000
Momentum [MeV/c]
$\mu^\pm$
$e^\pm$

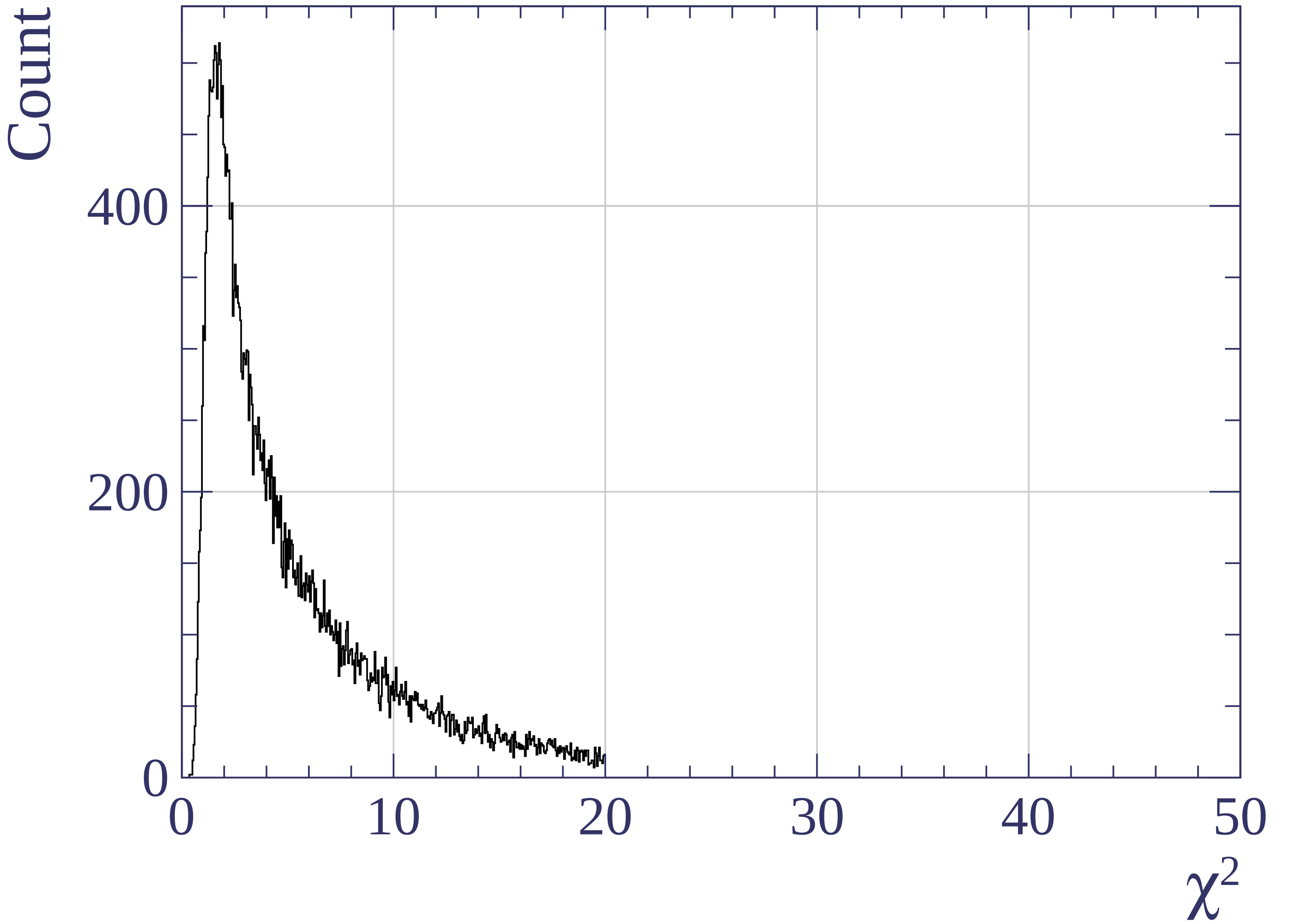


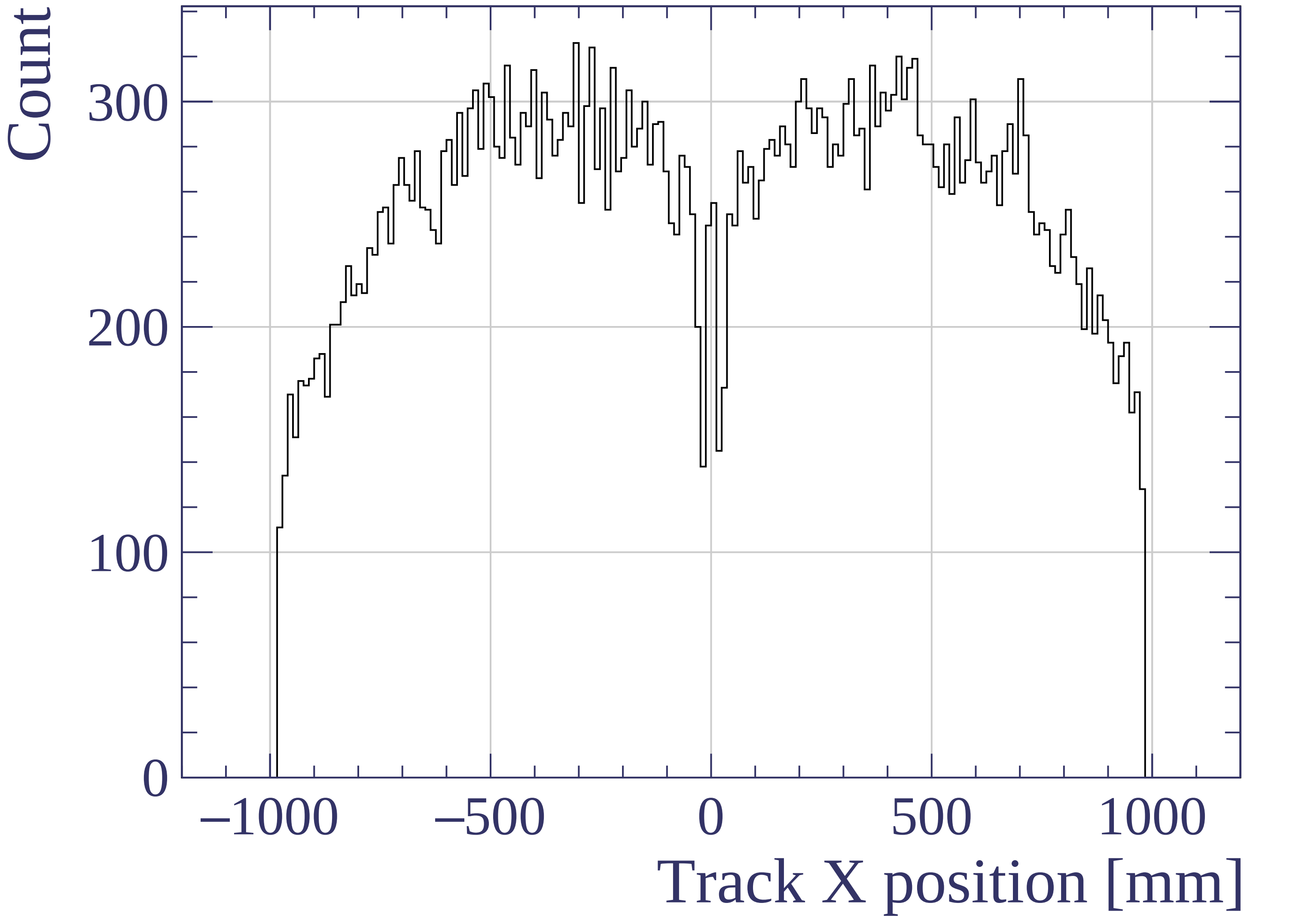
Pull e
20
10
0
−10
−20
−20
−10
0
10
20
Pull μ
×10³
30
20
10
0

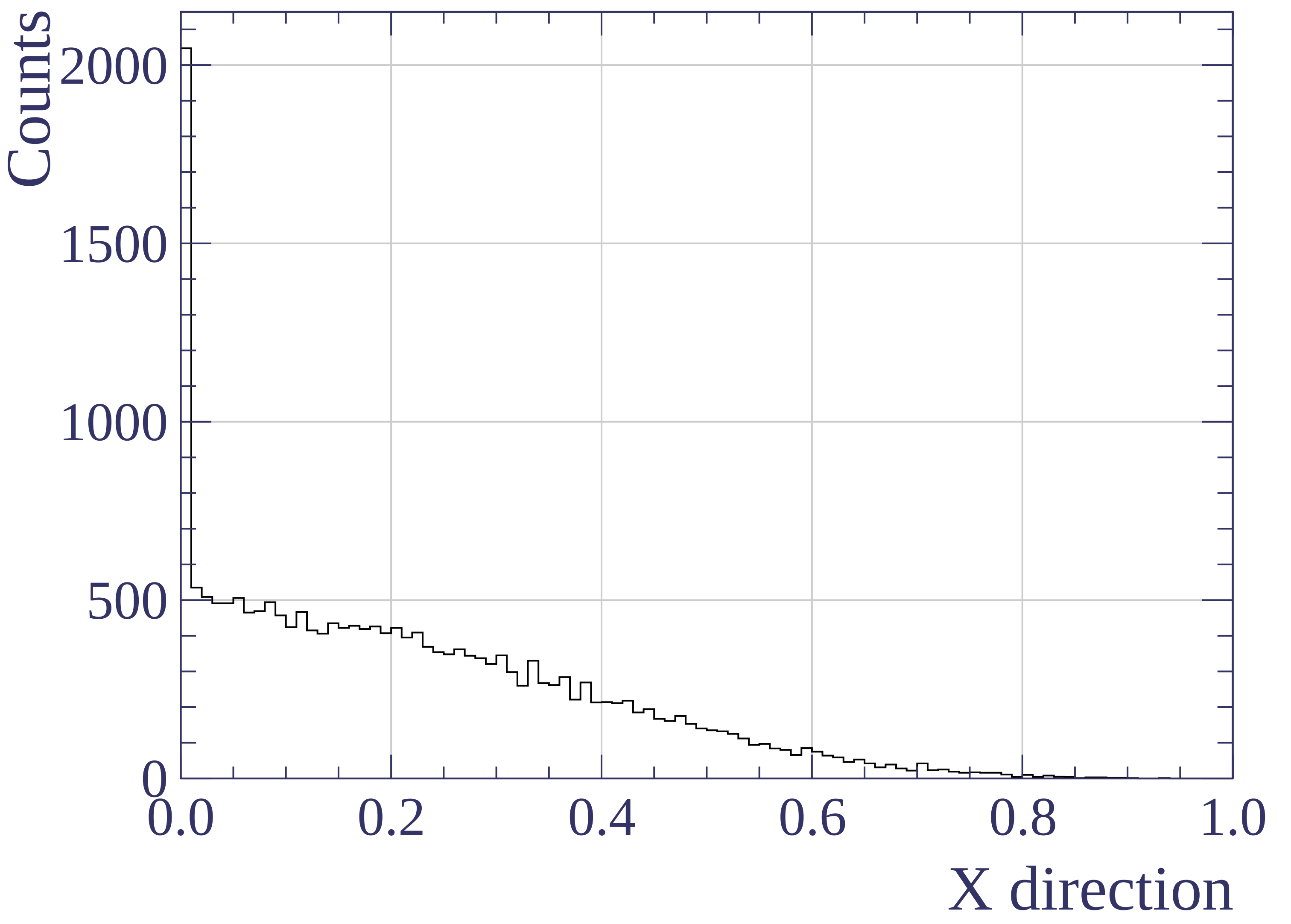


×10³
15
10
5
0
Count
μ?
e?
p?
μ_μ = -0.001 ± 0.004
σ_μ = 0.666 ± 0.003
μ_e = -3.433 ± 0.004
σ_e = 0.565 ± 0.004
S(μ/e) = 5.56 ± 0.02
−20
−10
0
10
20
Pull

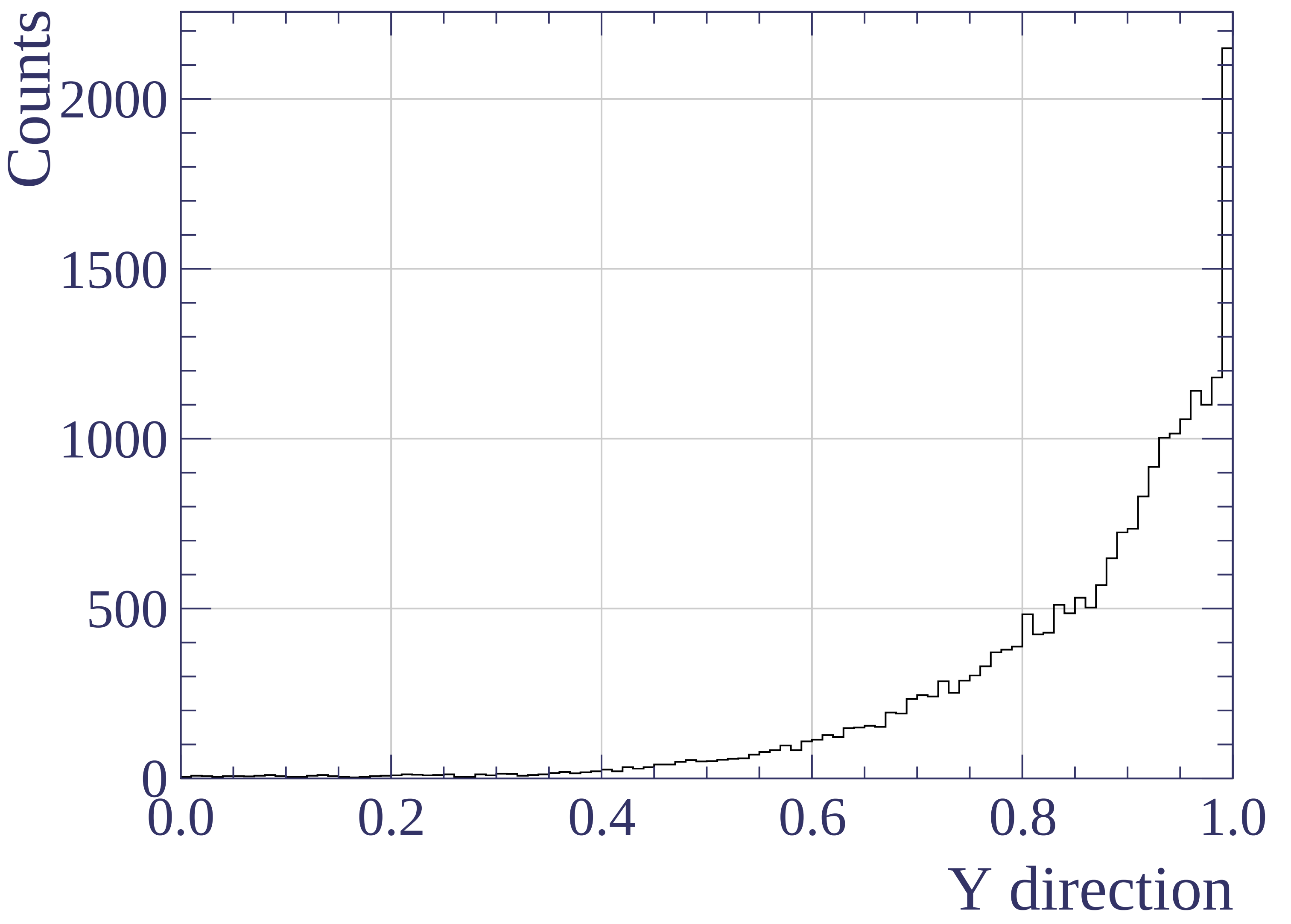

Count
4000
2000
0
0
50
100
150
Track length [cm]

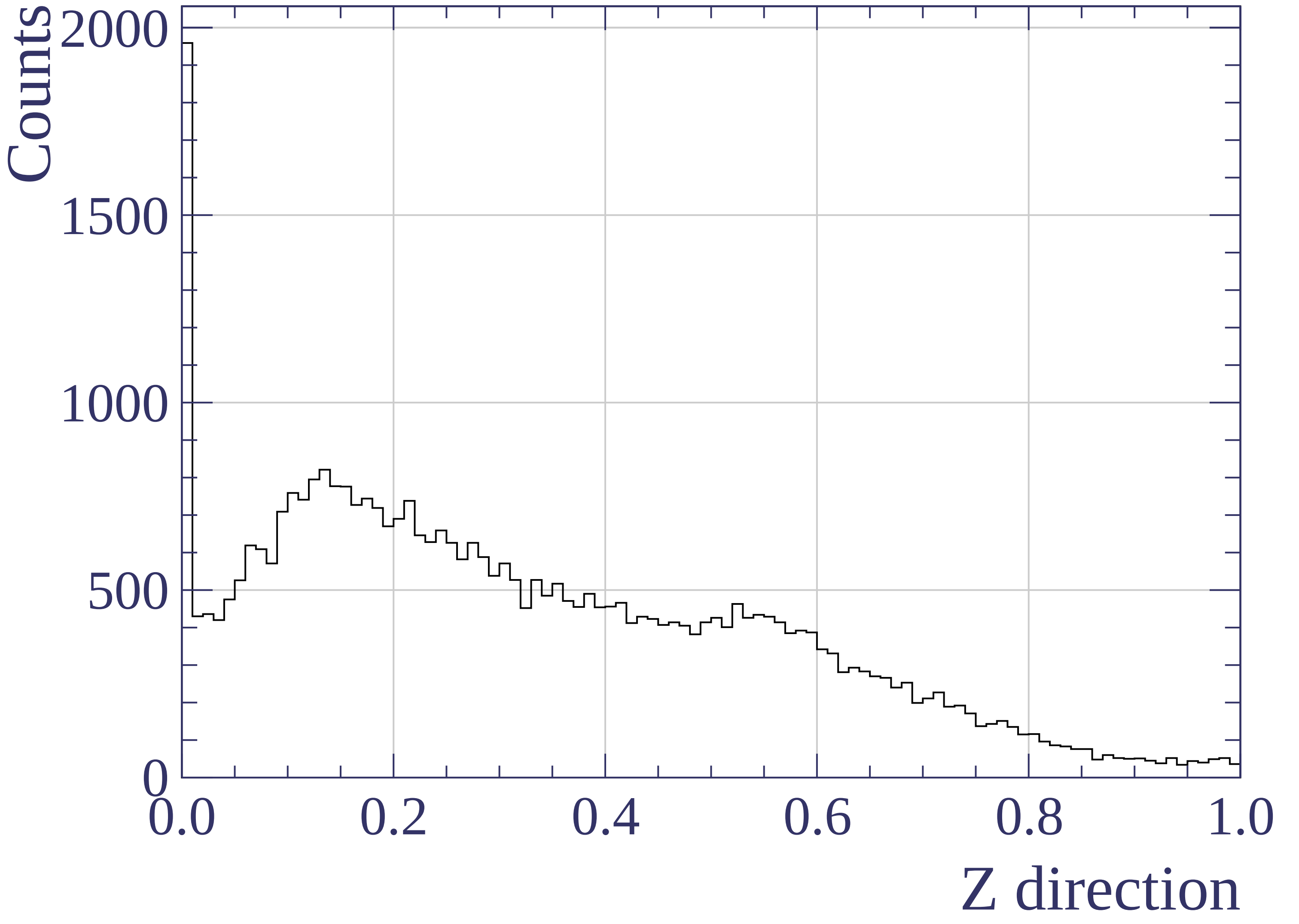

Count
400
200
0
0
10
20
30
40
50
χ^2


Count
300
200
100
0
−1000
−500
0
500
1000
Track X position [mm]

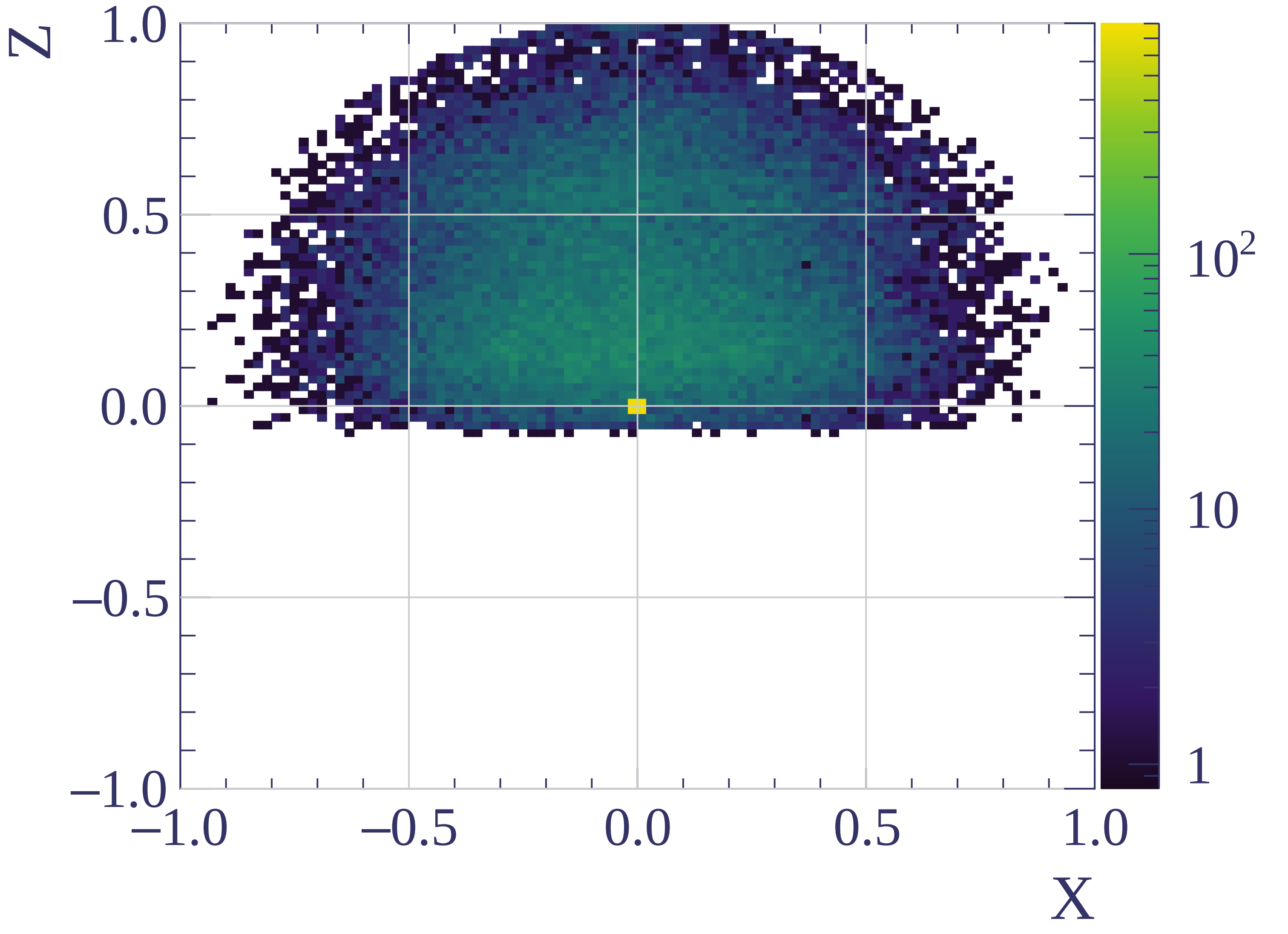


Counts
2000
1500
1000
500
0
0.0
0.2
0.4
0.6
0.8
1.0
X direction

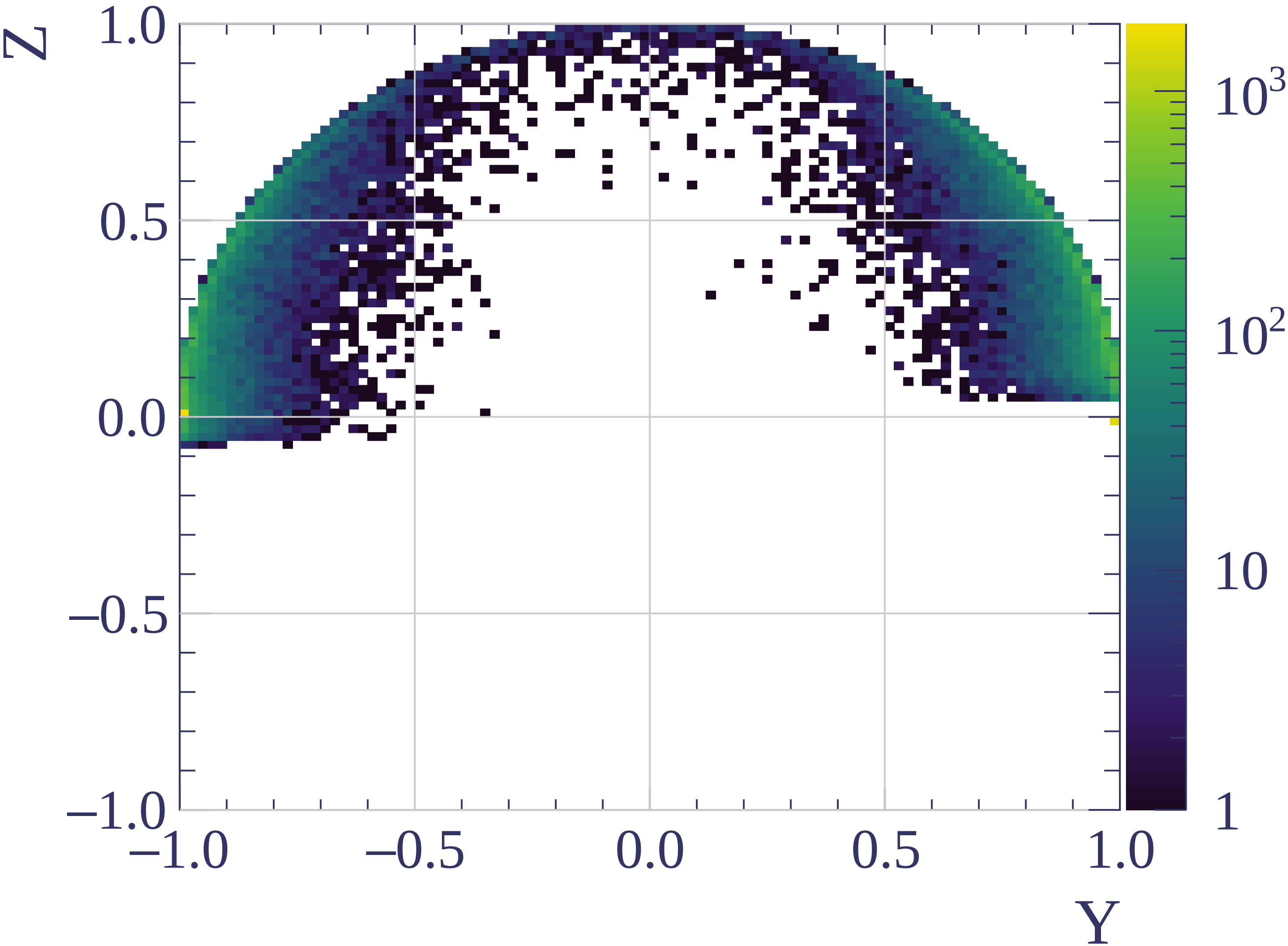

Counts
2000
1500
1000
500
0
0.0
0.2
0.4
0.6
0.8
1.0
Y direction



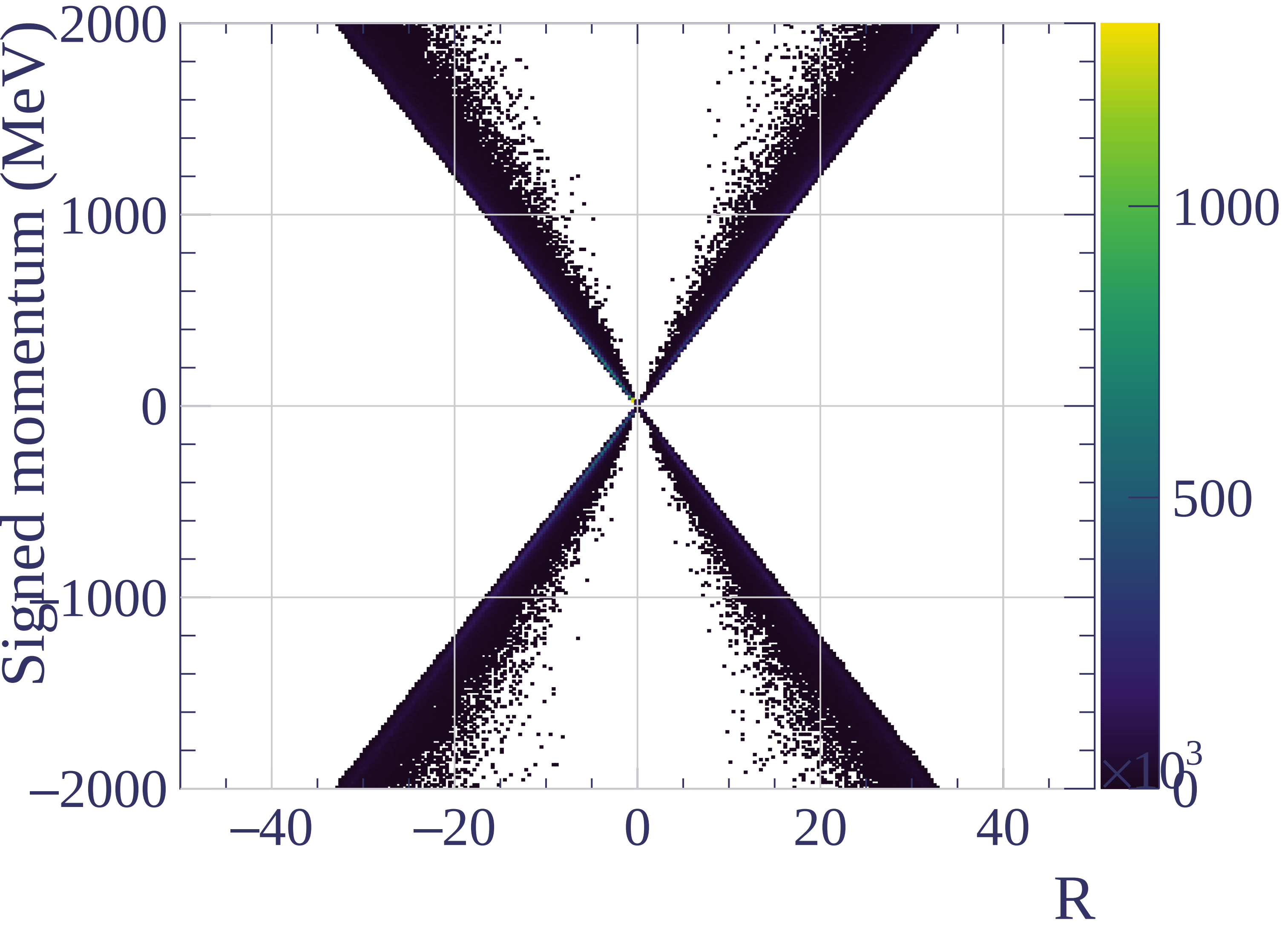
Counts
2000
1500
1000
500
0
0.0
0.2
0.4
0.6
0.8
1.0
Z direction



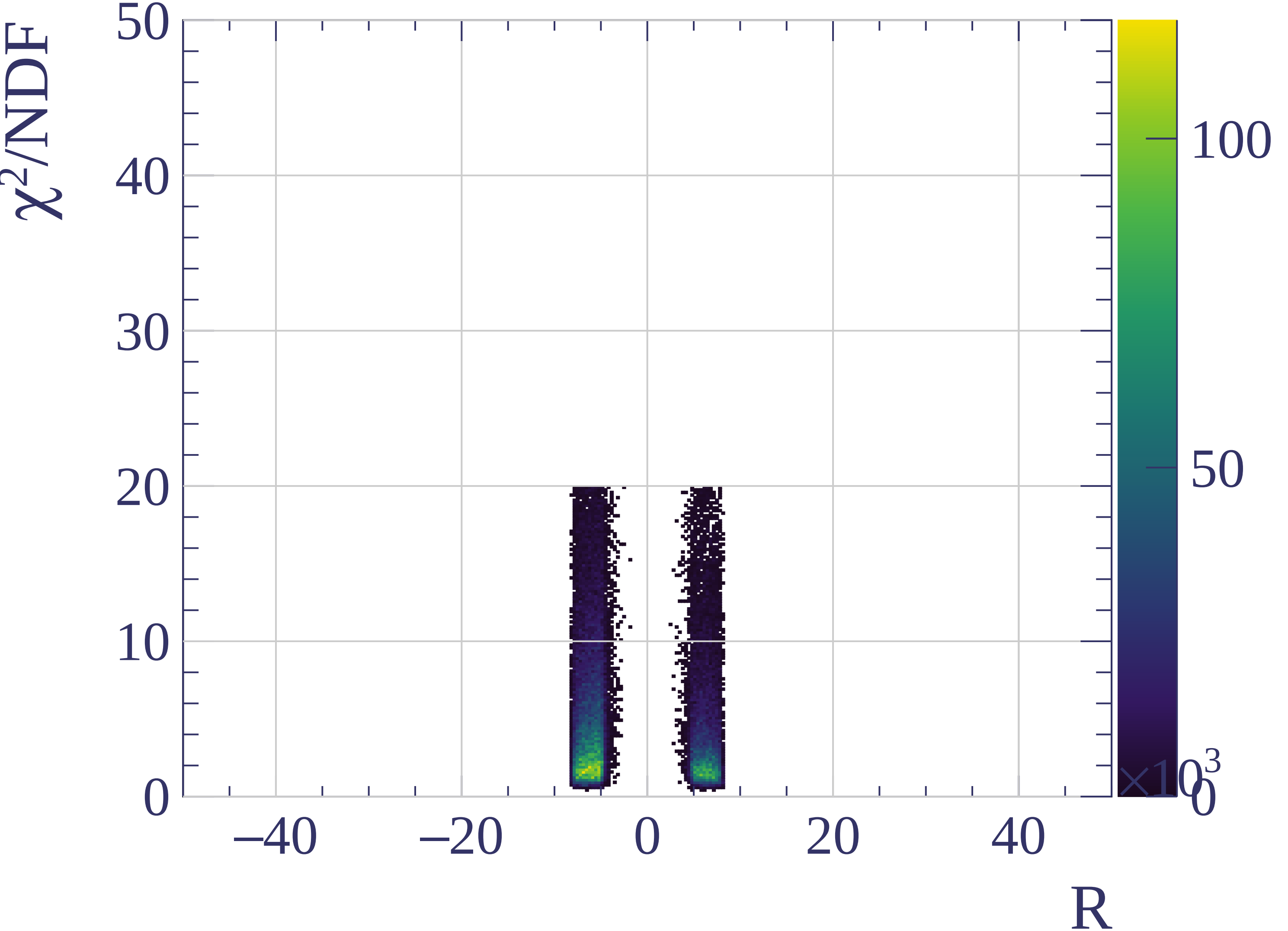
Z
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
X
10^2
10
1



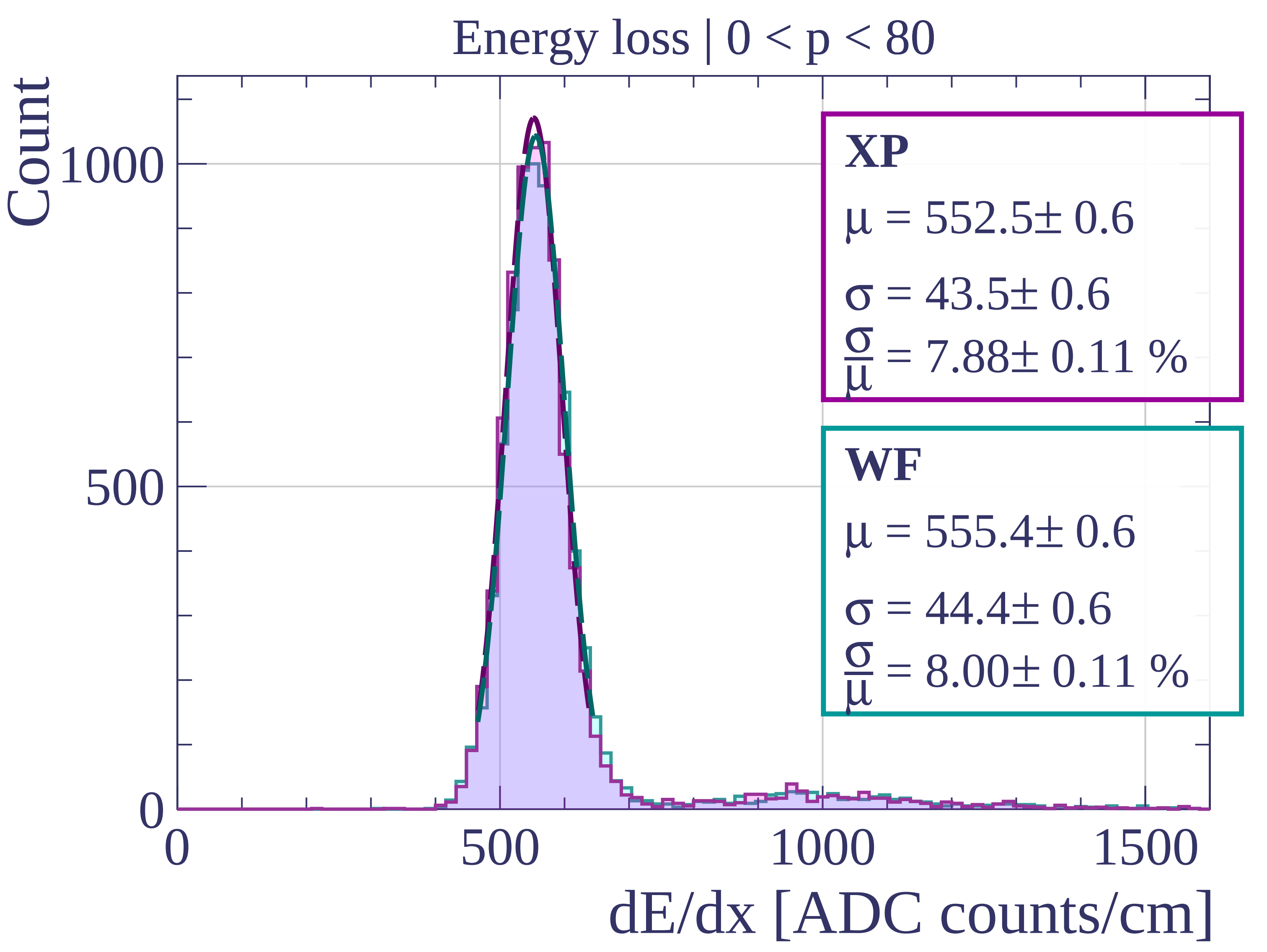
Z
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
Y
10^3
10^2
10
1

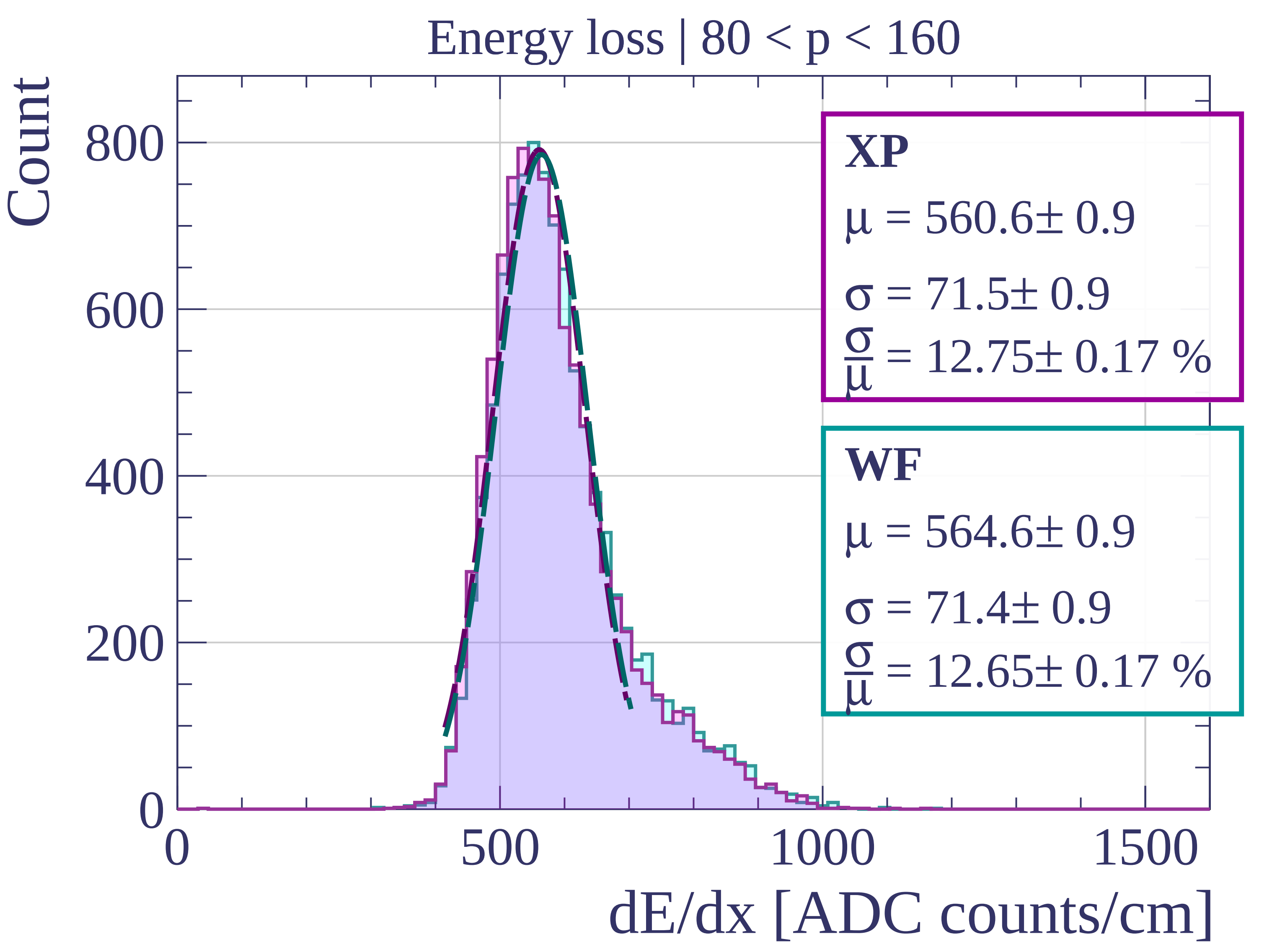


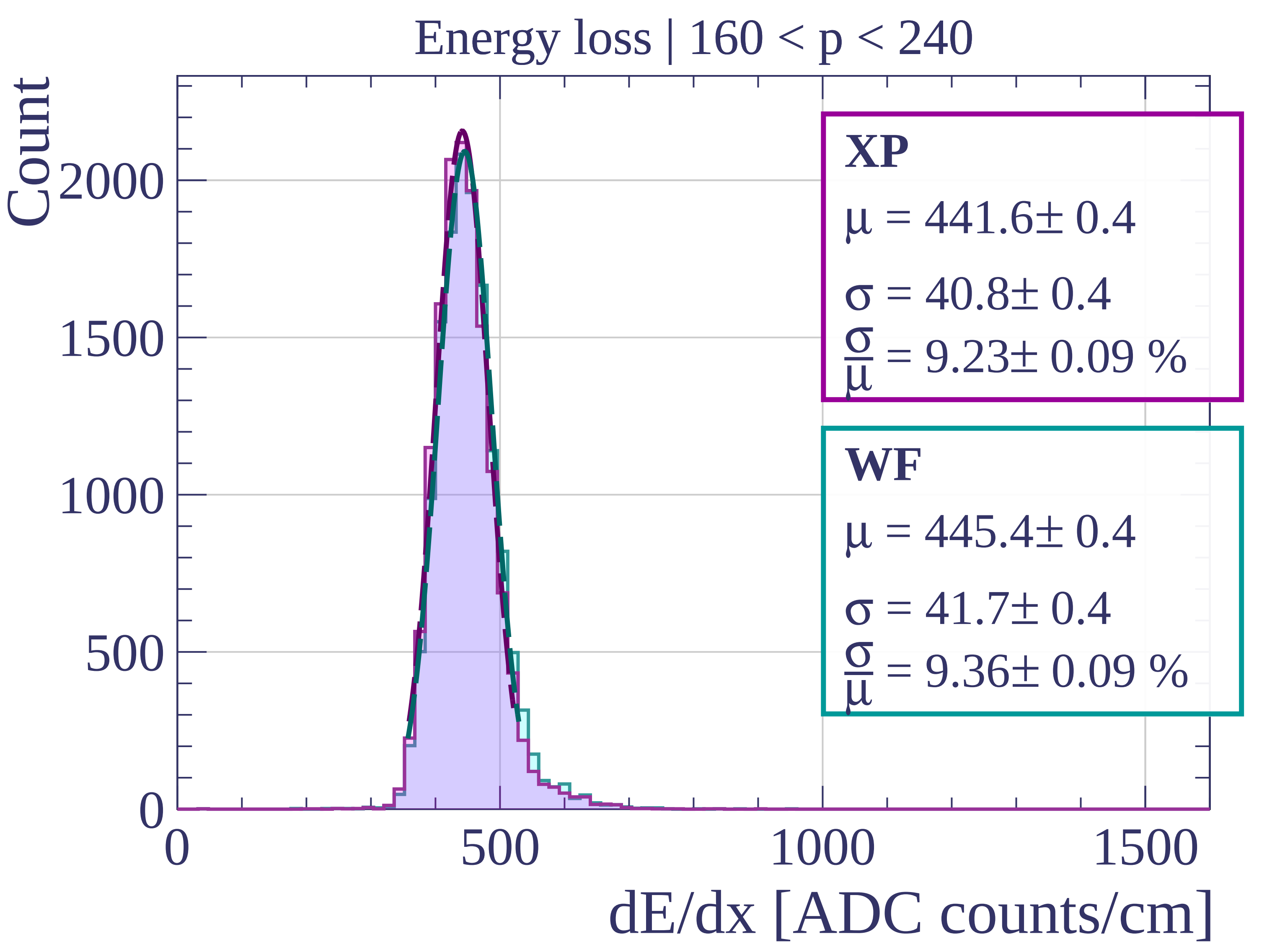
Signed momentum (MeV)
2000
1000
0
−1000
−2000
−40
−20
0
20
40
R
1000
500
0
×10³



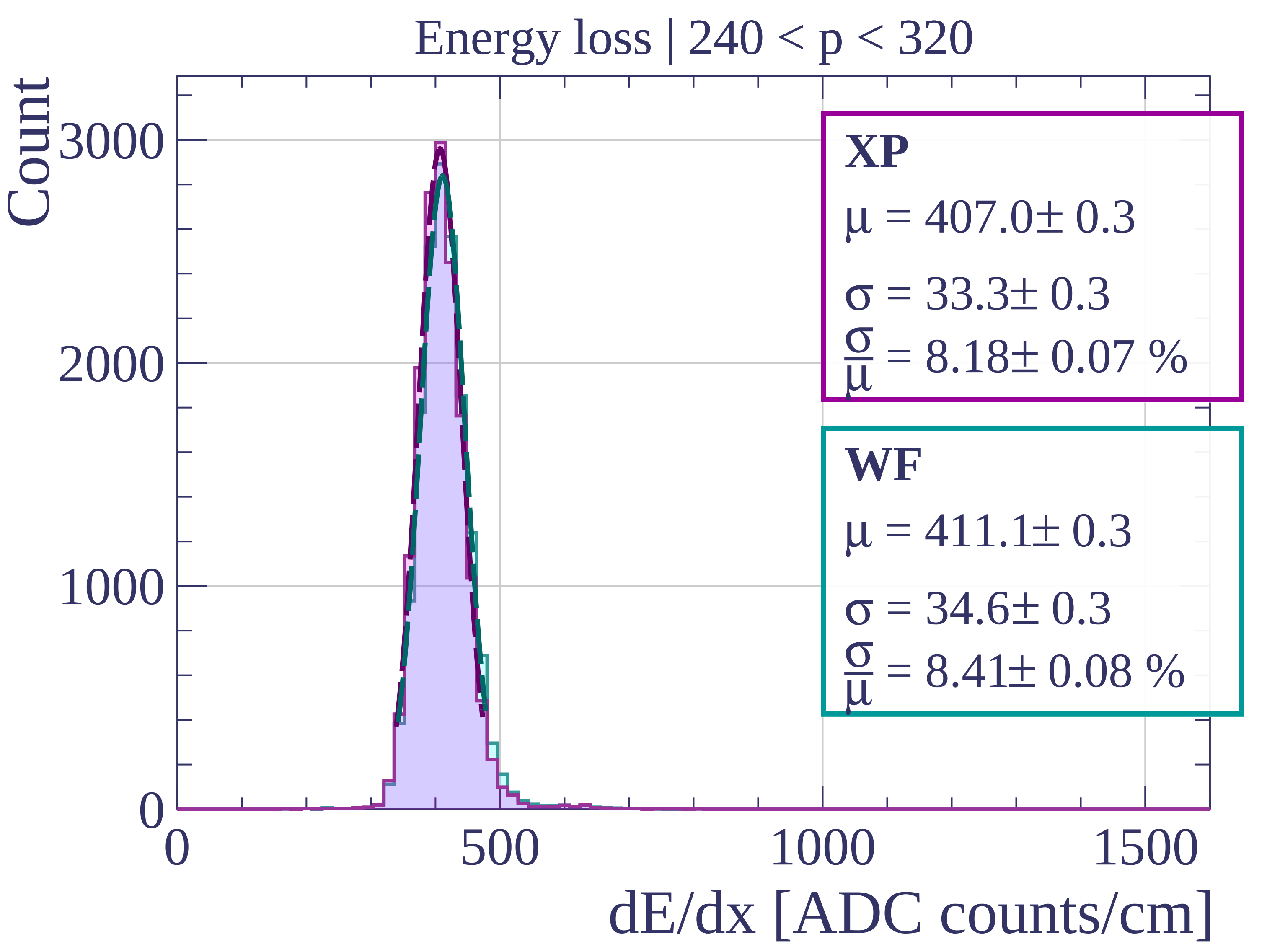
χ^2/NDF
50
40
30
20
10
0
−40
−20
0
20
40
R
100
50
0
$\times10^3$

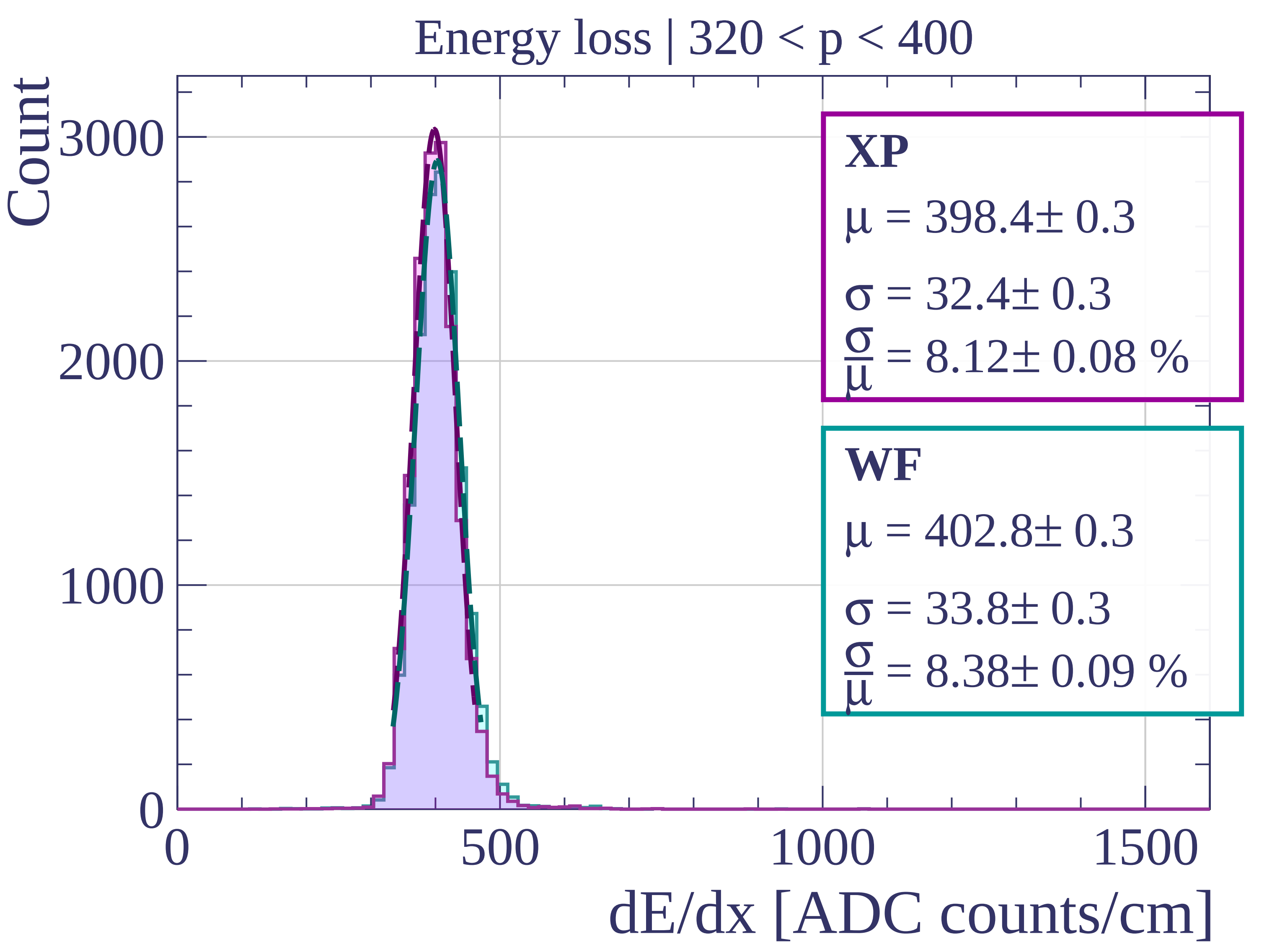

Energy loss | 0 < p < 80
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 552.5± 0.6
σ = 43.5± 0.6
σ/μ = 7.88± 0.11 %
WF
μ = 555.4± 0.6
σ = 44.4± 0.6
σ/μ = 8.00± 0.11 %


Energy loss | 80 < p < 160
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 560.6± 0.9
σ = 71.5± 0.9
σ/μ = 12.75± 0.17 %
WF
μ = 564.6± 0.9
σ = 71.4± 0.9
σ/μ = 12.65± 0.17 %

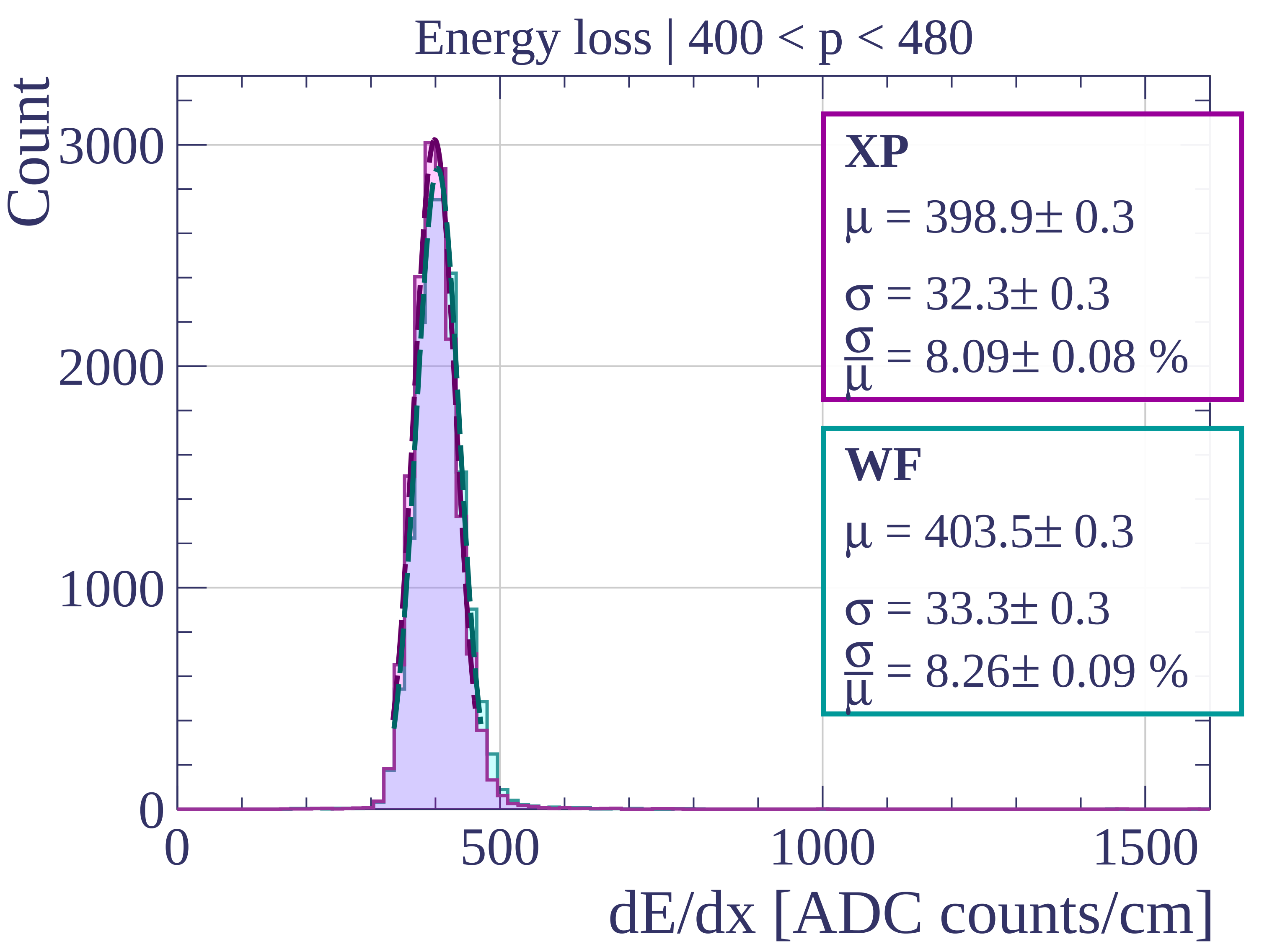


Energy loss | 160 < p < 240
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
2000
0
500
1000
1500
XP
μ = 441.6± 0.4
σ = 40.8± 0.4
σ/μ = 9.23± 0.09 %
WF
μ = 445.4± 0.4
σ = 41.7± 0.4
σ/μ = 9.36± 0.09 %

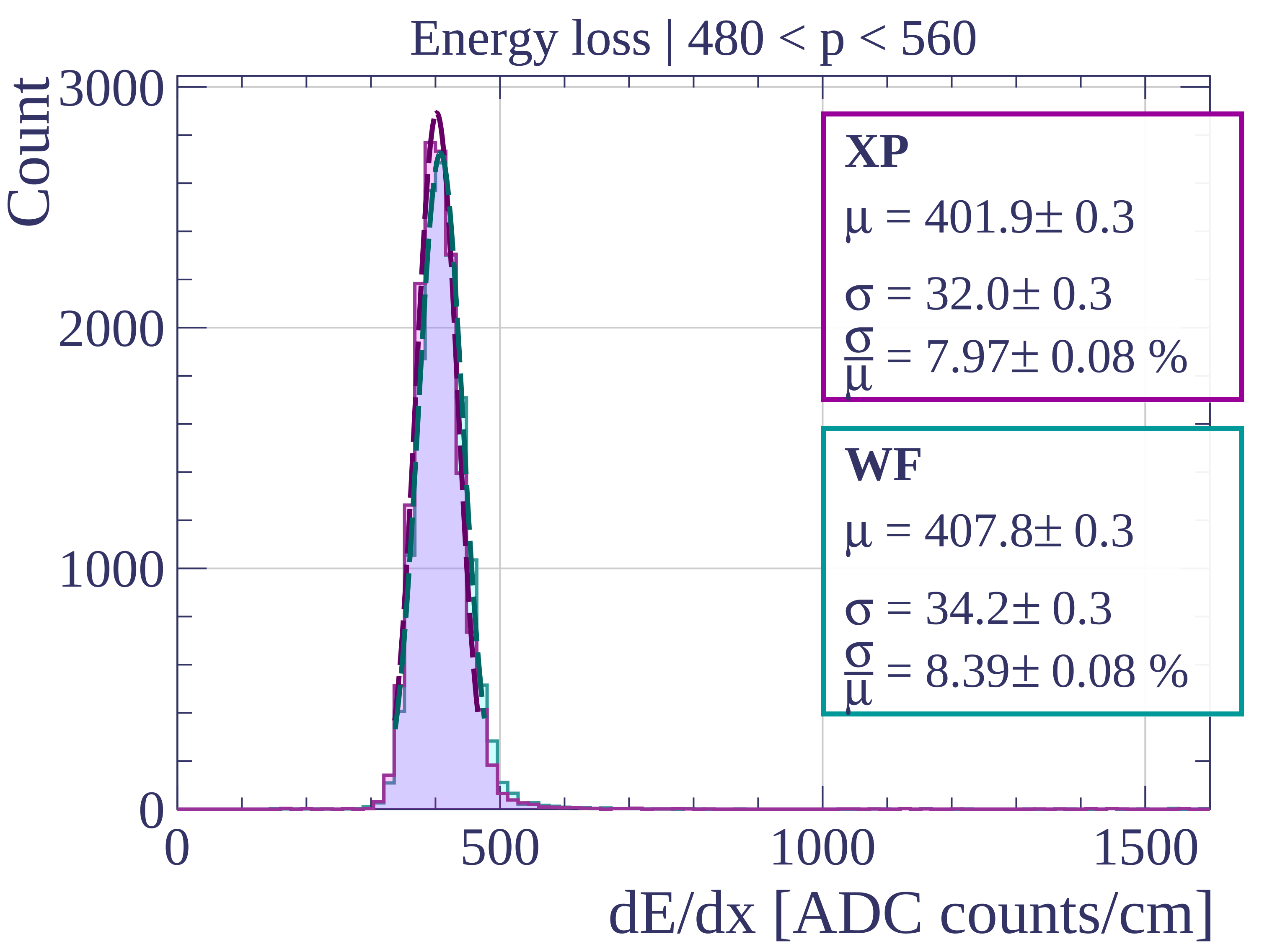

Energy loss | 240 < p < 320
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 407.0± 0.3
σ = 33.3± 0.3
σ/μ = 8.18± 0.07 %
WF
μ = 411.1± 0.3
σ = 34.6± 0.3
σ/μ = 8.41± 0.08 %



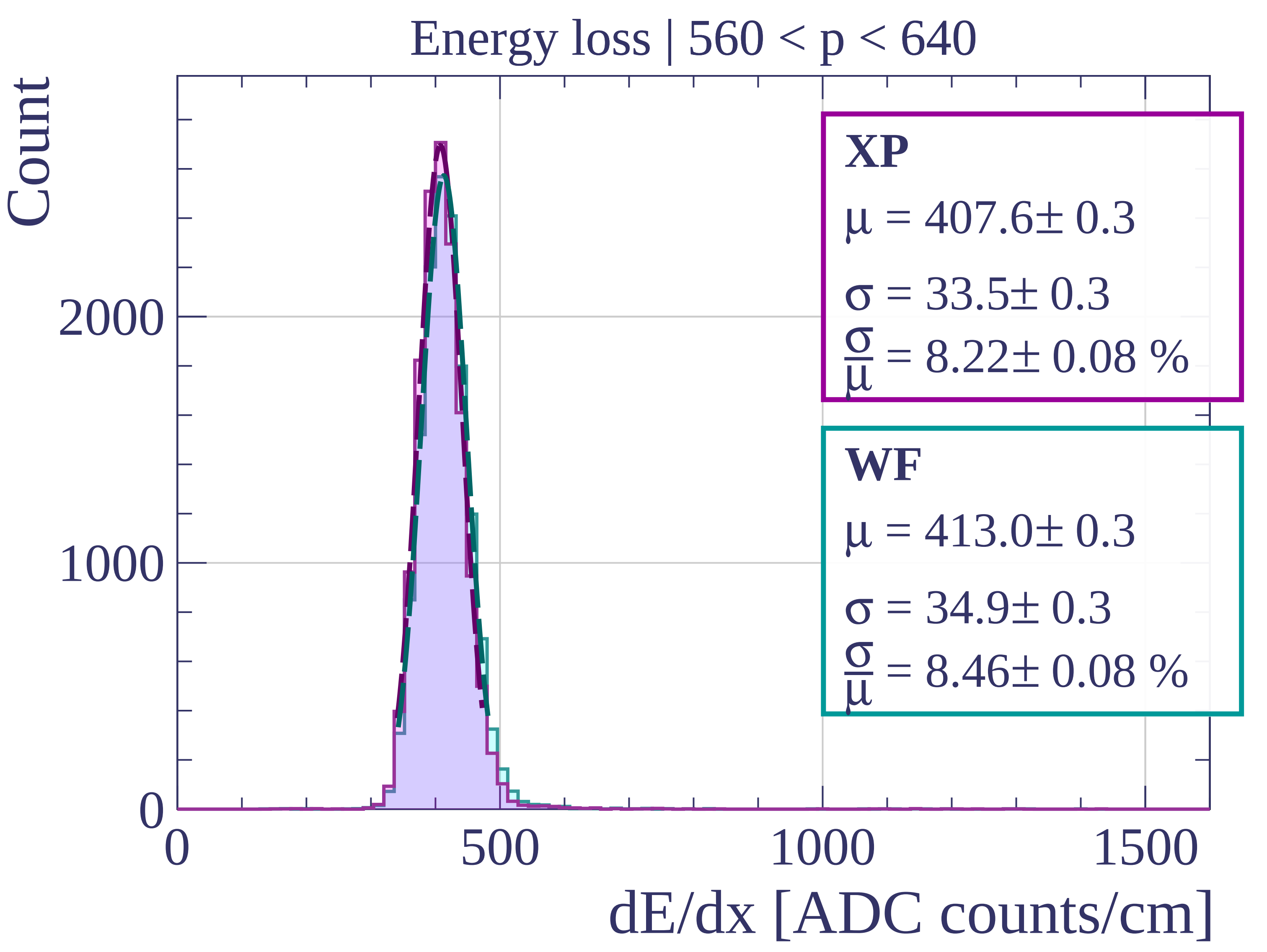
Energy loss | 320 < p < 400
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 398.4± 0.3
σ = 32.4± 0.3
σ/μ = 8.12± 0.08 %
WF
μ = 402.8± 0.3
σ = 33.8± 0.3
σ/μ = 8.38± 0.09 %



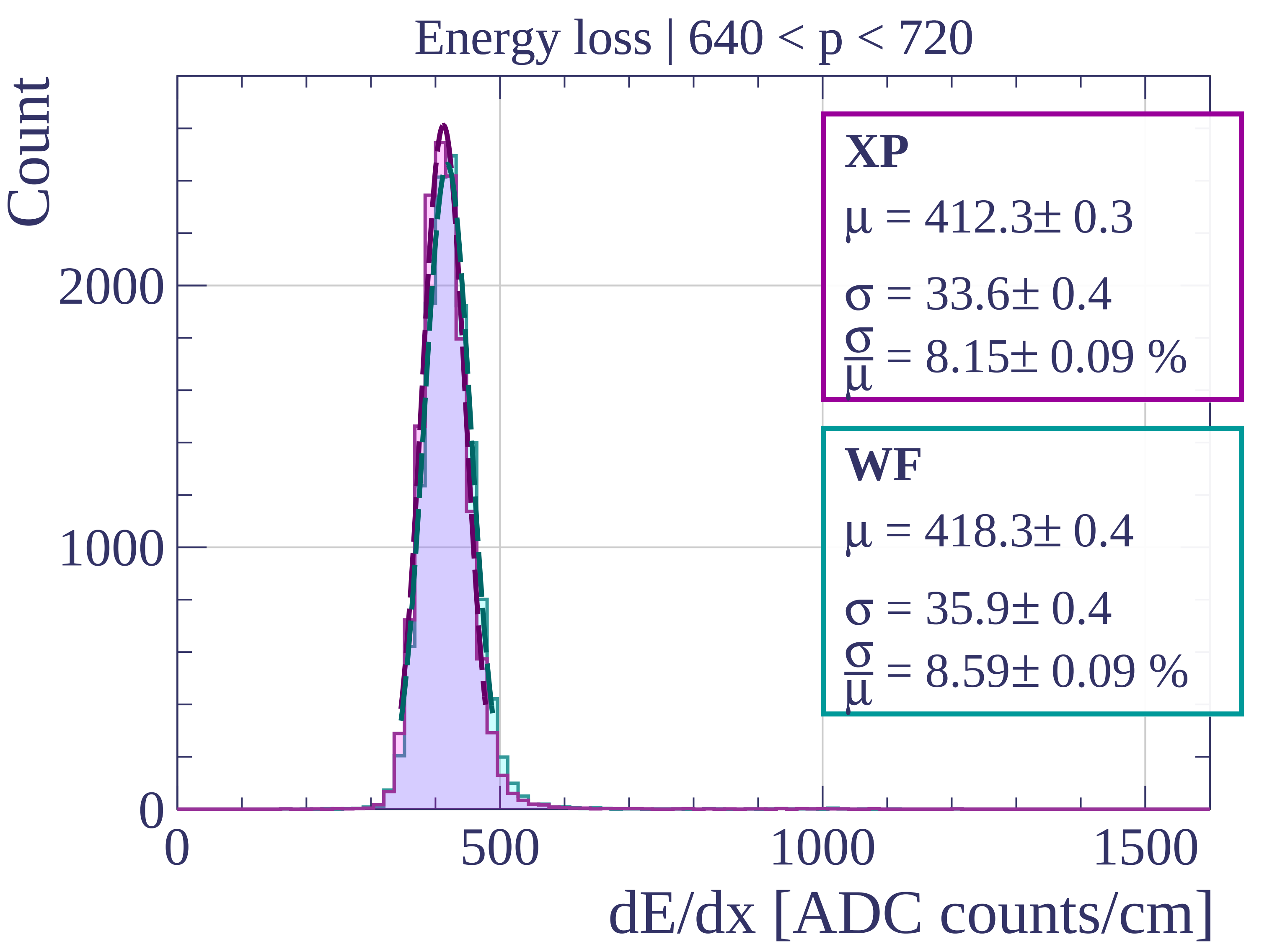
Energy loss | 400 < p < 480
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 398.9± 0.3
σ = 32.3± 0.3
σ/μ = 8.09± 0.08 %
WF
μ = 403.5± 0.3
σ = 33.3± 0.3
σ/μ = 8.26± 0.09 %



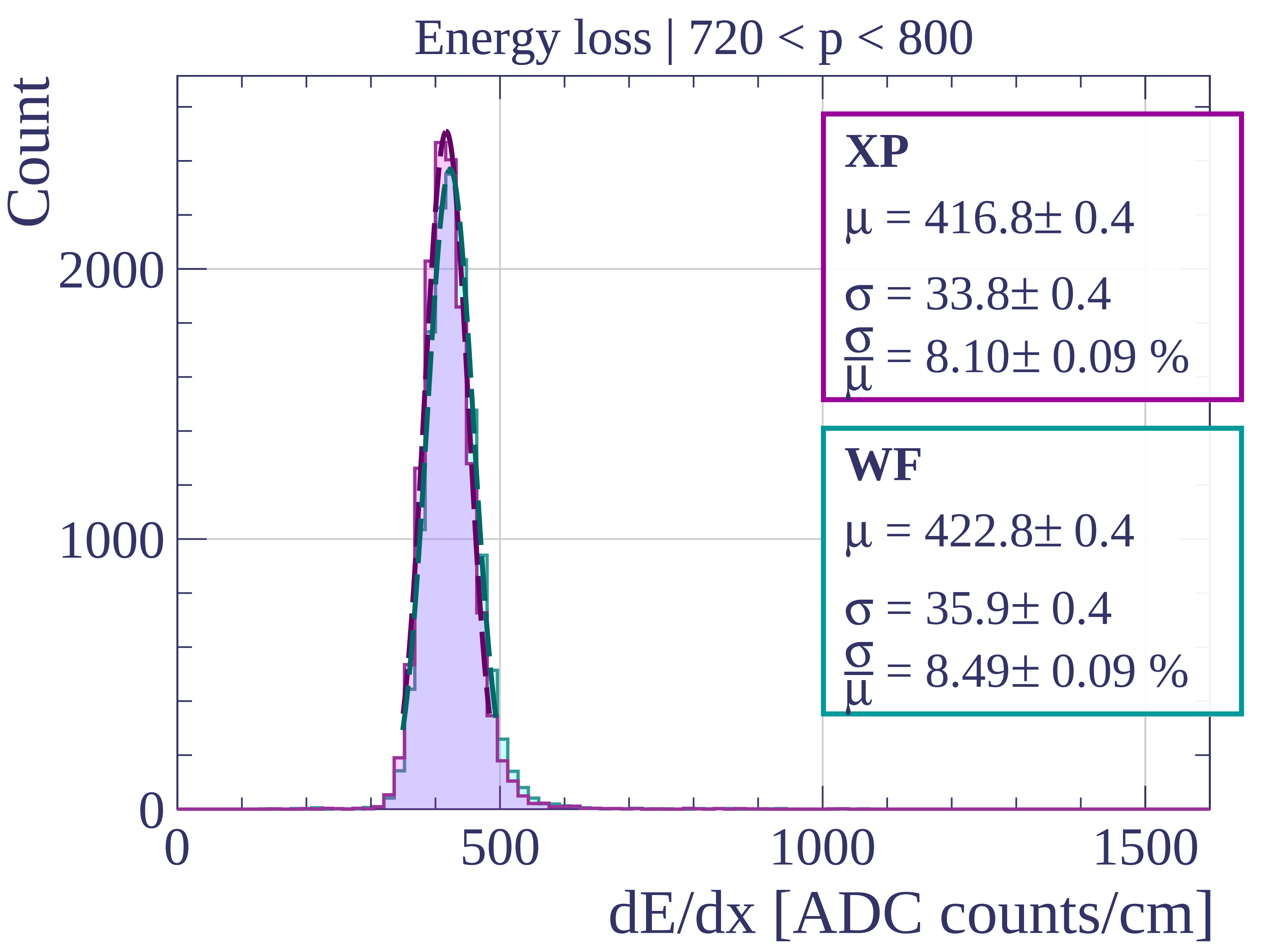
Energy loss | 480 < p < 560
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 401.9± 0.3
σ = 32.0± 0.3
σ/μ = 7.97± 0.08 %
WF
μ = 407.8± 0.3
σ = 34.2± 0.3
σ/μ = 8.39± 0.08 %

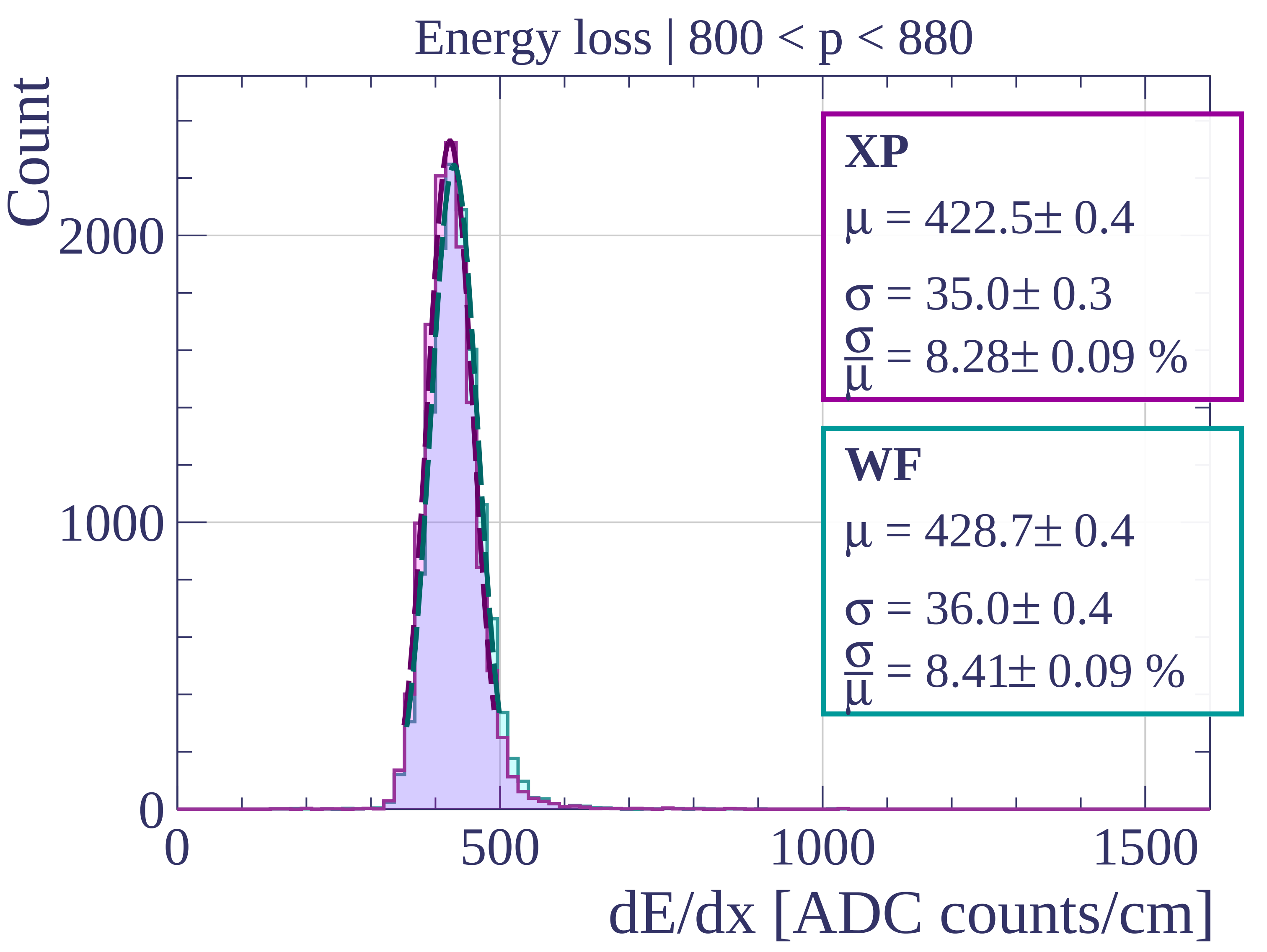


Energy loss | 560 < p < 640
Count
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 407.6± 0.3
σ = 33.5± 0.3
σ/μ = 8.22± 0.08 %
WF
μ = 413.0± 0.3
σ = 34.9± 0.3
σ/μ = 8.46± 0.08 %

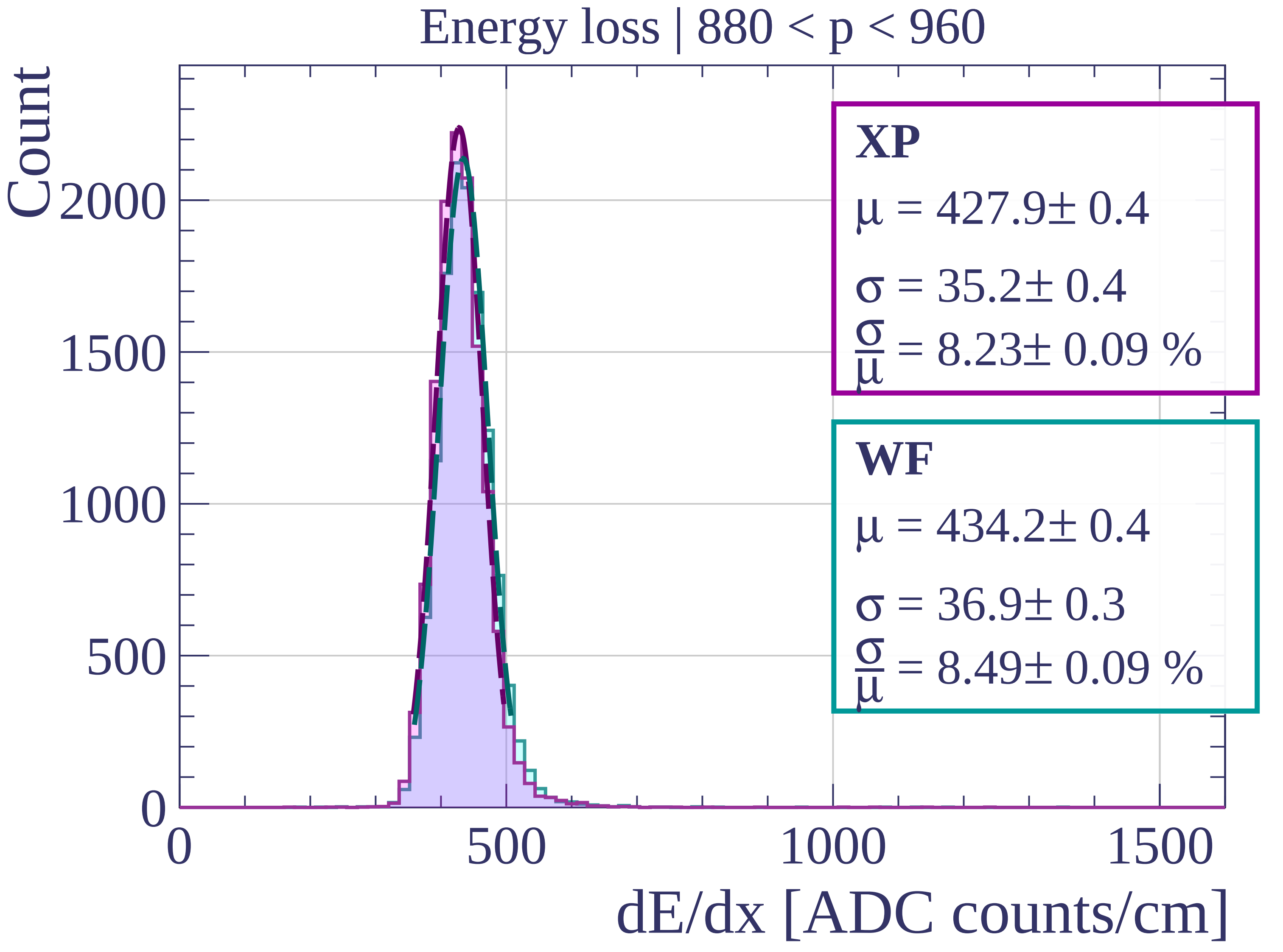


Energy loss | 640 < p < 720
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
1000
2000
XP
μ = 412.3± 0.3
σ = 33.6± 0.4
σ/μ = 8.15± 0.09 %
WF
μ = 418.3± 0.4
σ = 35.9± 0.4
σ/μ = 8.59± 0.09 %

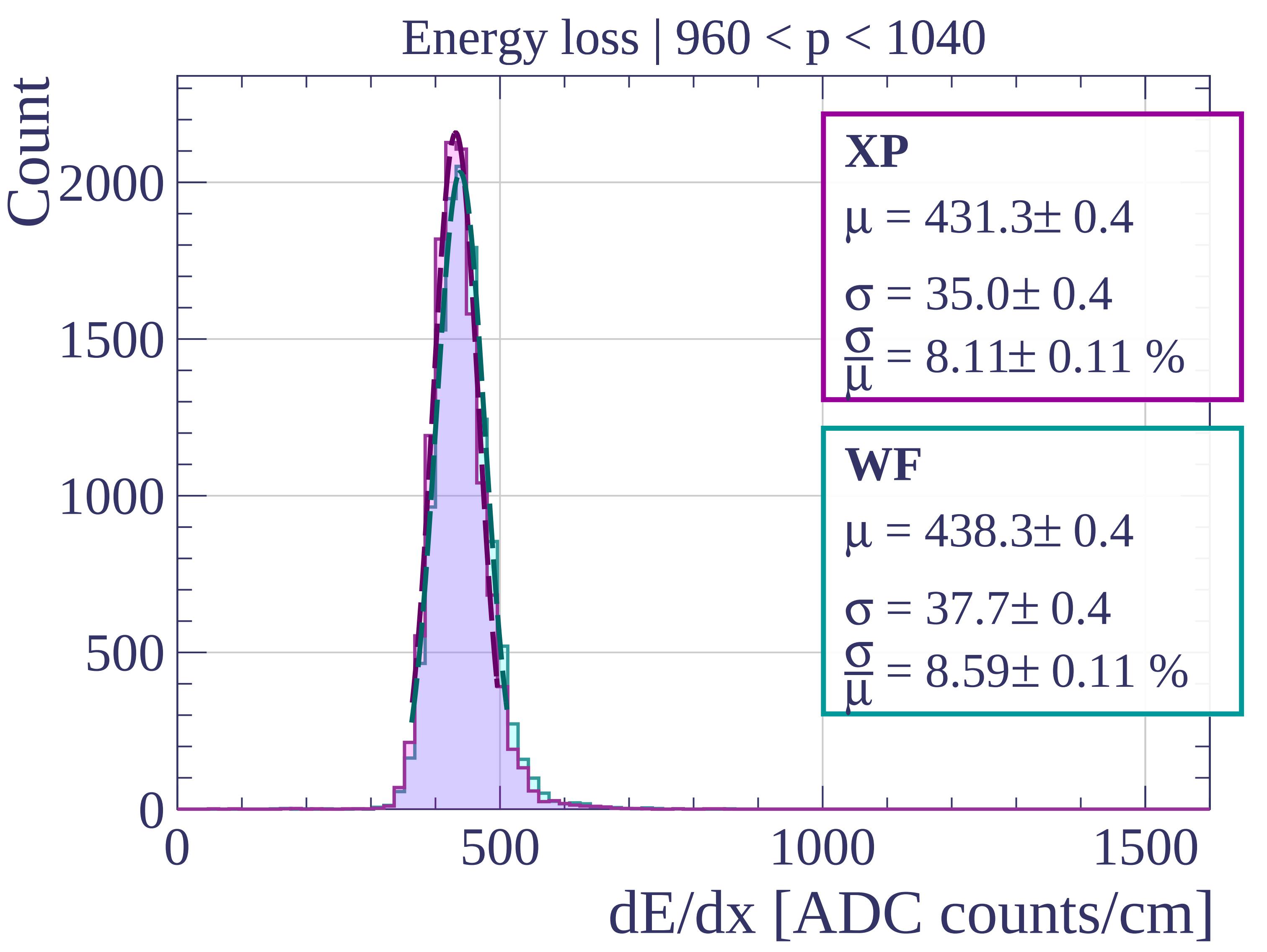

Energy loss | 720 < p < 800
Count
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 416.8± 0.4
σ = 33.8± 0.4
σ/μ = 8.10± 0.09 %
WF
μ = 422.8± 0.4
σ = 35.9± 0.4
σ/μ = 8.49± 0.09 %

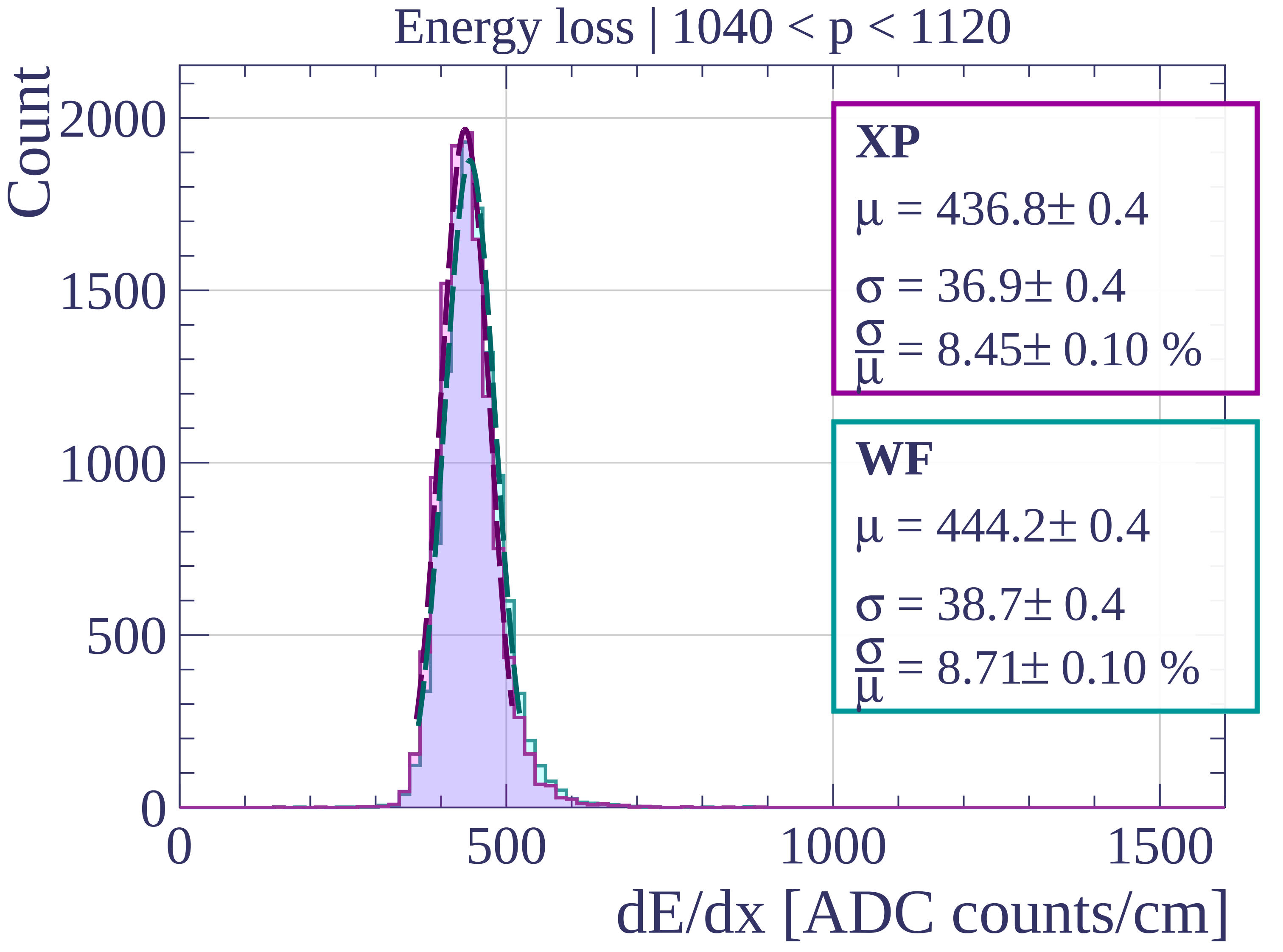


Energy loss | 800 < p < 880
Count
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 422.5± 0.4
σ = 35.0± 0.3
σ/μ = 8.28± 0.09 %
WF
μ = 428.7± 0.4
σ = 36.0± 0.4
σ/μ = 8.41± 0.09 %

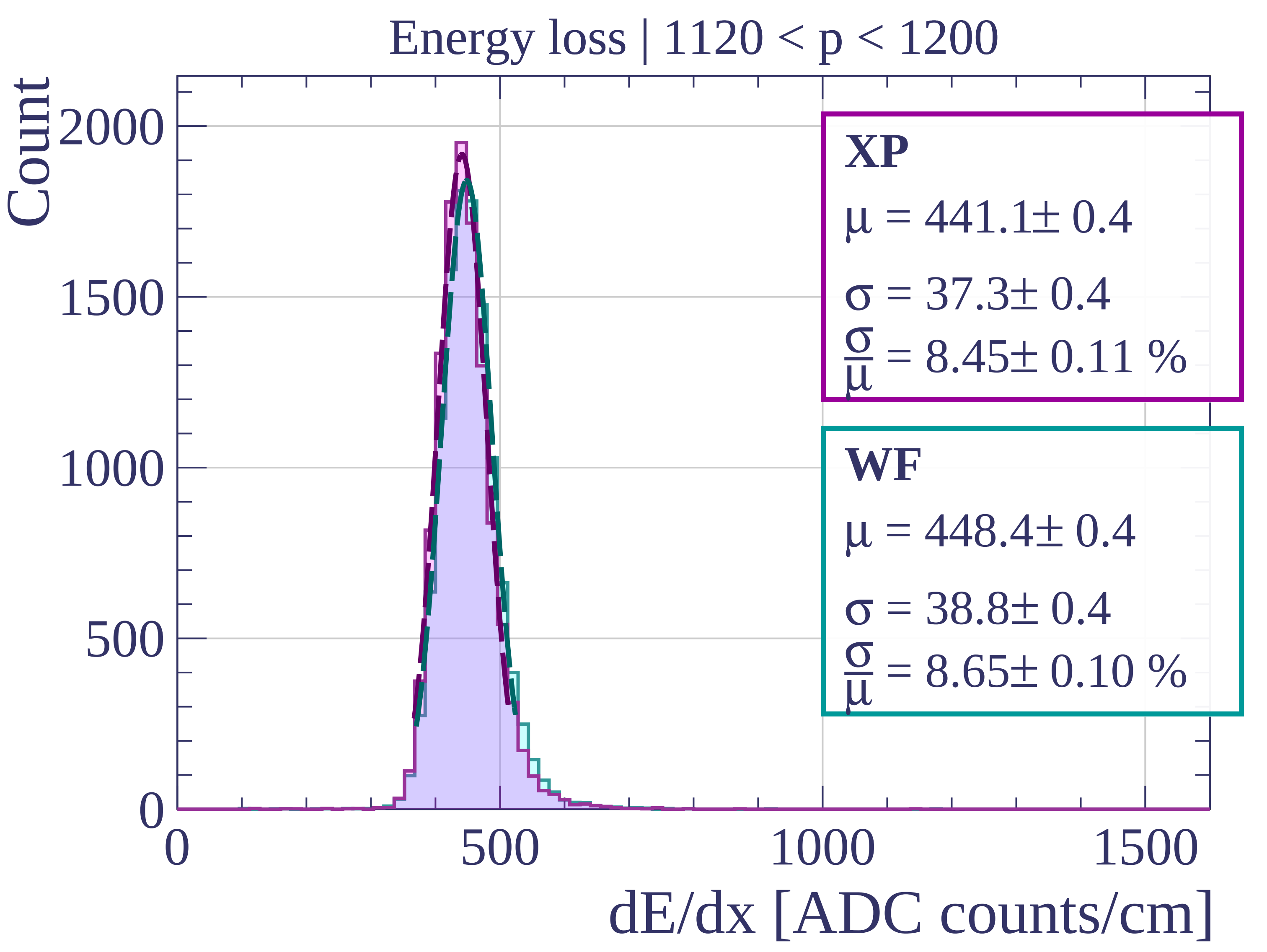


Energy loss | 880 < p < 960
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 427.9± 0.4
σ = 35.2± 0.4
σ/μ = 8.23± 0.09 %
WF
μ = 434.2± 0.4
σ = 36.9± 0.3
σ/μ = 8.49± 0.09 %

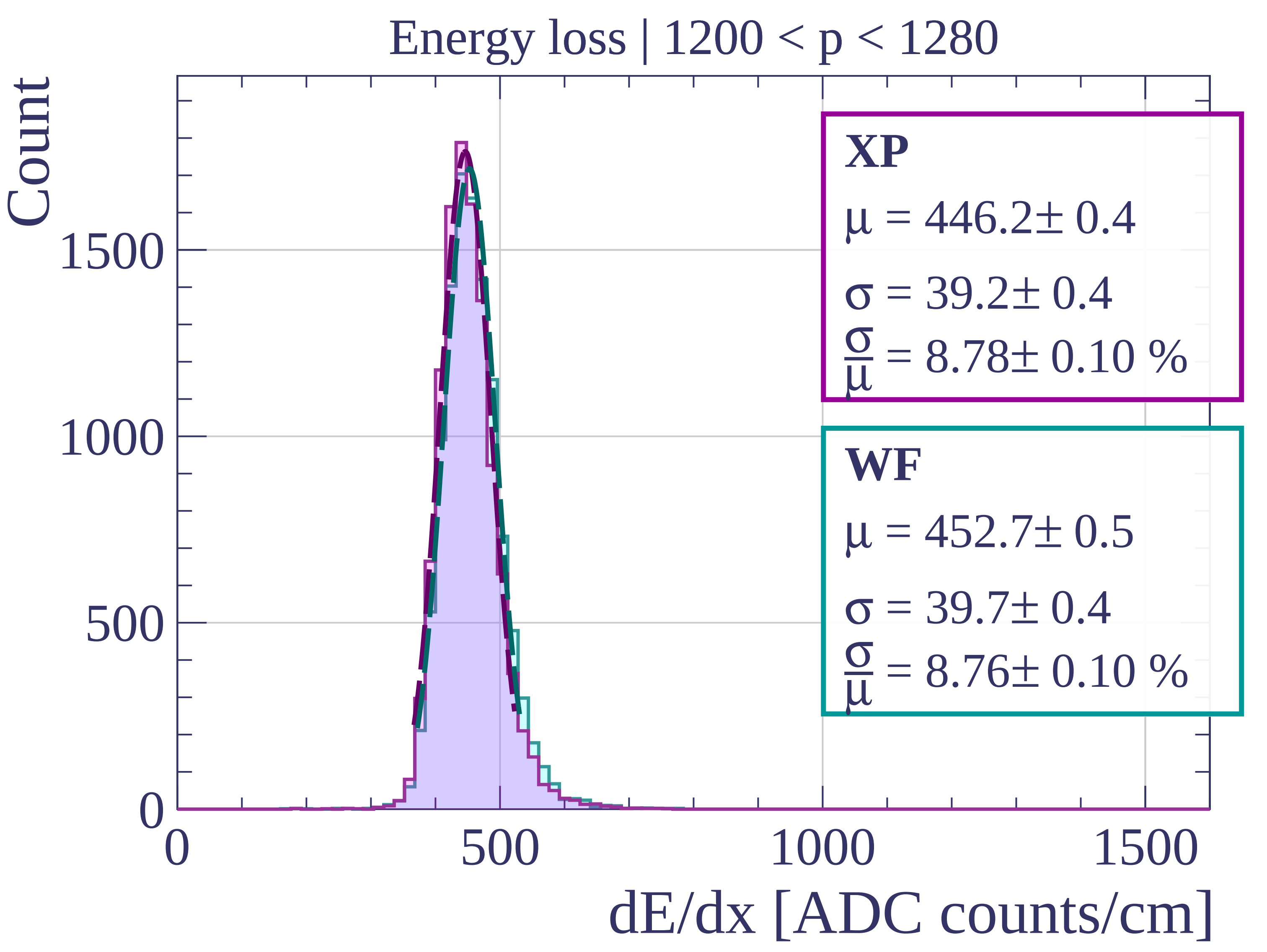

Energy loss | 960 < p < 1040
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 431.3± 0.4
σ = 35.0± 0.4
σ/μ = 8.11± 0.11 %
WF
μ = 438.3± 0.4
σ = 37.7± 0.4
σ/μ = 8.59± 0.11 %

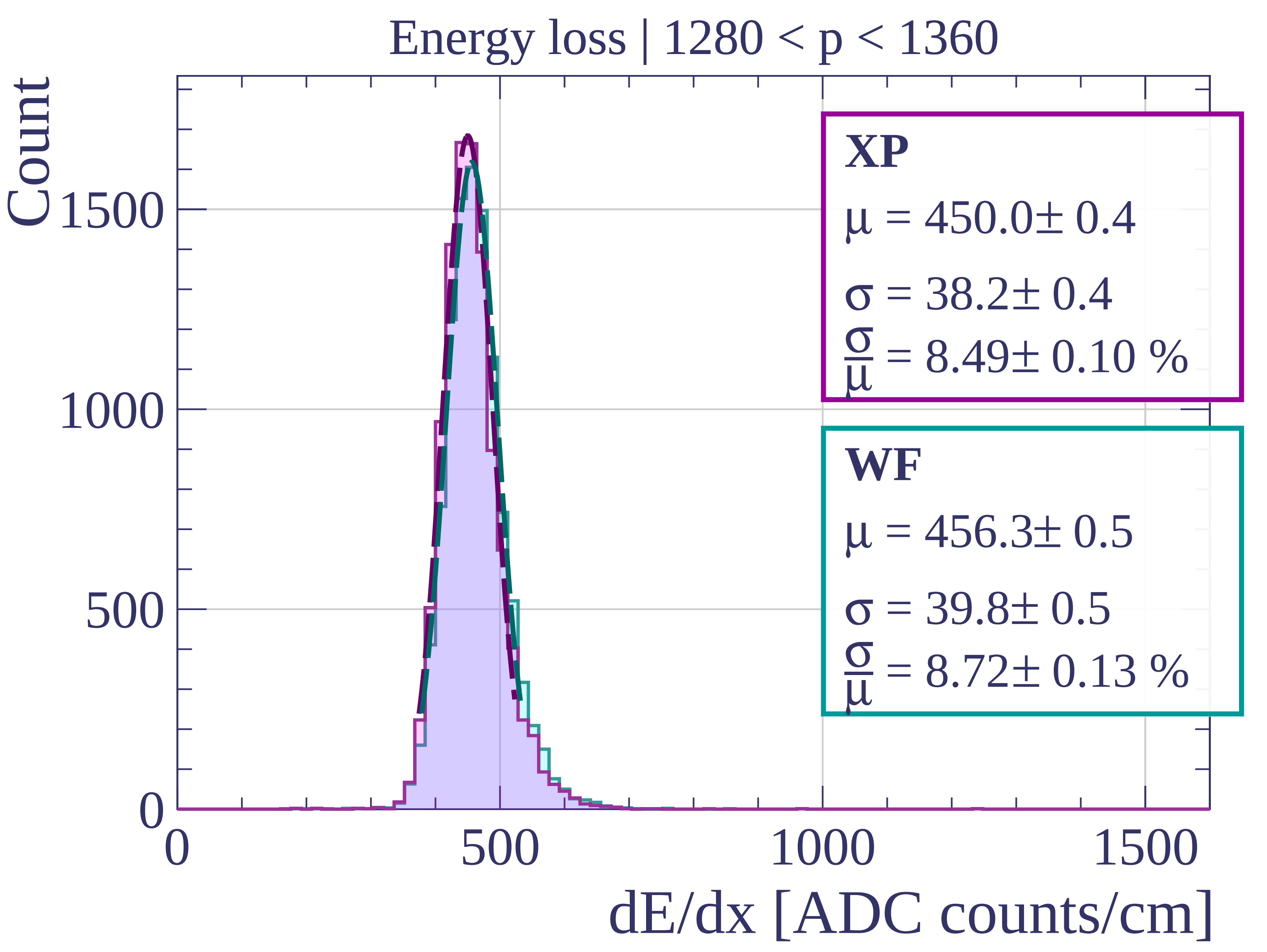


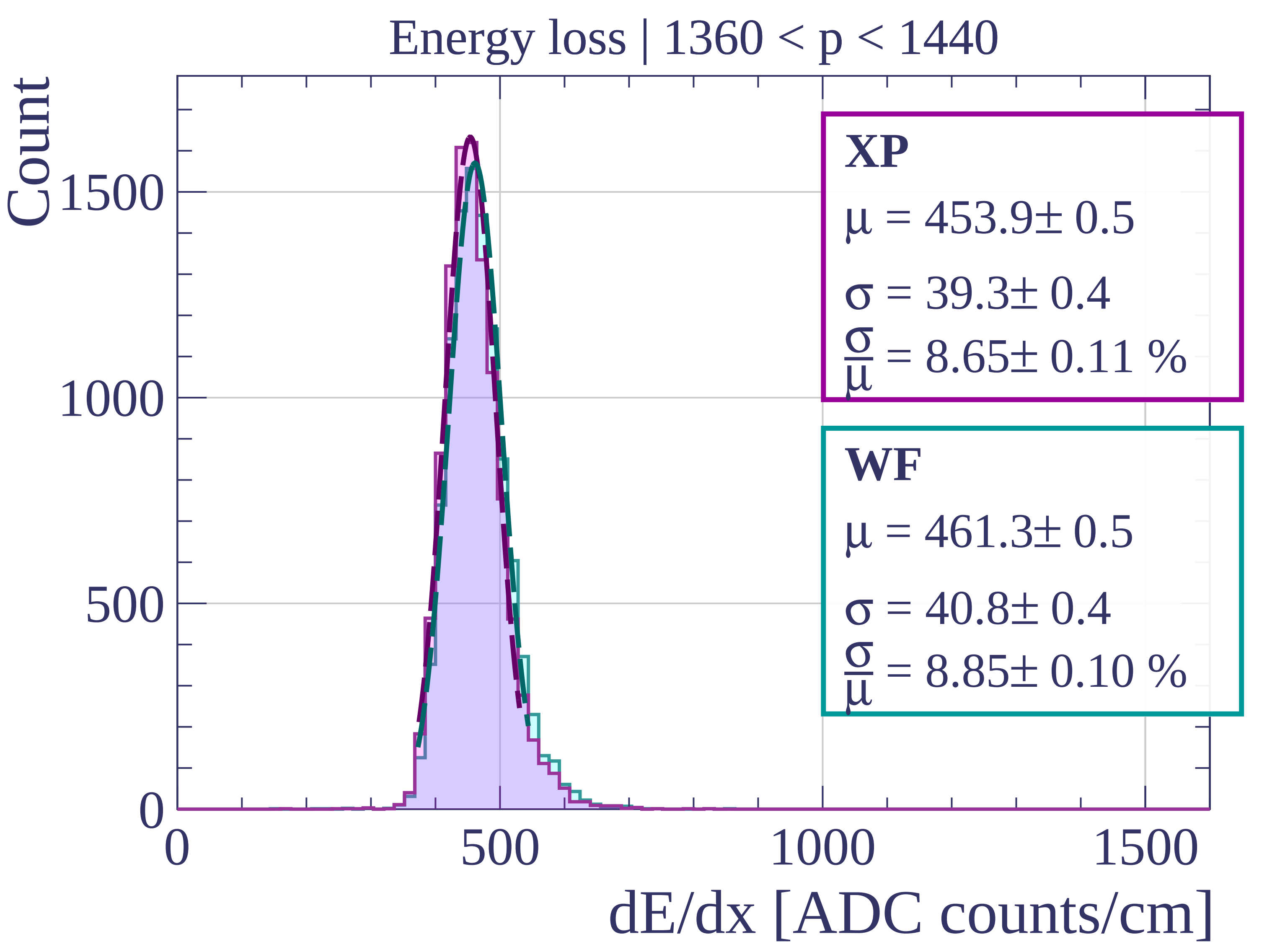
Energy loss | 1040 < p < 1120
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 436.8± 0.4
σ = 36.9± 0.4
σ/μ = 8.45± 0.10 %
WF
μ = 444.2± 0.4
σ = 38.7± 0.4
σ/μ = 8.71± 0.10 %

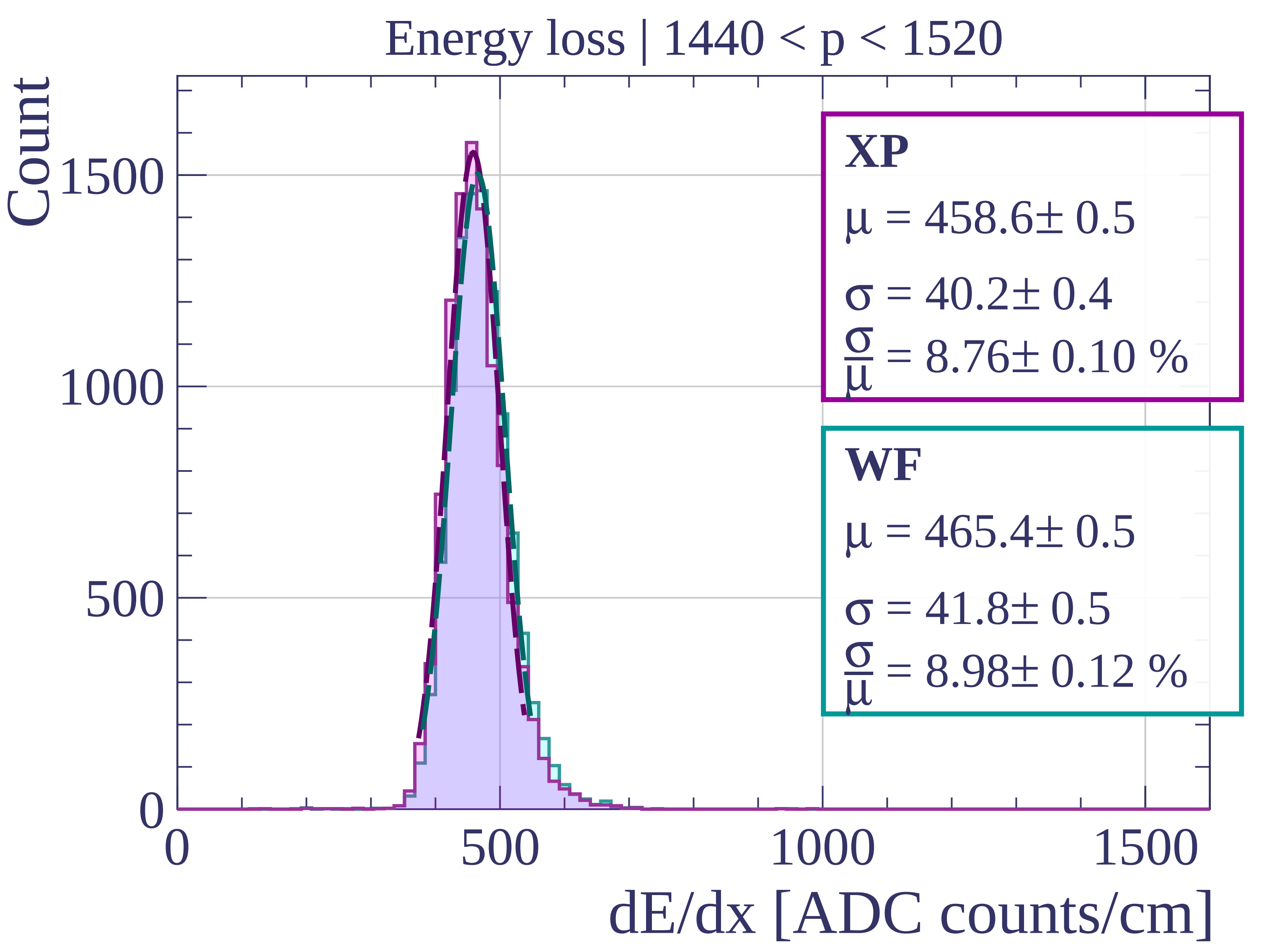


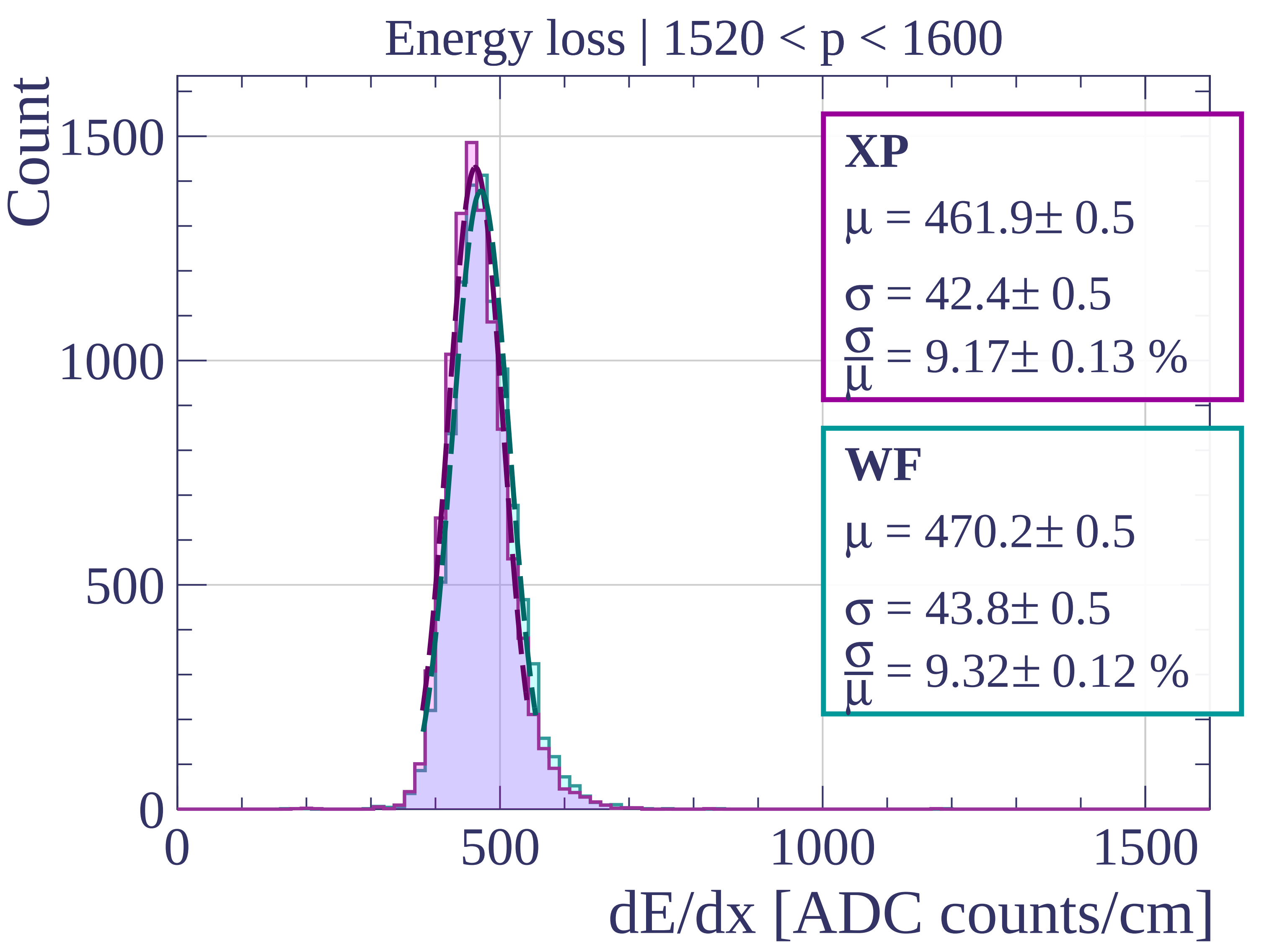
Energy loss | 1120 < p < 1200
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
2000
XP
μ = 441.1± 0.4
σ = 37.3± 0.4
σ/μ = 8.45± 0.11 %
WF
μ = 448.4± 0.4
σ = 38.8± 0.4
σ/μ = 8.65± 0.10 %

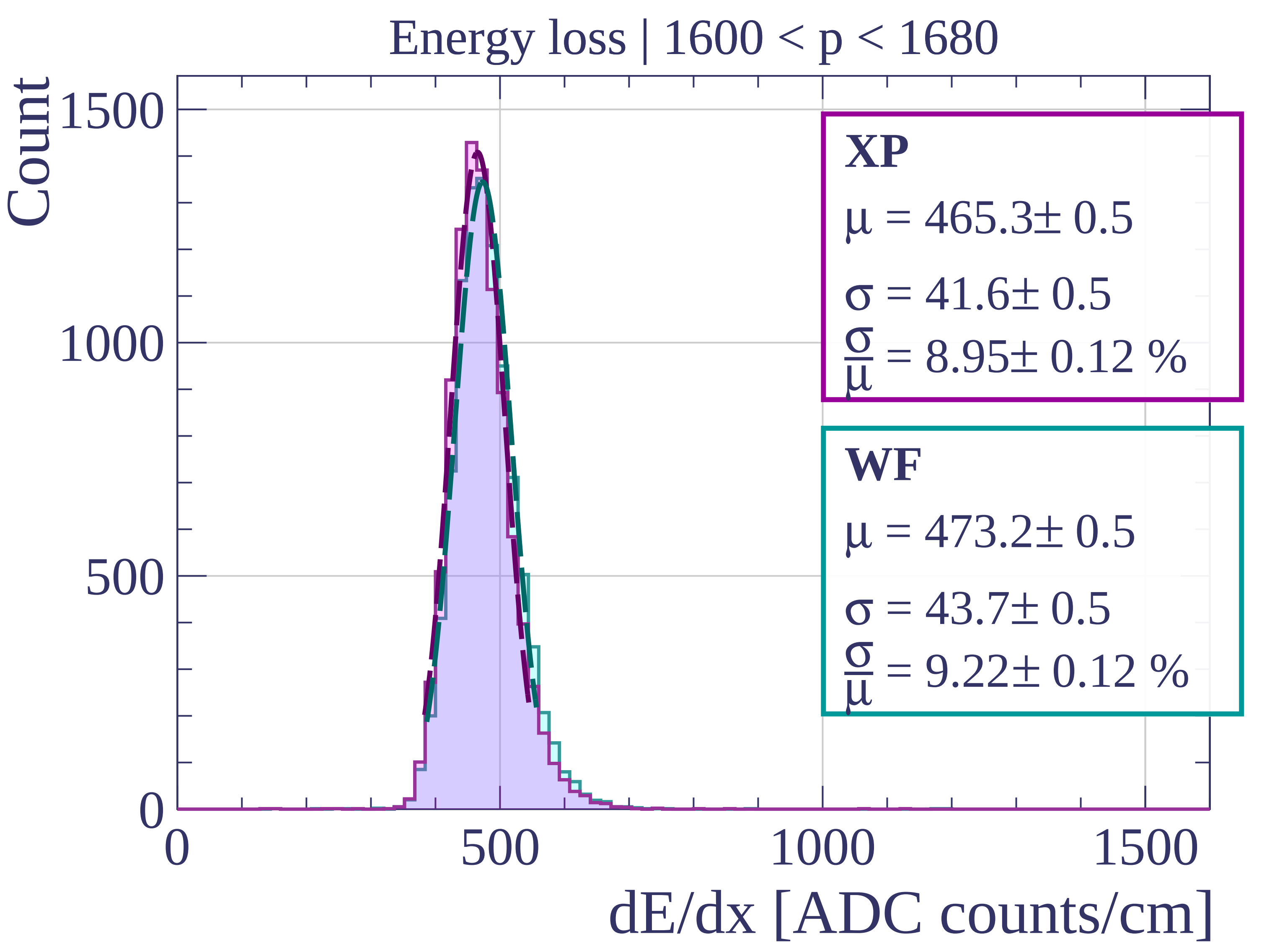

Energy loss | 1200 < p < 1280
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
XP
μ = 446.2± 0.4
σ = 39.2± 0.4
σ/μ = 8.78± 0.10 %
WF
μ = 452.7± 0.5
σ = 39.7± 0.4
σ/μ = 8.76± 0.10 %


Energy loss | 1280 < p < 1360
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
XP
μ = 450.0± 0.4
σ = 38.2± 0.4
σ/μ = 8.49± 0.10 %
WF
μ = 456.3± 0.5
σ = 39.8± 0.5
σ/μ = 8.72± 0.13 %

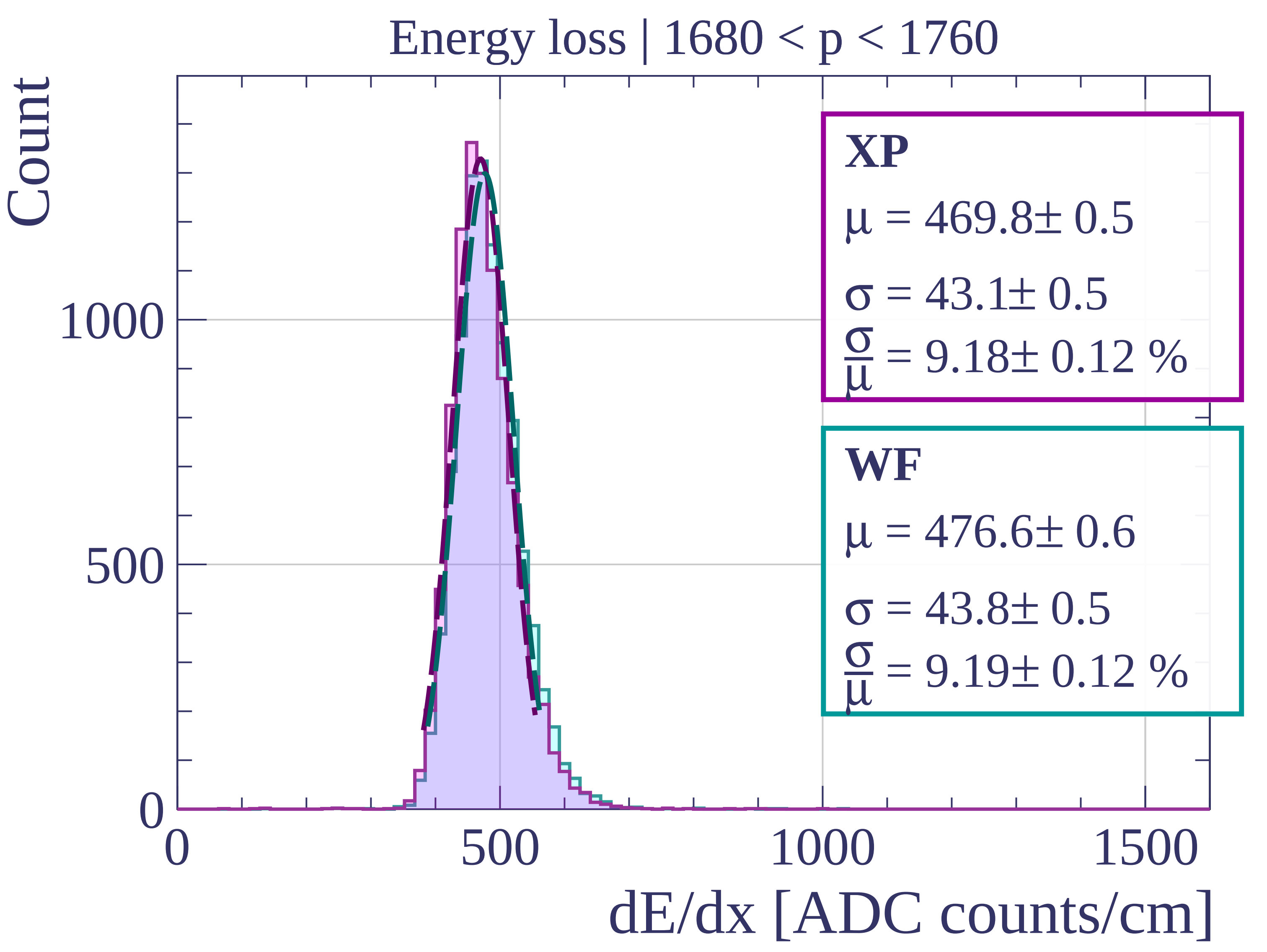

Energy loss | 1360 < p < 1440
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
XP
μ = 453.9± 0.5
σ = 39.3± 0.4
σ/μ = 8.65± 0.11 %
WF
μ = 461.3± 0.5
σ = 40.8± 0.4
σ/μ = 8.85± 0.10 %

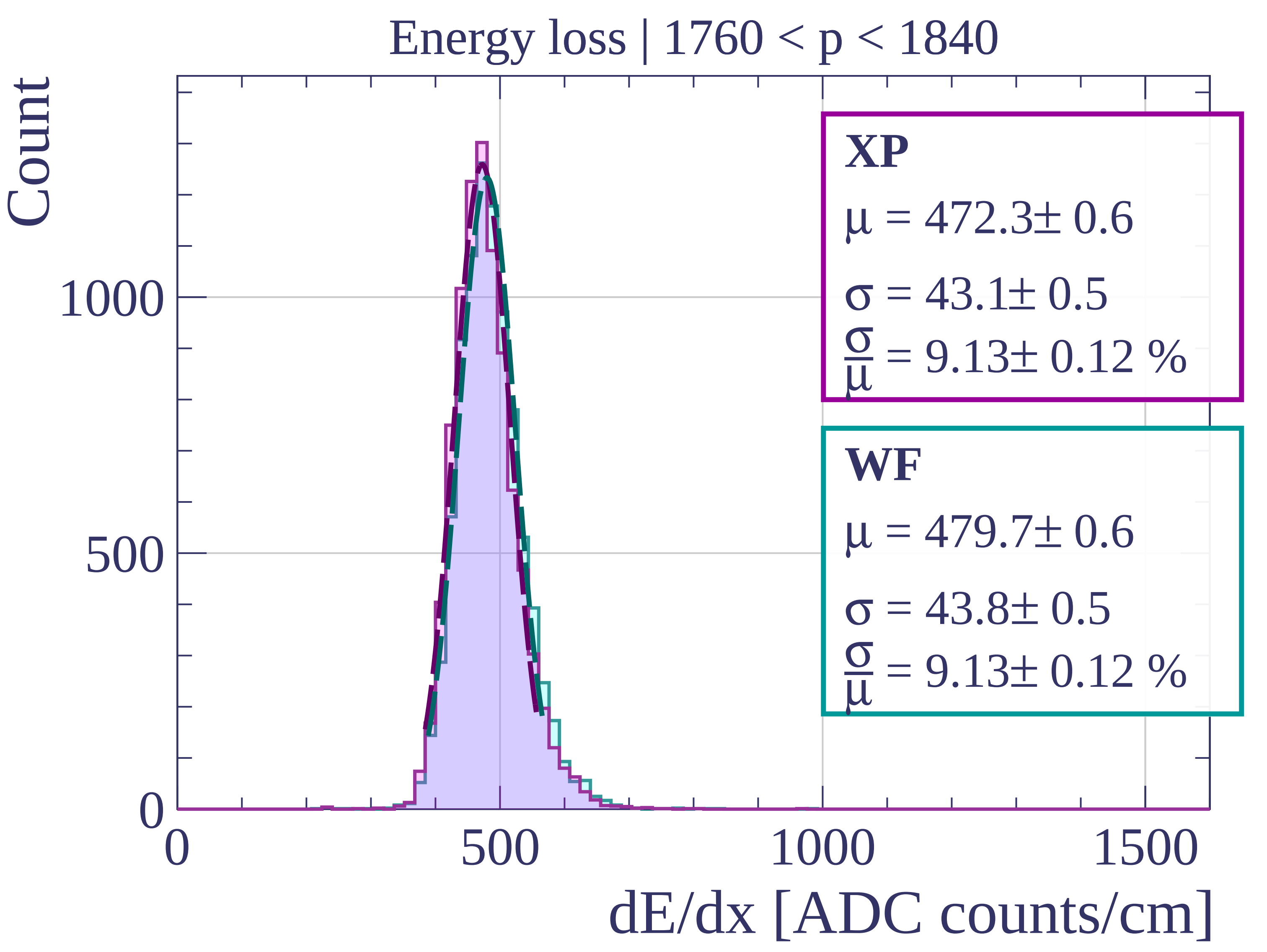

Energy loss | 1440 < p < 1520
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 458.6± 0.5
σ = 40.2± 0.4
σ/μ = 8.76± 0.10 %
WF
μ = 465.4± 0.5
σ = 41.8± 0.5
σ/μ = 8.98± 0.12 %


Energy loss | 1520 < p < 1600
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
XP
μ = 461.9± 0.5
σ = 42.4± 0.5
σ/μ = 9.17± 0.13 %
WF
μ = 470.2± 0.5
σ = 43.8± 0.5
σ/μ = 9.32± 0.12 %

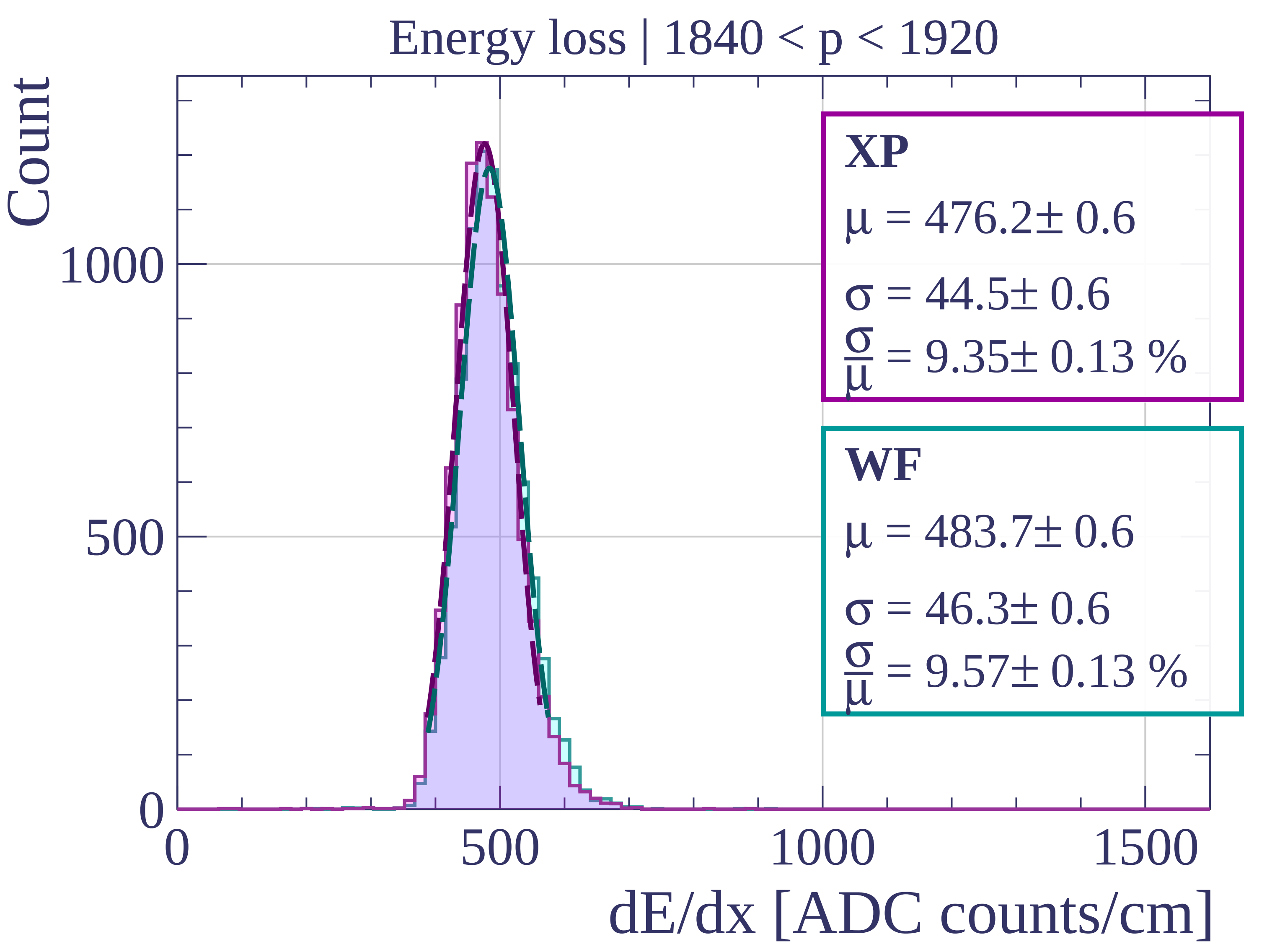


Energy loss | 1600 < p < 1680
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 465.3± 0.5
σ = 41.6± 0.5
σ/μ = 8.95± 0.12 %
WF
μ = 473.2± 0.5
σ = 43.7± 0.5
σ/μ = 9.22± 0.12 %

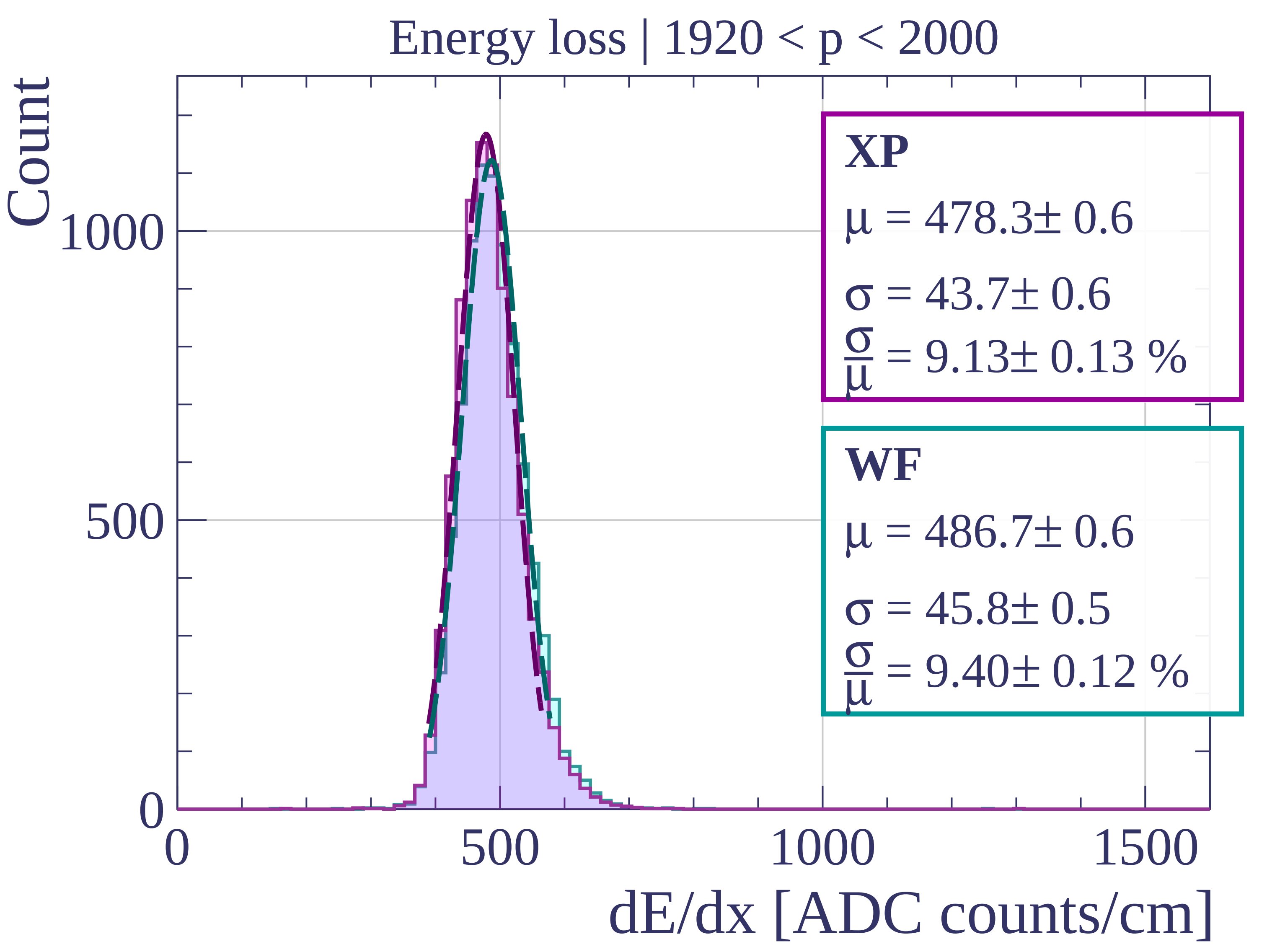

Energy loss | 1680 < p < 1760
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
XP
μ = 469.8± 0.5
σ = 43.1± 0.5
σ/μ = 9.18± 0.12 %
WF
μ = 476.6± 0.6
σ = 43.8± 0.5
σ/μ = 9.19± 0.12 %

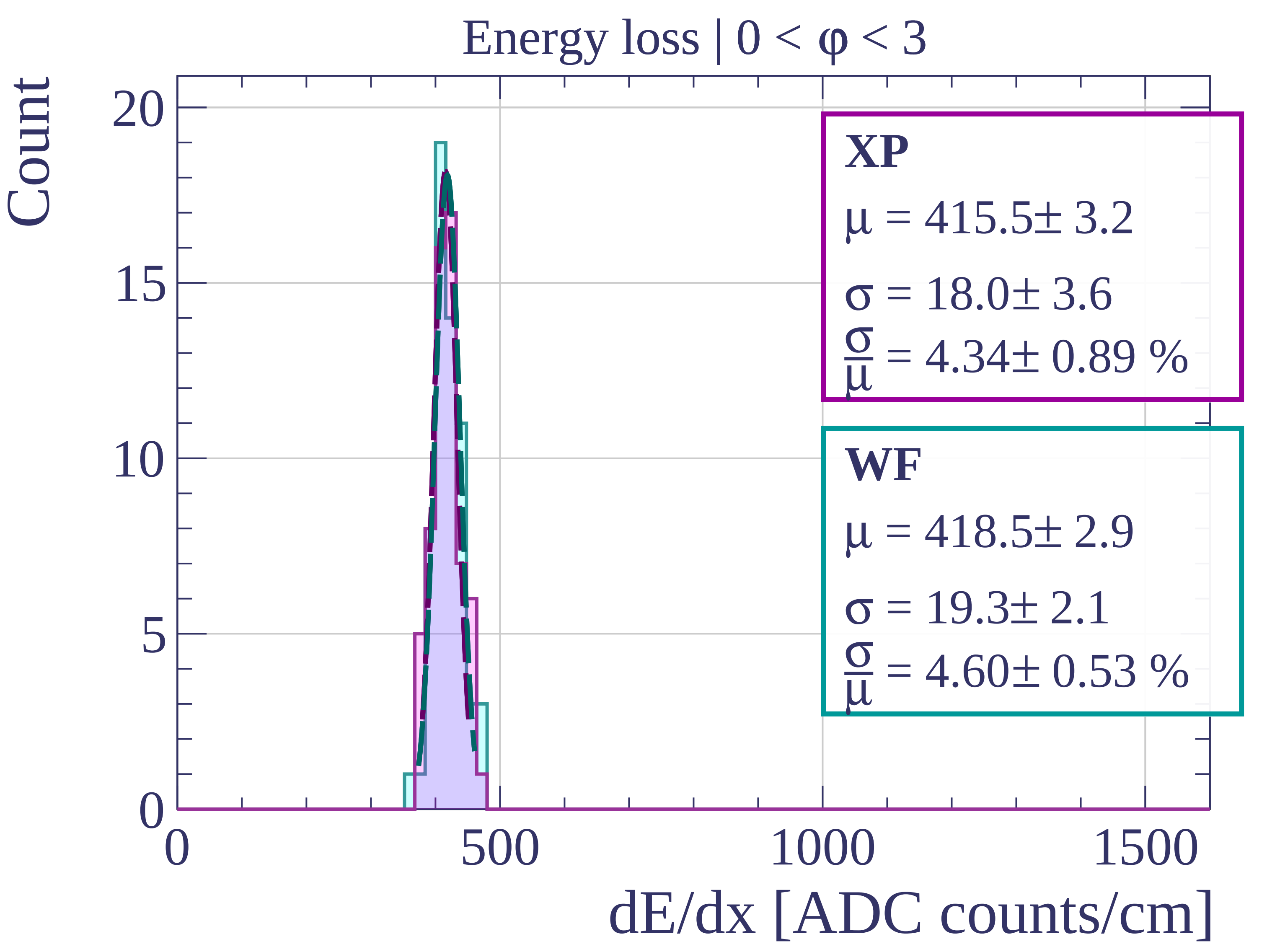


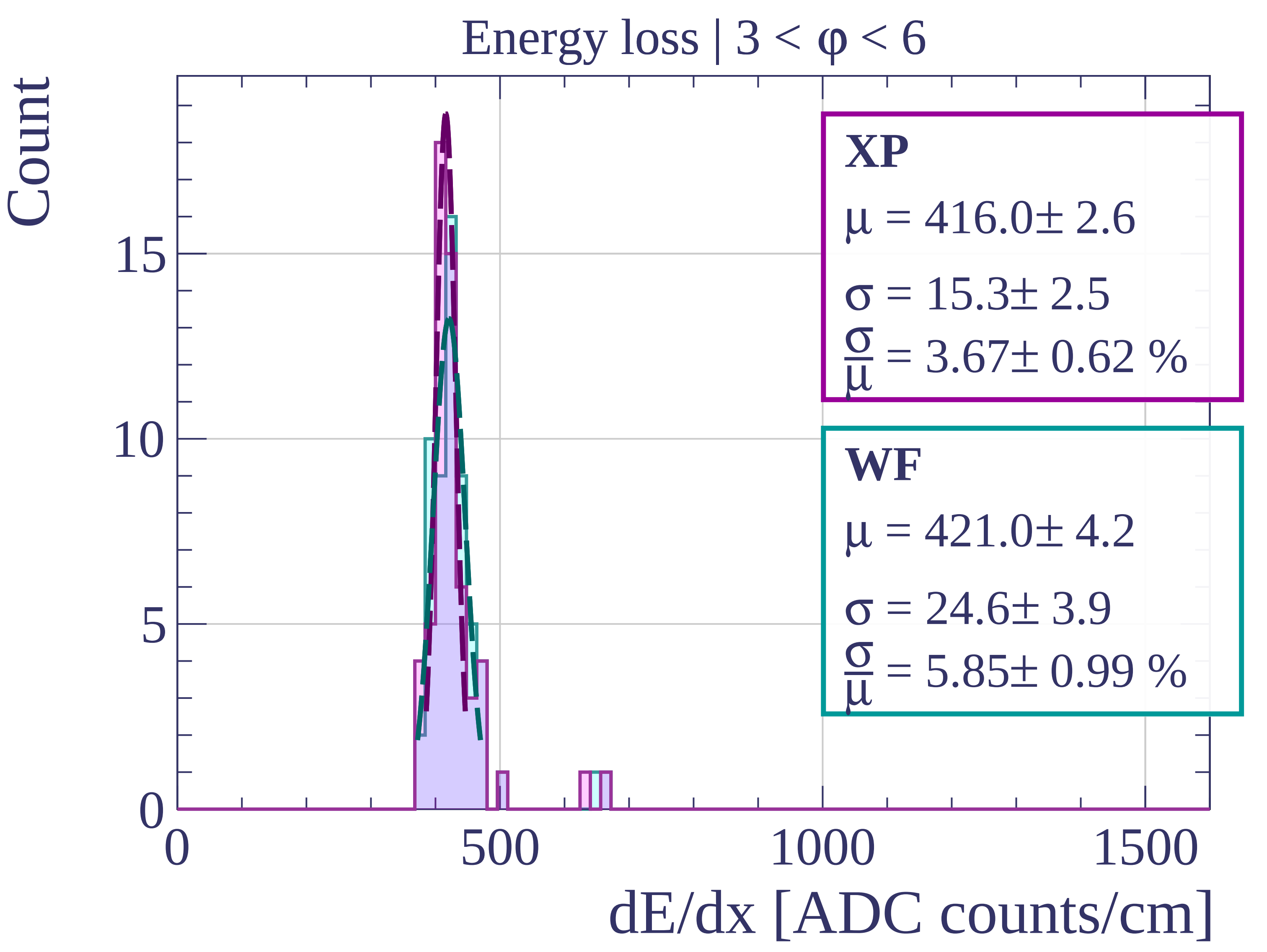
Energy loss | 1760 < p < 1840
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
XP
μ = 472.3± 0.6
σ = 43.1± 0.5
σ/μ = 9.13± 0.12 %
WF
μ = 479.7± 0.6
σ = 43.8± 0.5
σ/μ = 9.13± 0.12 %



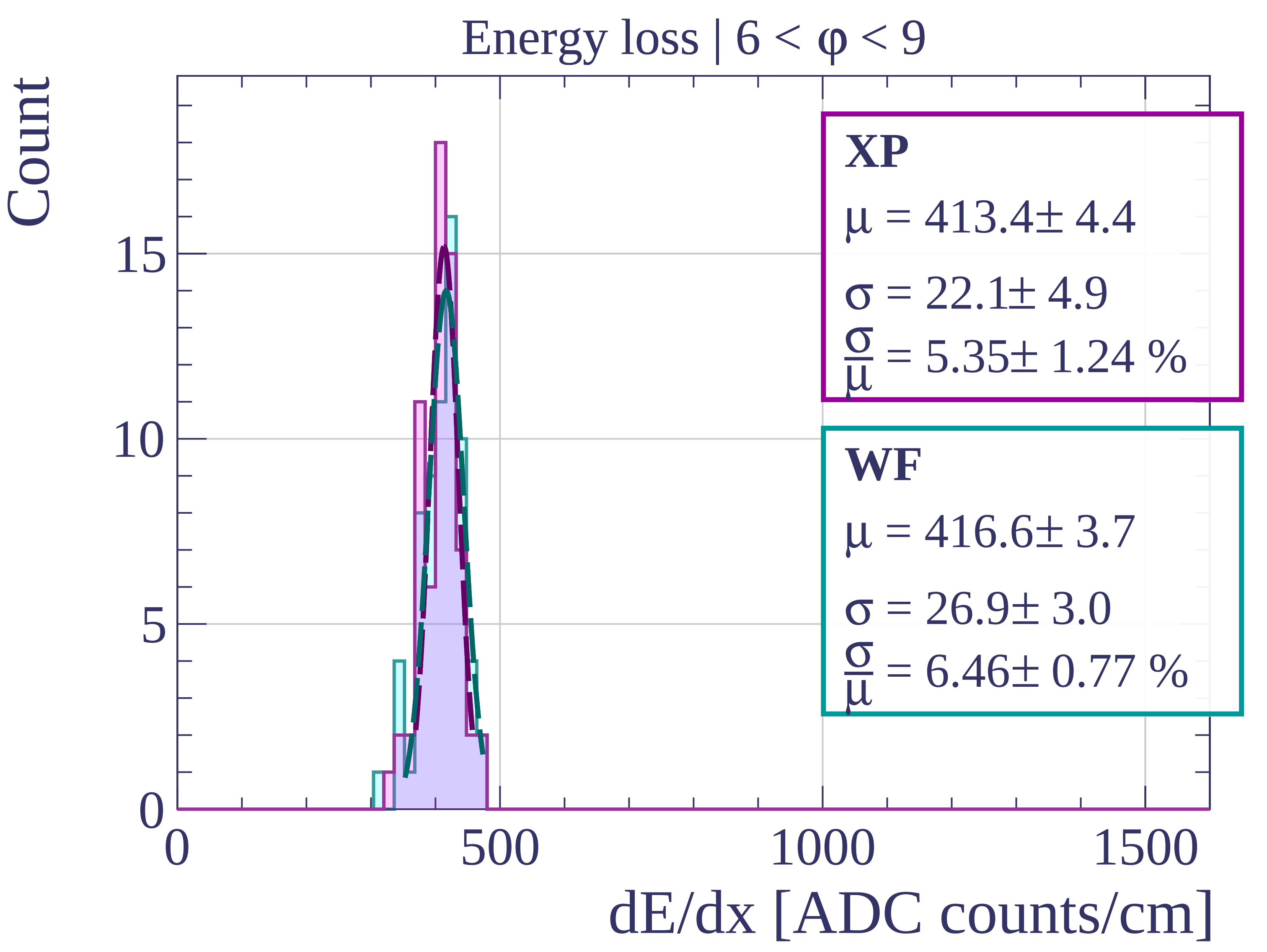
Energy loss | 1840 < p < 1920
Count
0
500
1000
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 476.2± 0.6
σ = 44.5± 0.6
σ/μ = 9.35± 0.13 %
WF
μ = 483.7± 0.6
σ = 46.3± 0.6
σ/μ = 9.57± 0.13 %


Energy loss | 1920 < p < 2000
Count
dE/dx [ADC counts/cm]
0
500
1000
0
500
1000
1500
XP
μ = 478.3± 0.6
σ = 43.7± 0.6
σ/μ = 9.13± 0.13 %
WF
μ = 486.7± 0.6
σ = 45.8± 0.5
σ/μ = 9.40± 0.12 %

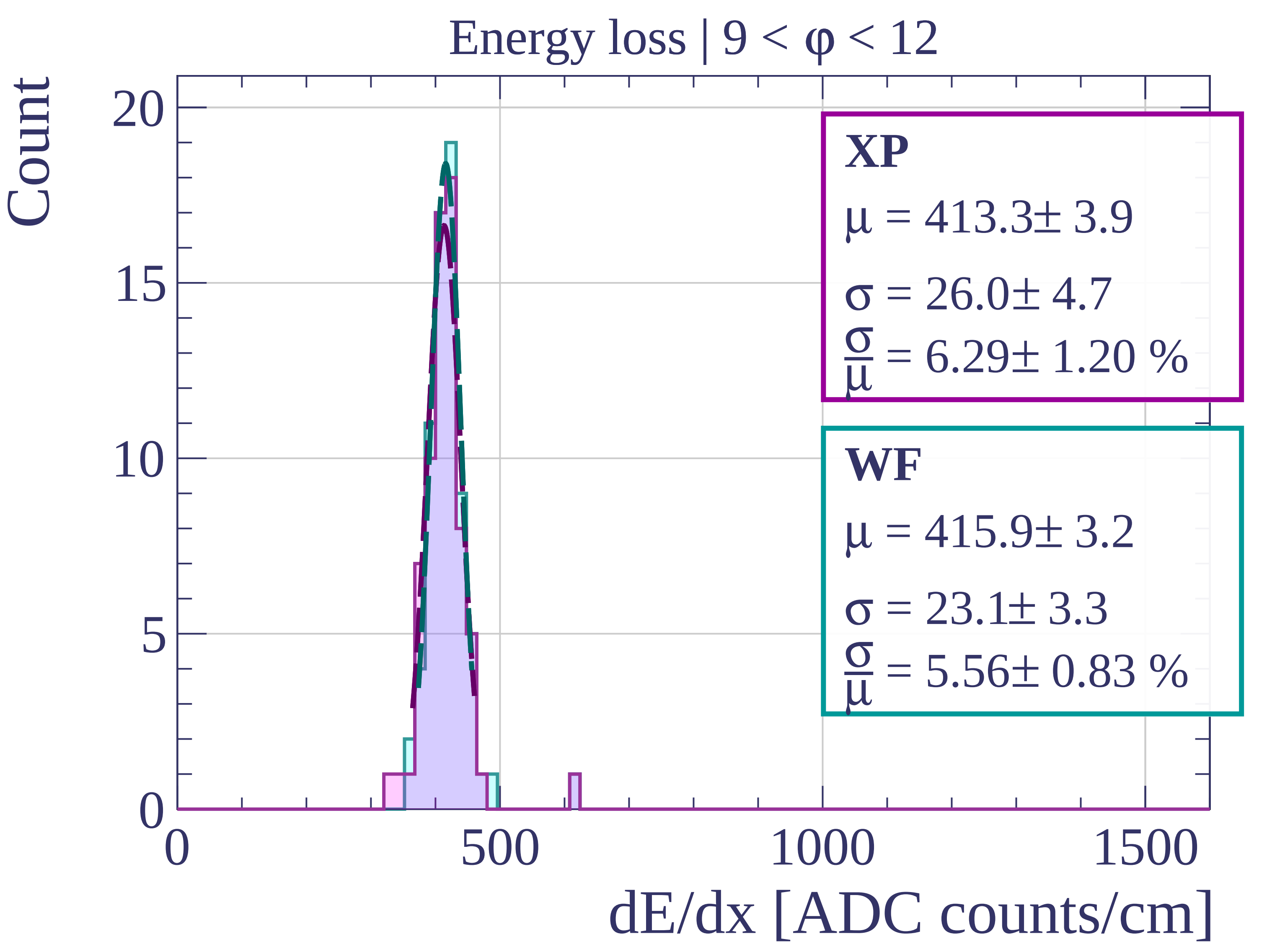

Energy loss | 0 < φ < 3
Count
dE/dx [ADC counts/cm]
0
5
10
15
20
0
500
1000
1500
XP
μ = 415.5± 3.2
σ = 18.0± 3.6
σ/μ = 4.34± 0.89 %
WF
μ = 418.5± 2.9
σ = 19.3± 2.1
σ/μ = 4.60± 0.53 %



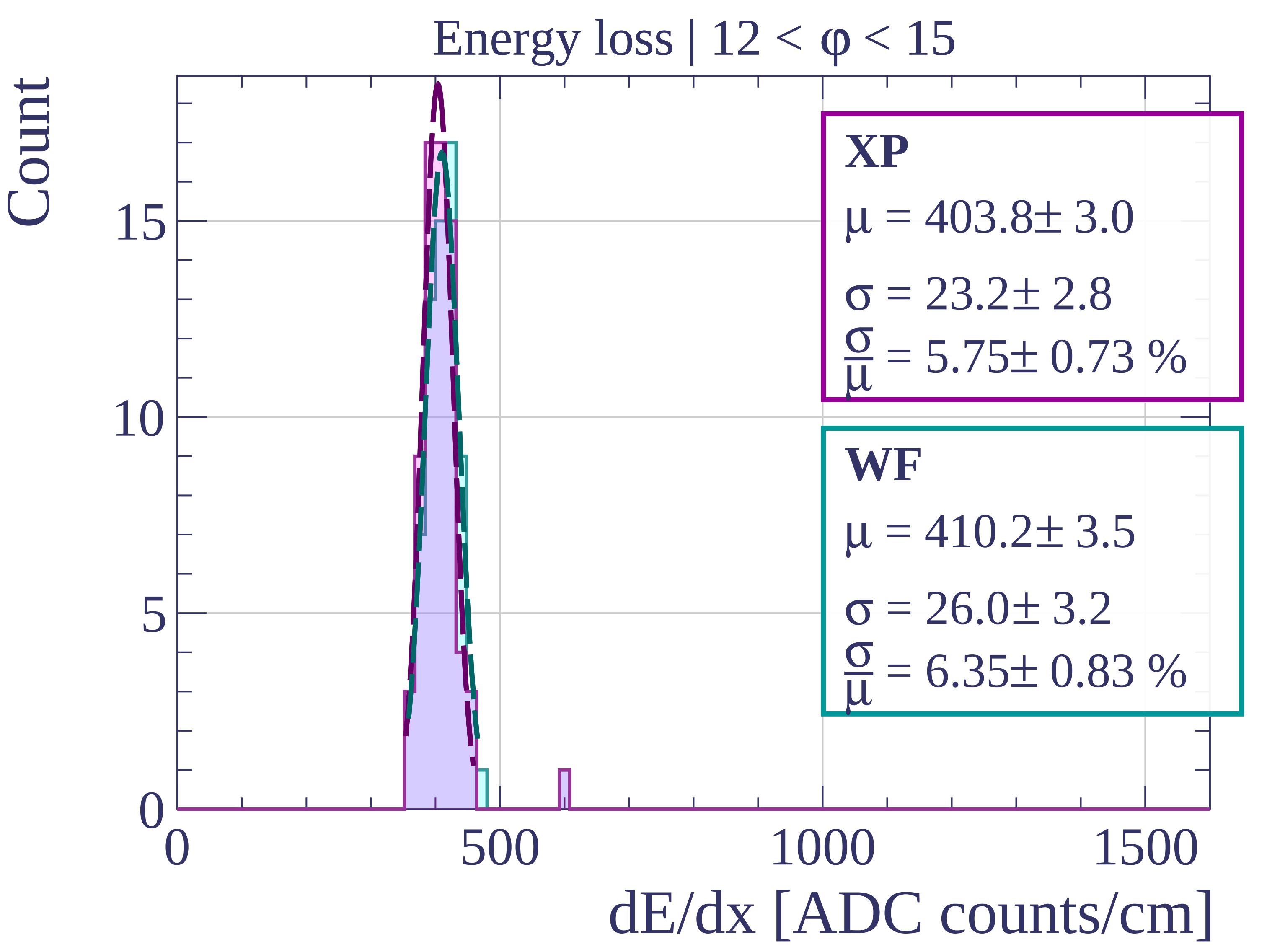
Energy loss | 3 < φ < 6
Count
dE/dx [ADC counts/cm]
0
5
10
15
0
500
1000
1500
XP
μ = 416.0± 2.6
σ = 15.3± 2.5
σ/μ = 3.67± 0.62 %
WF
μ = 421.0± 4.2
σ = 24.6± 3.9
σ/μ = 5.85± 0.99 %



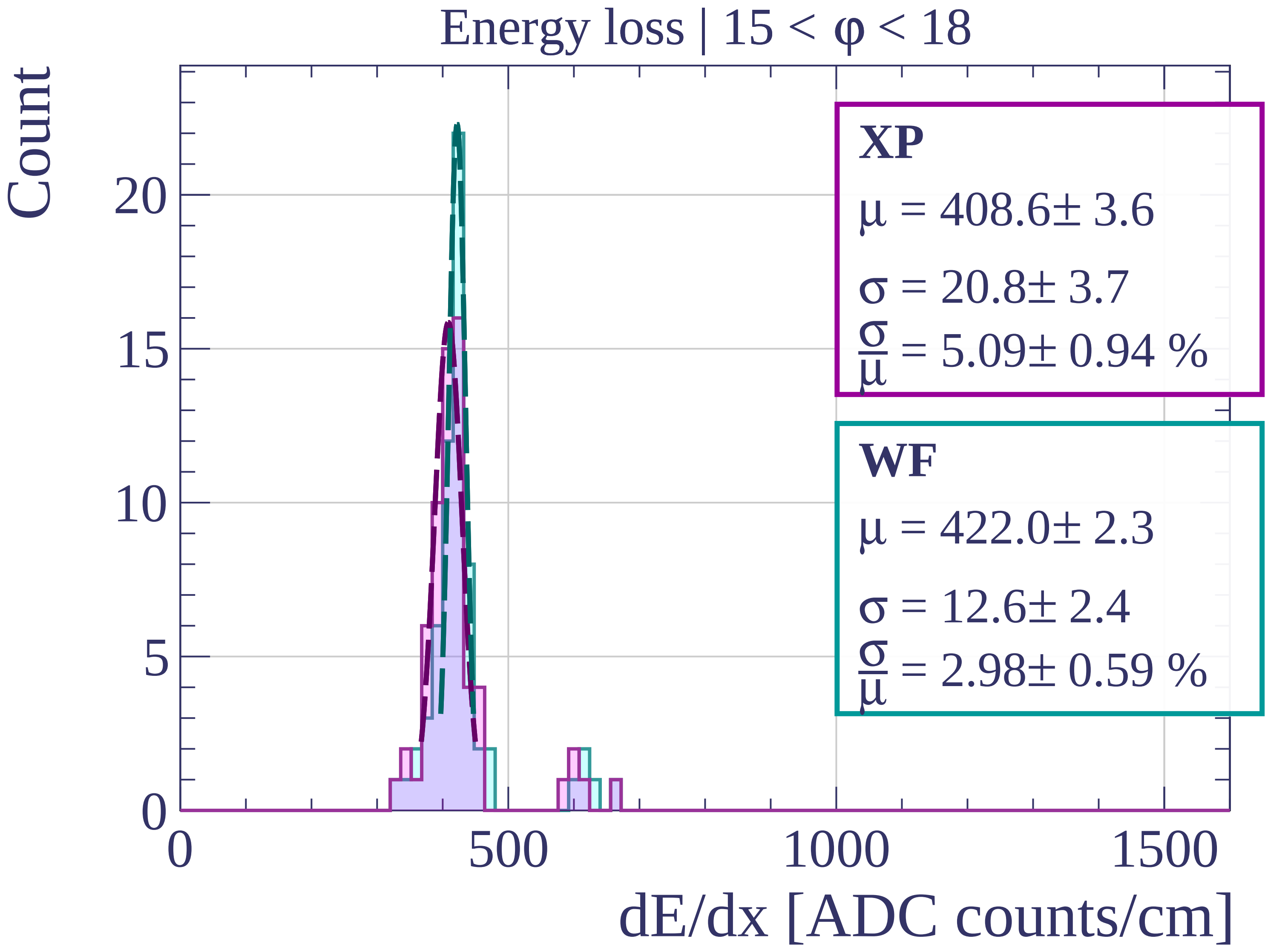
Energy loss | 6 < φ < 9
Count
dE/dx [ADC counts/cm]
0
5
10
15
0
500
1000
1500
XP
μ = 413.4± 4.4
σ = 22.1± 4.9
σ/μ = 5.35± 1.24 %
WF
μ = 416.6± 3.7
σ = 26.9± 3.0
σ/μ = 6.46± 0.77 %

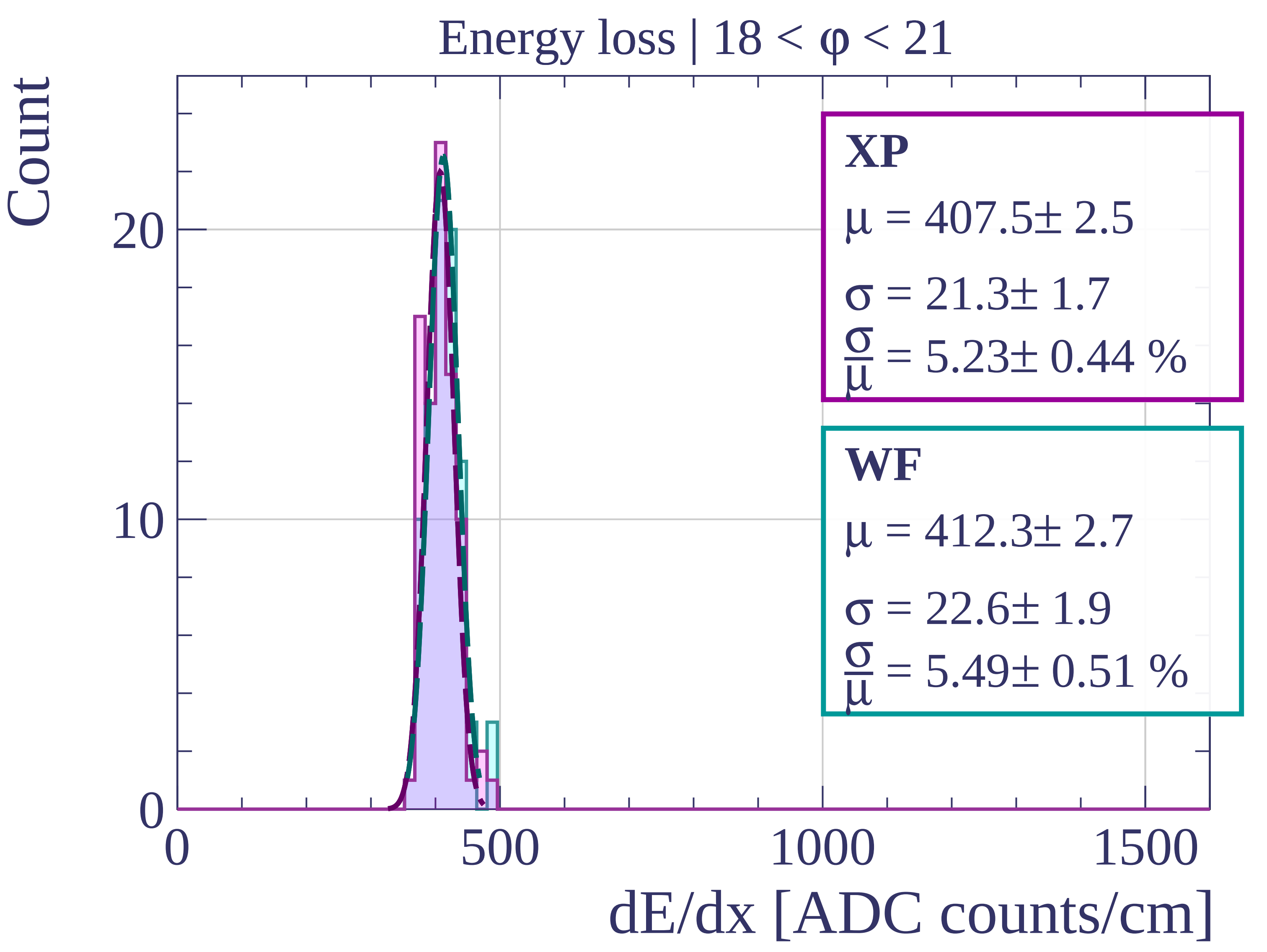


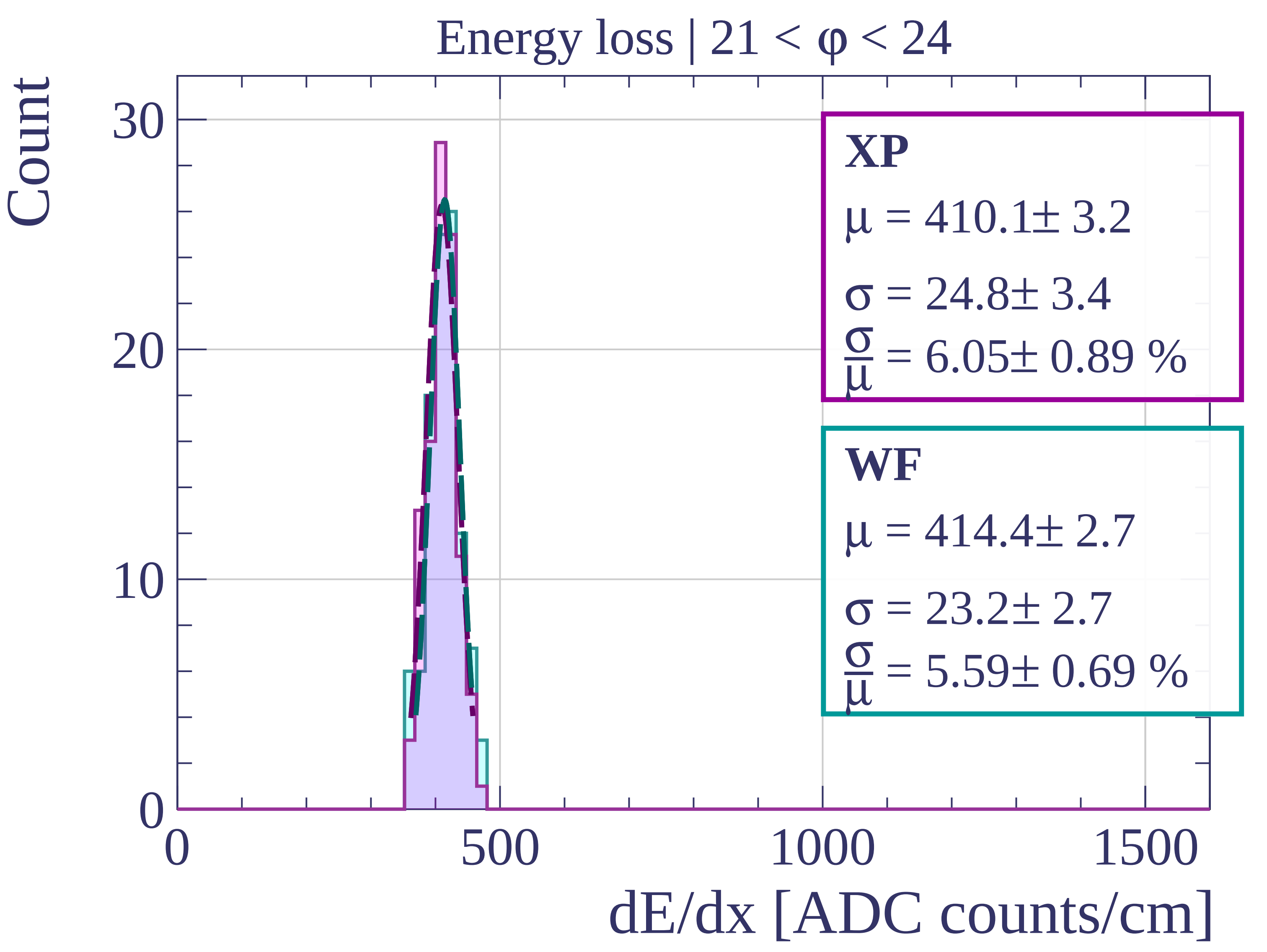
Energy loss | 9 < φ < 12
Count
dE/dx [ADC counts/cm]
0
5
10
15
20
0
500
1000
1500
XP
μ = 413.3± 3.9
σ = 26.0± 4.7
σ/μ = 6.29± 1.20 %
WF
μ = 415.9± 3.2
σ = 23.1± 3.3
σ/μ = 5.56± 0.83 %

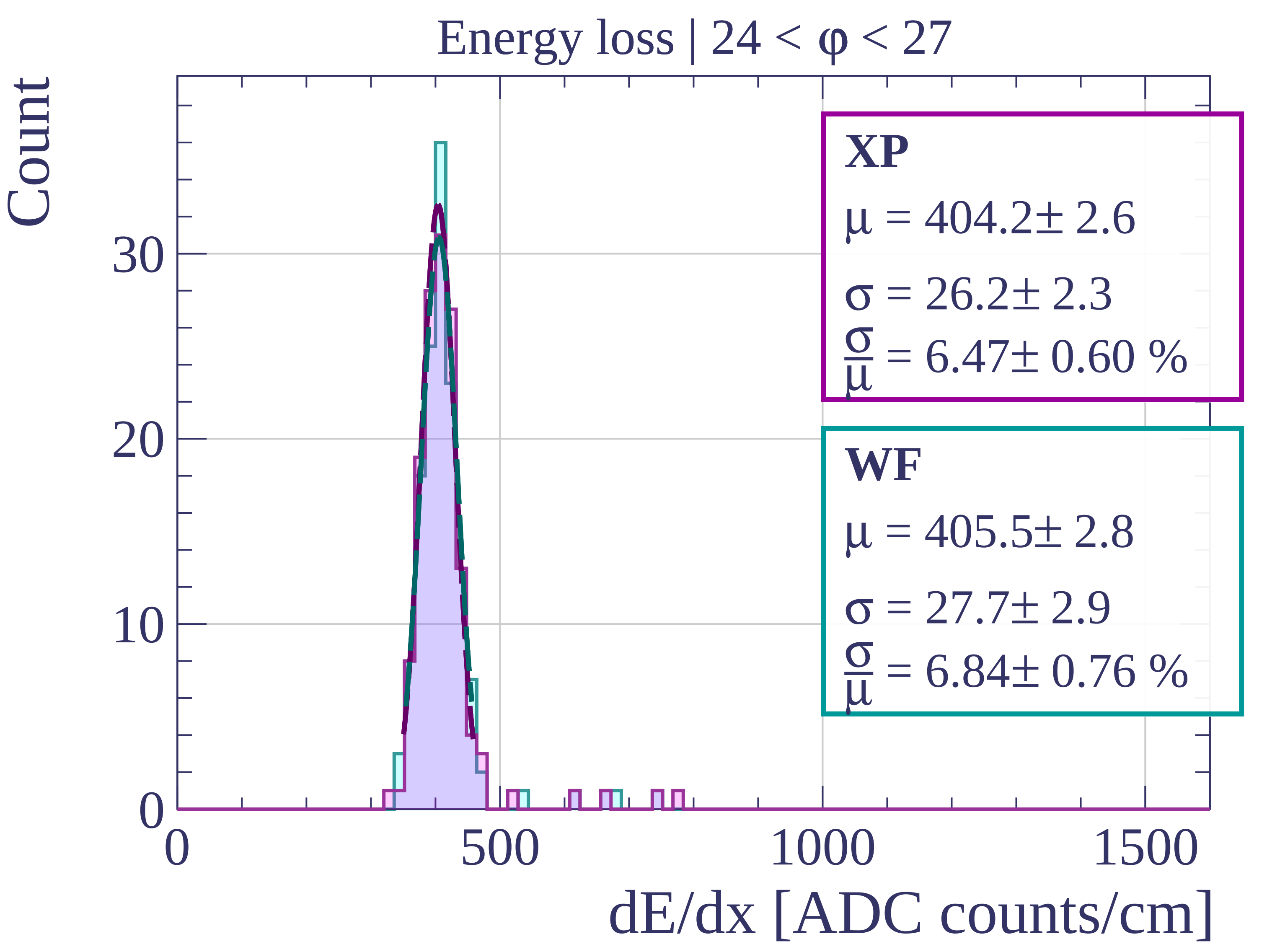


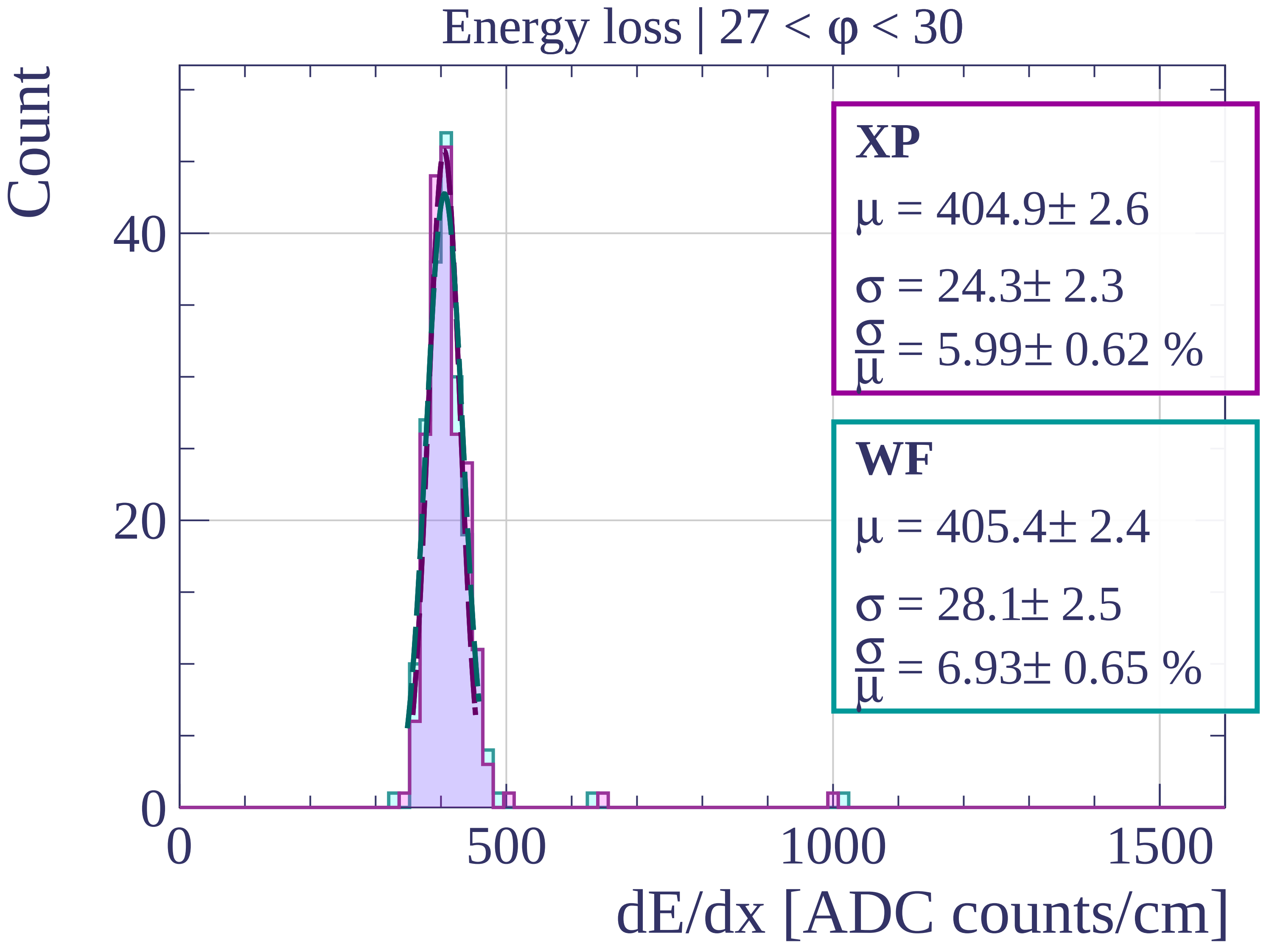
Energy loss | 12 < φ < 15
Count
dE/dx [ADC counts/cm]
0
5
10
15
0
500
1000
1500
XP
μ = 403.8± 3.0
σ = 23.2± 2.8
σ/μ = 5.75± 0.73 %
WF
μ = 410.2± 3.5
σ = 26.0± 3.2
σ/μ = 6.35± 0.83 %


Energy loss | 15 < φ < 18
Count
dE/dx [ADC counts/cm]
0
5
10
15
20
0
500
1000
1500
XP
μ = 408.6± 3.6
σ = 20.8± 3.7
σ/μ = 5.09± 0.94 %
WF
μ = 422.0± 2.3
σ = 12.6± 2.4
σ/μ = 2.98± 0.59 %

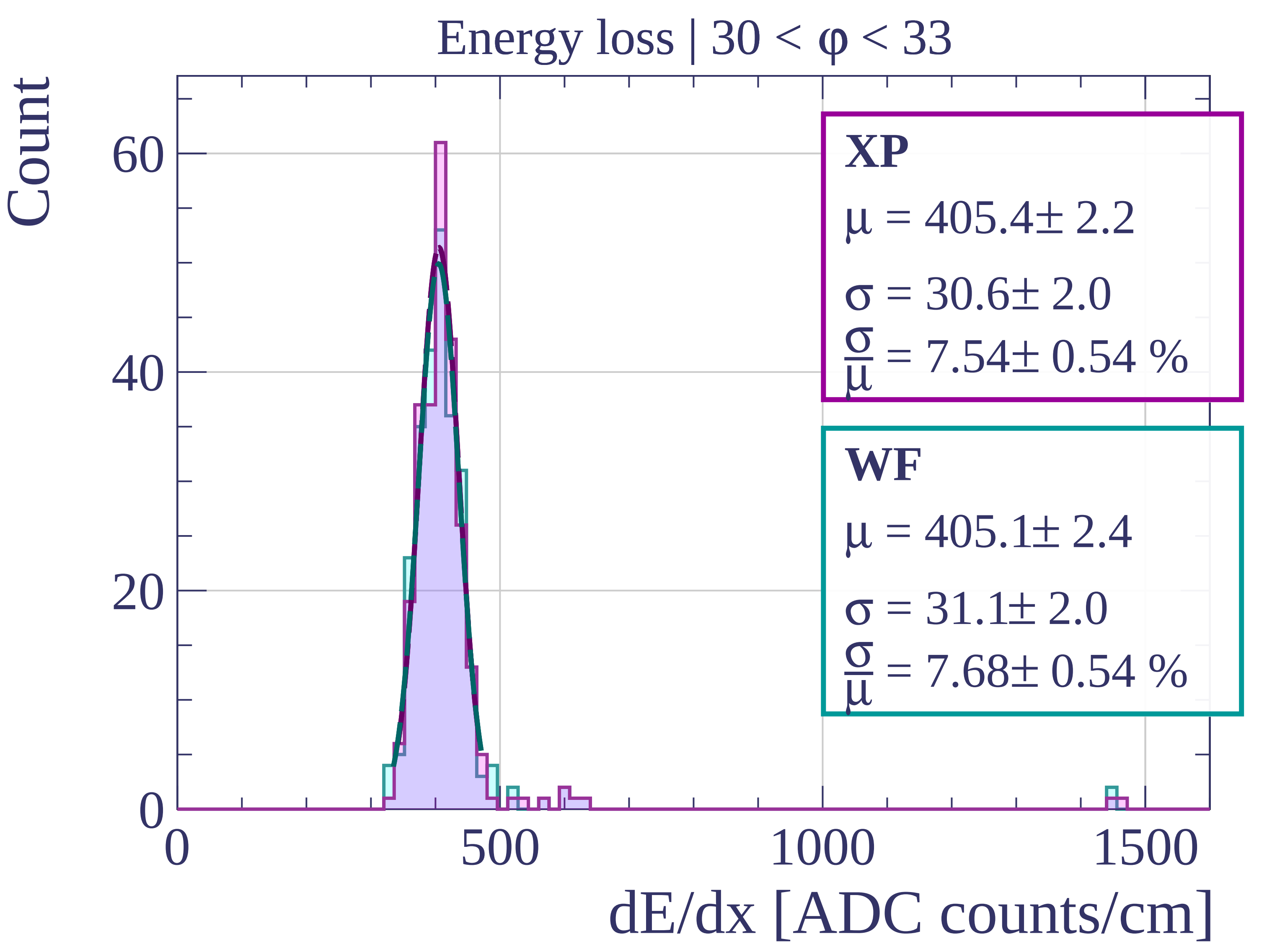

Energy loss | 18 < φ < 21
Count
dE/dx [ADC counts/cm]
0
10
20
0
500
1000
1500
XP
μ = 407.5± 2.5
σ = 21.3± 1.7
σ/μ = 5.23± 0.44 %
WF
μ = 412.3± 2.7
σ = 22.6± 1.9
σ/μ = 5.49± 0.51 %


Energy loss | 21 < φ < 24
Count
dE/dx [ADC counts/cm]
0
10
20
30
0
500
1000
1500
XP
μ = 410.1± 3.2
σ = 24.8± 3.4
σ/μ = 6.05± 0.89 %
WF
μ = 414.4± 2.7
σ = 23.2± 2.7
σ/μ = 5.59± 0.69 %

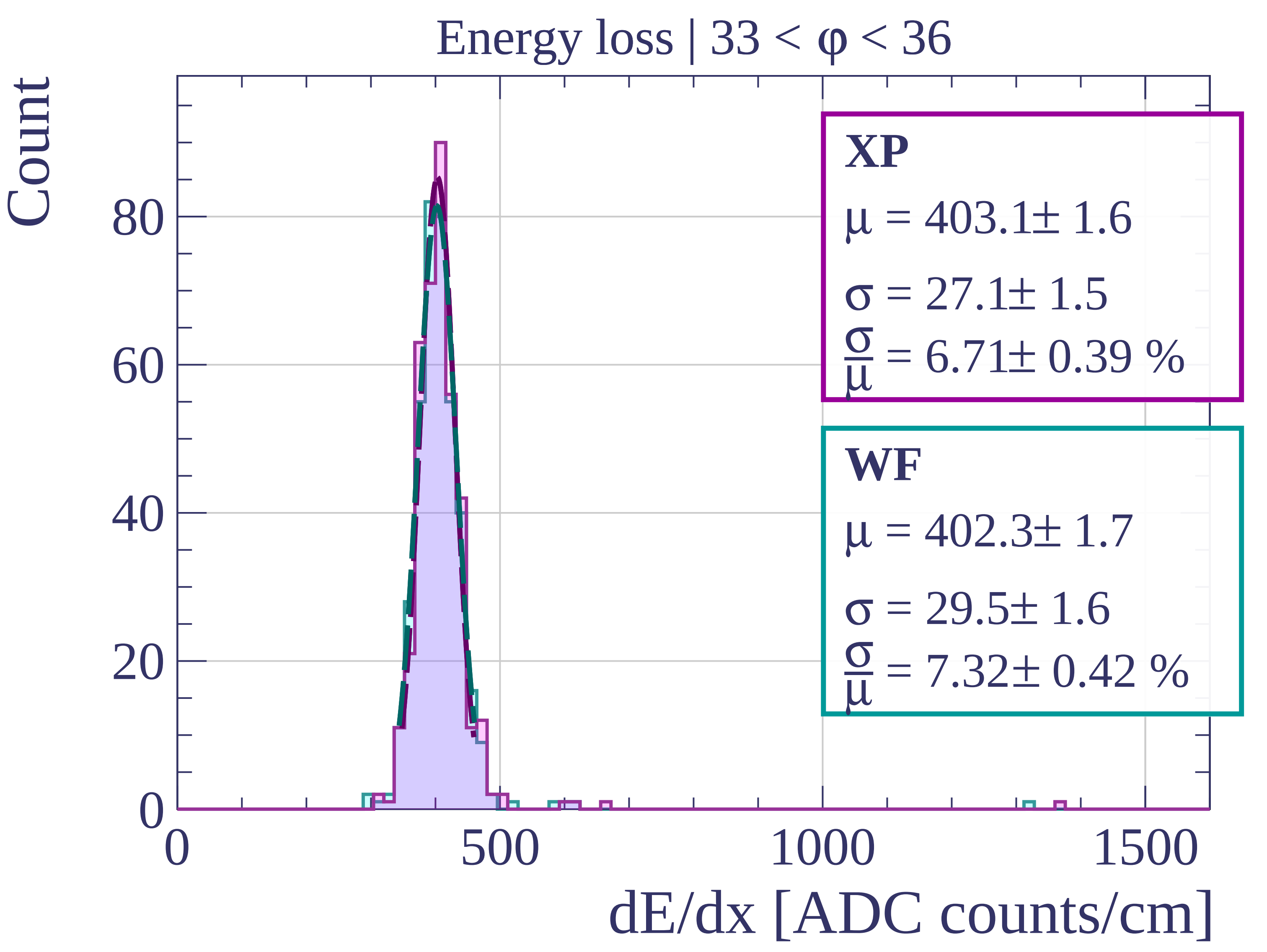

Energy loss | 24 < φ < 27
Count
dE/dx [ADC counts/cm]
0
10
20
30
0
500
1000
1500
XP
μ = 404.2± 2.6
σ = 26.2± 2.3
σ/μ = 6.47± 0.60 %
WF
μ = 405.5± 2.8
σ = 27.7± 2.9
σ/μ = 6.84± 0.76 %



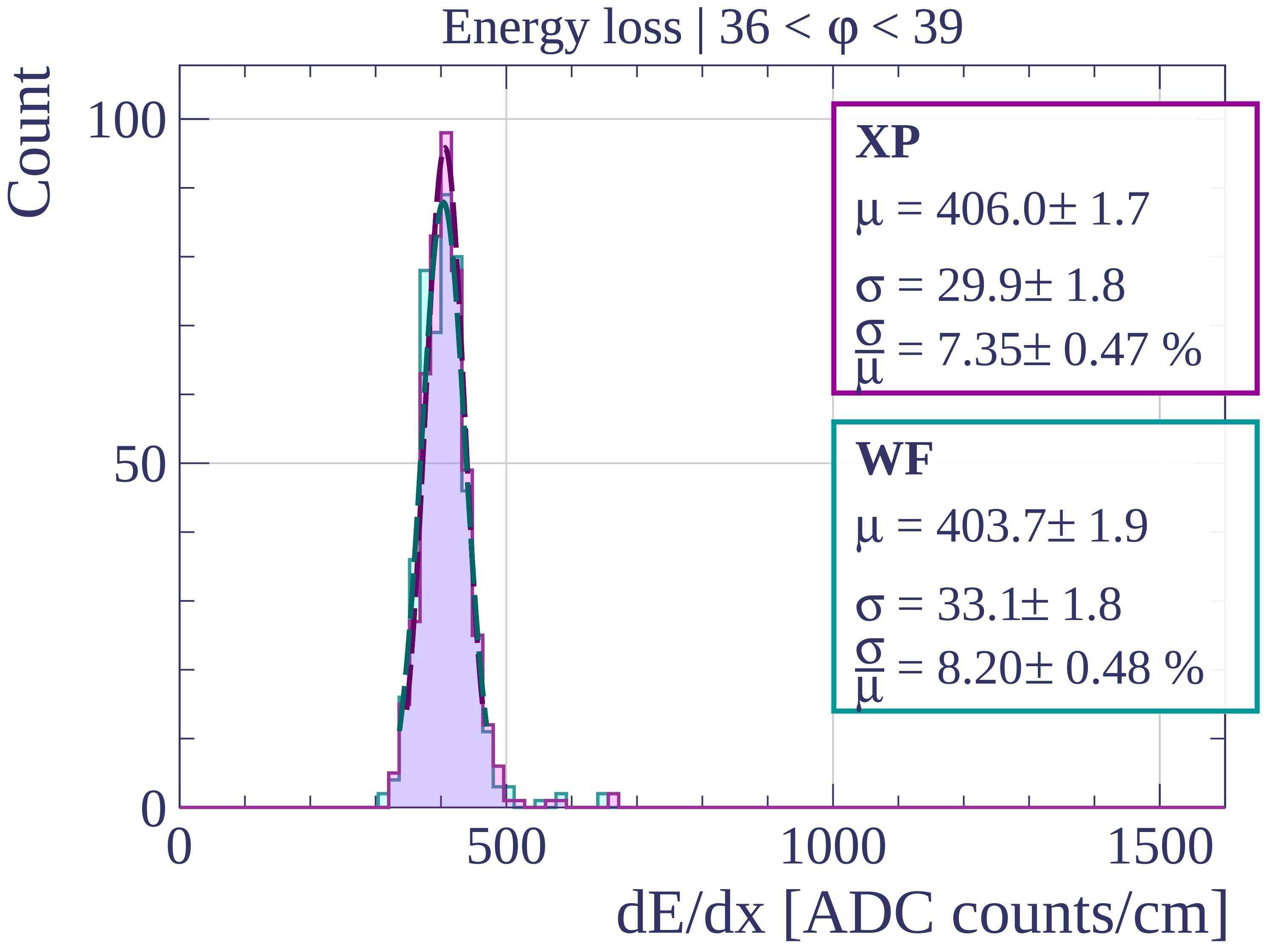
Energy loss | 27 < φ < 30
Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 404.9± 2.6
σ = 24.3± 2.3
σ/μ = 5.99± 0.62 %
WF
μ = 405.4± 2.4
σ = 28.1± 2.5
σ/μ = 6.93± 0.65 %



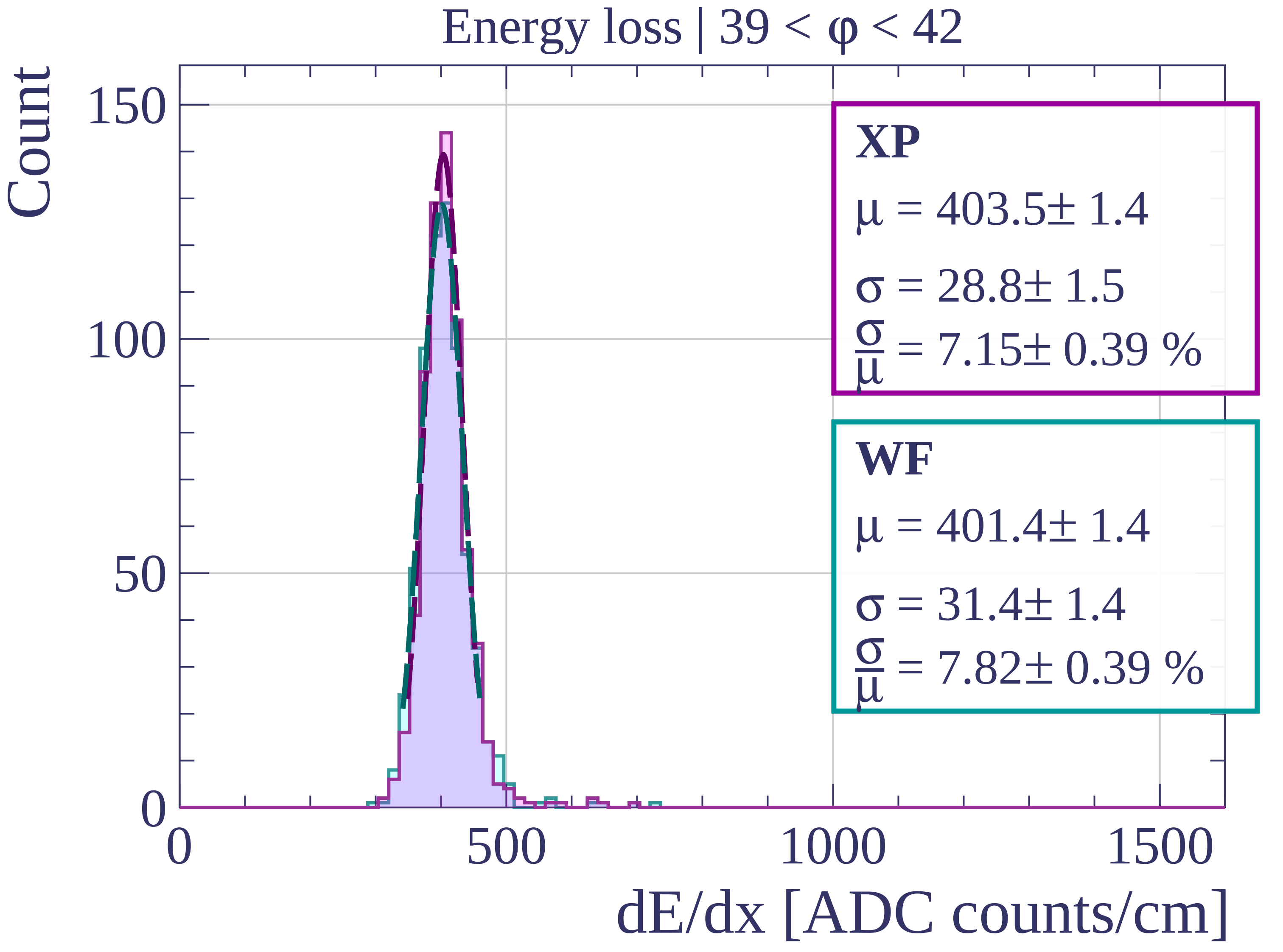
Energy loss | 30 < φ < 33
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 405.4± 2.2
σ = 30.6± 2.0
σ/μ = 7.54± 0.54 %
WF
μ = 405.1± 2.4
σ = 31.1± 2.0
σ/μ = 7.68± 0.54 %



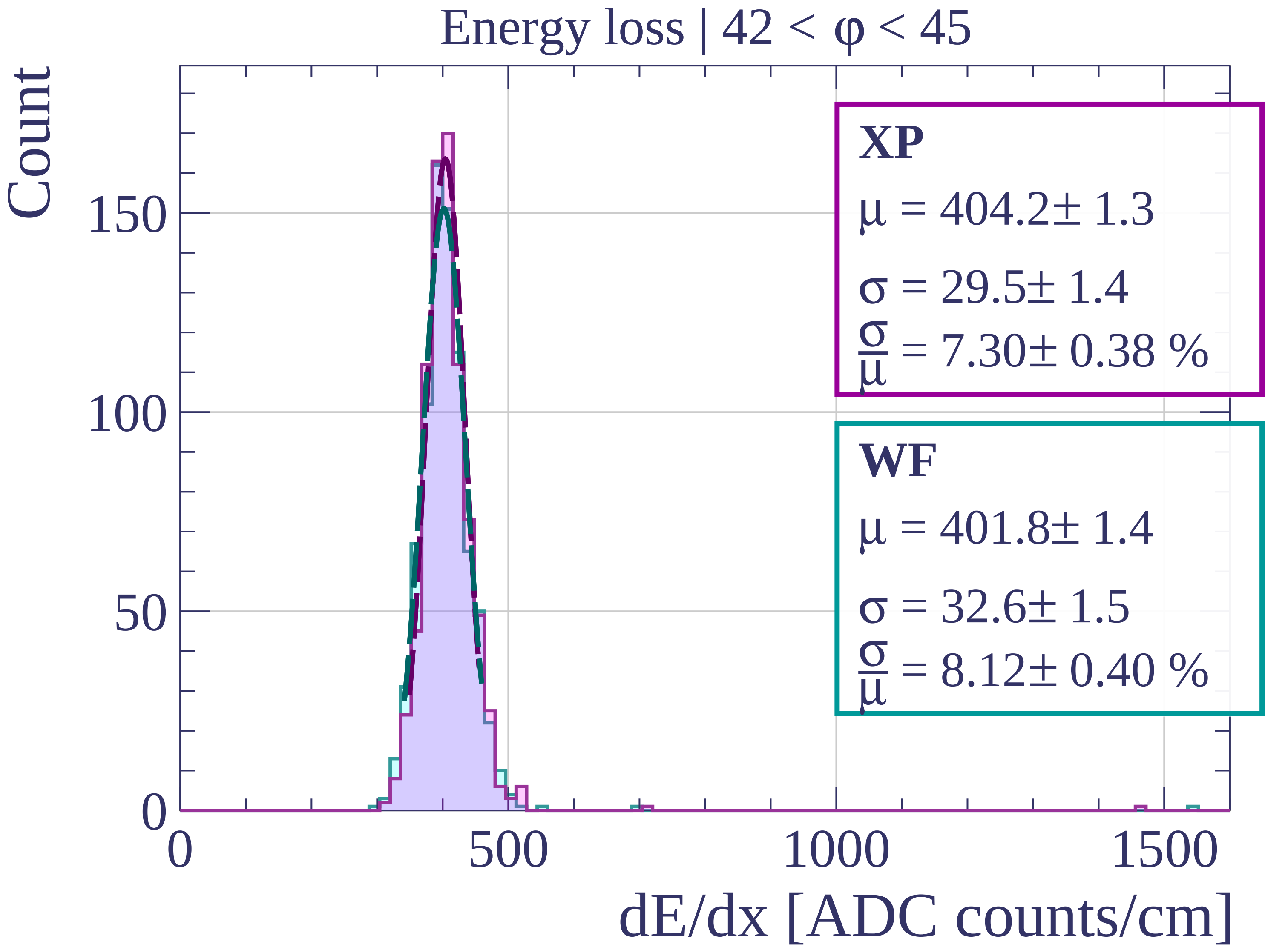
Energy loss | 33 < φ < 36
Count
0
20
40
60
80
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 403.1± 1.6
σ = 27.1± 1.5
σ/μ = 6.71± 0.39 %
WF
μ = 402.3± 1.7
σ = 29.5± 1.6
σ/μ = 7.32± 0.42 %

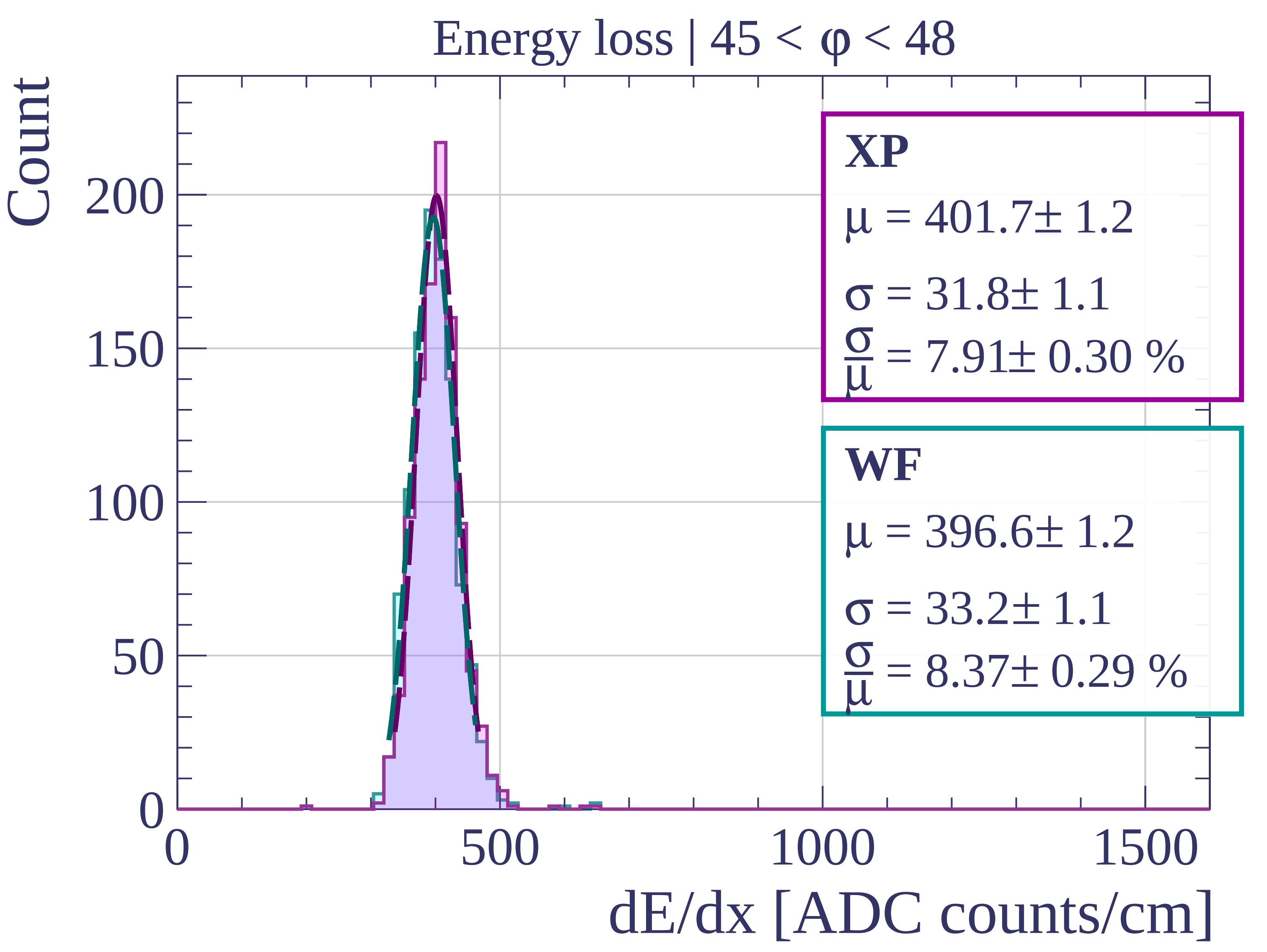


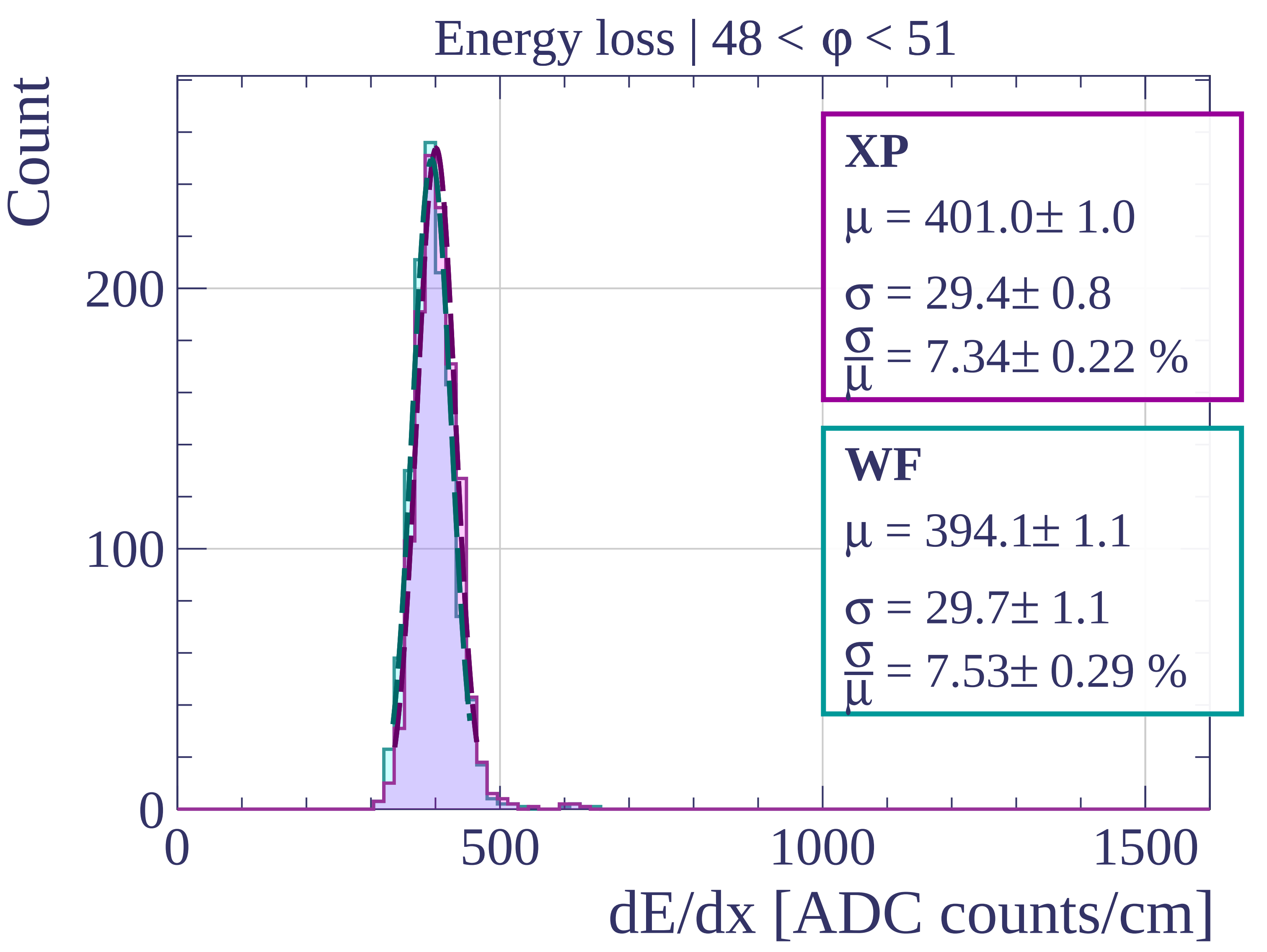
Energy loss | 36 < φ < 39
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 406.0± 1.7
σ = 29.9± 1.8
σ/μ = 7.35± 0.47 %
WF
μ = 403.7± 1.9
σ = 33.1± 1.8
σ/μ = 8.20± 0.48 %

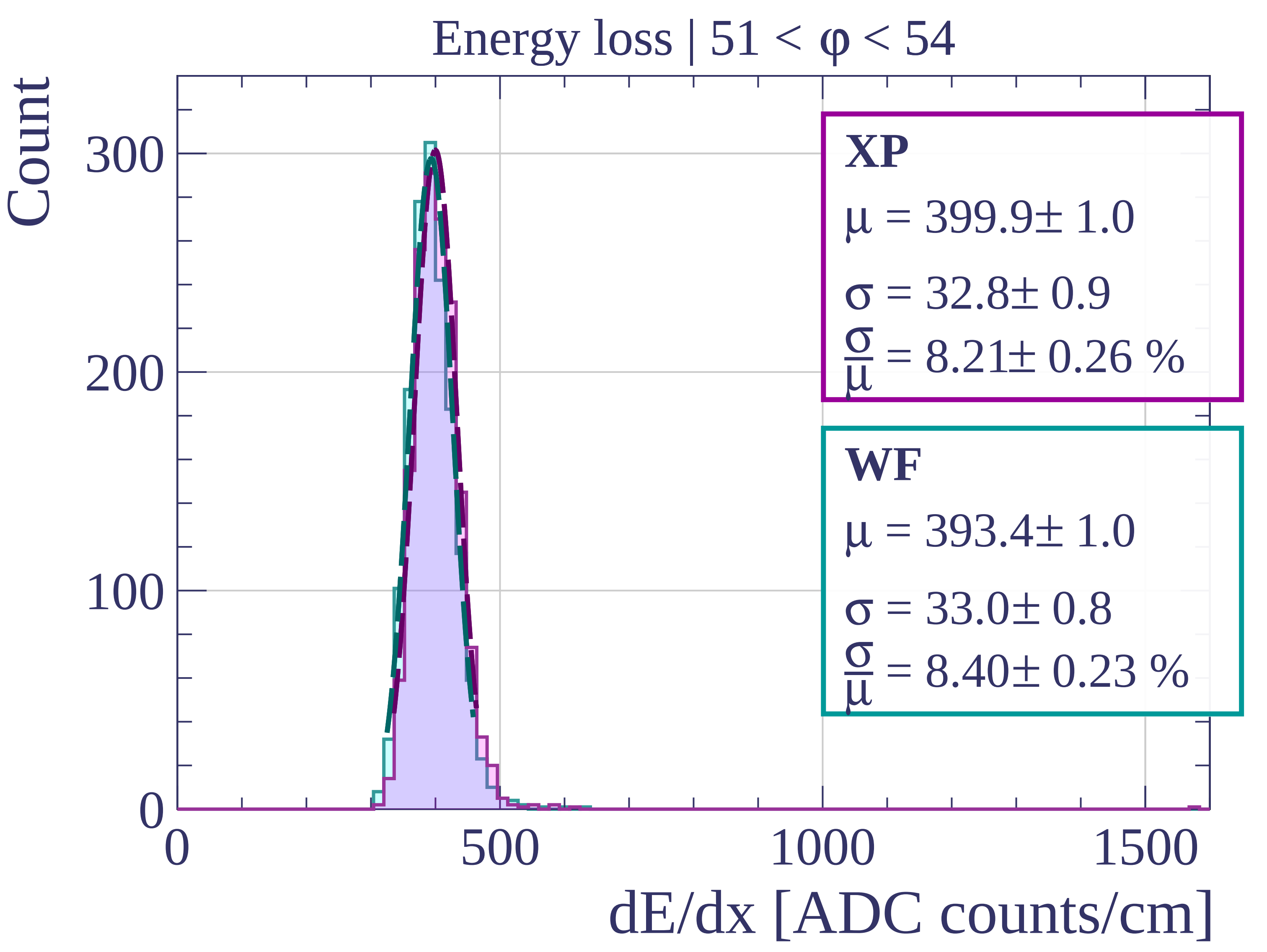


Energy loss | 39 < φ < 42
Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 403.5± 1.4
σ = 28.8± 1.5
σ/μ = 7.15± 0.39 %
WF
μ = 401.4± 1.4
σ = 31.4± 1.4
σ/μ = 7.82± 0.39 %

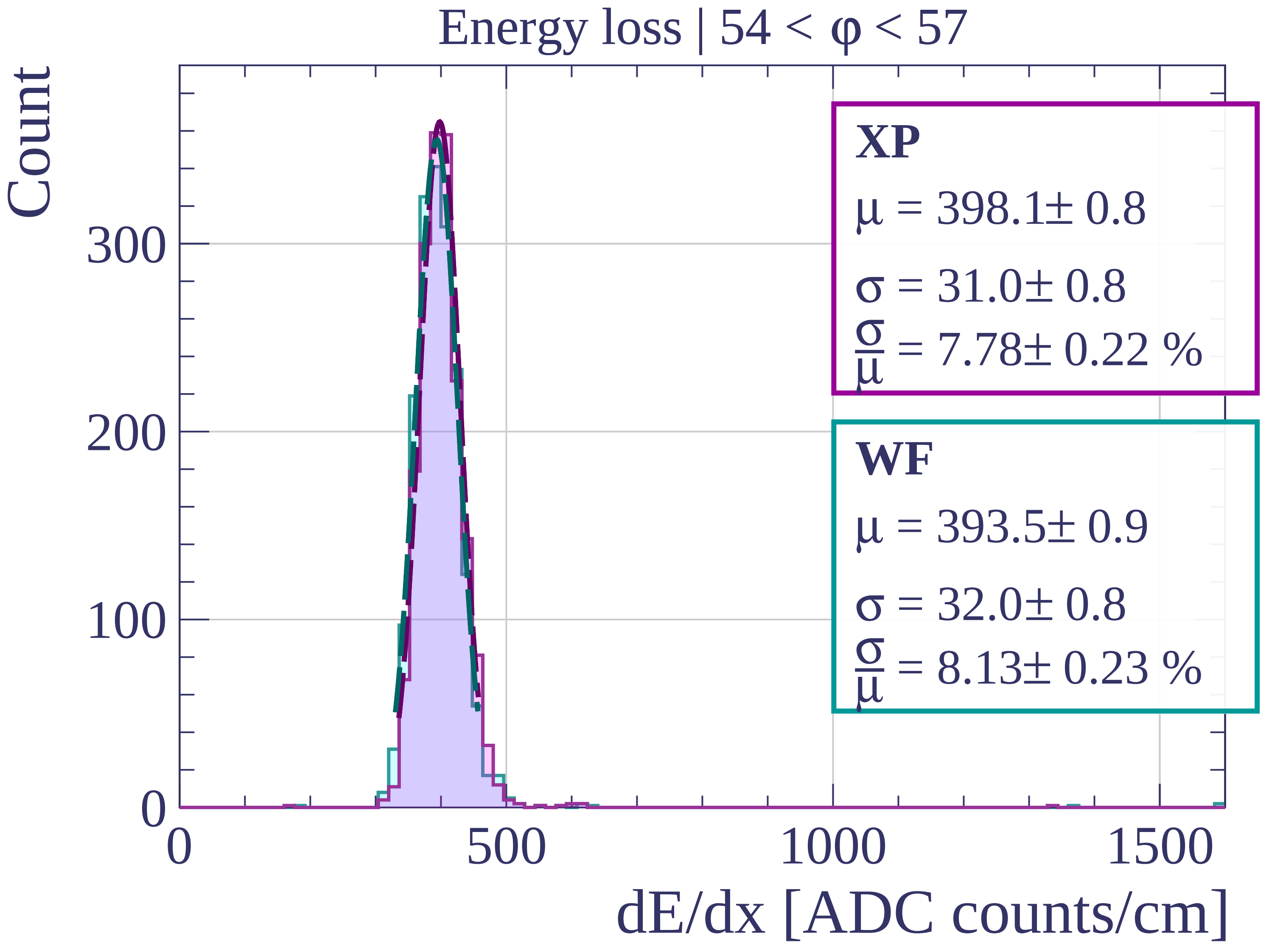

Energy loss | 42 < φ < 45
Count
dE/dx [ADC counts/cm]
0
50
100
150
0
500
1000
1500
XP
μ = 404.2± 1.3
σ = 29.5± 1.4
σ/μ = 7.30± 0.38 %
WF
μ = 401.8± 1.4
σ = 32.6± 1.5
σ/μ = 8.12± 0.40 %

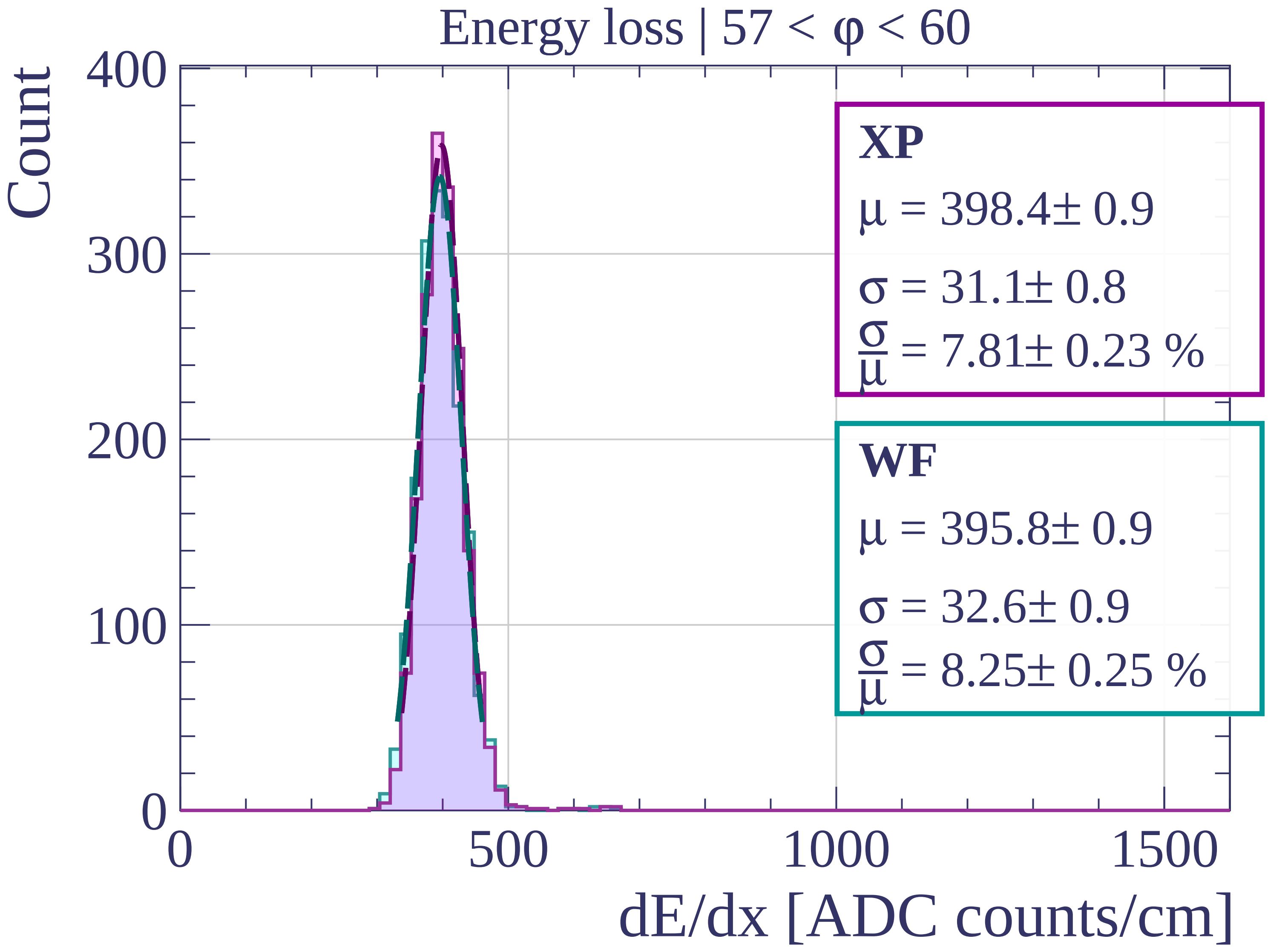

Energy loss | 45 < φ < 48
Count
dE/dx [ADC counts/cm]
0
50
100
150
200
0
500
1000
1500
XP
μ = 401.7± 1.2
σ = 31.8± 1.1
σ/μ = 7.91± 0.30 %
WF
μ = 396.6± 1.2
σ = 33.2± 1.1
σ/μ = 8.37± 0.29 %

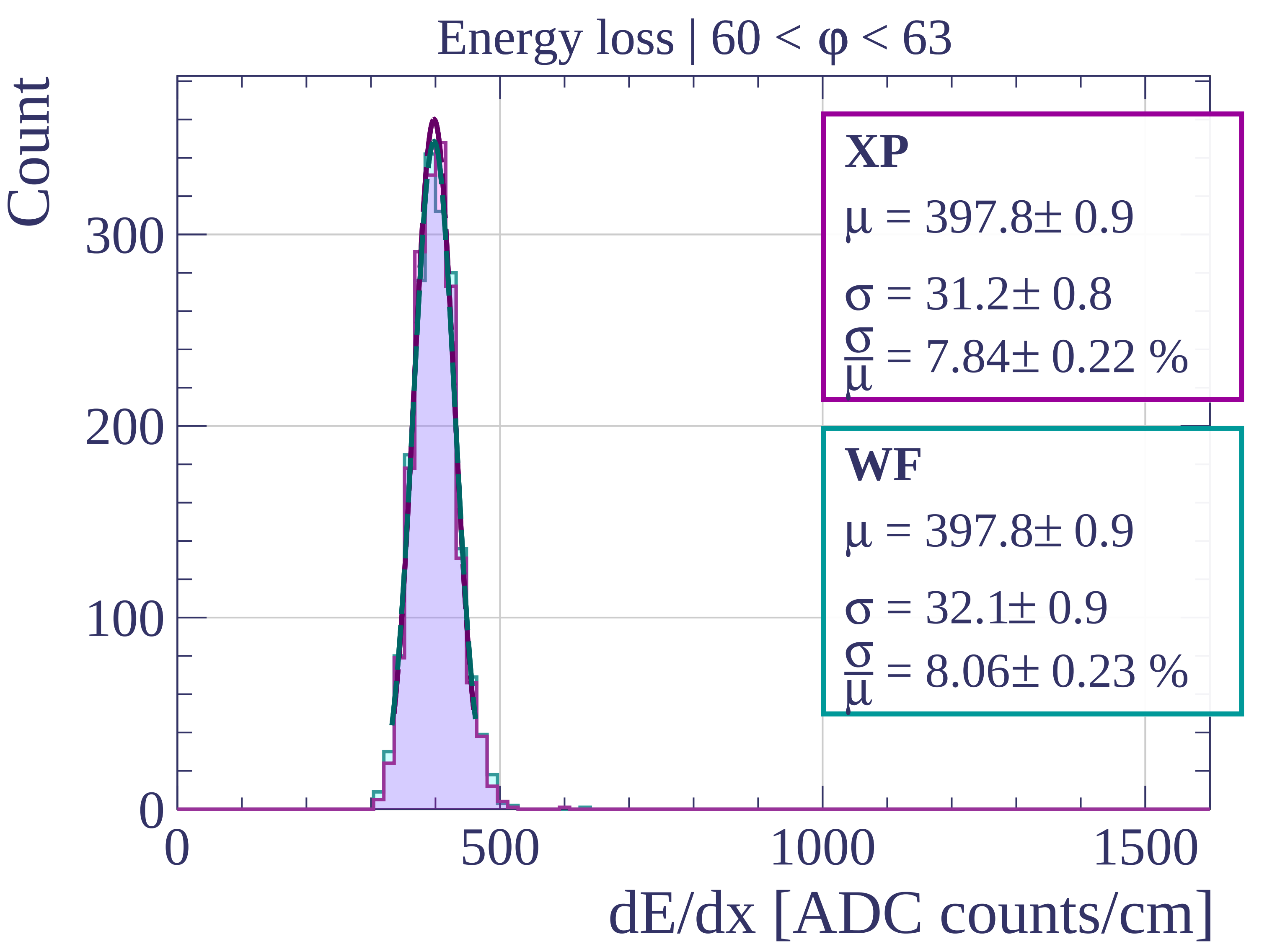

Energy loss | 48 < φ < 51
Count
0
100
200
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 401.0± 1.0
σ = 29.4± 0.8
σ/μ = 7.34± 0.22 %
WF
μ = 394.1± 1.1
σ = 29.7± 1.1
σ/μ = 7.53± 0.29 %

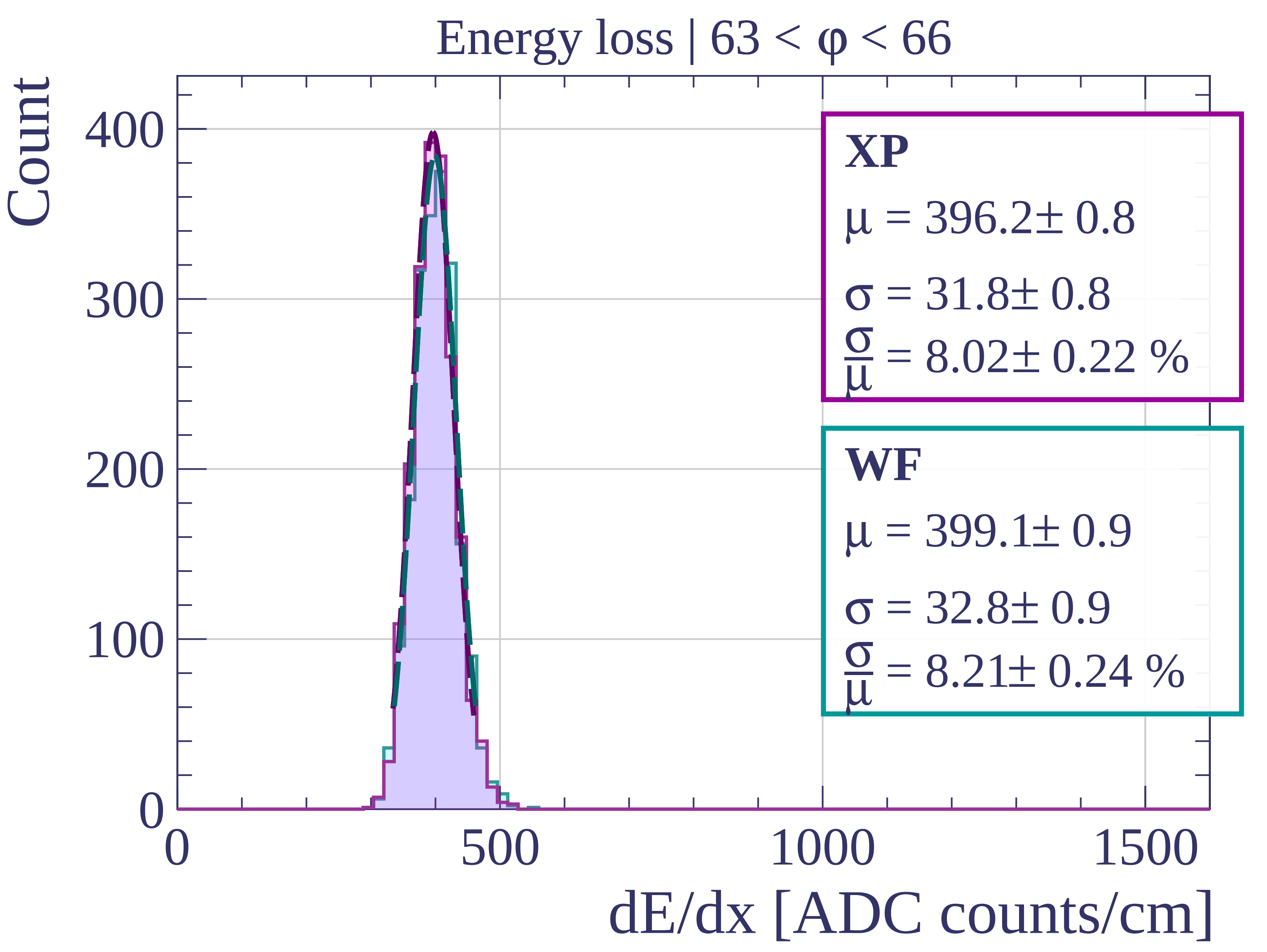


Energy loss | 51 < φ < 54
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 399.9± 1.0
σ = 32.8± 0.9
σ/μ = 8.21± 0.26 %
WF
μ = 393.4± 1.0
σ = 33.0± 0.8
σ/μ = 8.40± 0.23 %

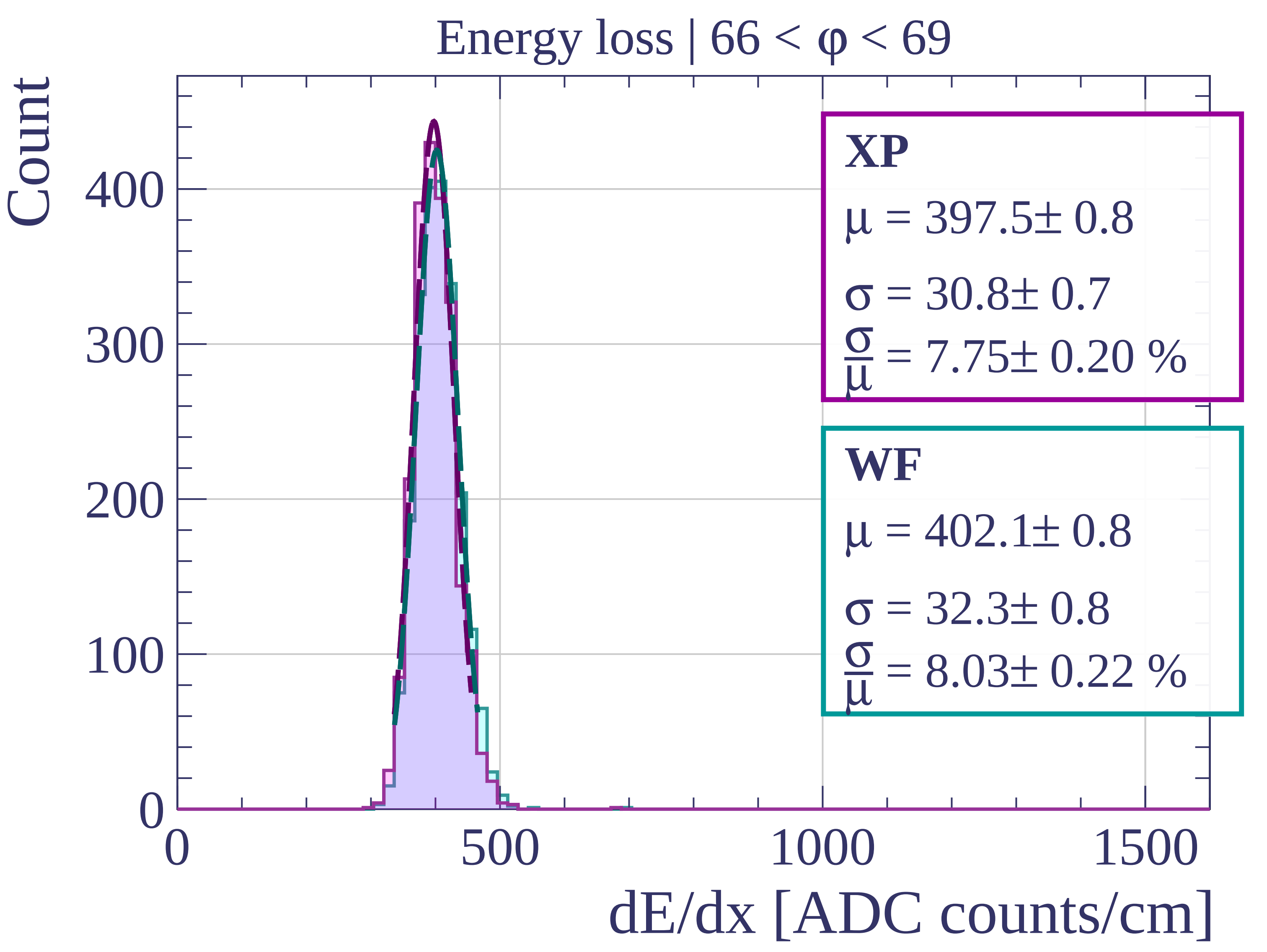

Energy loss | 54 < φ < 57
Count
dE/dx [ADC counts/cm]
0
100
200
300
0
500
1000
1500
XP
μ = 398.1± 0.8
σ = 31.0± 0.8
σ/μ = 7.78± 0.22 %
WF
μ = 393.5± 0.9
σ = 32.0± 0.8
σ/μ = 8.13± 0.23 %

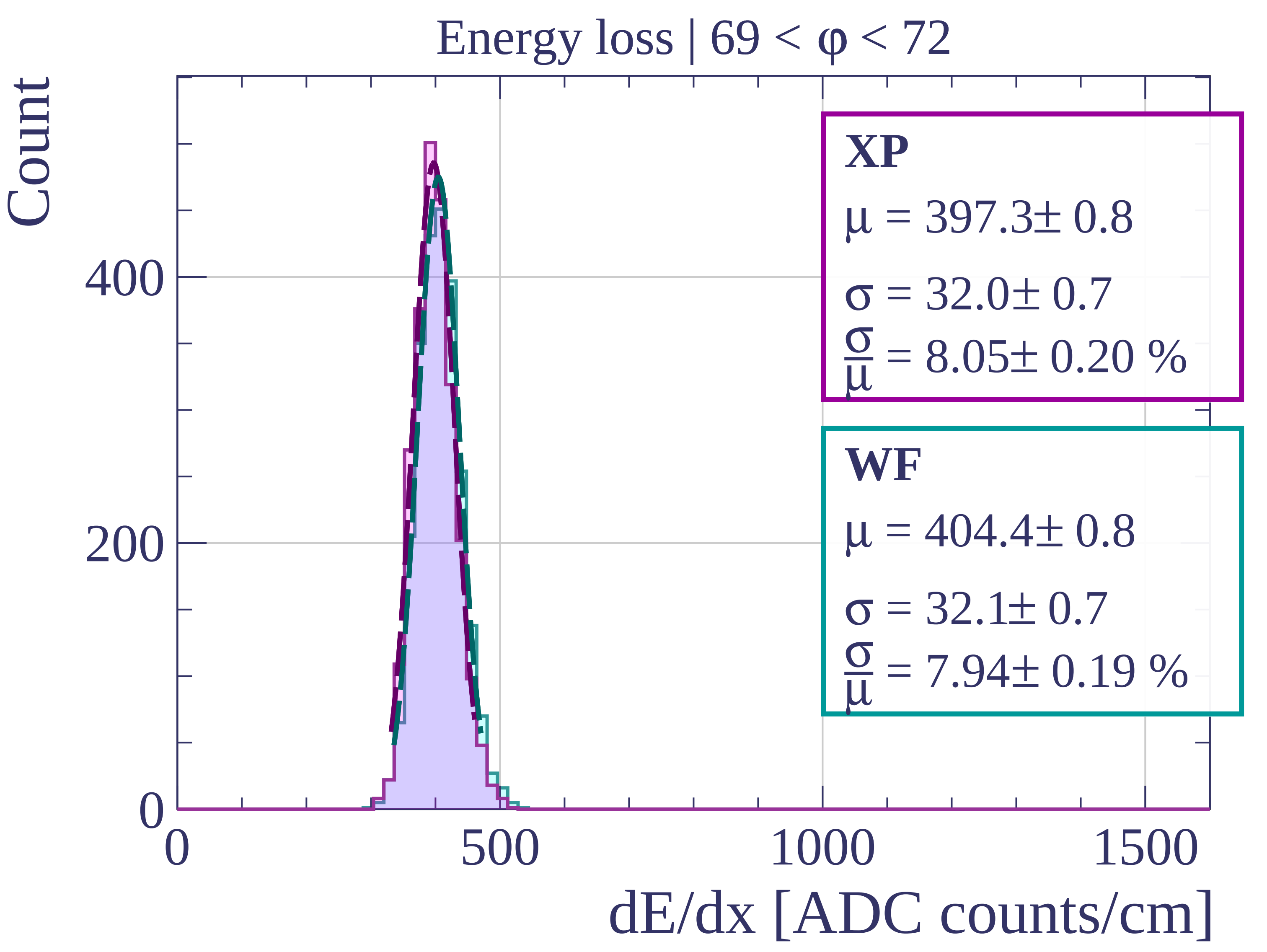

Energy loss | 57 < φ < 60
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 398.4± 0.9
σ = 31.1± 0.8
σ/μ = 7.81± 0.23 %
WF
μ = 395.8± 0.9
σ = 32.6± 0.9
σ/μ = 8.25± 0.25 %


Energy loss | 60 < φ < 63
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 397.8± 0.9
σ = 31.2± 0.8
σ/μ = 7.84± 0.22 %
WF
μ = 397.8± 0.9
σ = 32.1± 0.9
σ/μ = 8.06± 0.23 %

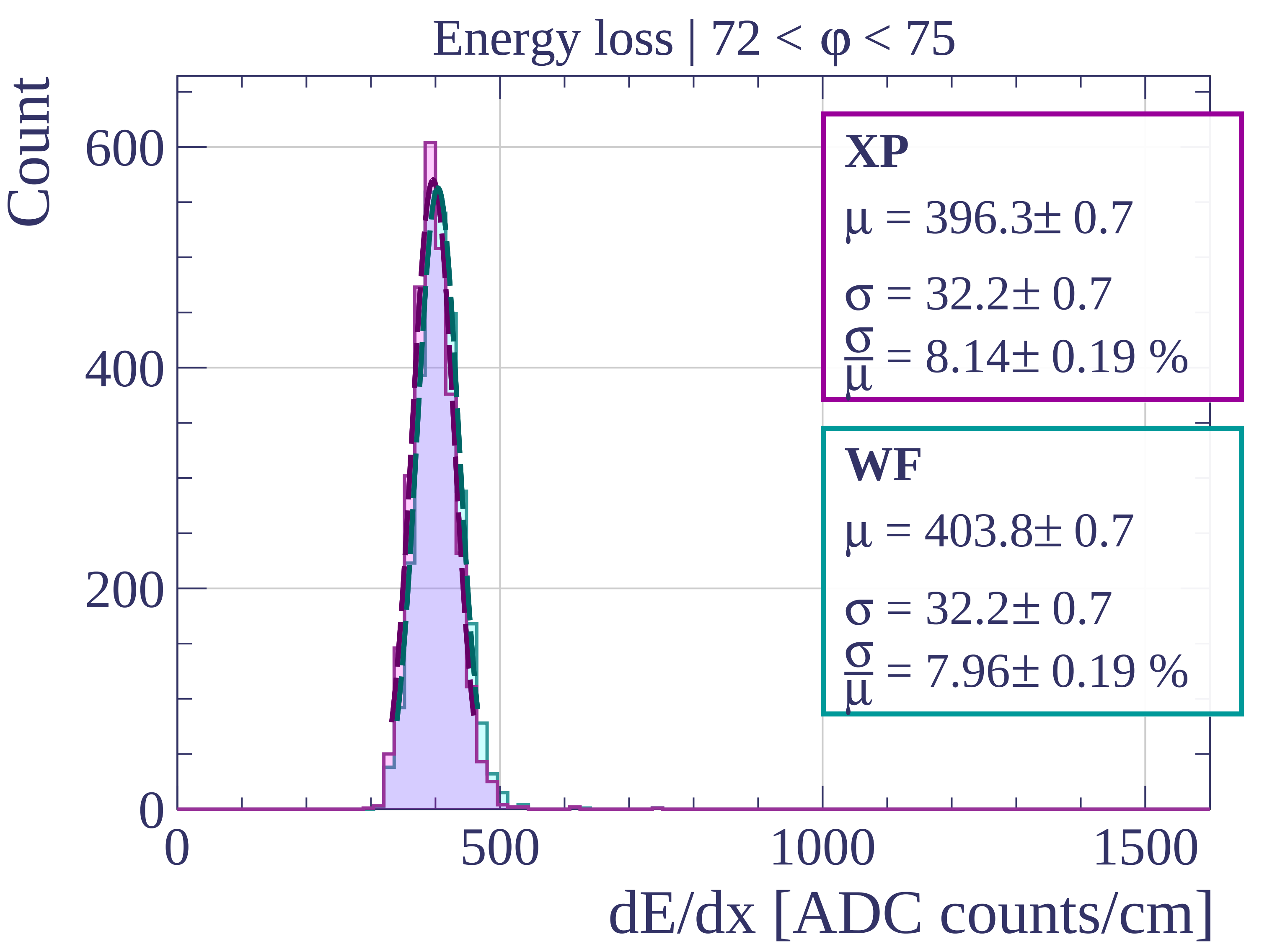


Energy loss | 63 < φ < 66
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 396.2± 0.8
σ = 31.8± 0.8
σ/μ = 8.02± 0.22 %
WF
μ = 399.1± 0.9
σ = 32.8± 0.9
σ/μ = 8.21± 0.24 %

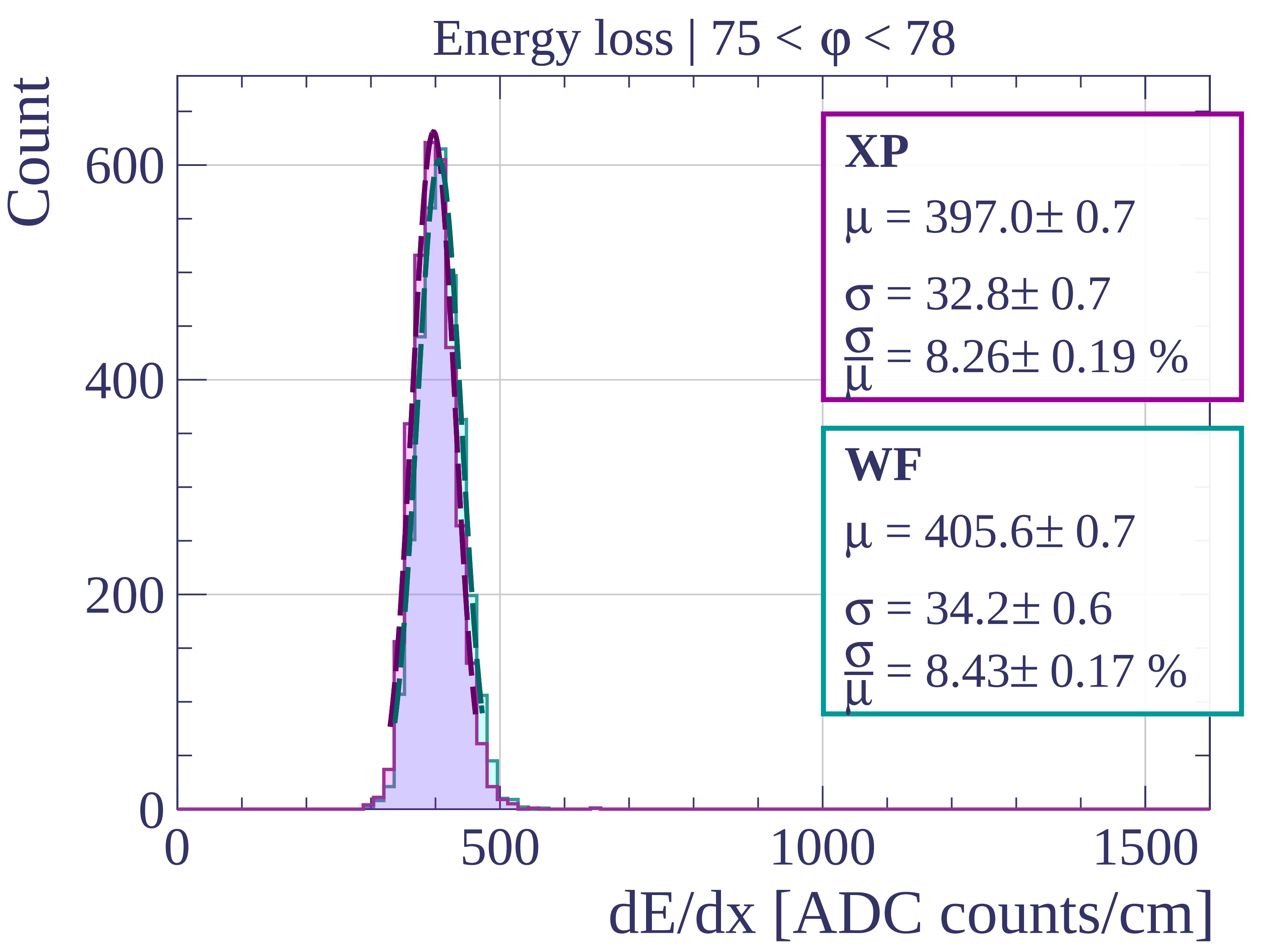

Energy loss | 66 < φ < 69
Count
dE/dx [ADC counts/cm]
0
100
200
300
400
0
500
1000
1500
XP
μ = 397.5± 0.8
σ = 30.8± 0.7
σ/μ = 7.75± 0.20 %
WF
μ = 402.1± 0.8
σ = 32.3± 0.8
σ/μ = 8.03± 0.22 %



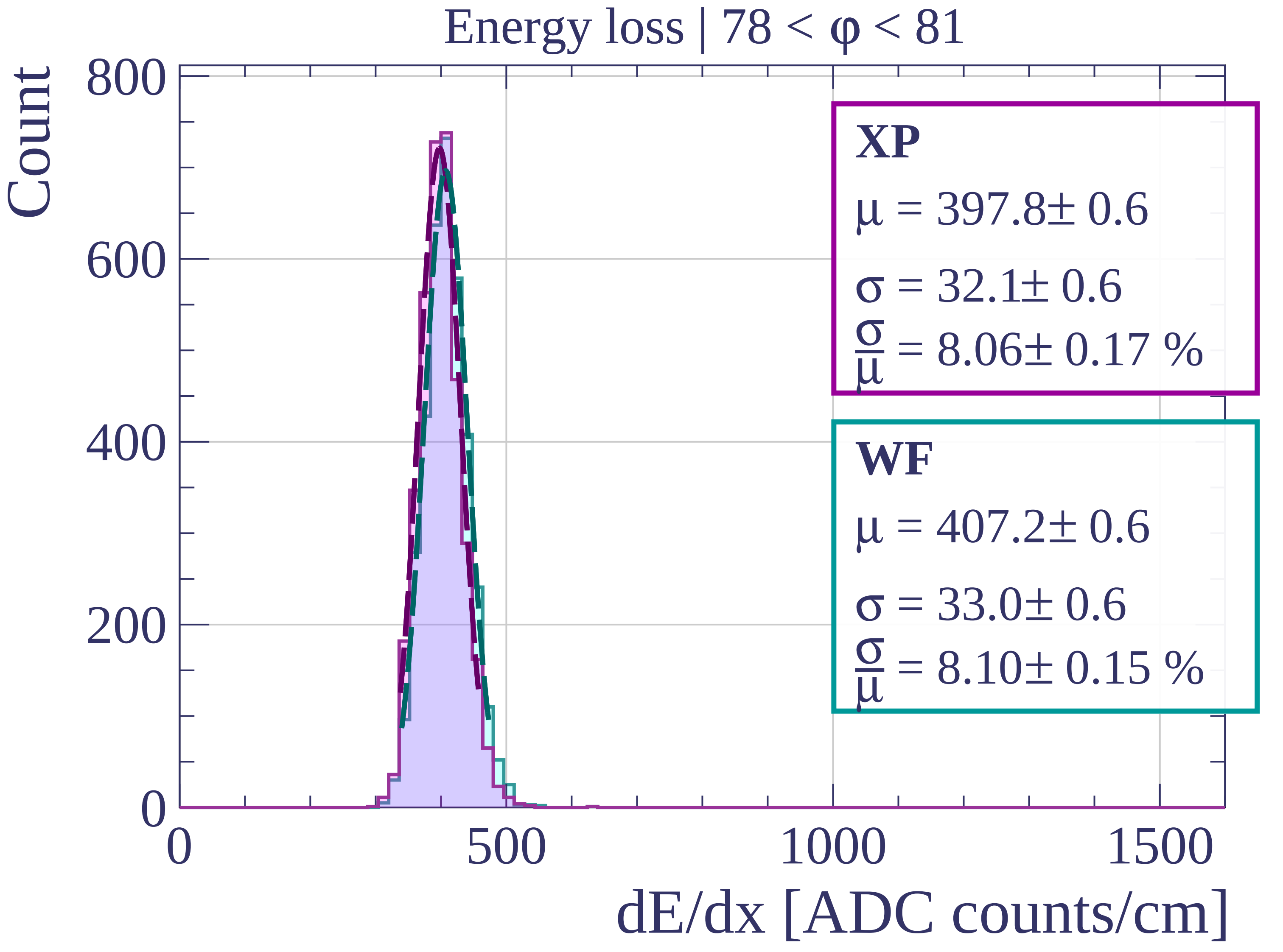
Energy loss | 69 < φ < 72
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 397.3± 0.8
σ = 32.0± 0.7
σ/μ = 8.05± 0.20 %
WF
μ = 404.4± 0.8
σ = 32.1± 0.7
σ/μ = 7.94± 0.19 %



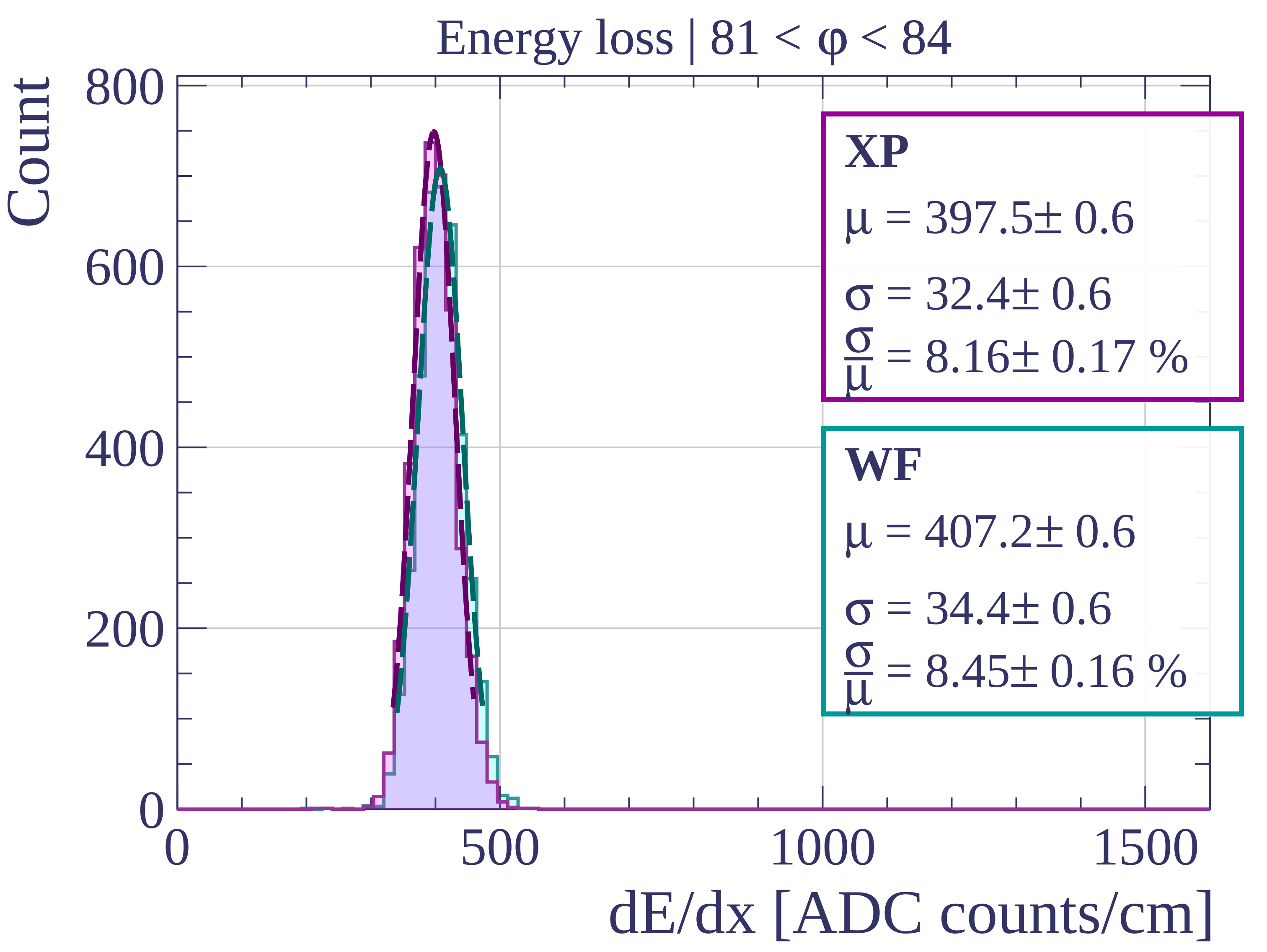
Energy loss | 72 < φ < 75
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 396.3± 0.7
σ = 32.2± 0.7
σ/μ = 8.14± 0.19 %
WF
μ = 403.8± 0.7
σ = 32.2± 0.7
σ/μ = 7.96± 0.19 %



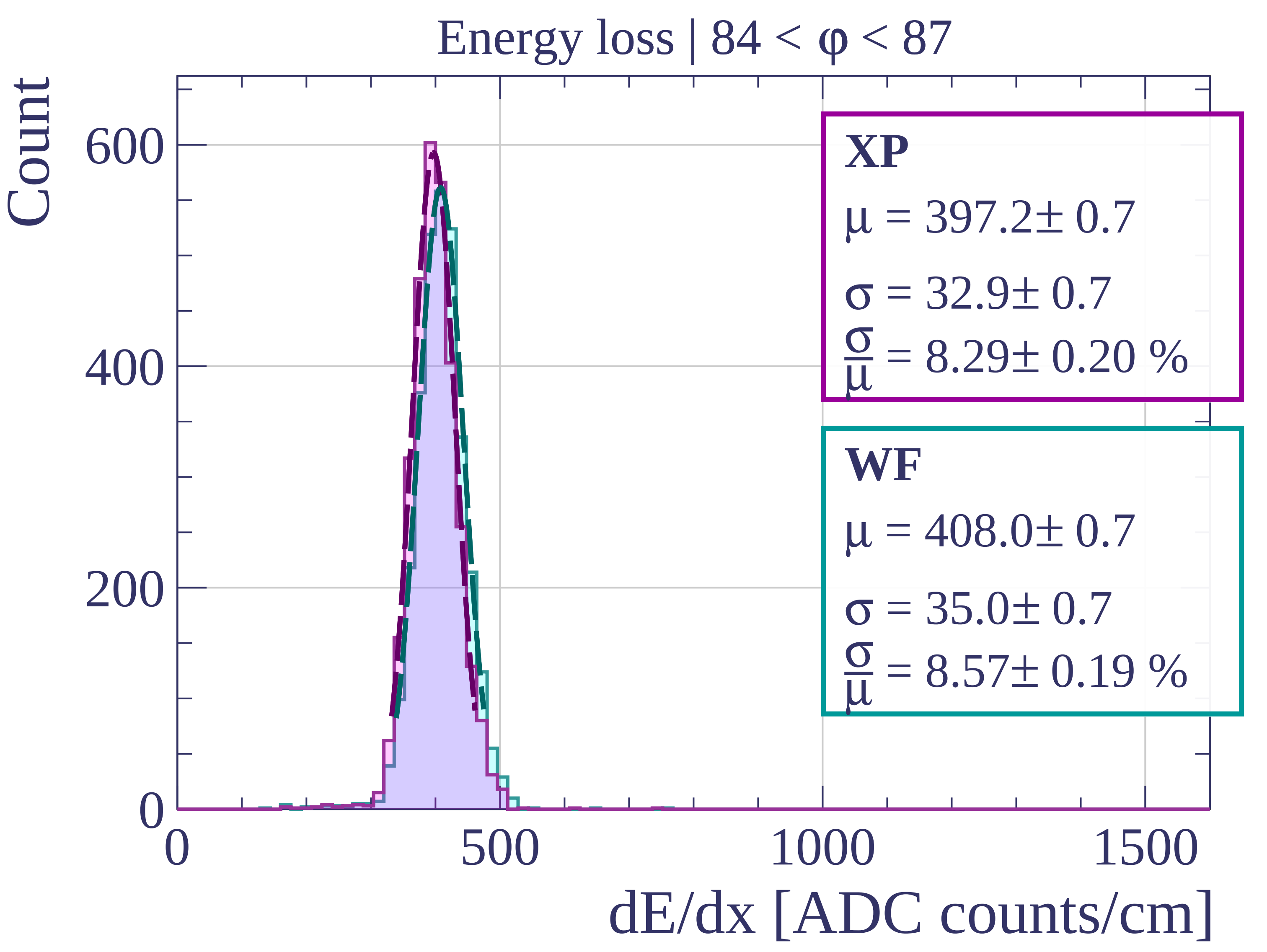
Energy loss | 75 < φ < 78
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 397.0± 0.7
σ = 32.8± 0.7
σ/μ = 8.26± 0.19 %
WF
μ = 405.6± 0.7
σ = 34.2± 0.6
σ/μ = 8.43± 0.17 %



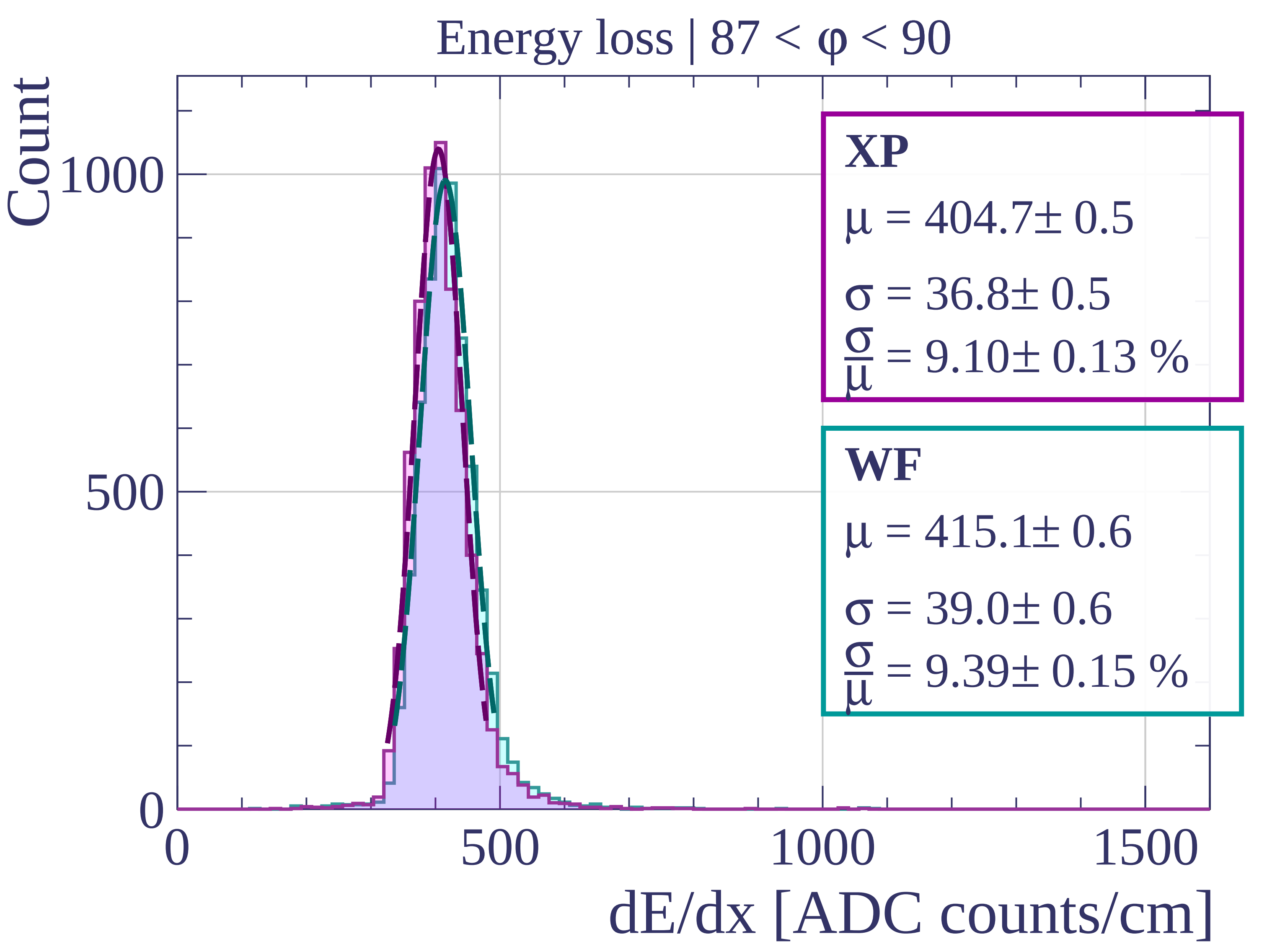
Energy loss | 78 < φ < 81
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 397.8± 0.6
σ = 32.1± 0.6
σ/μ = 8.06± 0.17 %
WF
μ = 407.2± 0.6
σ = 33.0± 0.6
σ/μ = 8.10± 0.15 %



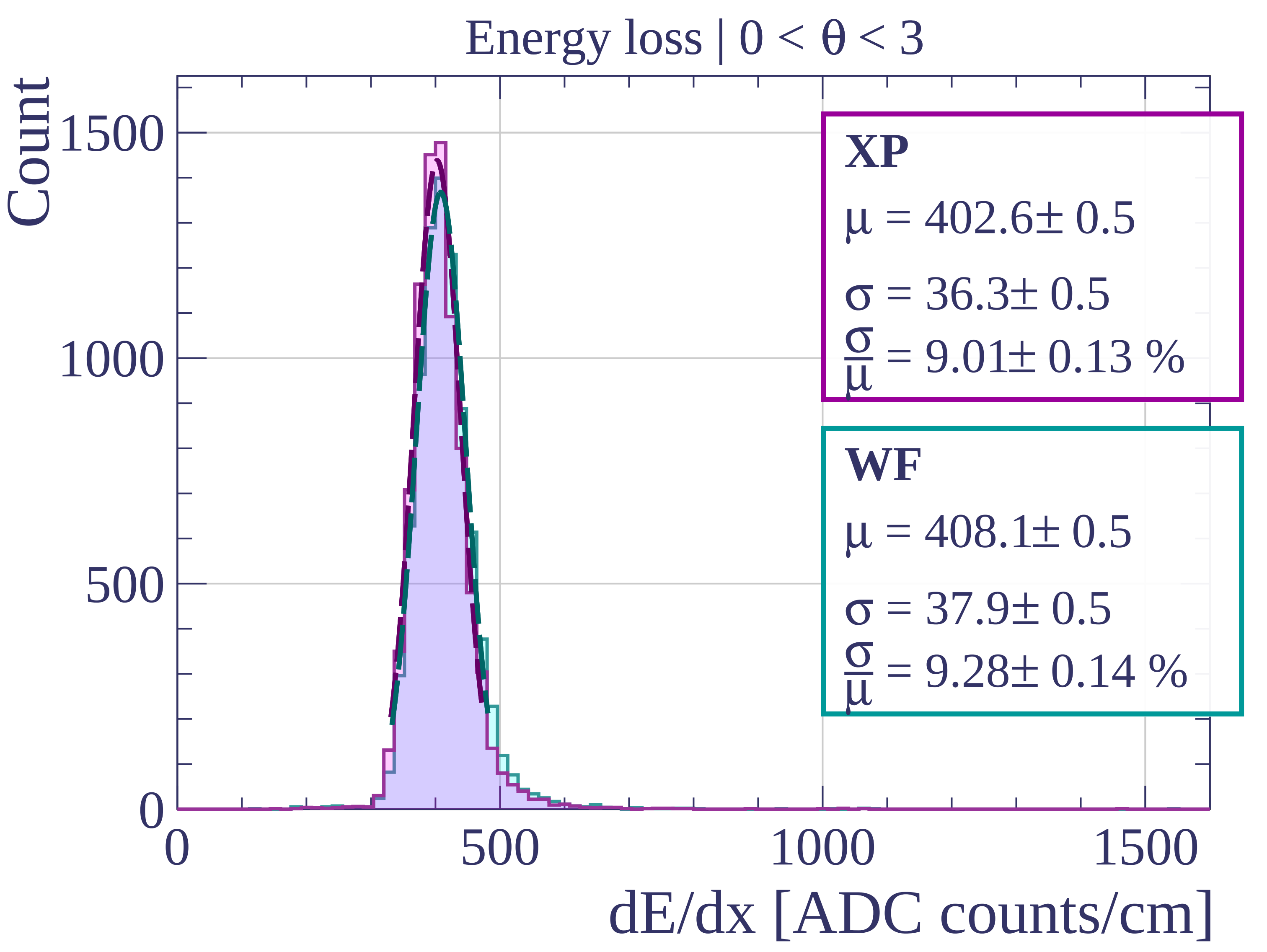
Energy loss | 81 < φ < 84
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 397.5± 0.6
σ = 32.4± 0.6
σ/μ = 8.16± 0.17 %
WF
μ = 407.2± 0.6
σ = 34.4± 0.6
σ/μ = 8.45± 0.16 %

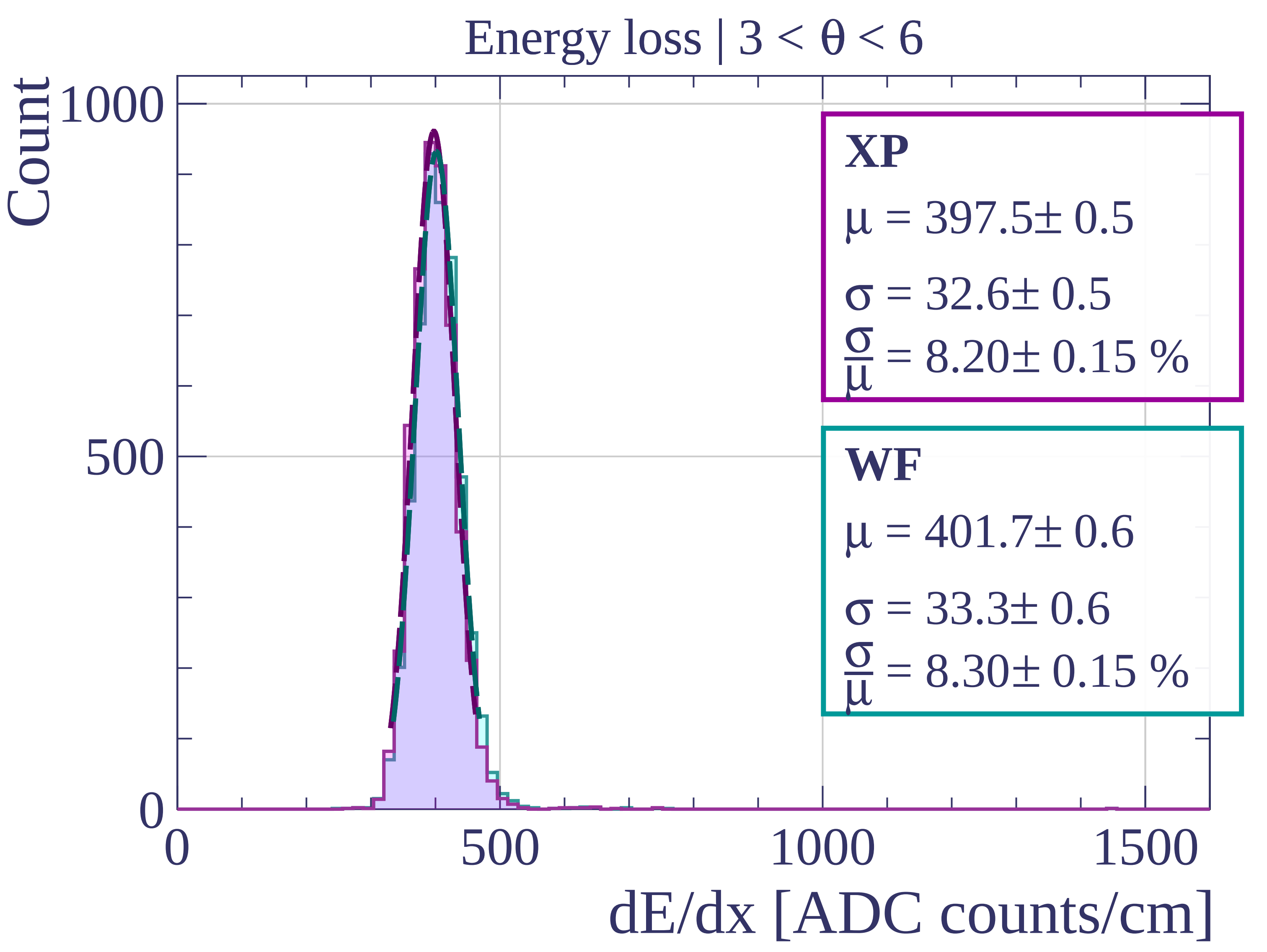


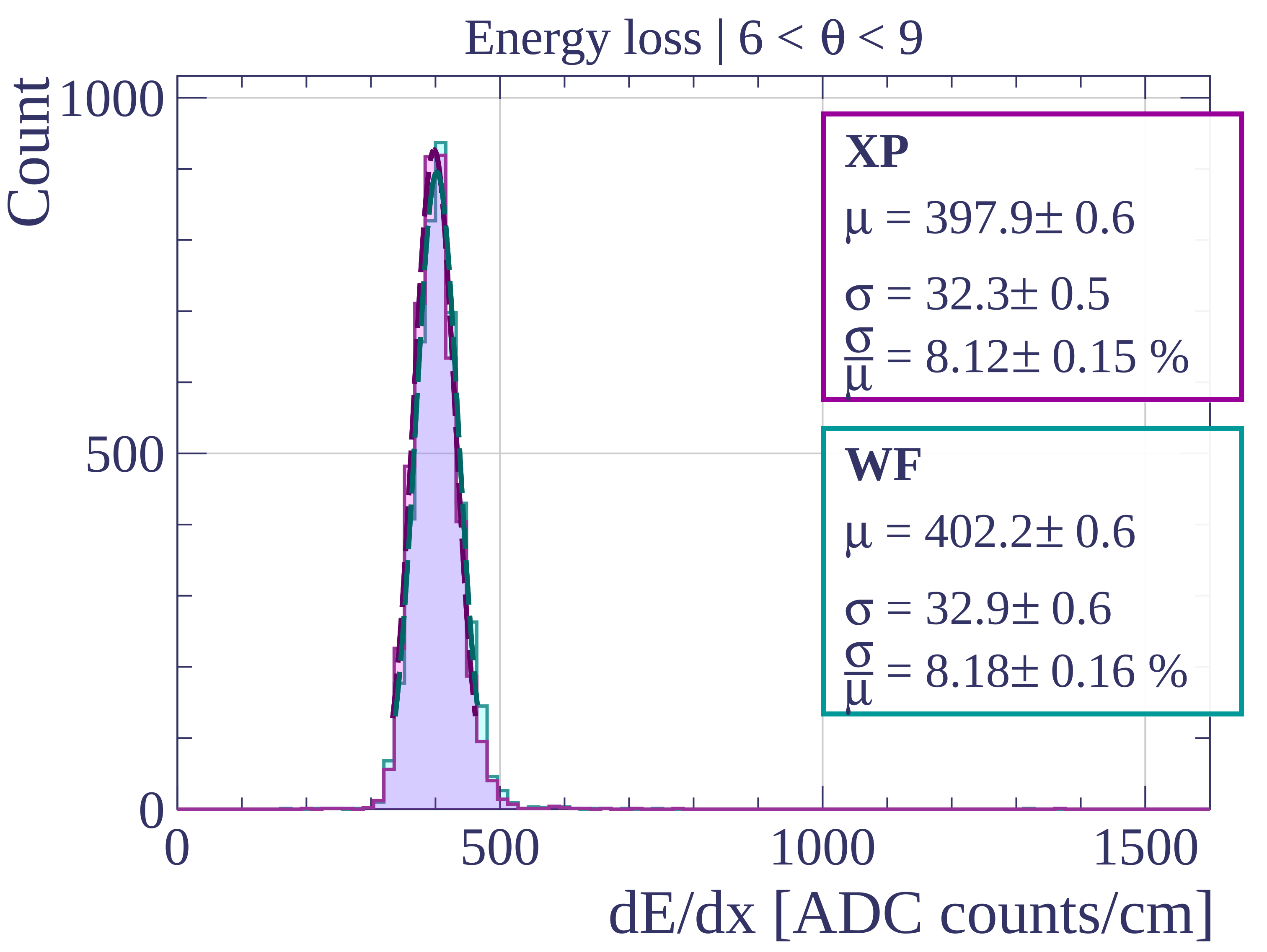
Energy loss | 84 < φ < 87
Count
0
200
400
600
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 397.2± 0.7
σ = 32.9± 0.7
σ/μ = 8.29± 0.20 %
WF
μ = 408.0± 0.7
σ = 35.0± 0.7
σ/μ = 8.57± 0.19 %



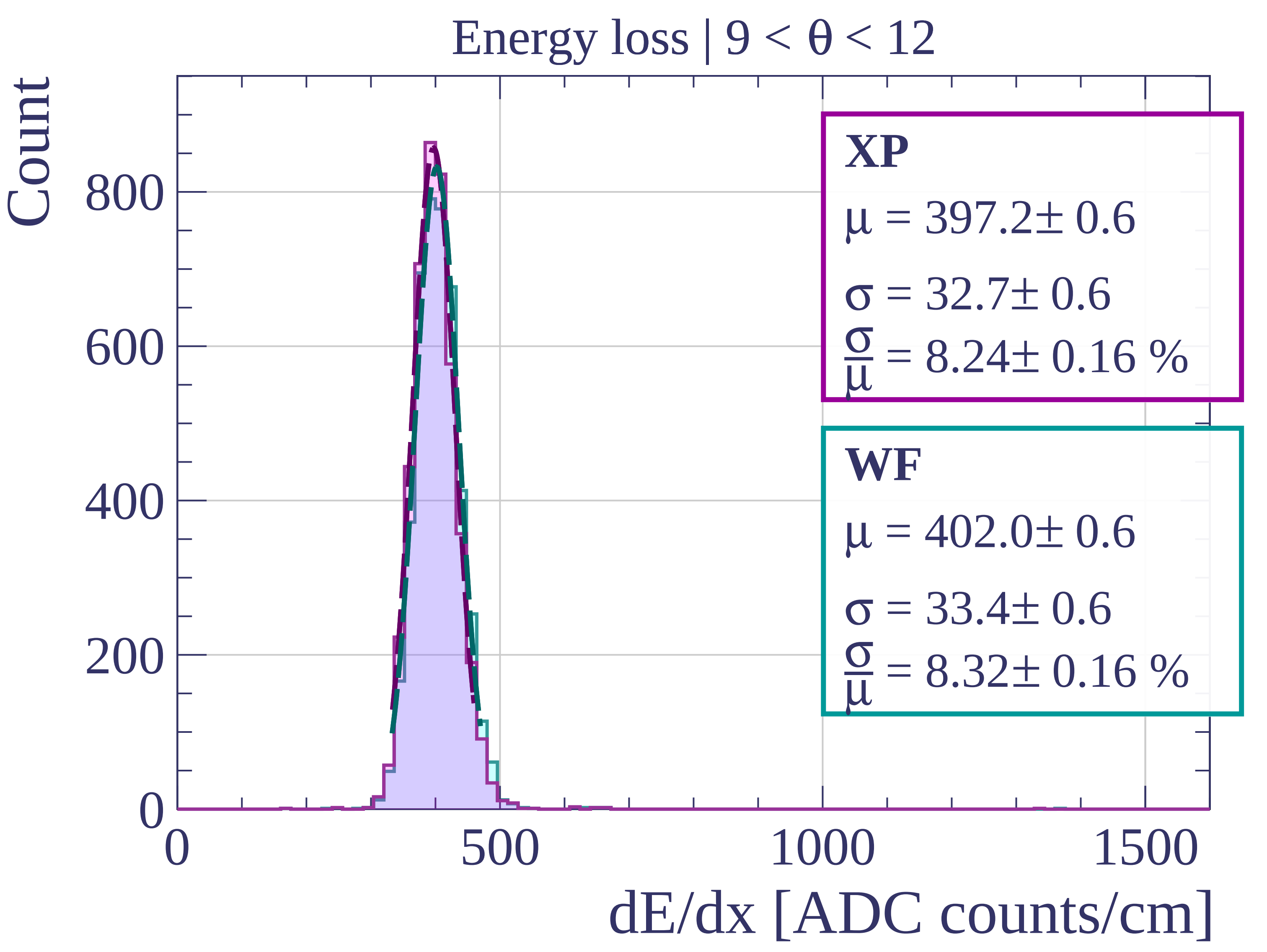
Energy loss | 87 < φ < 90
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
XP
μ = 404.7± 0.5
σ = 36.8± 0.5
σ/μ = 9.10± 0.13 %
WF
μ = 415.1± 0.6
σ = 39.0± 0.6
σ/μ = 9.39± 0.15 %


Energy loss | 0 < θ < 3
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
XP
μ = 402.6± 0.5
σ = 36.3± 0.5
σ/μ = 9.01± 0.13 %
WF
μ = 408.1± 0.5
σ = 37.9± 0.5
σ/μ = 9.28± 0.14 %

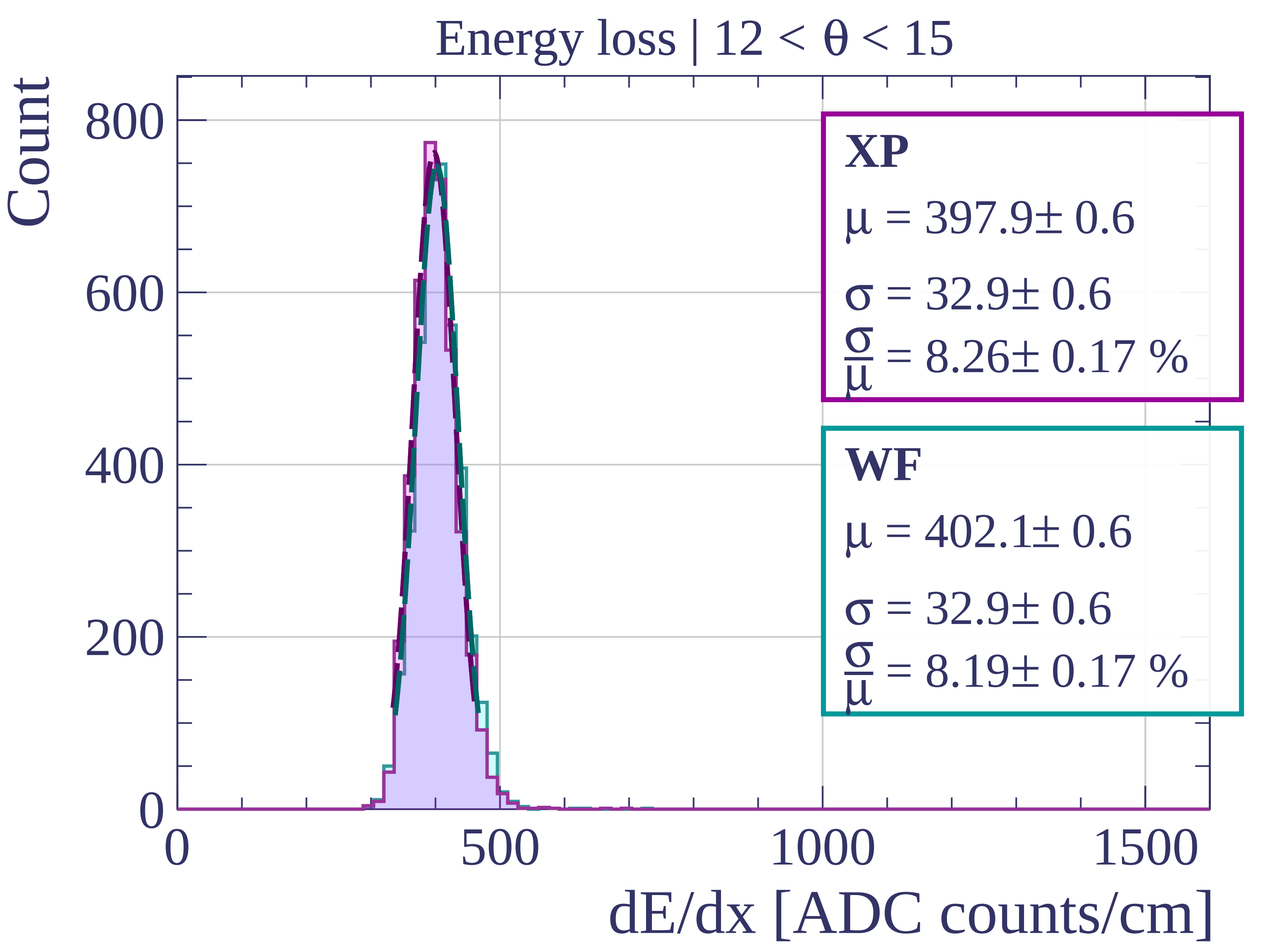

Energy loss | 3 < θ < 6
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 397.5± 0.5
σ = 32.6± 0.5
σ/μ = 8.20± 0.15 %
WF
μ = 401.7± 0.6
σ = 33.3± 0.6
σ/μ = 8.30± 0.15 %

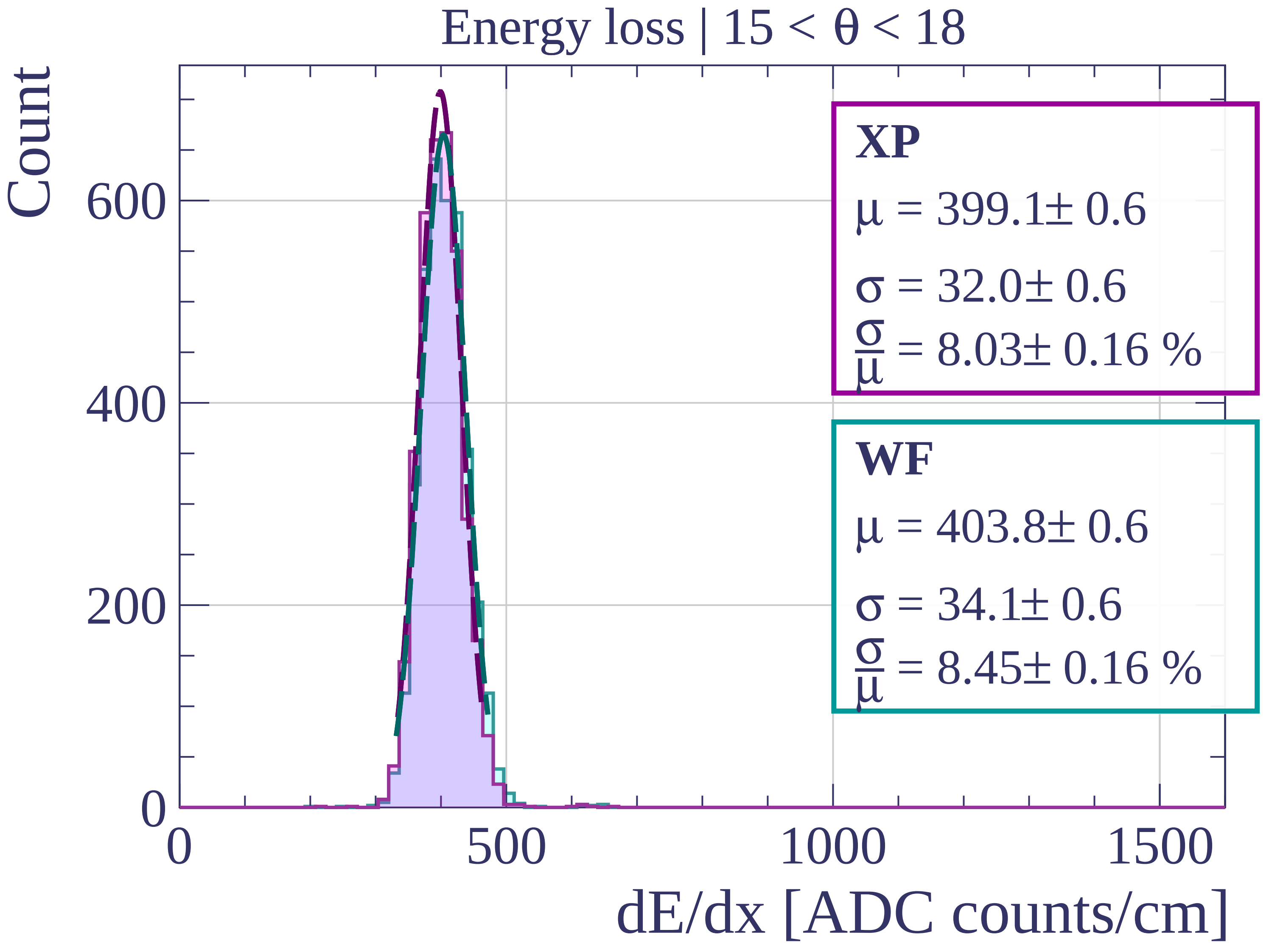


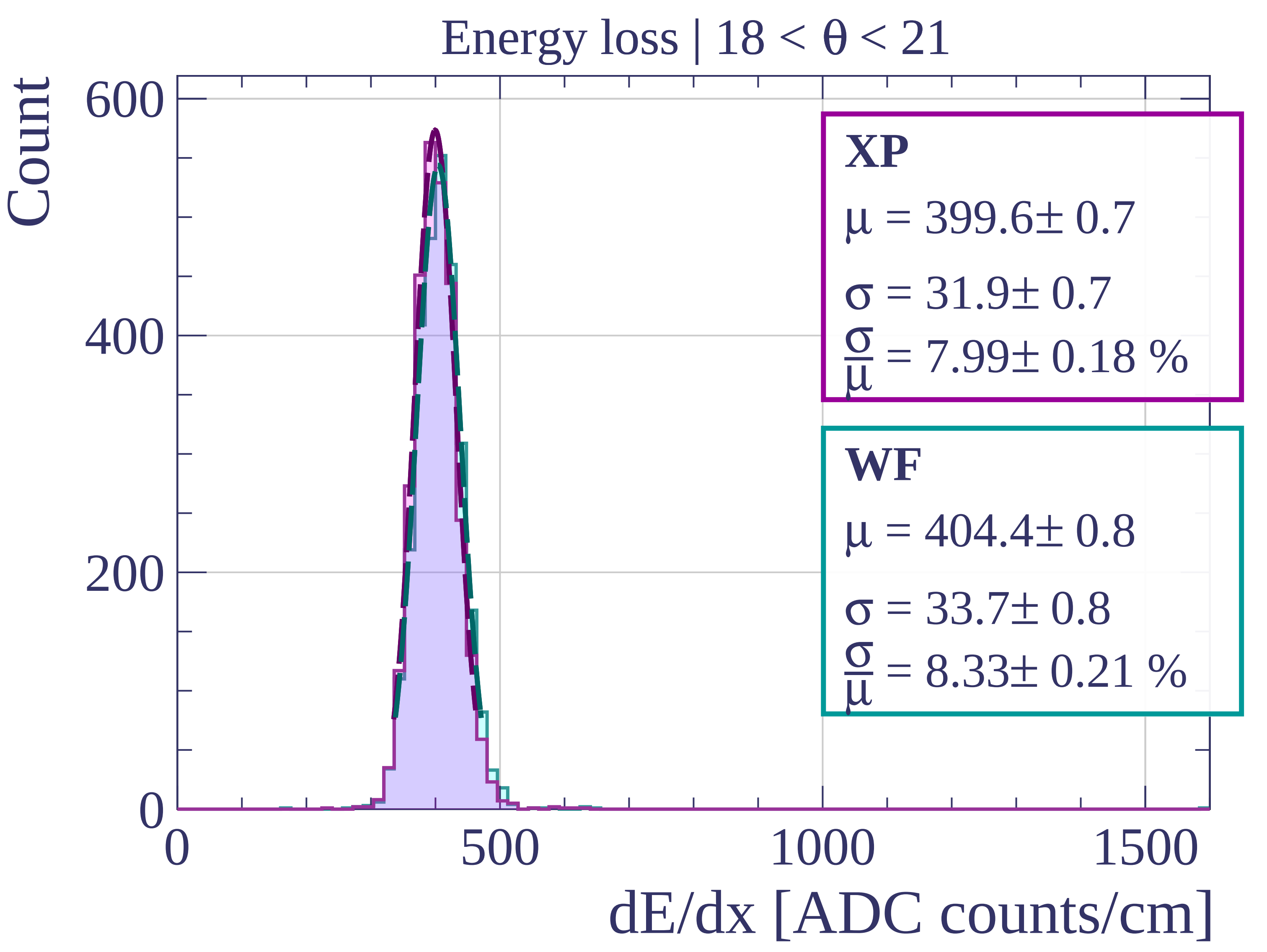
Energy loss | 6 < θ < 9
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 397.9± 0.6
σ = 32.3± 0.5
σ/μ = 8.12± 0.15 %
WF
μ = 402.2± 0.6
σ = 32.9± 0.6
σ/μ = 8.18± 0.16 %

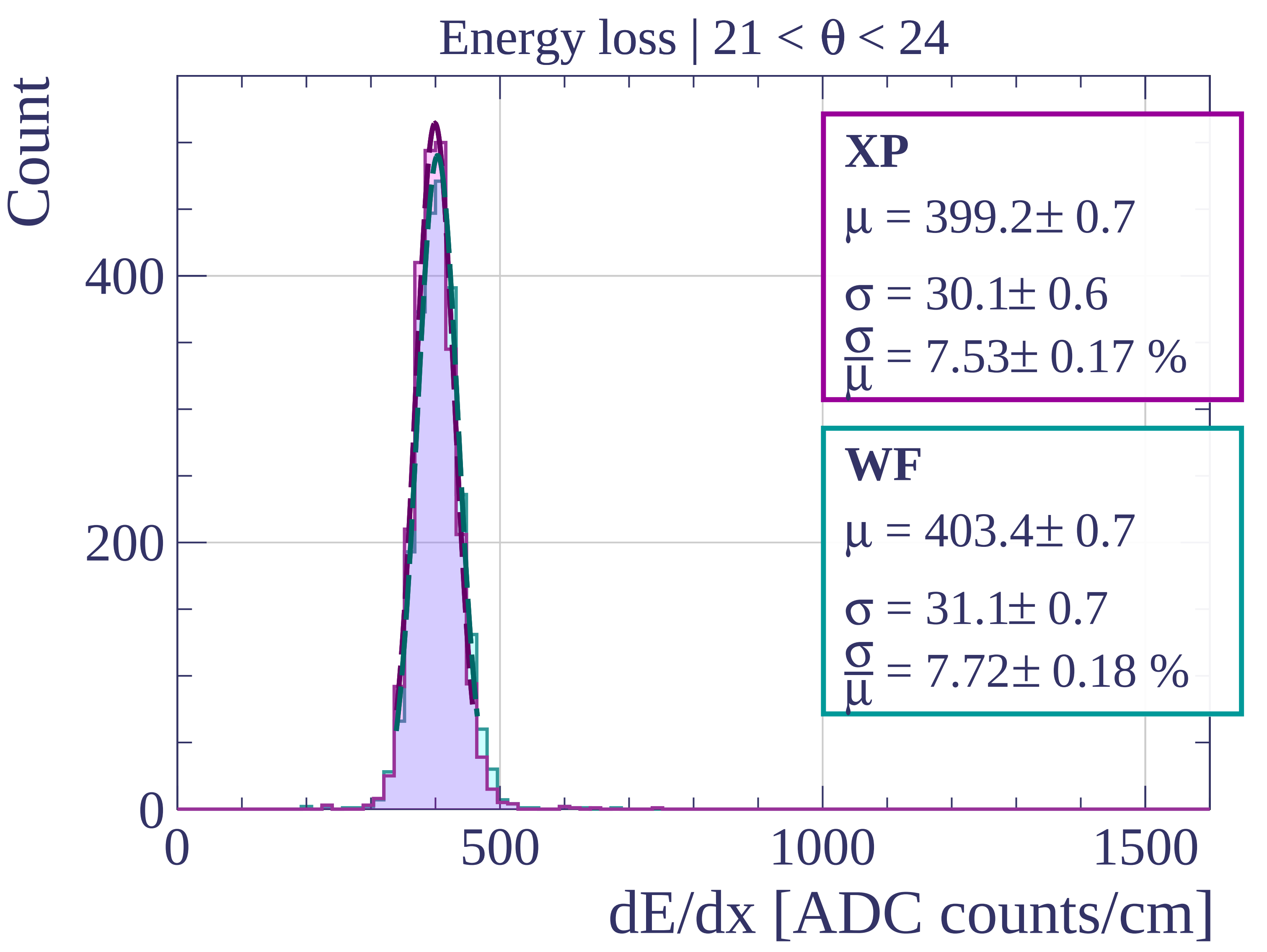


Energy loss | 9 < θ < 12
Count
dE/dx [ADC counts/cm]
0
200
400
600
800
0
500
1000
1500
XP
μ = 397.2± 0.6
σ = 32.7± 0.6
σ/μ = 8.24± 0.16 %
WF
μ = 402.0± 0.6
σ = 33.4± 0.6
σ/μ = 8.32± 0.16 %

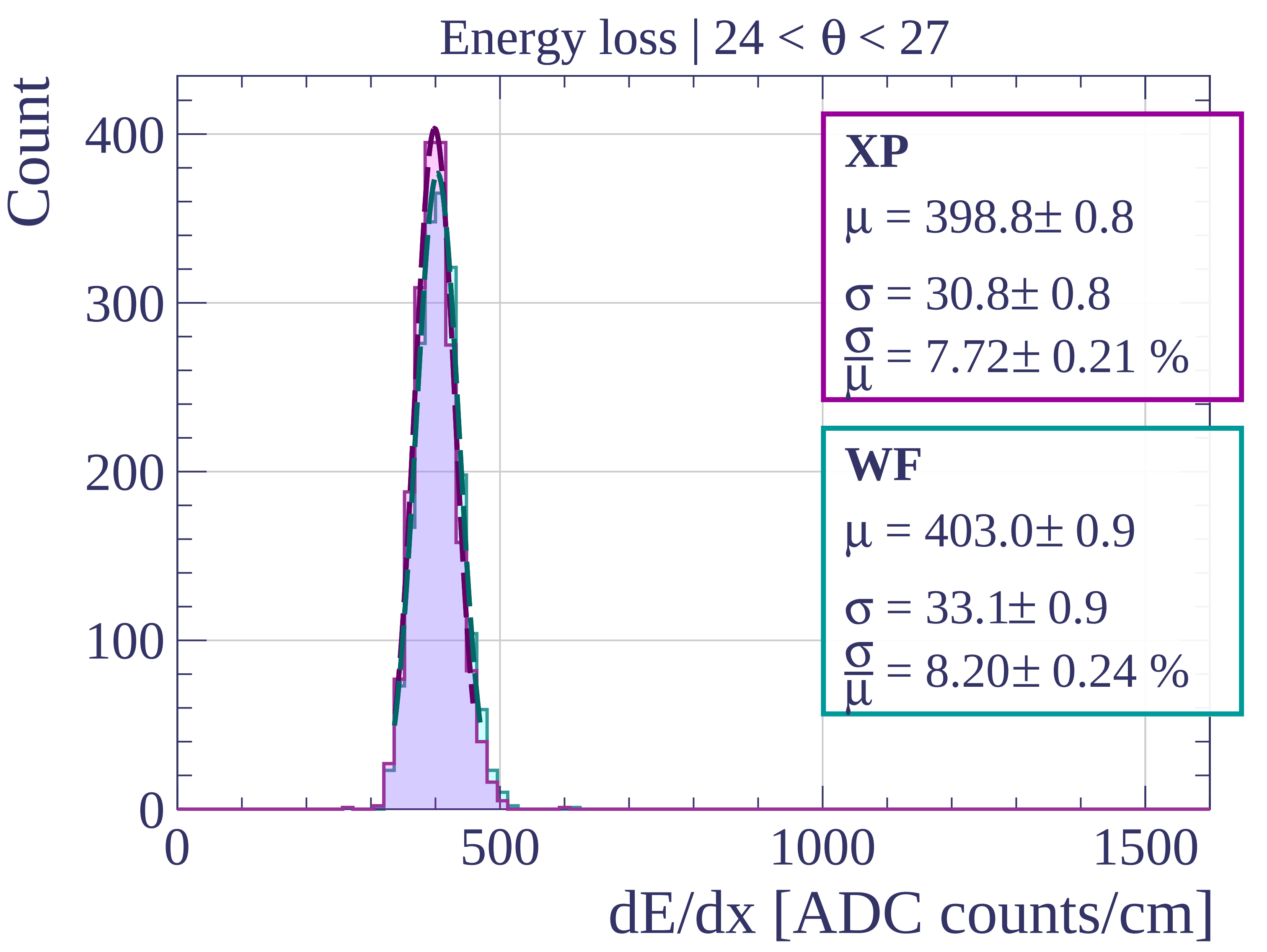

Energy loss | 12 < θ < 15
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 397.9± 0.6
σ = 32.9± 0.6
σ/μ = 8.26± 0.17 %
WF
μ = 402.1± 0.6
σ = 32.9± 0.6
σ/μ = 8.19± 0.17 %

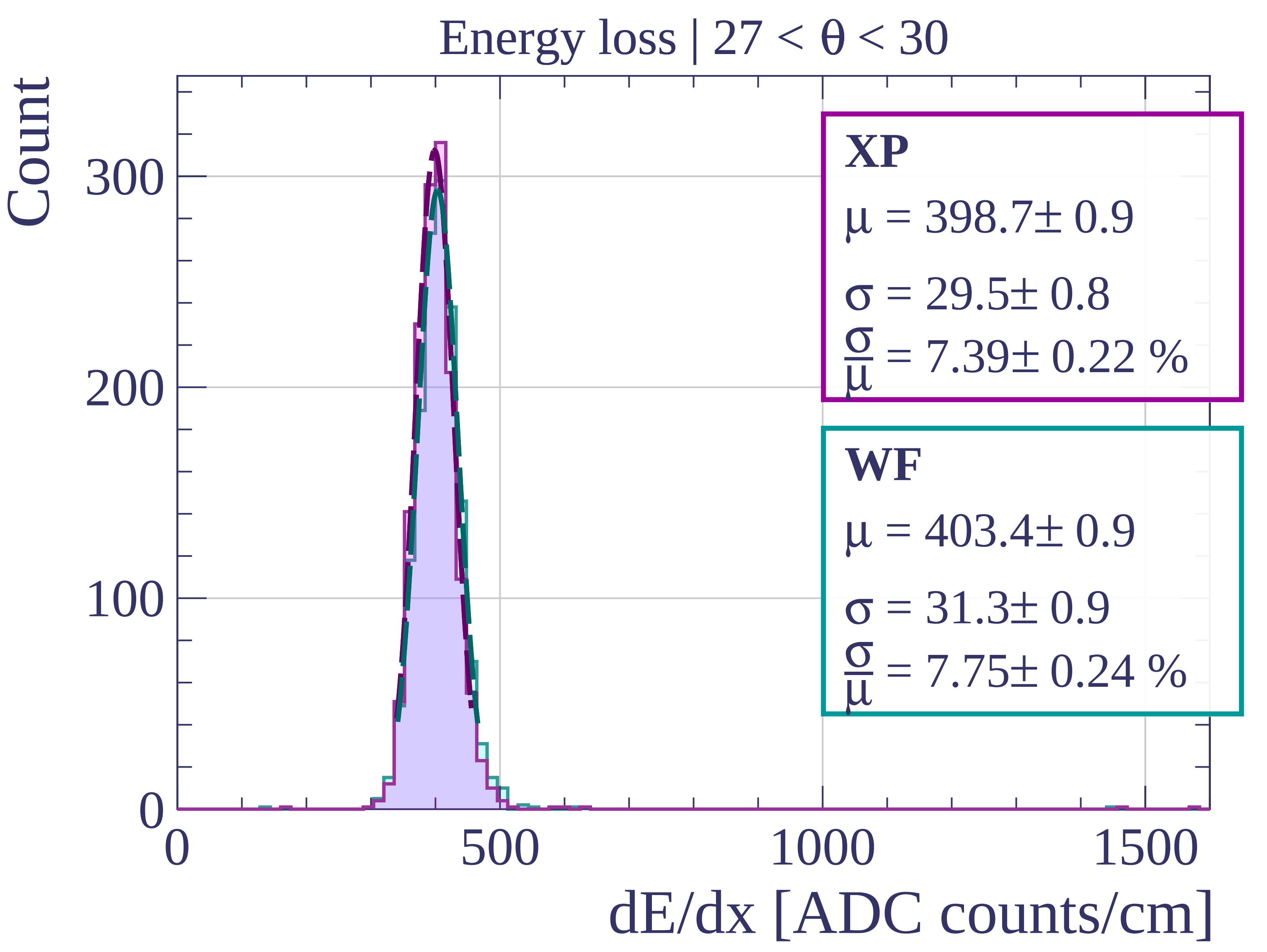

Energy loss | 15 < θ < 18
Count
dE/dx [ADC counts/cm]
0
200
400
600
0
500
1000
1500
XP
μ = 399.1± 0.6
σ = 32.0± 0.6
σ/μ = 8.03± 0.16 %
WF
μ = 403.8± 0.6
σ = 34.1± 0.6
σ/μ = 8.45± 0.16 %


Energy loss | 18 < θ < 21
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 399.6± 0.7
σ = 31.9± 0.7
σ/μ = 7.99± 0.18 %
WF
μ = 404.4± 0.8
σ = 33.7± 0.8
σ/μ = 8.33± 0.21 %

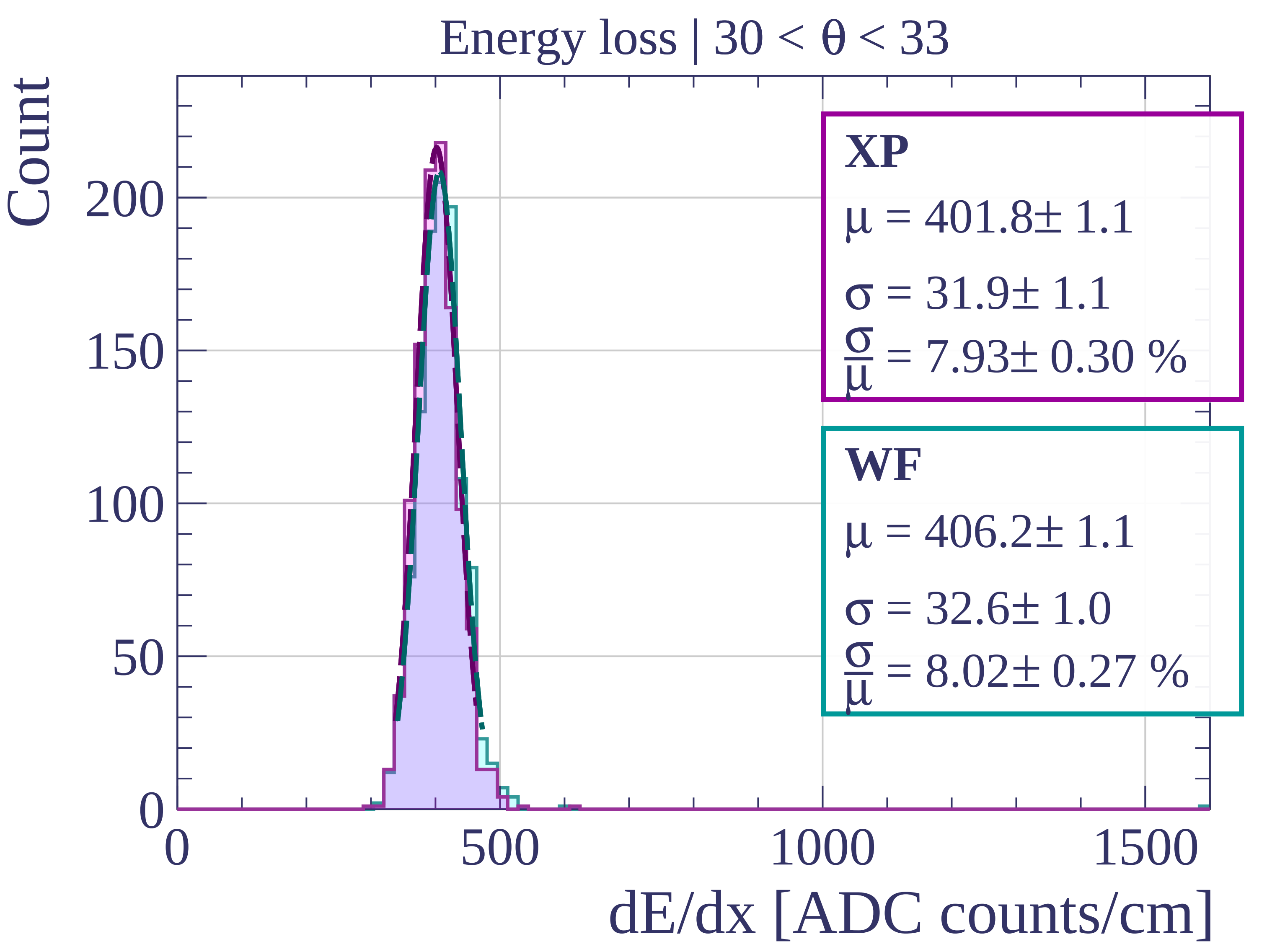


Energy loss | 21 < θ < 24
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 399.2± 0.7
σ = 30.1± 0.6
σ/μ = 7.53± 0.17 %
WF
μ = 403.4± 0.7
σ = 31.1± 0.7
σ/μ = 7.72± 0.18 %

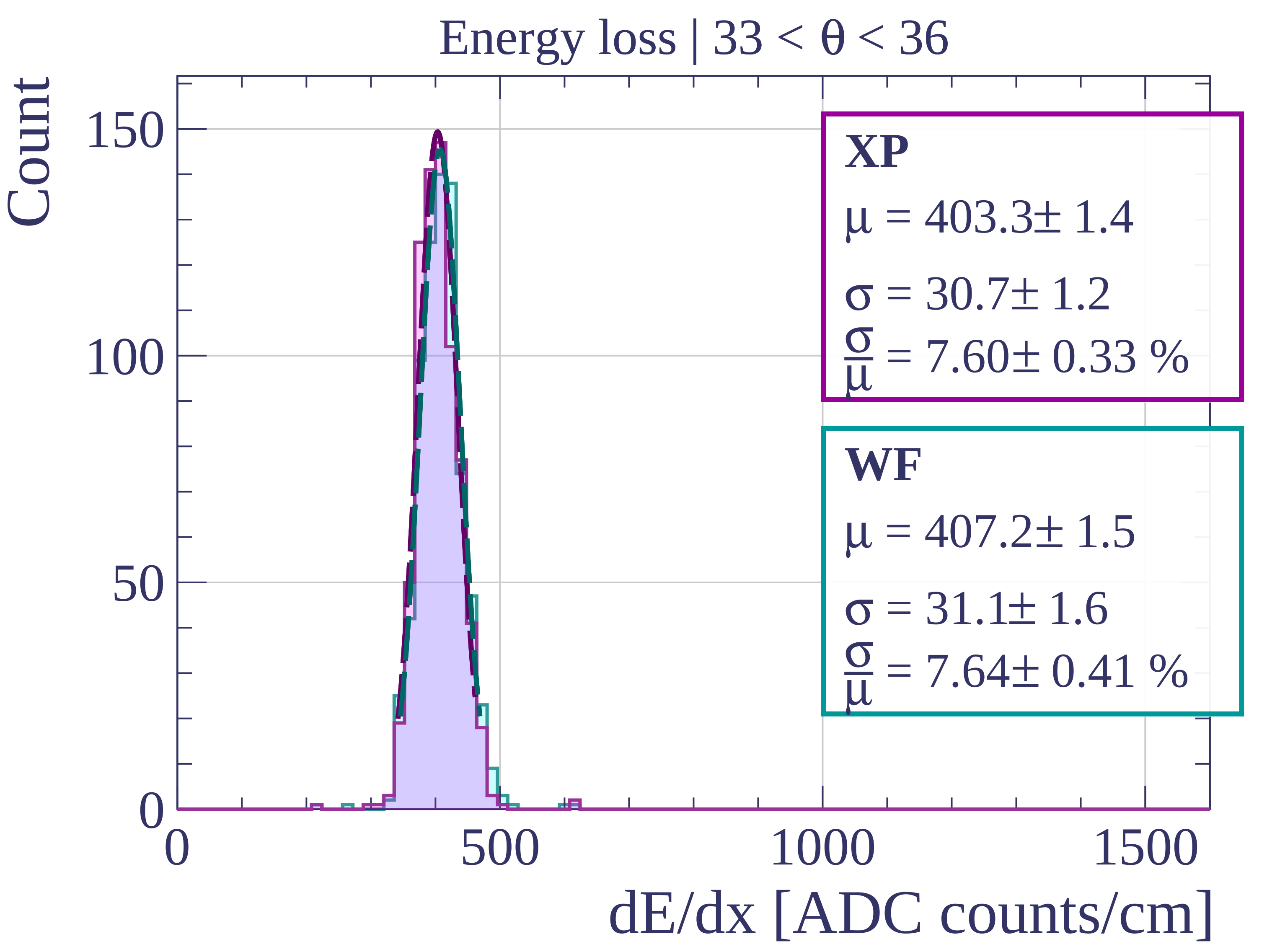

Energy loss | 24 < θ < 27
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 398.8± 0.8
σ = 30.8± 0.8
σ/μ = 7.72± 0.21 %
WF
μ = 403.0± 0.9
σ = 33.1± 0.9
σ/μ = 8.20± 0.24 %

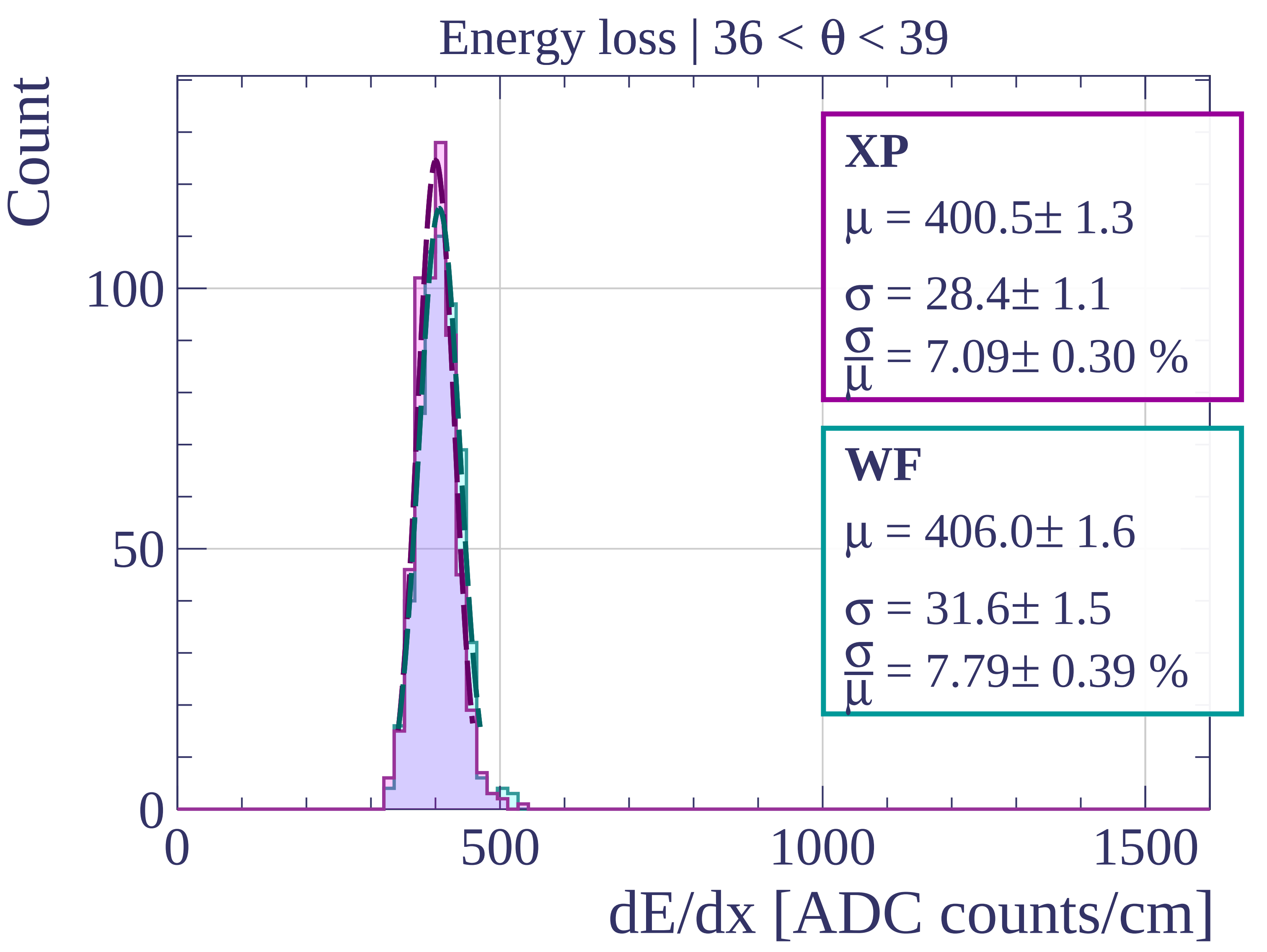


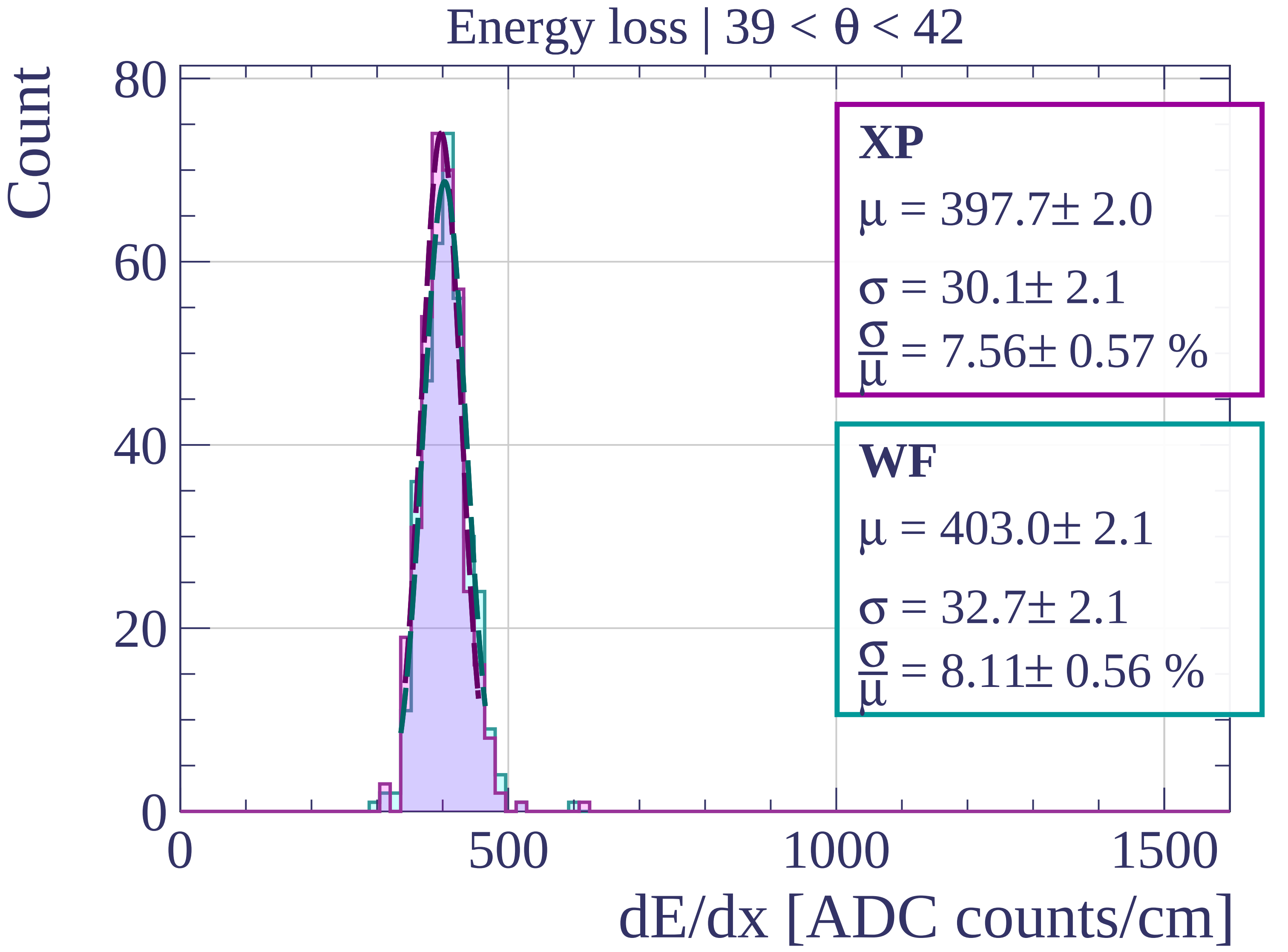
Energy loss | 27 < θ < 30
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 398.7± 0.9
σ = 29.5± 0.8
σ/μ = 7.39± 0.22 %
WF
μ = 403.4± 0.9
σ = 31.3± 0.9
σ/μ = 7.75± 0.24 %



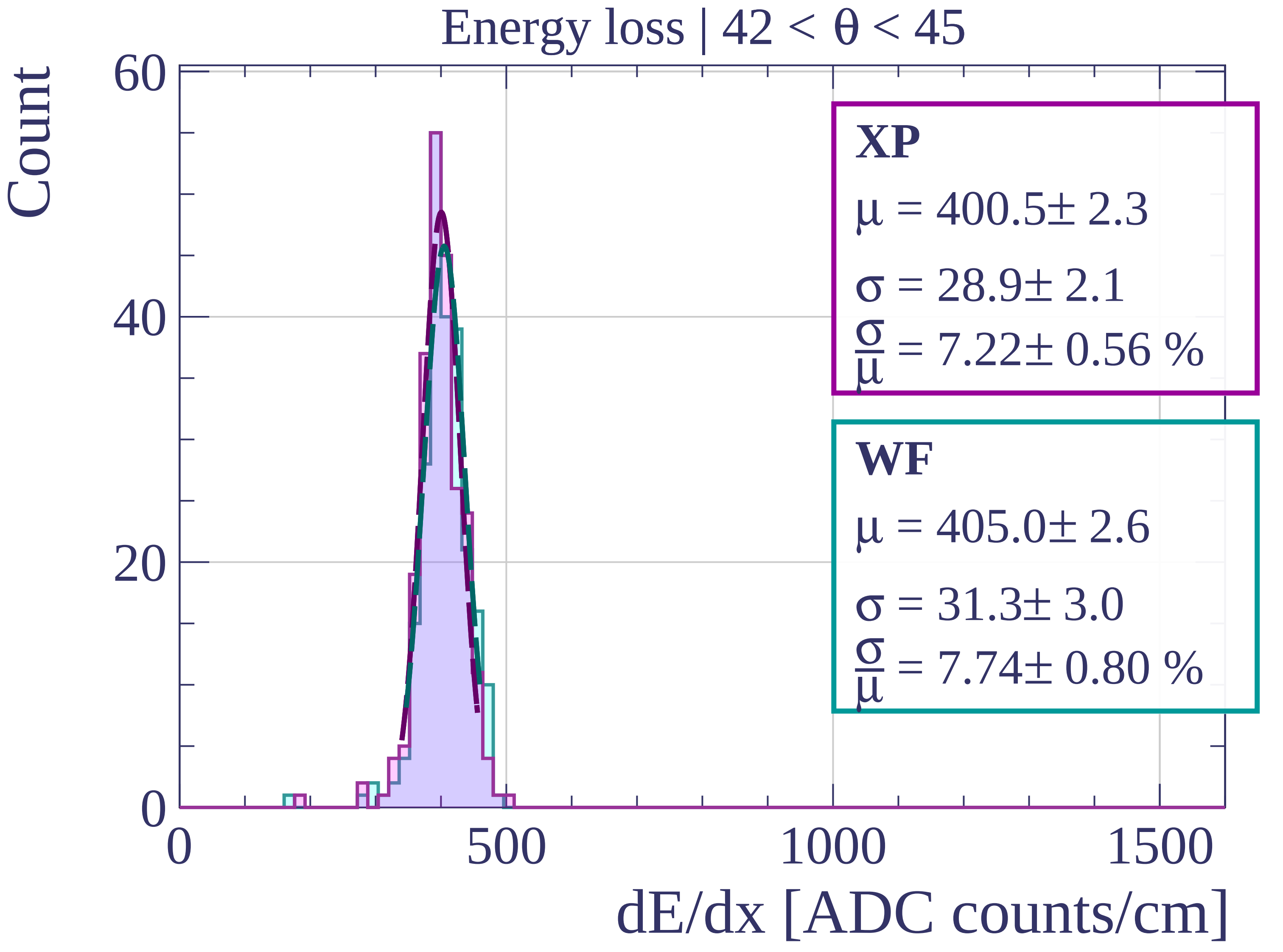
Energy loss | 30 < θ < 33
Count
dE/dx [ADC counts/cm]
0
50
100
150
200
0
500
1000
1500
XP
μ = 401.8± 1.1
σ = 31.9± 1.1
σ/μ = 7.93± 0.30 %
WF
μ = 406.2± 1.1
σ = 32.6± 1.0
σ/μ = 8.02± 0.27 %

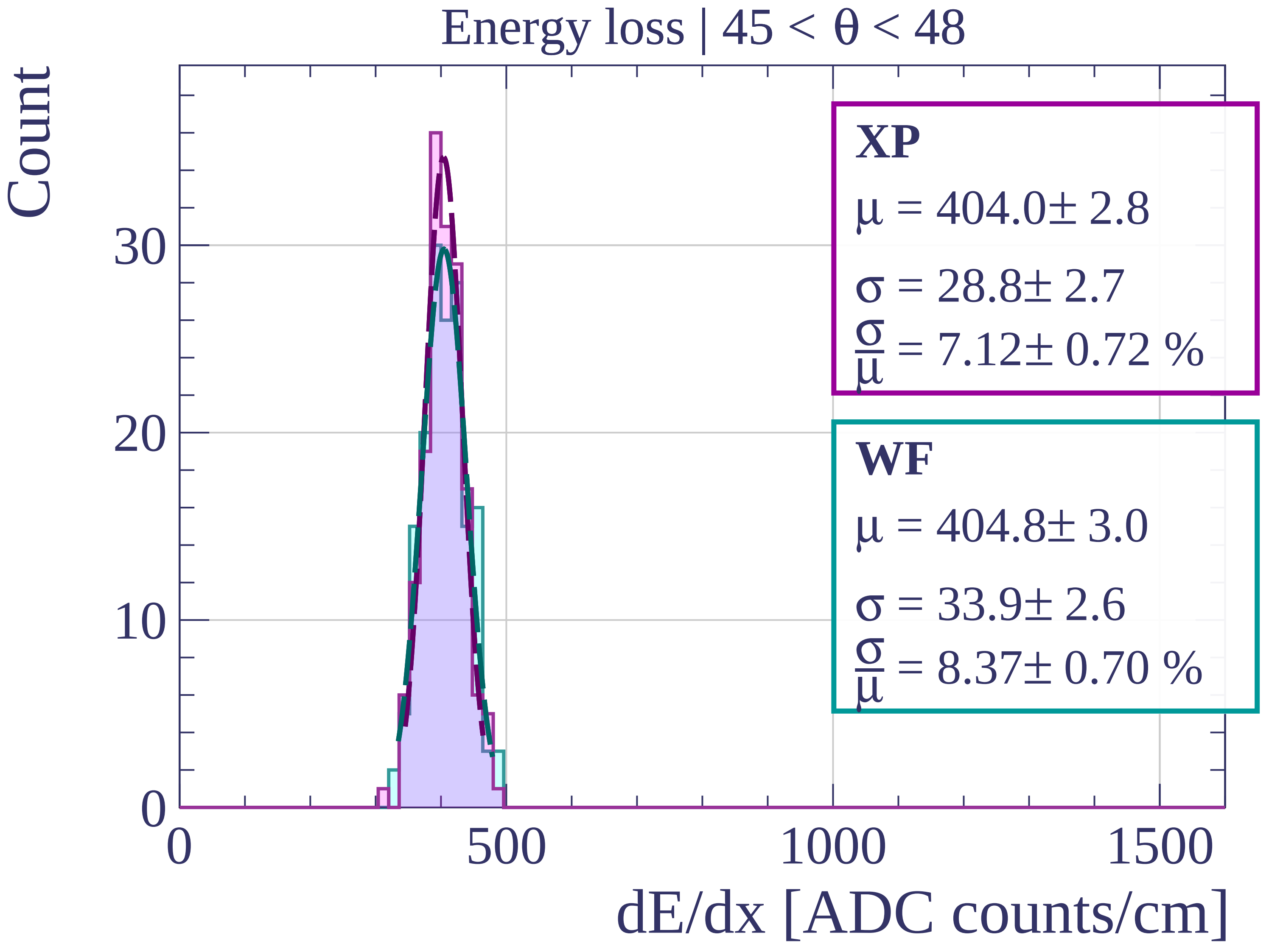

Energy loss | 33 < θ < 36
Count
dE/dx [ADC counts/cm]
0
50
100
150
0
500
1000
1500
XP
μ = 403.3± 1.4
σ = 30.7± 1.2
σ/μ = 7.60± 0.33 %
WF
μ = 407.2± 1.5
σ = 31.1± 1.6
σ/μ = 7.64± 0.41 %


Energy loss | 36 < θ < 39
Count
0
50
100
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 400.5± 1.3
σ = 28.4± 1.1
σ/μ = 7.09± 0.30 %
WF
μ = 406.0± 1.6
σ = 31.6± 1.5
σ/μ = 7.79± 0.39 %

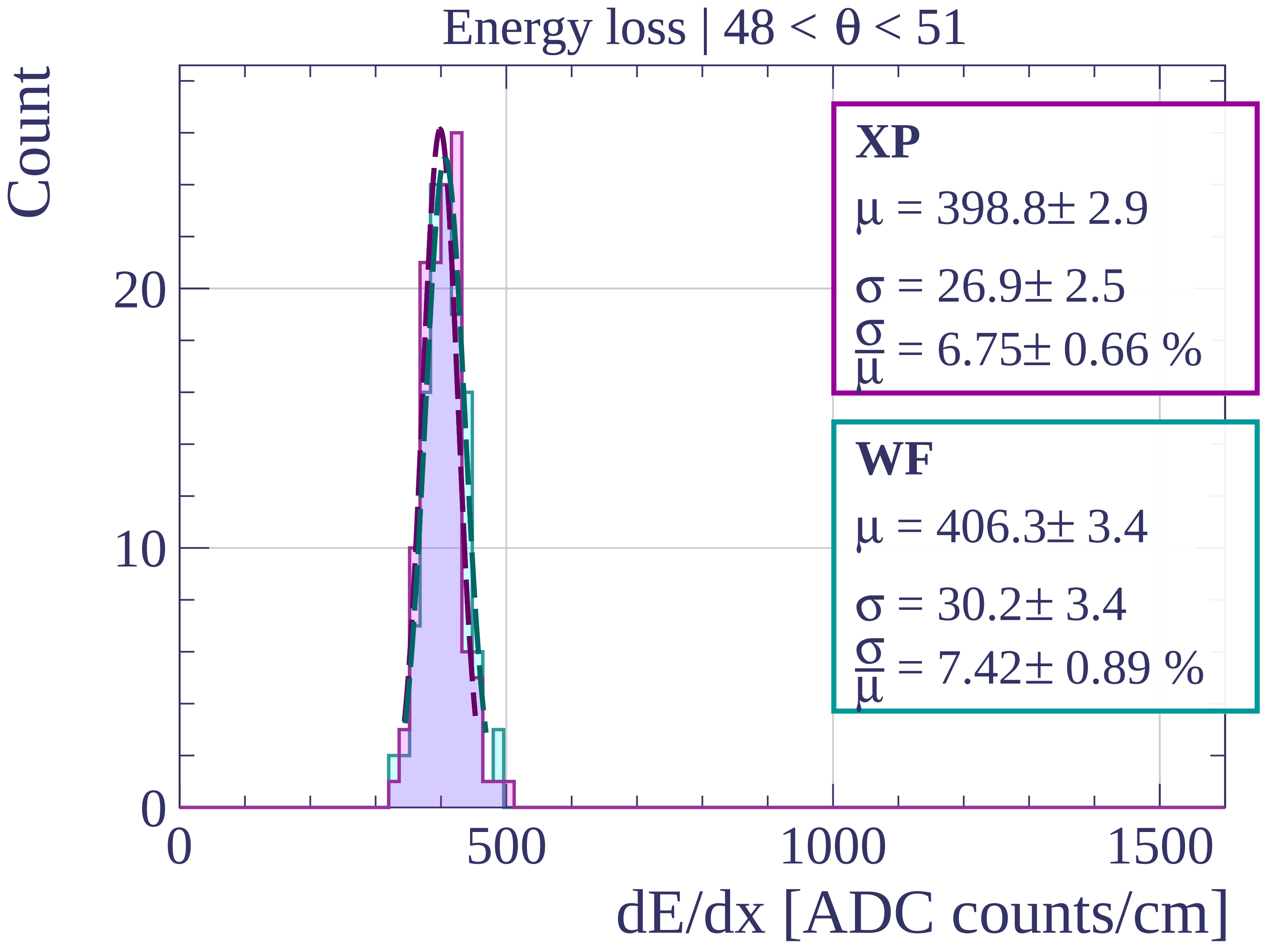


Energy loss | 39 < θ < 42
Count
0
20
40
60
80
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 397.7± 2.0
σ = 30.1± 2.1
σ/μ = 7.56± 0.57 %
WF
μ = 403.0± 2.1
σ = 32.7± 2.1
σ/μ = 8.11± 0.56 %

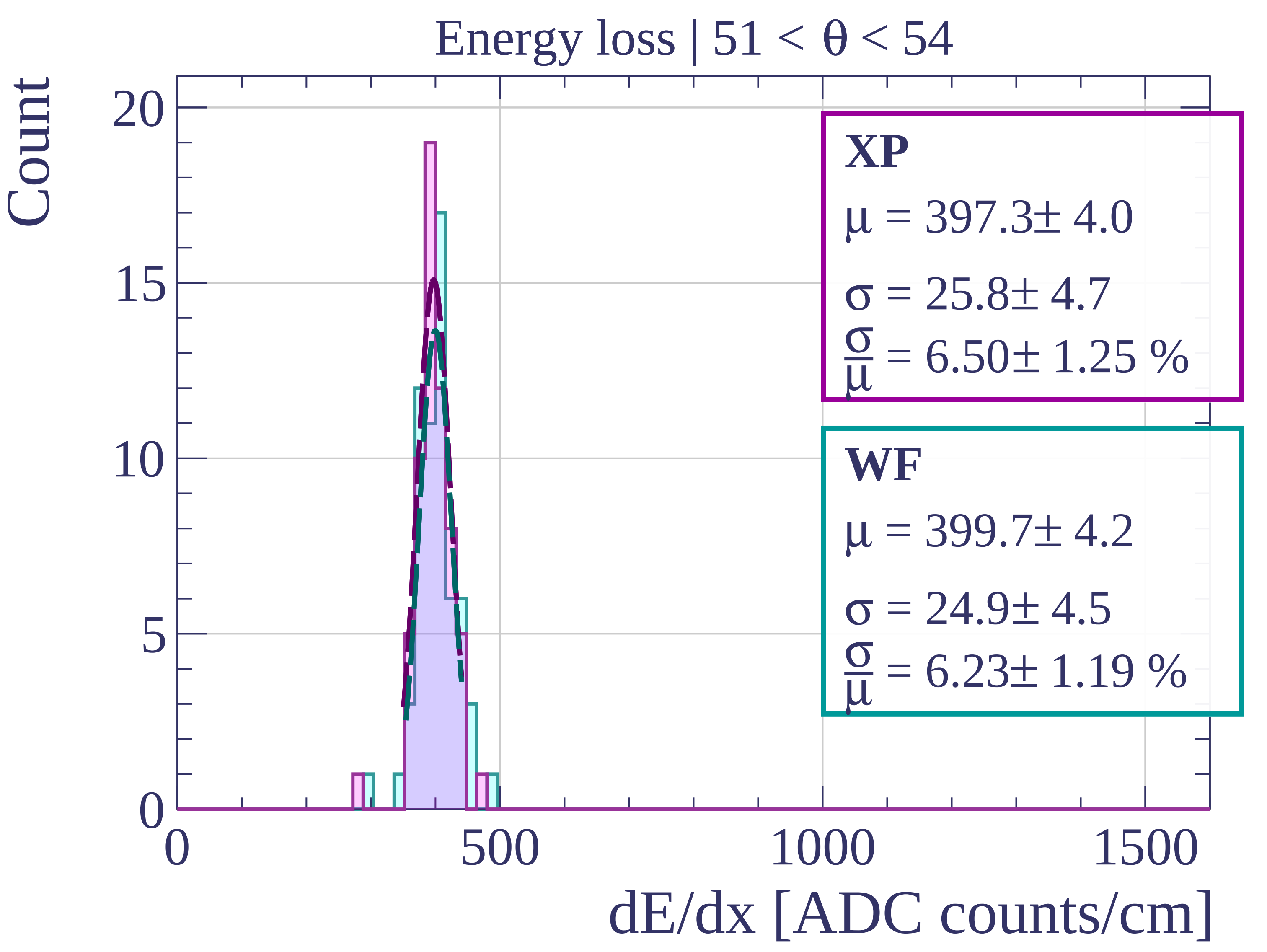

Energy loss | 42 < θ < 45
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 400.5± 2.3
σ = 28.9± 2.1
σ/μ = 7.22± 0.56 %
WF
μ = 405.0± 2.6
σ = 31.3± 3.0
σ/μ = 7.74± 0.80 %



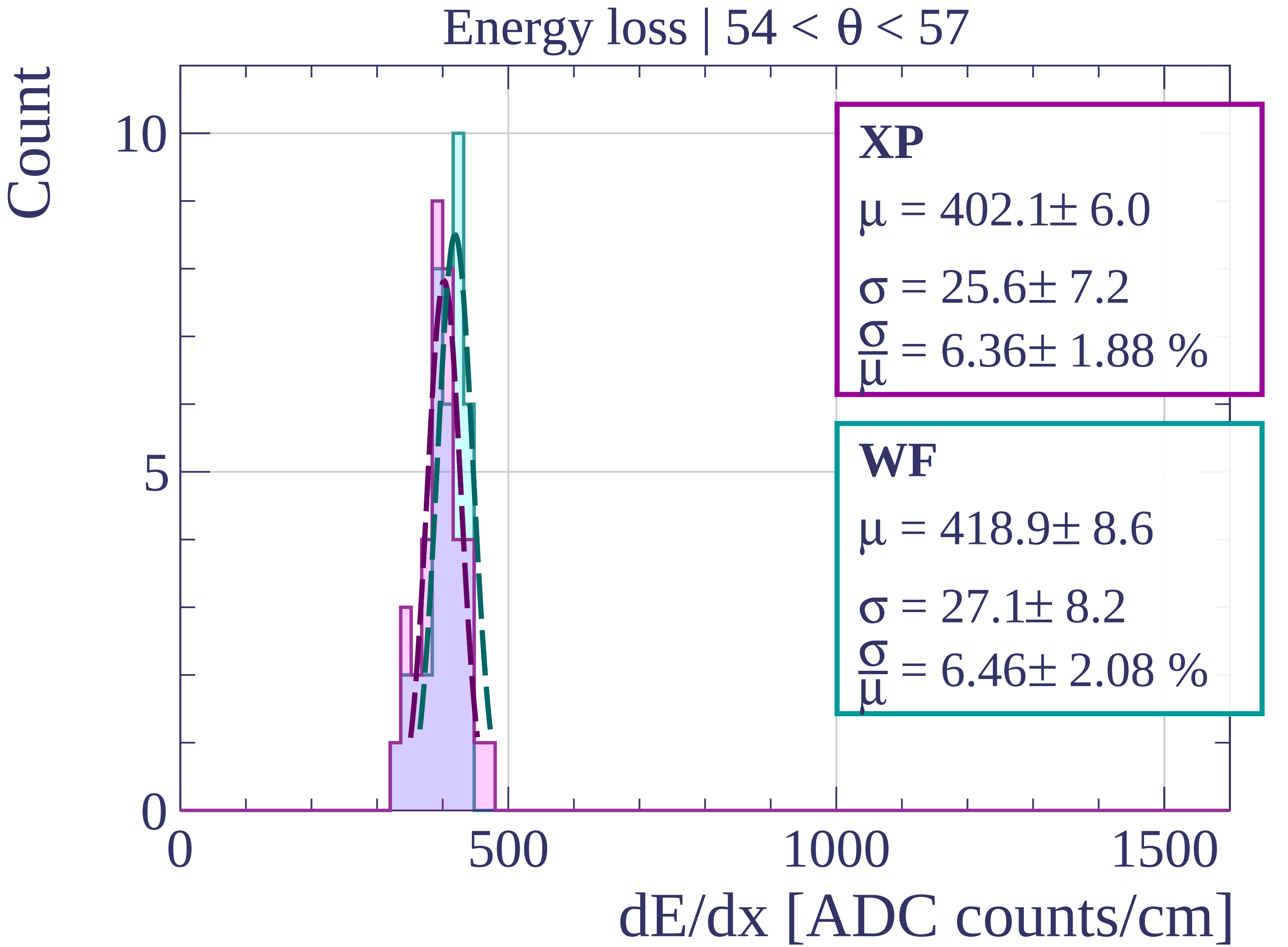
Energy loss | 45 < θ < 48
Count
dE/dx [ADC counts/cm]
0
10
20
30
0
500
1000
1500
XP
μ = 404.0± 2.8
σ = 28.8± 2.7
σ/μ = 7.12± 0.72 %
WF
μ = 404.8± 3.0
σ = 33.9± 2.6
σ/μ = 8.37± 0.70 %



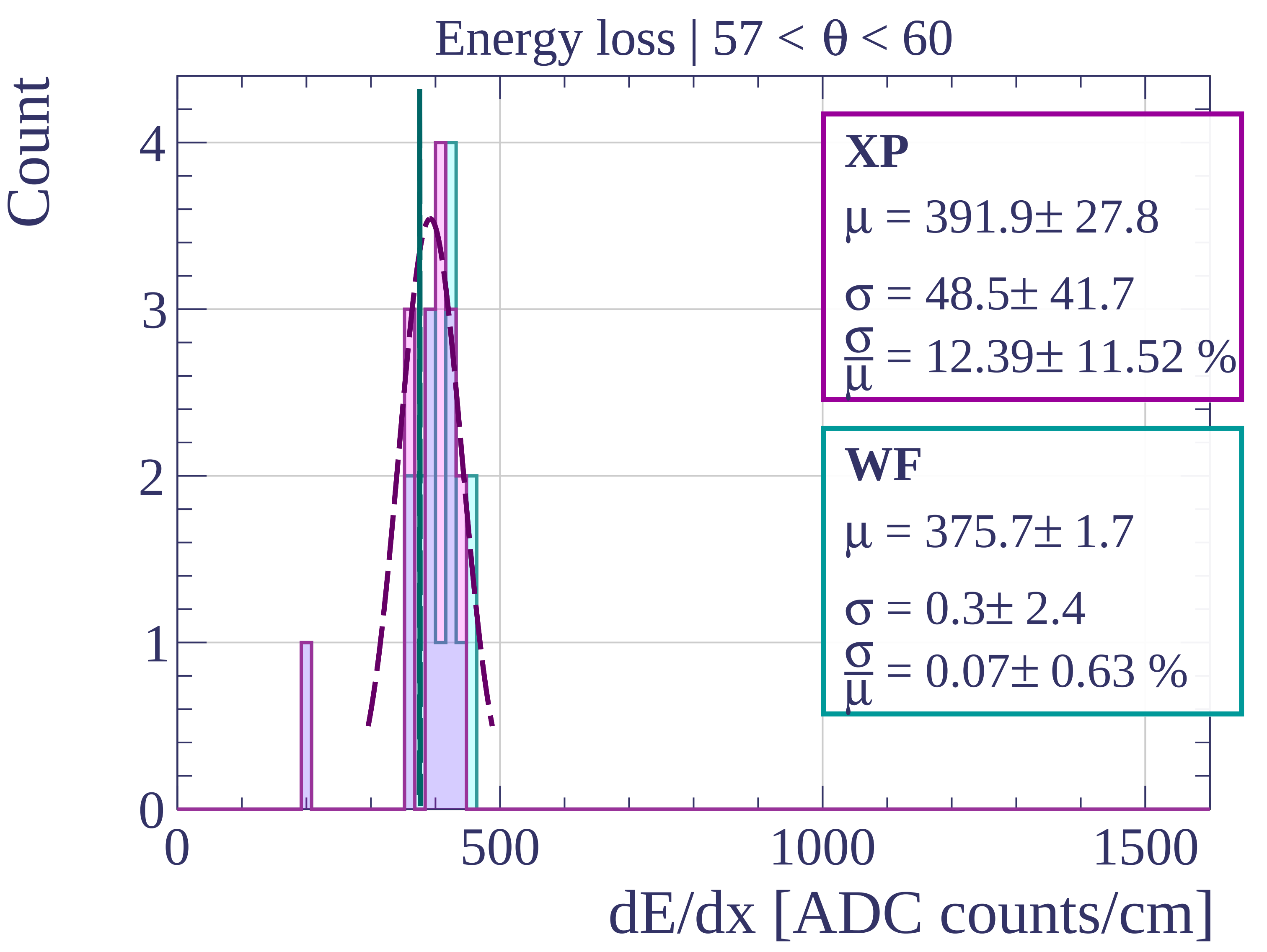
Energy loss | 48 < θ < 51
Count
0
10
20
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 398.8± 2.9
σ = 26.9± 2.5
σ/μ = 6.75± 0.66 %
WF
μ = 406.3± 3.4
σ = 30.2± 3.4
σ/μ = 7.42± 0.89 %



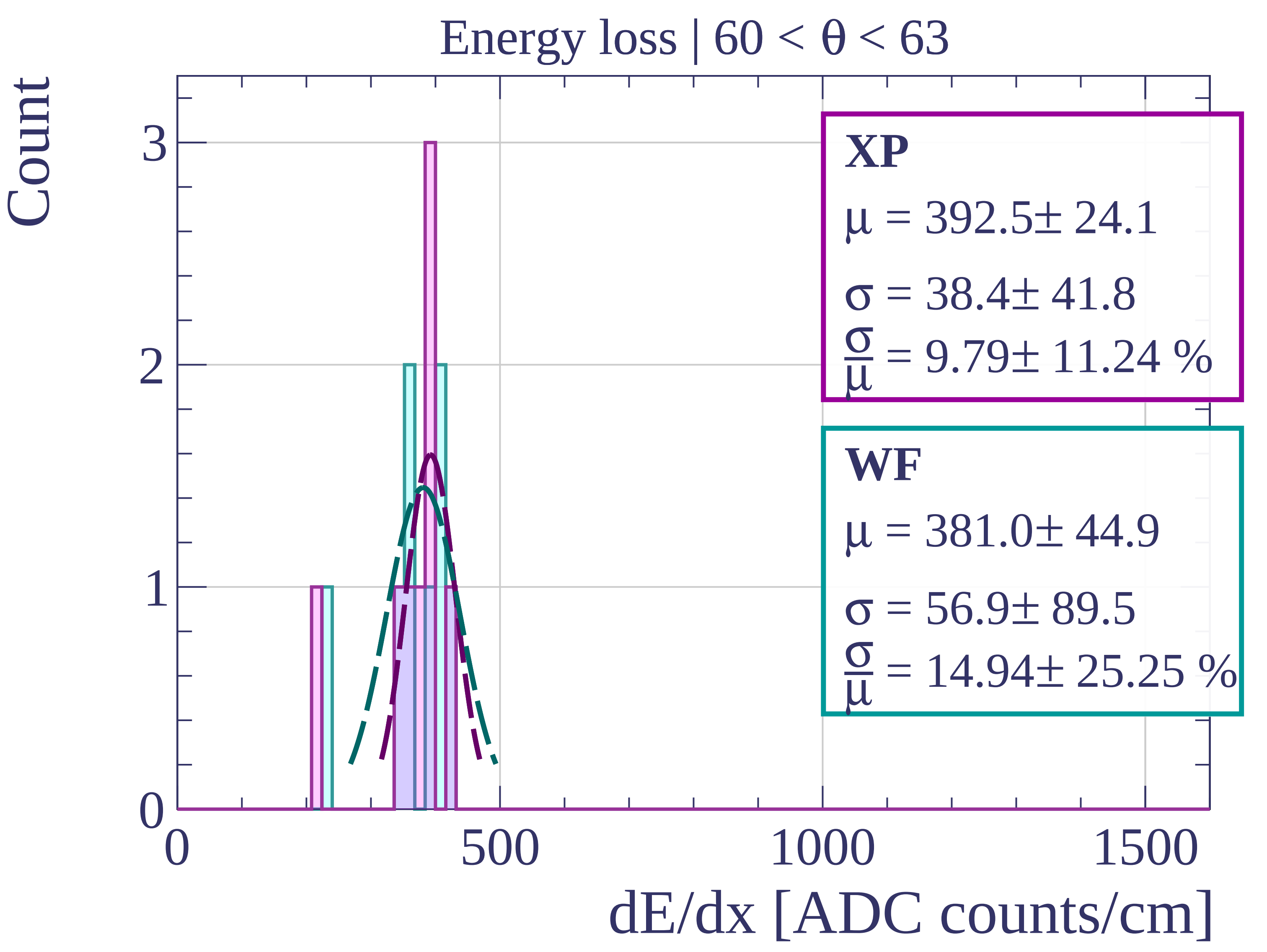
Energy loss | 51 < θ < 54
Count
20
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 397.3± 4.0
σ = 25.8± 4.7
σ/μ = 6.50± 1.25 %
WF
μ = 399.7± 4.2
σ = 24.9± 4.5
σ/μ = 6.23± 1.19 %



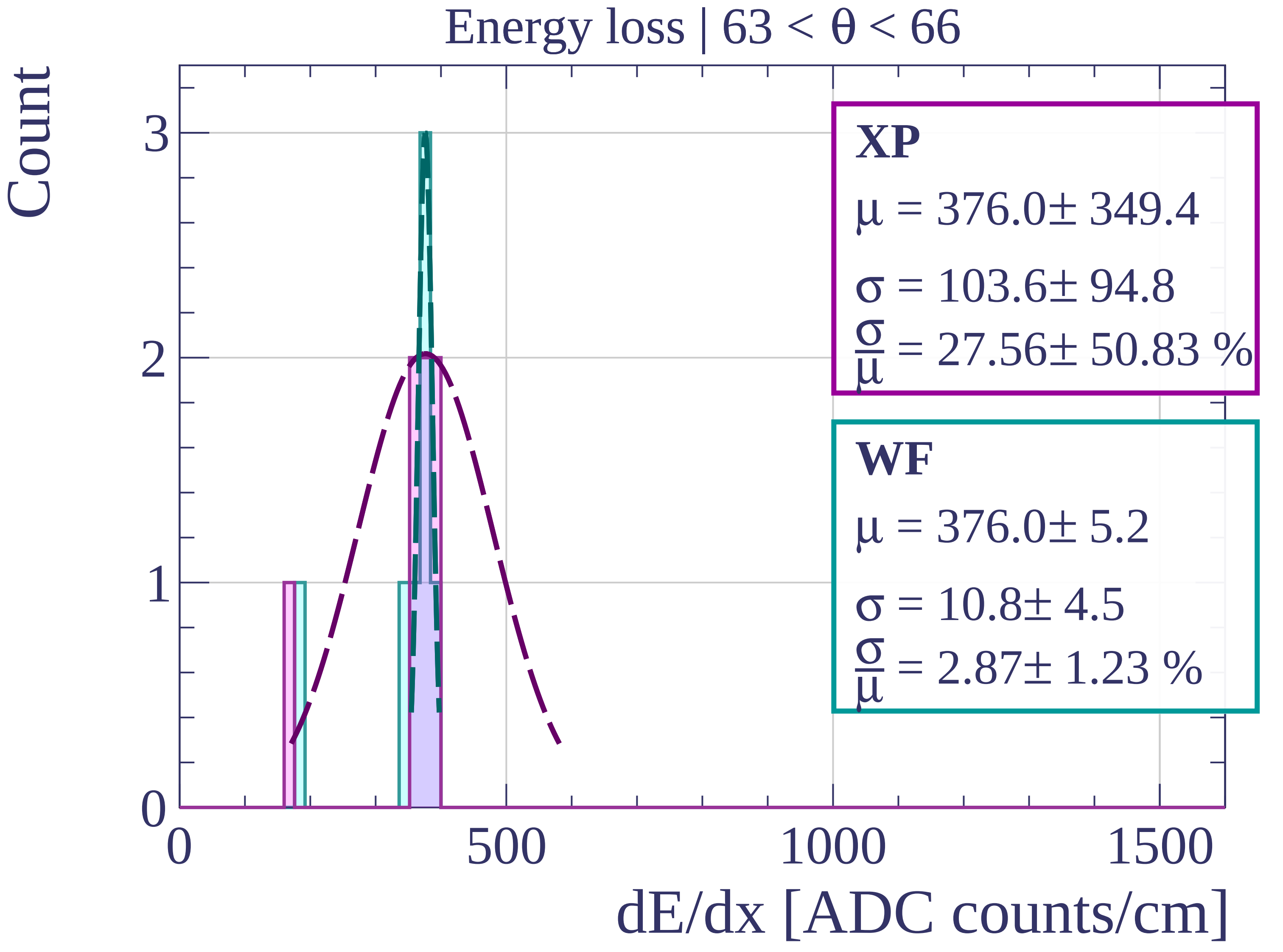
Energy loss | 54 < θ < 57
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 402.1± 6.0
σ = 25.6± 7.2
σ/μ = 6.36± 1.88 %
WF
μ = 418.9± 8.6
σ = 27.1± 8.2
σ/μ = 6.46± 2.08 %

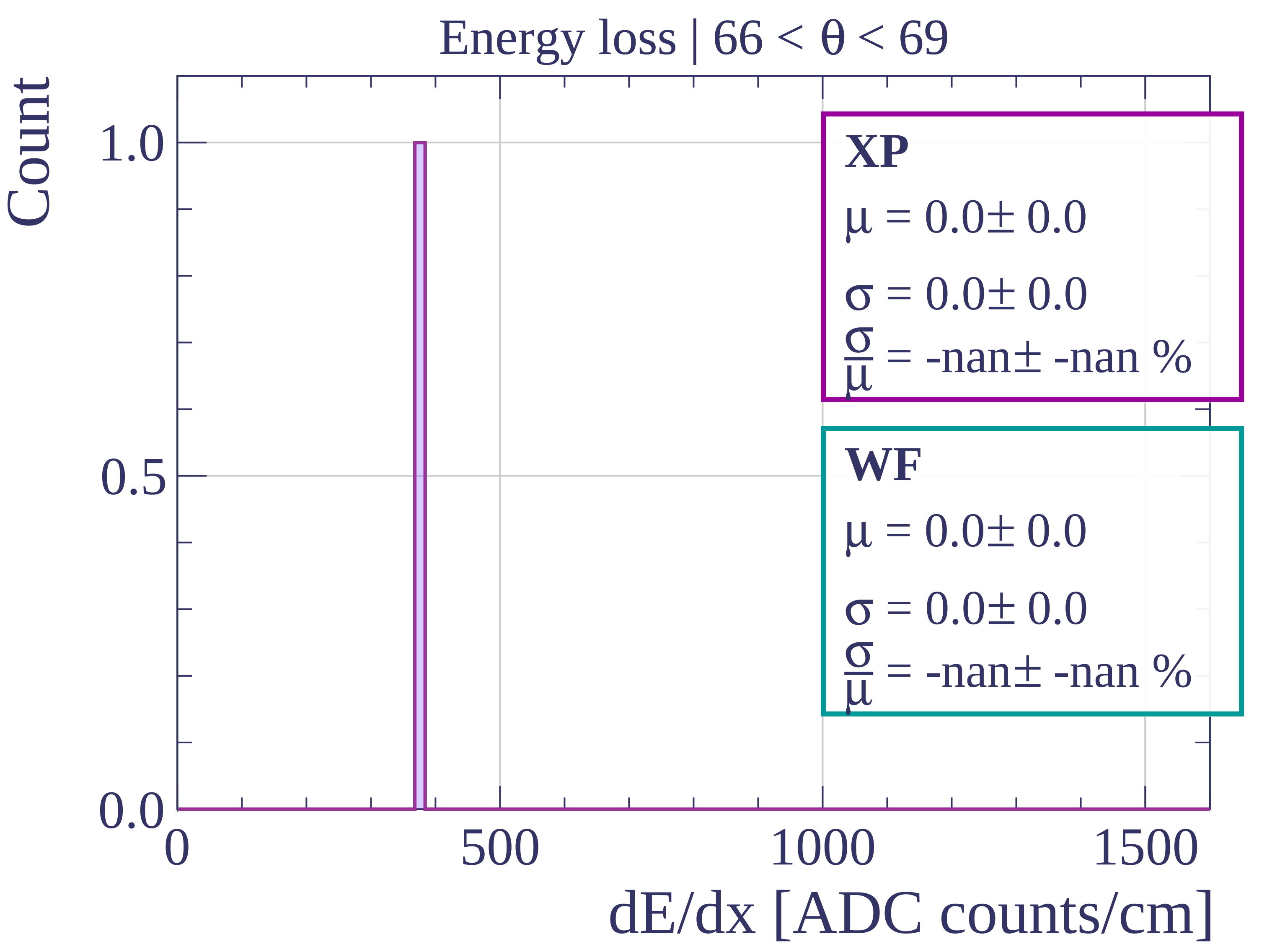


Energy loss | 57 < θ < 60
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
1
2
3
4
XP
μ = 391.9± 27.8
σ = 48.5± 41.7
σ/μ = 12.39± 11.52 %
WF
μ = 375.7± 1.7
σ = 0.3± 2.4
σ/μ = 0.07± 0.63 %

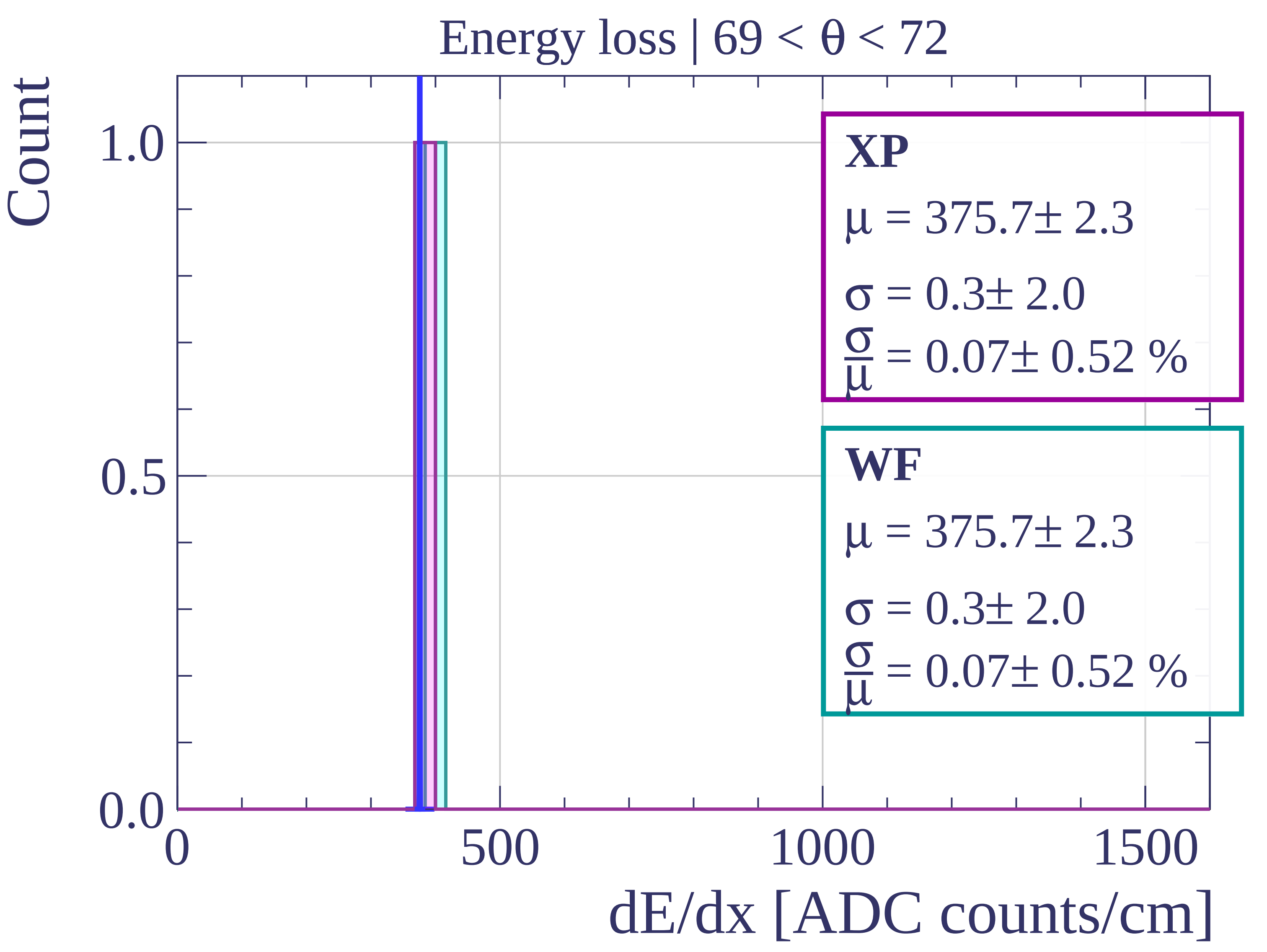


Energy loss | 60 < θ < 63
Count
0
1
2
3
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 392.5± 24.1
σ = 38.4± 41.8
σ/μ = 9.79± 11.24 %
WF
μ = 381.0± 44.9
σ = 56.9± 89.5
σ/μ = 14.94± 25.25 %

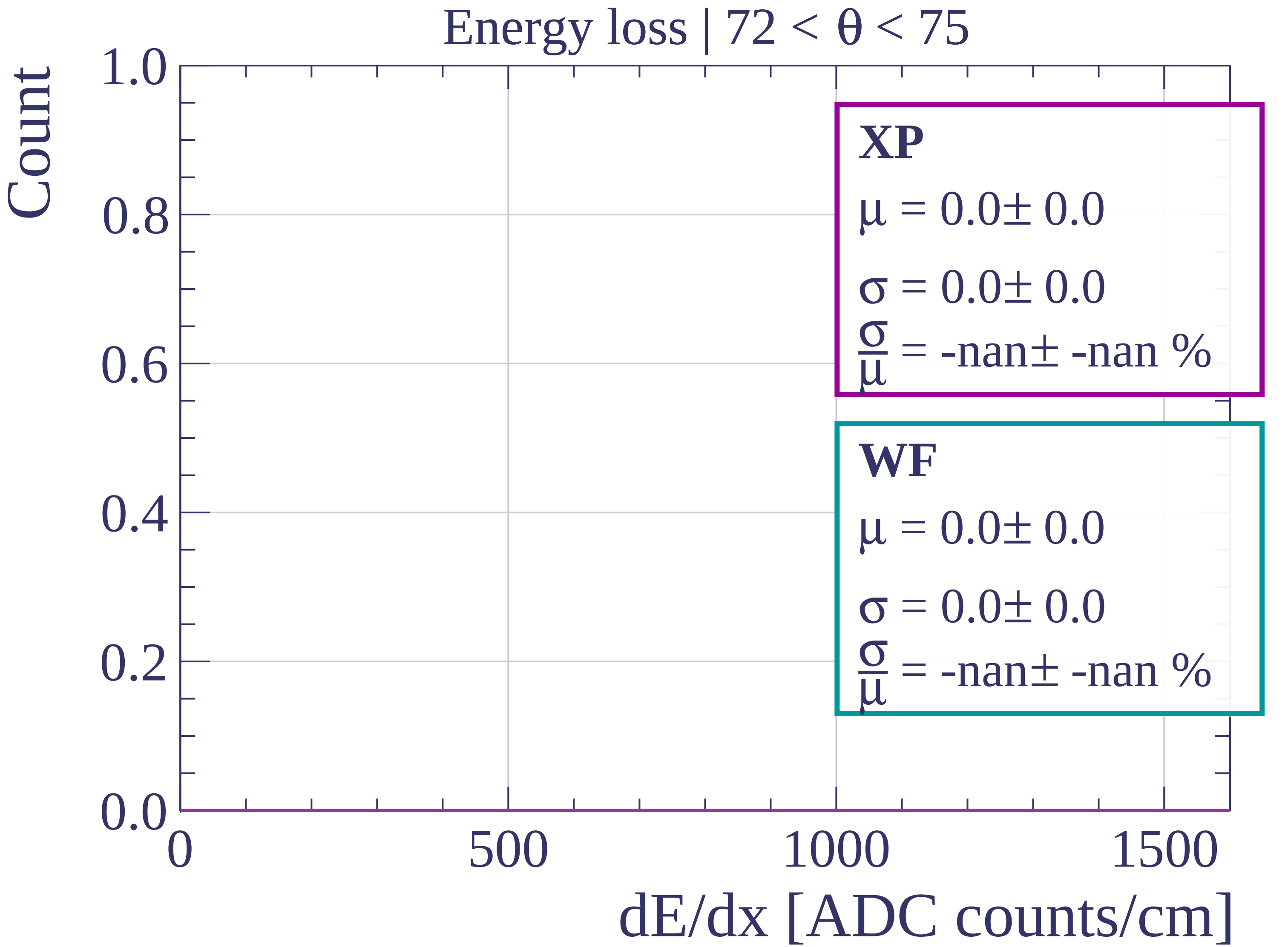

Energy loss | 63 < θ < 66
Count
dE/dx [ADC counts/cm]
0
1
2
3
0
500
1000
1500
XP
μ = 376.0± 349.4
σ = 103.6± 94.8
σ/μ = 27.56± 50.83 %
WF
μ = 376.0± 5.2
σ = 10.8± 4.5
σ/μ = 2.87± 1.23 %

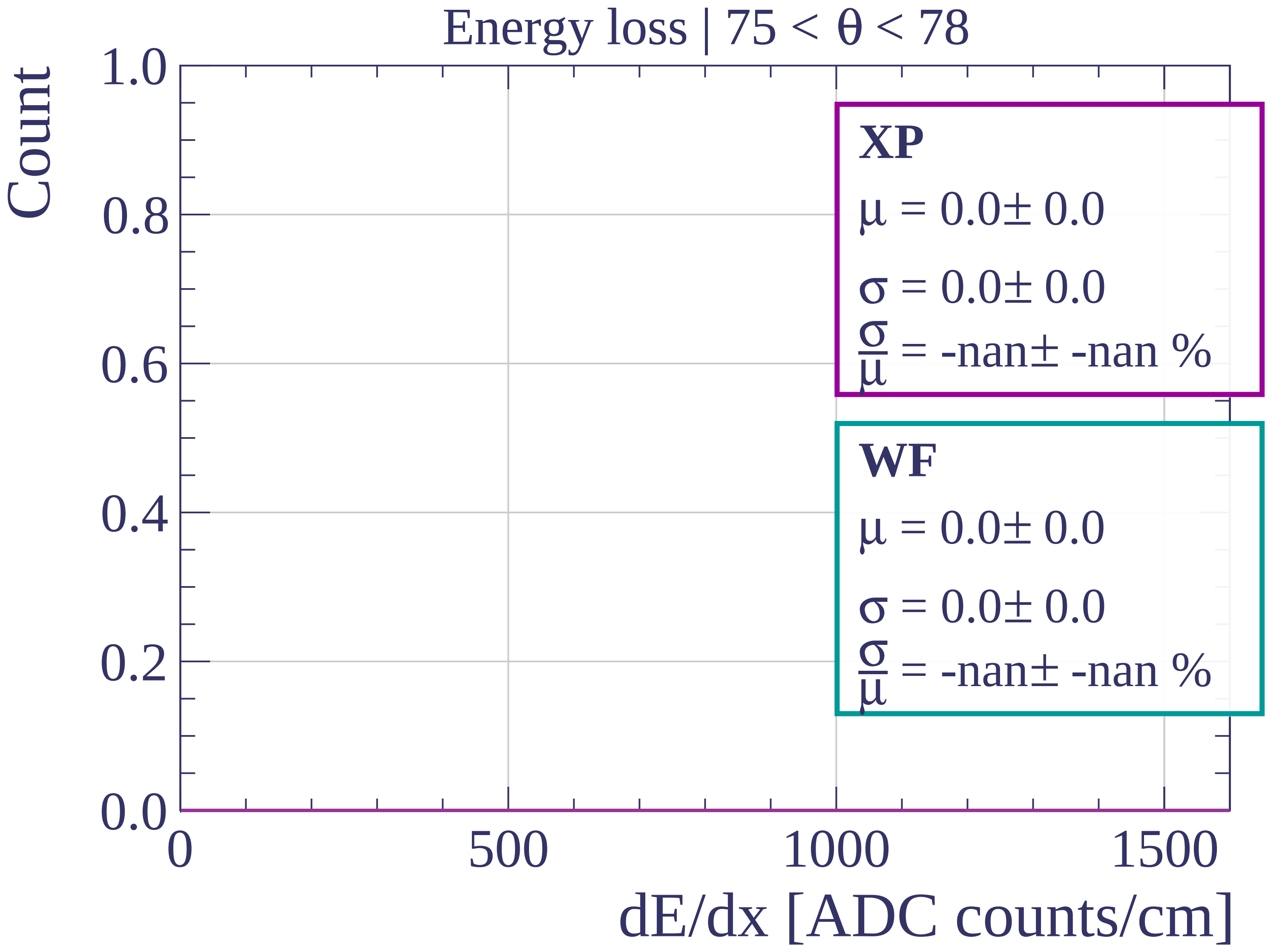


Energy loss | 66 < θ < 69
Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 0.0± 0.0
σ = 0.0± 0.0
σ/μ = -nan± -nan %
WF
μ = 0.0± 0.0
σ = 0.0± 0.0
σ/μ = -nan± -nan %

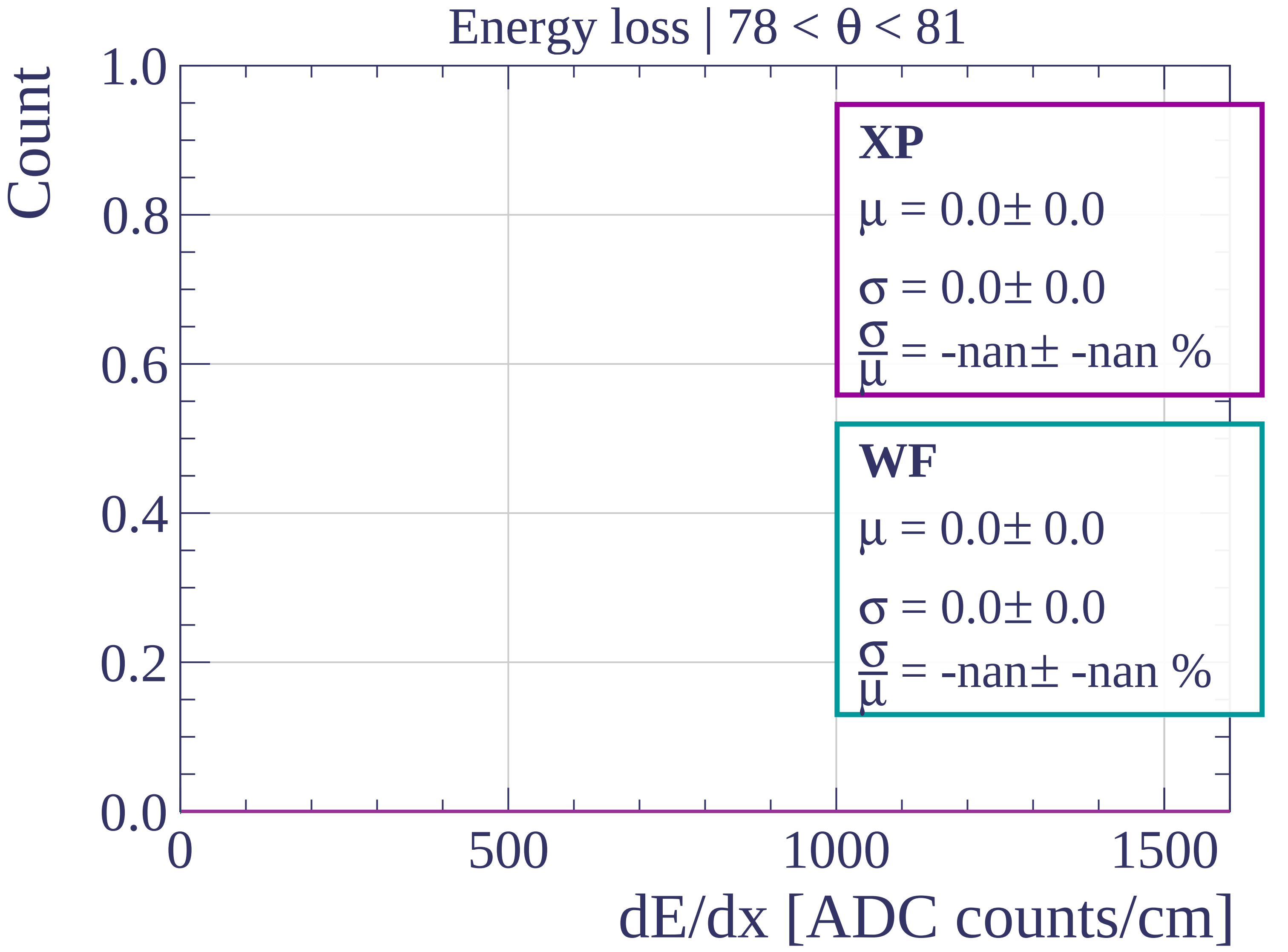


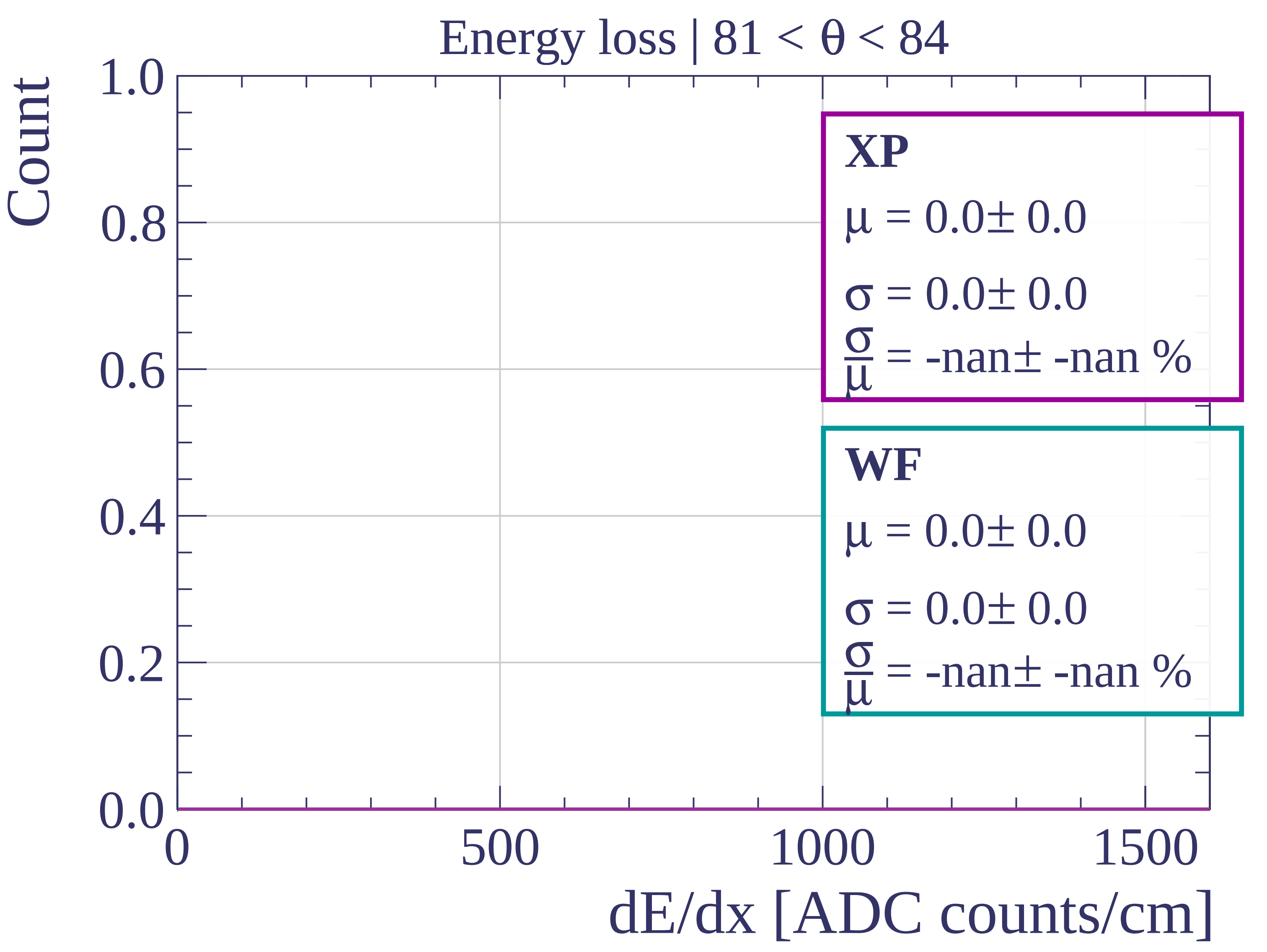
Energy loss | 69 < θ < 72
Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 375.7± 2.3
σ = 0.3± 2.0
σ/μ = 0.07± 0.52 %
WF
μ = 375.7± 2.3
σ = 0.3± 2.0
σ/μ = 0.07± 0.52 %


Energy loss | 72 < θ < 75
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 0.0± 0.0
σ = 0.0± 0.0
σ/μ = -nan± -nan %
WF
μ = 0.0± 0.0
σ = 0.0± 0.0
σ/μ = -nan± -nan %

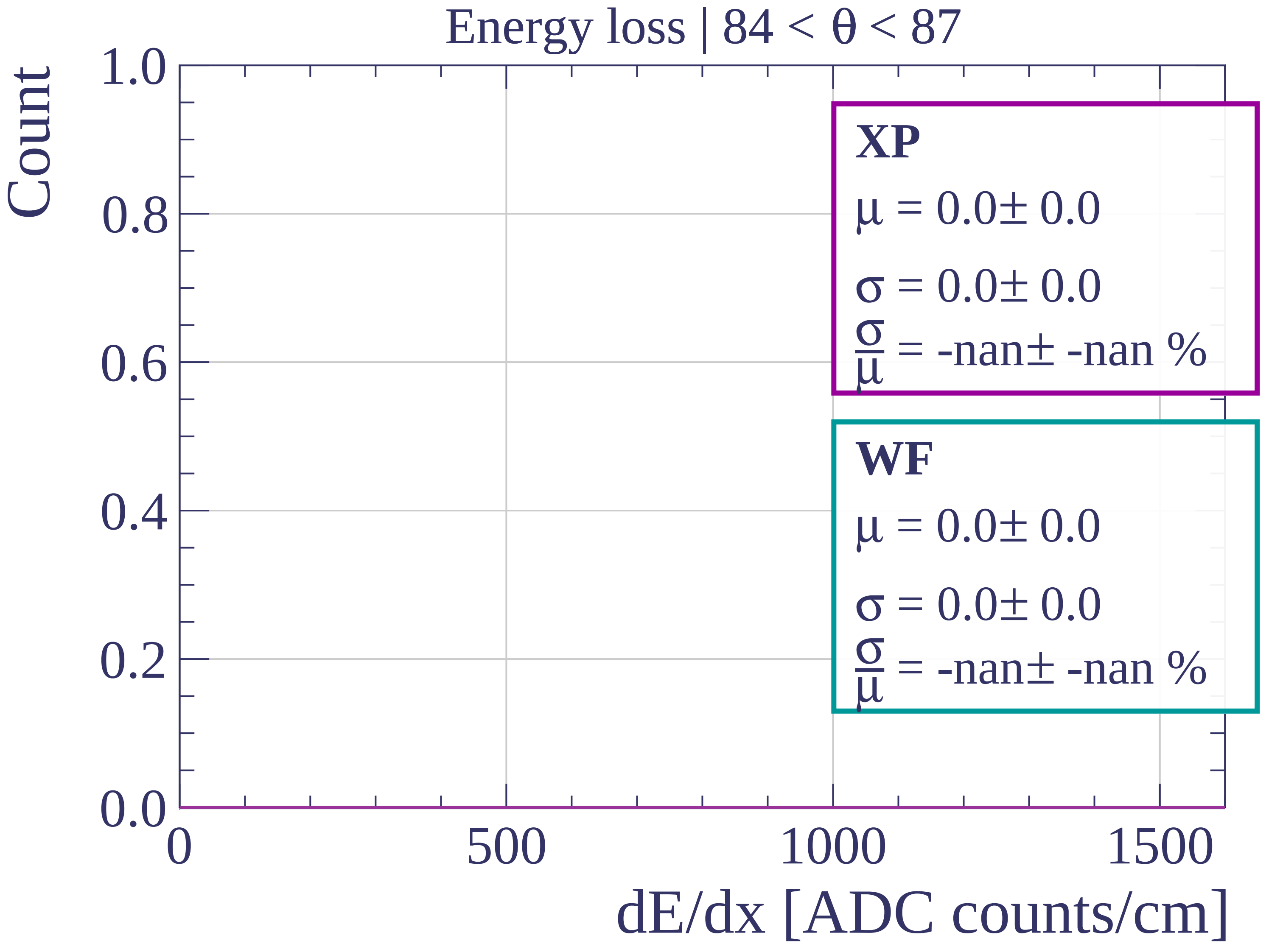


Energy loss | 75 < θ < 78
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 0.0± 0.0
σ = 0.0± 0.0
σ/μ = -nan± -nan %
WF
μ = 0.0± 0.0
σ = 0.0± 0.0
σ/μ = -nan± -nan %

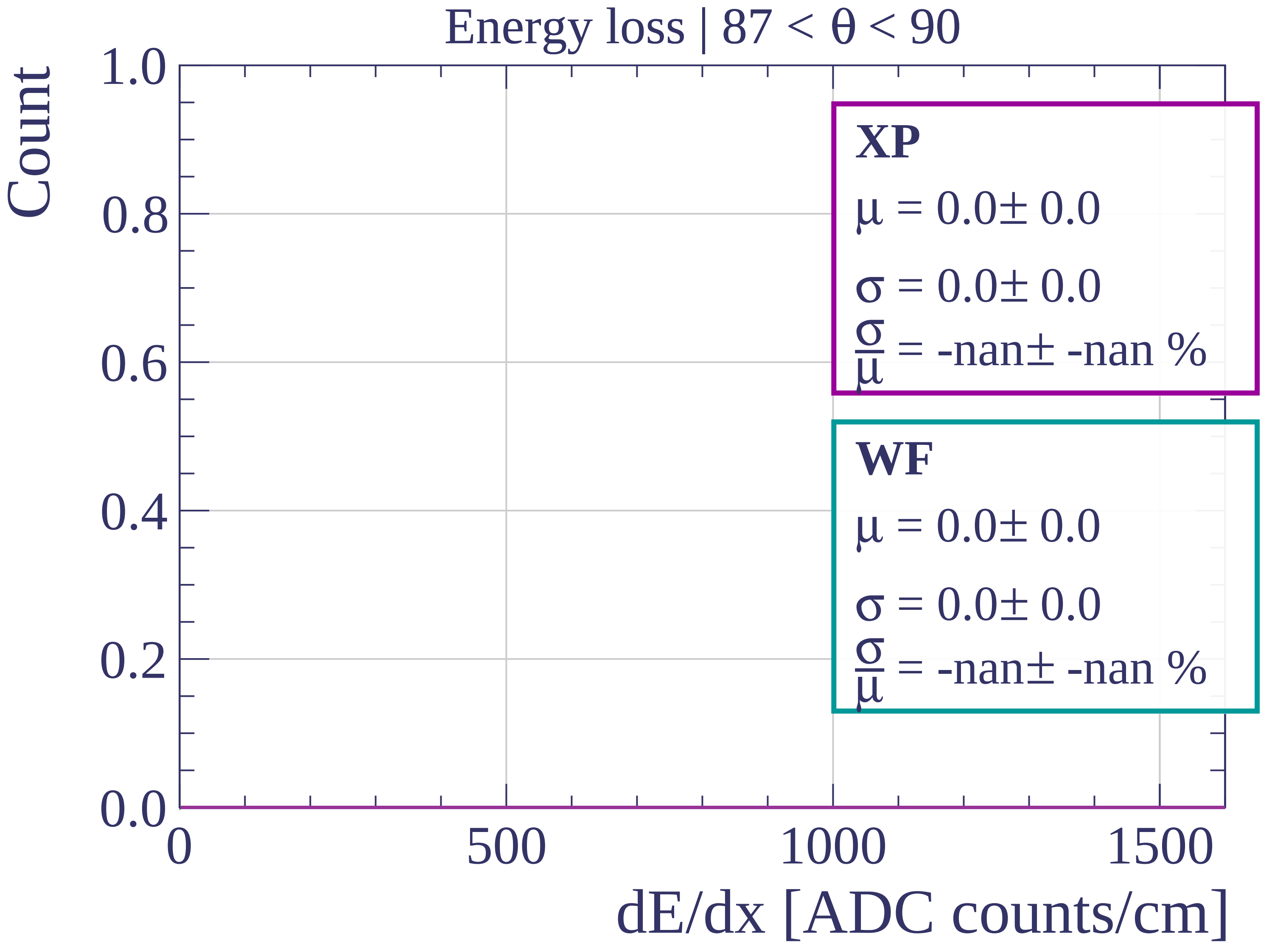

Energy loss | 78 < θ < 81
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 0.0± 0.0
σ = 0.0± 0.0
σ/μ = -nan± -nan %
WF
μ = 0.0± 0.0
σ = 0.0± 0.0
σ/μ = -nan± -nan %



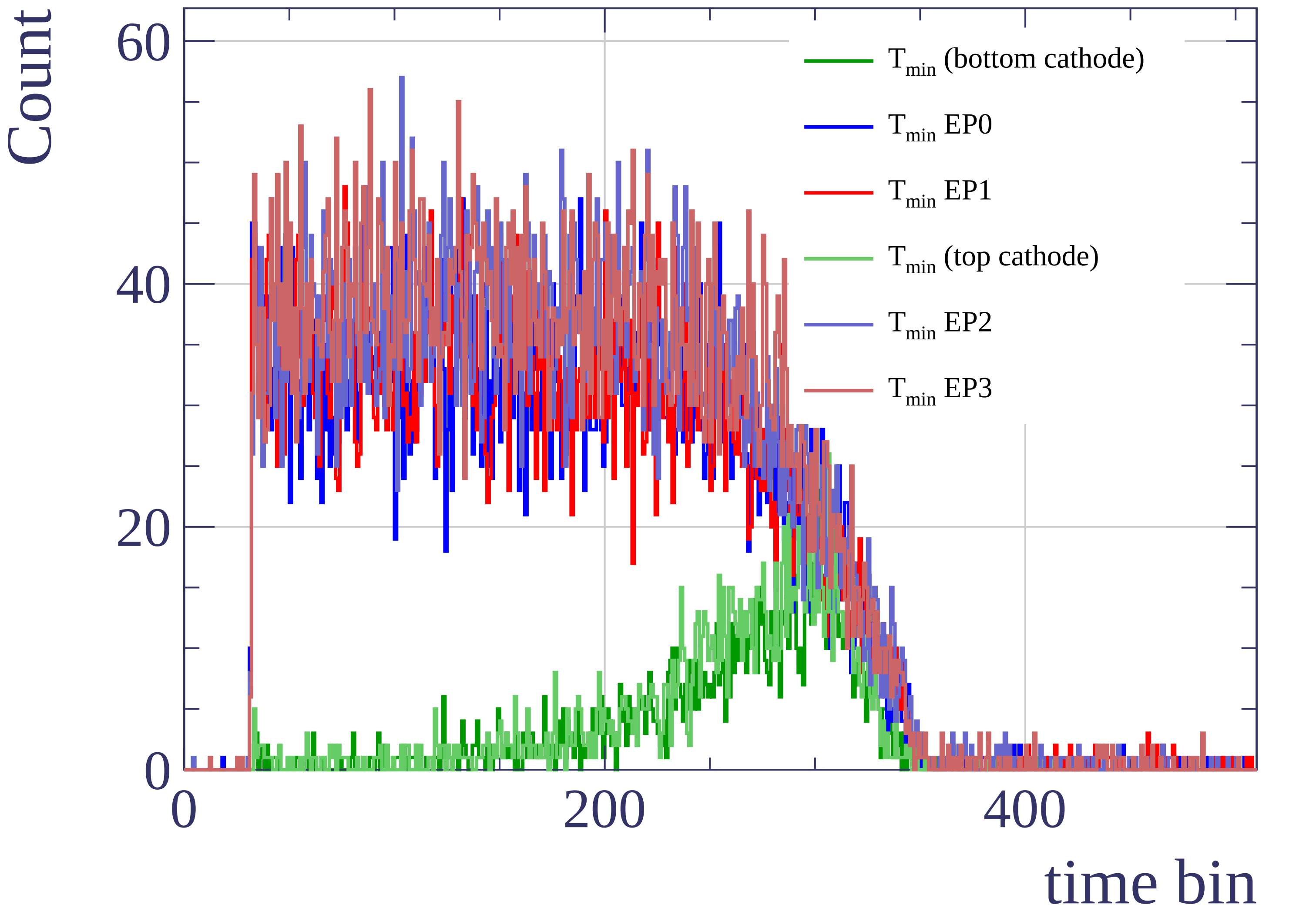
Energy loss | 81 < θ < 84
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 0.0± 0.0
σ = 0.0± 0.0
σ/μ = -nan± -nan %
WF
μ = 0.0± 0.0
σ = 0.0± 0.0
σ/μ = -nan± -nan %



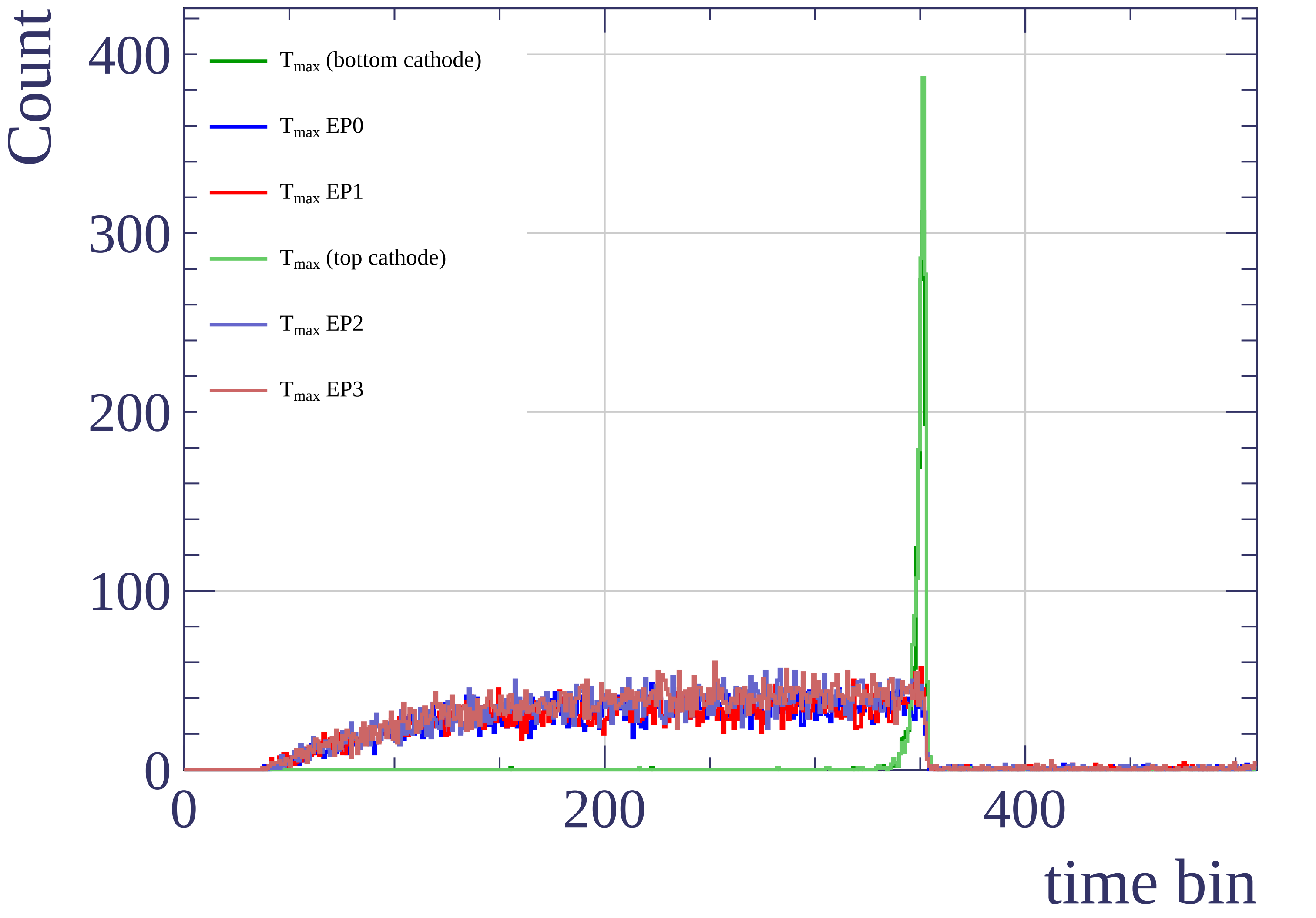
Energy loss | 84 < θ < 87
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 0.0± 0.0
σ = 0.0± 0.0
σ/μ = -nan± -nan %
WF
μ = 0.0± 0.0
σ = 0.0± 0.0
σ/μ = -nan± -nan %



Energy loss | 87 < θ < 90
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 0.0± 0.0
σ = 0.0± 0.0
σ/μ = -nan± -nan %
WF
μ = 0.0± 0.0
σ = 0.0± 0.0
σ/μ = -nan± -nan %



Count
60
40
20
0
0
200
400
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3



Count
400
300
200
100
0
0
200
400
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3